Bending — Plan

English: all seven planning documents (design, animation, looks, UI, audio, tech, build plan)

Hebrew: build plan (PLAN.he.md) and suggestions for Eilon (SUGGESTIONS.md)

1. **DESIGN**: Game design and balance tables
2. **ANIMATION**: Animation and effects bible
3. **LOOKS**: Character looks
4. **UI**: Screens and HUD
5. **AUDIO**: Sounds and music
6. **TECH**: Technical architecture
7. **PLAN**: Build plan with lane briefs

DESIGN.md — Game design and balance tables

Bending — Full Game Design

Bending (mod id `bending`) is a Minecraft mod in the world of *Avatar: The Last Airbender*, *The Legend of Korra* and *Avatar: Seven Havens*. A player bends one element — Earth, Air, Water or Fire — earns skill points by beating monsters with their bending, unlocks abilities on a skill tree, masters each ability by using it, picks a combat loadout, and wears the look of a bender from the shows. It is built on the foundations of **Enhanced** (`yuvalkolodkingal/Enhanced`): the same platform layer, the same hero plug-in shape, the same gesture and FX systems, the same HUD, config and test kit, adapted for elements instead of heroes (see `TECH.md` for exactly what is copied).

Original design: **Eilon Kolodkin-Gal**. Everything Eilon asked for is kept. Where his design leaves a gap (a number, a key, a Sneak variant, an edge case), it is filled in here and marked **(added)**; where something is only an idea for him to decide, it is marked **(suggestion)**. The Hebrew summary of every decision he has to make is `SUGGESTIONS.md`.

Conventions in this document:

- **1 heart = 2 damage.** Every damage number is in hearts.
- Ticks: 20 ticks = 1 second. Cooldowns are given in seconds (the config stores ticks).
- **Mastery** is 1 to 10. "M" in a formula is the ability's mastery level. "Element strength" (ES) is defined in 1.4.
- Every number below is a **server config default** (`world/serverconfig/bending-server.toml`, sections named in `TECH.md`). Nothing is hard-coded.
- Keys are the **defaults**; everything is rebindable under *Options → Controls → Bending*.

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0. Rules for every bender

Rule	Detail
One element per player	A player bends exactly one of Earth, Air, Water or Fire, or none. The Avatar (all four) is **not** in the design (see 8.1, a later idea).
Permanent	The element, the skill points, the unlocked abilities and every mastery level survive death, logout and dimension change (saved on the player, kept on death, `TECH.md` 4). Only an operator command removes them (`/bending element set <player> none`), or the optional change rule of 1.6.
Combat keys **(added)**	Four **loadout slots**, on **X, C, Left Alt, K** by default (1.5), each with a plain and a **Sneak** use: up to 8 abilities ready at once. Loadout, skill tree and looks live on one screen, **O**.
Sneak variants	Every ability has a Sneak variant (its own name, its own numbers, the same cooldown slot unless stated). The server reads the Sneak flag the client sent with the key, as Enhanced does (`AbilityKeyPayload.sneak`).
Hold, press, toggle	An ability is *press* (once per press, then its cooldown), *hold* (while the key is down; stops on key up, death, screen, hunger 0, Spectator), or *on/off*. Every hold ability is released by `stopHold` when the player dies or loses the element.

Rule	Detail
Hunger	Bending costs exhaustion, as Enhanced's powers do: a press costs `pressExhaustion` 0.6 food-exhaustion, a hold `holdExhaustionPerSecond` 0.8. At 0 hunger nothing but Rock Throw, Air Blast, Water Shot and Fireball work (the "free" starters).
Creative and Spectator	Creative benders have no cooldowns and no costs, but every protection and rights rule still holds. Spectators bend nothing.
Who an area attack may hurt	One rule for every wave, ring, burst and tornado: Enhanced's `AreaTargets.mayHurt` copied as `bending.ability.AreaTargets` — never the caster, their pets, their team, Creative or Spectator players, armour stands; other players only with PvP on; villagers, golems, animals and others' pets only with the element's `hurtsPassives` option (**false** by default).
What a bending power may change in the world	One rule book, `BendingRights` (Enhanced's `BlockRights` adapted, `TECH.md` 6): spawn protection, the world border, Adventure mode, claim mods (through the loader's block events), Enhanced's own walls if present, and the `bending:unbendable` tag. Every moved, raised or melted block is **temporary** and reverts (1.8).
Looks	Each element has its character looks (`LOOKS.md`), each a full skin drawn for the mod plus `ModelPart` layers (hair, robes, plates). A look changes only how you look and the colour of your element (Azula: blue fire). Every element also has *Your own skin*.
Languages	Everything is in English and Hebrew, the Hebrew in the dub's words: כשף אדמה / כשפות אדמה, כשף אוויר / כשפות אוויר, כשף מים / כשפות מים, כשף אש / כשפות אש, כשפות דם, כשפות ברק.

1. Core systems

1.1 Becoming a bender

Default (Eilon's open question 1): the Element Scroll.

- **Element Scroll** (item `bending:element_scroll`): crafted from 4 paper, 1 ink sac, 1 stick and **one item of each element** in the corners: a block of stone (Earth), a feather (Air), a water bucket (Water, returned) and a blaze powder (Fire). Cheap on purpose: Eilon wants everyone to be a bender early. **(added)**
- Right-click with no element: the **Element Choice screen** opens (`UI.md` 2.1): four cards with the element's symbol, colours, three sample abilities and the looks; a 3D preview of you in the element's first look. Choosing plays the **awakening**: a title ("You are an Earthbender" / "אתם כשפי אדמה"), the element's awakening sound, a burst of its particles, and a chat line to the server ("Eilon has become an Earthbender!"). The scroll is used up.
- With an element already: the scroll says so in the action bar and does nothing (the change rule is 1.6).
- Operators: `/bending element set <players> <earth|air|water|fire|none>`; `/bending points give <players> <n>`; `/bending mastery set <player> <ability> <1..10>`; `/bending unlock <player> <ability|all>`; `/bending look <players> <look>`.
- **(suggestion)** Alternative Eilon may prefer: a **spirit at first join** (a small glowing fox-spirit that appears once near a new player and offers the same screen). Not built unless chosen; the scroll stays available either way.

1.2 Skill points

Skill points (SP) are earned by defeating monsters **with your bending** and spent on the skill tree.

- A kill counts when the killing blow is a bending damage type (tag `bending:bending_damage`) **or** the monster was hit by your bending within the last **5 seconds** (`assistWindowTicks` 100) — so a rock that softens a zombie and a sword that finishes it still counts. A kill with no bending in it gives nothing.
- Points per monster (`[progression]`, every row configurable; a mob not listed uses the formula):

Monster	SP	Notes
Zombie, Skeleton, Spider, Creeper, Drowned, Husk, Stray, Cave Spider, Silverfish, Slime, Zombified Piglin, Pillager, Phantom	1	
Enderman, Blaze, Witch, Piglin Brute, Guardian, Vindicator, Magma Cube, Hoglin, Zoglin, Bogged, Breeze	2	
Evoker, Ravager, Shulker, Illusioner, Ghast	4	
Elder Guardian	10	
Warden	25	
Wither	40	
Ender Dragon	60	once per dragon
Players	0	never: no PvP farming
Passive animals, villagers, golems	0	
Any other hostile mob (other mods)	`round(maxHealth / 20)`, at least 1, at most 50	bosses of other mods by their health
(suggestion) The element bosses of 8.5	30 each	

- The **first 3 SP of each element's tree are free** (given with the element), so a new bender is never stuck with one ability.
- The HUD shows your SP; the skill tree shows every ability's cost.
- No cap on SP. A full tree (every ability unlocked, every mastery bought to 10) costs about 160 SP per element (table in 7.3), which is roughly a Wither and two hundred zombies.

1.3 Mastery

Every unlocked ability has a **mastery level 1 to 10**. It rises in two ways:

1. **Use.** Each *successful* use (a hit, a wall that stopped something, a heal that healed, a ride that moved you) gives **1 mastery XP**; an unsuccessful one (a rock thrown into the void) gives 0, so spamming into the air teaches nothing. Levels need cumulative XP: 2: 10, 3: 25, 4: 50, 5: 90, 6: 150, 7: 230, 8: 340, 9: 480, 10: 660 (`masteryXp` list). A training dummy (8.2) gives XP at half rate.
2. **Points.** On the skill tree, one level of mastery costs **SP = the level you are buying, halved and rounded up** (level 2: 1 SP ... level 10: 5 SP), bought one level at a time. Buying is capped at **level 7**: levels 8 to 10 come only from use. ("The more skilled you are, the more earth you lift": the last levels are earned.)

What mastery scales is listed per ability (section 2 to 5); in general: **size** (rocks, walls, tornadoes, whips), **count** (rocks and knives at once), **range**, **duration**, and a little **damage** (never more than +50% at 10), and some abilities gain a **new form** at a threshold (fire breath at 7, dual-hand lightning at 10, the small fireball option at 6).

1.4 Comparing benders: element strength (Eilon's "one rule")

Element strength (ES) of a player = **the sum of the mastery levels of every unlocked ability of their element** (an unlocked ability counts at least 1). A fresh earthbender has ES 1; a full tree has ES 100. The HUD shows it next to the SP.

Used by: **Quicksand** (fails on an earthbender whose ES ≥ yours), **Blood Bending** (fails on a waterbender whose ES ≥ yours), **Bending Severance** (duration by the target's ES, 4.5), and the **(suggestion)** duel ranks of 8.6. A tie always favours the defender. ES is also what the full-moon-free rule of blood bending reads (4.5: "fully mastered" = Blood Bending at mastery 10).

1.5 Combat loadout and keys

- **Four slots**, default keys **X**, **C**, **Left Alt**, **K**; each slot holds one ability, whose Sneak variant comes with it on **Sneak + key**. Eight abilities ready at once; the rest wait on the skill tree and are swapped in on the **Loadout tab** (`UI.md` 2.3), also by dragging in the inventory-style grid. Swapping has no cost and no cooldown, but a swapped-out ability keeps its running cooldown.
- **Why these keys (added).** Enhanced owns **R, G, V, B, N, H, Y, Z** (its five ability keys, Change Look, Build Magic, the Vision order) and **J** (Stone Wheel / Web Assist); vanilla owns W A S D, E, Q, F, T, L, P, Tab, 1–9, Shift, Ctrl, Space. Left of the keyboard that leaves **X** and **C** (vanilla's Creative toolbar save/load, which almost nobody uses; Enhanced's Build Magic uses them only with **Ctrl** held and only in build mode, so there is no clash), **Left Alt** (Enhanced's "second knob" uses it only inside build mode), and no fourth: **K** is the nearest free letter. Eilon chooses (SUGGESTIONS 3); alternatives are **I** for slot 4, or three slots only.
- **O** opens the **Bending screen** (tabs: Skill Tree, Loadout, Looks, and the Water Skin's level for waterbenders). **Sneak + O** cycles the look at once (like Enhanced's H). No other keys: the glider, the water skin and the scroll are items used with right-click.
- Pressing a slot key with nothing in it says "Empty slot: press O" once in the action bar.
- A held slot key shows the element's **ready stance** (`ANIMATION.md` 9) for as long as it is held with a *press* ability that is on cooldown; this is how Eilon's "idle stance while a bending key is held" is reached without a separate key.

1.6 Changing element (Eilon's open question 7)

Default: no. The element is for keeps, like an Enhanced power; an operator can reset it. **(suggestion)** If Eilon wants a way: **Spirit Water** (a rare item from the Spirit Oasis structure of 8.4, or 1 per world from the Water boss), used on yourself, removes the element and **every point and level with it**, so changing is a real restart; a server option `allowElementChange` (false) turns the item on. The scroll then works again.

1.7 Element matchups

Eilon's two rules, then the **(added)** rest, every factor in config `[matchups]`:

Attacker's element → defender / effect	Rule
Water → Fire	Any water ability that hits a **firebender** deals **×1.5**; water that touches a fire ability's effect **puts it out** (a Fire Whip or Flamethrower stream crossed by a Water Shot ends early; burning targets stop burning; a fireball hit by a water shot bursts harmlessly). Water Shot and Water Whip never miss a Fire Circle: they cut a gap in it.

Attacker's element → defender / effect	Rule
Fire → Ice	Any fire ability deals **×1.5** to a target that is frozen (Freeze), and **melts** ice walls, ice knives in flight and frozen water on contact (an Ice Wall takes 1 fireball per 4 blocks). Water (the liquid abilities) is *not* weak to fire.
Earth → Air (added)	A rock in flight passes through an Air Blast and an Air Shield absorbs rocks only up to its strength (2.3); Earth Wall and Steel Wall stop Air Blades and Tornadoes completely.
Air → Fire (added)	An Air Blast or Tornado pushes a Fireball off course (it keeps flying in the new direction) and snuffs a Fire Circle within its radius; a firebender standing in a Tornado cannot start a hold ability.
Lightning → Water (added)	Lightning deals **×1.5** to a target that is in water, wet (rain, a Water Shot in the last 3 s) or a waterbender with an octopus or arms up; a Water Spout that lightning strikes collapses.
Lightning → Metal (added)	Lightning deals **×1.5** to Steel Armour and Iron Golems; **Earth Armour** (stone) takes **×0.5**.
Earth ↔ Earth	Rocks shatter on Earth Wall; a Rock Throw at an Earth Armour wearer deals ×0.5.
Fire → Fire	Firebenders take ×0.5 from fire (the passive 5.0).

1.8 Temporary blocks, lag and griefing (every element)

Block-moving abilities never make permanent changes to the world:

- **Every block an ability raises, moves, melts or freezes is a *temporary block* (`TempBlocks`, `TECH.md` 6):** the original state is recorded, the new one placed, and a revert is scheduled. On revert the original comes back unless a player stands inside it (then the revert waits up to 10 s and finally pushes the player up). Containers, block entities, bedrock, portals, spawners, reinforced deepslate and the tag `bending:unbendable` are never touched; a wall grows around them.
- **Rights.** Before any cell changes, `BendingRights.mayChange(player, level, pos)` asks: spawn protection, the world border, Adventure/Spectator, the loader's block-place and block-break events (claim mods answer there), Enhanced's Hex and boss arenas if Enhanced is loaded (optional hook), and the dimension rules of 1.9. A refused cell is skipped; if more than half of a wall's cells are refused the wall is refused (the action bar says "Protected ground").
- **Budgets (per player, `[limits]`):** at most **1 wall**, **1 pillar**, **1 lava path**, **3 ice walls**, **40 temporary blocks** standing at once (a bigger wall replaces the older one); server-wide **2,000** temporary blocks (a new one beyond that reverts the oldest). Projectile entities: at most **6 rocks**, **8 knives**, **2 tornadoes**, **1 spout**, **1 scooter** per player. A fight of **4 benders** therefore never exceeds ~160 temp blocks and ~60 entities; every entity is `MobCategory.MISC`, `noSave()` where it may be, and discarded on logout.
- **Particles** are client-side with budgets per ability (`ANIMATION.md`, each ability's Effects row; the client option *Effect density* 0.25–1.0 scales them, and nothing is spawned beyond 48 blocks).
- **Crash safety.** Temporary blocks are journaled in the level's `bending:temp_blocks` attachment; on server start, every entry still standing is reverted (the lesson of Enhanced's `ArenaLoans`).
- **Griefing.** Because nothing is permanent, the only lasting change bending can make is damage to entities and the fires lit by fire abilities (**only** with gamerule `mobGriefing` on and `[fire] setsFires` true, on blocks the rights allow, at most 12 fire blocks per ability use, never in a claim).

1.9 Dimensions, water, lava and other mods' blocks

Where	What changes
Water (the bender is in it)	Earth: no rocks (no earth under you unless you stand on the floor); Earth Armour crumbles when fully submerged; Earth Pillar works from the floor. Air: Air Bubble is the point; Air Blast works at half range; Scooter becomes a water-skimming ride at the surface. Water: everything works and the water supply is infinite. Fire: Fireball, Flamethrower and Fire Whip **do not work under water**; Fire Punch works at ×0.5; Lightning works and hurts the bender too (1 heart) unless they are an Azula/Ozai look... no: hurts everyone equally (added for fairness).
Lava	Earth: Lava Bending draws from real lava too (a Lava Path next to lava is free of cost); the lava resistance passive (2.7) applies. Water: a Water Shot into lava makes obsidian/cobble as vanilla does, temporary; healing works. Fire: full lava immunity only for Ozai at mastery 10 (**no**: no look changes rules; drop) — fire passive applies to lava damage ×0.5. Air: nothing special.
The Nether	Water: **no water source** except the Water Skin (it evaporates ×2 faster: 1 charge per 20 s) and the Nether's plants (warped fungus, nether wart, at high mastery); blood bending's full moon is **never** (no moon): only at mastery 10. Fire: all fire abilities deal **×1.25** (`netherFireBonus`). Earth: netherrack, soul sand, basalt, blackstone and magma are earth; soul sand is "sand" for sand bending; Lava Path is free beside lava. Air: nothing special; Spiritual Projection cannot cross dimensions.
The End	Earth: end stone is earth (no sand, no metal unless carried). Air: the void is the void: Scooter and Glider do not save you below y=0; Spiritual Projection may not fly below y=0. Water: the Skin only; the moon rule reads the Overworld's moon phase through the server clock (a full moon in the Overworld counts). Lightning: ×1.0.
Rain and thunderstorms	Waterbenders: rain is a water source (any ability, half cost). Lightning: ×1.25 in a thunderstorm; Lightning Redirection may redirect vanilla lightning. Fire: Fireball and Flamethrower deal ×0.8 in rain.
Other mods' blocks	A block is earth if it is in `bending:earth` (the mod's tag: stone, dirt, gravel, deepslate, clay, terracotta, mud, netherrack, basalt, blackstone, end stone, tuff, calcite, dripstone, cobblestone, all `minecraft:base_stone_*`, `c:stones`, `c:gravels`, `c:dirt`...), sand if in `bending:sand` (sand, red sand, soul sand, soul soil, suspicious sand, `c:sands`), metal if in `bending:metal` (iron/gold/copper/netherite blocks, anvils, chains, iron bars, rails, hoppers, cauldrons, lanterns, ores of iron/gold/copper, raw blocks, `c:ores`, `c:storage_blocks/*metal*`), a plant if in `bending:plants`. Other mods add to the tags with data packs; anything not tagged is not bent.
Enhanced present (optional)	Bending checks Enhanced's `Shelters`/`BossFights` through a soft hook if the class is present (`TECH.md` 9): no bending into a Hex that keeps you out, nothing bent in a boss arena. An Enhanced hero can also be a bender (two mods, two keysets); the design does not forbid it, a server option `allowWithEnhancedPower` (true) can.

1.10 PvP, mobs and bosses in general

- **PvP off**: no bending damage to players, no quicksand, no blood bending, no suffocation on players; pushes (Air Blast) still push (they are not damage) unless `pushPlayersWithoutPvp` is false.
- **Bosses** (`minecraft:ender_dragon`, `wither`, `warden`, and the tag `bending:bosses`, which Enhanced's bosses join through a data pack): never lifted, never frozen in place, never blood-bent, never quicksanded, never suffocated; pushes are ×0.2; paralysis from lightning is 0.5 s; damage is full.
- **Iron golems, Enhanced's heroes in metal, armour-wearing mobs** count as metal for Metal Grip and Lightning ×1.5.

- **Players riding or in vehicles** are dismounted by a successful Air Blast or Water Spout push.

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2. Earth — כשפות אדמה (release 1.0, the first element)

Theme. Hung Gar: rooted, low, patient, then explosive. Earthbending is **neutral jing**: wait and listen, then strike. Colour: ochre and moss green `#7A8F3A` / `#C9B47A`, Toph's green and gold.

Earth source. Every Earth ability except Earth Armour's shed and the metal abilities needs **earth within reach**: a block in `bending:earth` (or `bending:sand` for sand abilities, `bending:metal` for metal) within `earthReach` 6 blocks of the bender's feet or of the aimed point. Standing on a wooden floor over stone is fine (the stone is within 6); flying high is not. The HUD dims Earth icons when no earth is in reach.

Passive (added, free with the element): Rooted. While standing on earth and not moving, knockback taken is ×0.5.

2.1 Rock Throw — זריקת סלע (free; the starter)

Key	Slot ability; **press**. Sneak: **Hold Rock** (אחיזת סלע)
Unlock	Free with the element
What	A rock tears out of the nearest earth in front of you (the ground under your aim, else under your feet) and flies where you look. It hurts the first thing it hits and shatters into gravel-coloured dust; the hole it left fills back in after `rockHoleRevert` 3 s (temporary block).
Damage	**2 hearts** +0.2 per mastery level above 1 (3.8 at 10). Knockback 1.0. A rock that hits a block breaks nothing.
Mastery	**Count and size.** M1–3: one 1-block rock. M4–6: **two** rocks (both at once, a half-tick apart, slightly spread). M7–9: **three**. M10: one **boulder** (2×2×2 blocks, 6 hearts, knockback 2.0, 60-tick cooldown) **or** three rocks: Sneak-free choice by tapping twice within 10 ticks (a double tap gives the boulder). Range 24 + M. Speed 1.6 blocks/tick.
Cooldown	1.5 s (30 ticks); boulder 3 s
Sneak: Hold Rock	Lift one rock and hold it in the air where you look (like Enhanced's telekinesis: `Telekinesis.hold`), move it with your look, press again to throw it (same damage), Sneak + press to set it down gently (it becomes a block where `BendingRights` allows, else drops as its item... no: a temporary rock never drops; it crumbles back to where it came from). Holding costs 0.1 exhaustion/s. From M5 you hold **two** rocks, from M8 **three**, orbiting you.
Counters	An Earth Wall or Steel Wall stops it; an Air Shield absorbs it (eating 1/3 of its strength per rock); Water Octopus arms slap one rock per 2 s; a Fire Whip at M7+ cuts a rock in half (1 heart). Blocked by any shield of Enhanced the same way as a thrown block.
Edge cases	No earth in reach: "No earth here" in the action bar, no cooldown (a move with nothing to act on starts no cooldown, so no gesture lies). Under water: no. In a claim: the rock is lifted only from cells you may change; otherwise refused. Mobs: full damage. Bosses: full damage, knockback ×0.2. A rock never breaks glass or crops (it is an entity that stops at a block).

2.2 Earth Armour — שריון אדמה

Key	Slot ability; **on/off** (press forms it, press again sheds it). Sneak: **Shed Blast** (ניפוץ השריון)
Unlock	4 SP (needs Rock Throw M2)

What	The earth within 3 blocks rises and wraps your whole body, leaving a slit for the eyes (the look: `LOOKS.md` 1.1, the first-person slit overlay: `ANIMATION.md` 2.2). It is a shell with its **own health**; blows hit the shell first.
Defence	Shell health **12 hearts** +1.5 per M (25.5 at 10); every hit on you is taken by the shell at ×1 until it breaks; while whole it also gives **+8 armour**, **+4 toughness**, knockback resistance 0.6, fire damage ×0.5 and lightning ×0.5 (1.7). It does not regenerate; shed and re-form to refresh.
Downside	Movement speed **×0.7** (×0.8 from M6, ×0.85 at M10); no sprinting at M1–3; you sink in water (and the shell crumbles when your head is under water); no Scooter-like speed.
Cost	8 earth blocks within 3 (they are *not* removed: the armour is drawn from the ground's "spirit", so nothing is bent out of the world; **(added)** keeps claims and lag simple). Forming takes **20 ticks** (you may not move while it forms; crumbling is instant).
Duration	Until shed, broken or the player dies. `armourMaxSeconds` 0 = unlimited (a server may set a limit).
Cooldown	20 s after it breaks or is shed
Mastery	Shell health (above); speed penalty eases; at M8 the slit's edges glow faintly and **Seismic Sense** (added): while standing still in armour, hostile mobs within 16 glow through walls for you only. At M10 the armour **re-forms itself once** after breaking if earth is in reach (another 20 s cooldown after that).
Sneak: Shed Blast	Burst the shell outward: a ring of stone shards to 4 blocks, **1.5 hearts**, knockback 1.5 (`AreaTargets`), and the armour is gone (cooldown as shed).
Counters	Lightning ×0.5 (good), Fire ×0.5, **Water Shot ×1.5 to the shell** (added: water erodes stone) and a frozen shell (Freeze) halves its armour until thawed; Metal Grip cannot grab stone. Air Blast pushes a wearer ×0.5 (rooted).
Edge cases	Forming while flying (Creative flight): refused. Under water: crumbles. Nether: netherrack counts. Bosses: the Warden's sonic boom ignores the shell (bypasses armour: `bypasses_armor` damage goes to you). PvP off: still protects from mobs. Other players see the full stone body; the wearer's own skin is hidden under it.

2.3 Earth Pillar — עמוד אדמה

Key	Slot; **press**. Sneak: **Earth Launch** (זינוק אדמה)
Unlock	2 SP
What	The ground under your feet rises as a 1×1 column of the earth you stood on (temporary), lifting you with it. Press again while on top to lower it.
Height	**3 blocks** +0.5 per M (up to 7.5 → 8 at M10); rise speed 0.5 block/tick (the player is carried smoothly, `ANIMATION.md` 2.3).
Duration	The pillar stands **15 s** (`pillarSeconds`), then sinks back at the same speed, carrying you down. If you step off it stays its full time (a bridge for a friend).
Cooldown	3 s
Mastery	Height (above). M5: you may raise a pillar **under an aimed spot** within 12 blocks (an enemy standing there is lifted and takes 1 heart; a friend is lifted for free). M8: the pillar is **2×2**. M10: two pillars per use (one under you, one aimed).
Sneak: Earth Launch	The pillar fires you up like a catapult: a 2-block stub rises in 4 ticks and you are thrown upward with speed **1.2** +0.08 per M (about 8 blocks at M1, 14 at M10), no fall damage for the next landing on earth (the ground catches you with a soft dip). Cooldown 4 s.
Counters	A pillar is stone: Rock Throw breaks one block of it (the rider falls); fire does nothing; water erodes it (a Water Shot takes 5 s off its life).

Edge cases	Not standing on earth: refused. Something above you (a ceiling within height): the pillar stops under it, never crushes you. Another player standing on the same block: both ride. A mob: rides and is not hurt (except the aimed pillar). Under water: works from the floor and lifts you out. Claim: refused if the top cells are protected. Nether roof: stops at y=127.

2.4 Earth Wall — קיר אדמה (the shield)

Key	Slot; **press**. Sneak: **Wall Push** (דחיפת קיר)
Unlock	3 SP
What	A wall of the ground rises in **4 ticks** 3 blocks in front of you, facing you (snapped to the nearest of 8 directions), of the blocks under it (temporary). It is real blocks: arrows, rocks, fireballs, Air Blades and mobs stop at it.
Size	**3 wide × 2 high** at M1; +1 wide per 2 M, +1 high per 3 M: 5×3 at M5, 7×4 at M10. Thickness 1 (2 at M10).
Duration	**10 s** (Eilon), `wallSeconds` 200 ticks; then it sinks back.
Cooldown	8 s (the HUD shows it; a new wall while one stands replaces it).
Mastery	Size (above). M6: the wall takes two rock hits per block instead of one. M10: thickness 2 and a parapet (the top row has gaps to shoot through).
Sneak: Wall Push	If your wall stands: it **slides forward** 6 blocks over 12 ticks (the temp blocks move one cell at a time), pushing creatures and players ahead of it (**1.5 hearts**, knockback 1.5, `AreaTargets`), then crumbles. If no wall stands: raises one and pushes at once (one cooldown of 10 s).
Counters	Rock Throw breaks one wall block per rock; a boulder breaks 4. Fire does nothing. Water Shot erodes (5 s off). Air Blast nothing. Lightning nothing. Metal Knives stick in it. A Lava Shuriken melts one block. Enhanced: a Chaos Beam breaks a block per second (through Enhanced's block-break event, which `TempBlocks` listens to: a broken temp block simply reverts early).
Edge cases	No earth under the wall's footprint: the wall is as wide as the earth found (at least 1 block, else refused). Blocks the wall would replace (grass, flowers, snow layers, water) are recorded and restored. A player or mob **inside** the footprint is pushed to the near side first (never suffocated). Claims: cells refused are left as gaps; more than half refused: no wall. Under water: a wall of the floor rises through the water (it is earth). Bosses: the Wither's skulls break it; the Dragon flies over.

2.5 Sand Bending — כשפות חול

Key	Slot; **press**: **Sand Stream** (זרם חול). Sneak: **Sand Wall** (קיר חול)
Unlock	3 SP (needs Rock Throw M3 and Earth Wall)
Needs	`bending:sand` within reach (a desert, a beach, soul sand in the Nether, or **a stack of sand in your inventory**: 4 sand per use, returned... no: carried sand is used up: 4 per use **(added)**).
Sand Stream	Lift sand into the air and fling it in a cone: 12 + 2M pellets (entities pooled: one `SandStreamEntity` that hurts in a cone, drawn as many grains) to 10 blocks, **1 heart** total per target + **Blindness 2 s** (3 s at M6, 4 s at M10) + Slowness I 2 s. Cooldown 2.5 s. Mastery: more grains, longer cone (10 → 16), wider (20° → 35°).
Sand Wall	Like Earth Wall but of sand: rises in 3 ticks, same sizes, lasts **6 s**, and is **see-through from your side** (gaps every other block: you can shoot through it, so can they, but it stops rocks and arrows 70% of the time by random cell). Cooldown 6 s.
Counters	Water turns a Sand Wall to mud: a Water Shot ends it. Fire does nothing (glass is a **(suggestion)**: a Fireball turns a Sand Wall cell to glass, which stays as a temp block). Air Blast blows a Sand Wall away (ends it).
Edge cases	Soul sand works in the Nether. Suspicious sand is never bent (archaeology). Red sand gives red grains. A Sand Stream never blinds a boss or a player with PvP off.

2.6 Quicksand — חול טובעני

Key	Slot; **press**. Sneak: **Sand Cloud** (ענן חול)
Unlock	4 SP (needs Sand Bending M3)
What	The sand (or gravel, soul sand, mud, dirt at M8+) under an enemy within **12 blocks** turns to quicksand: a 3×3 patch of temporary **suspicious-sand-like** state (the mod's own `quicksand` block: a sand look, `SoulSand`-style sinking, no jumping) for **4 s** +0.4 per M (7.6 at 10). Whoever stands in it sinks to the chest, cannot jump, moves at ×0.2, and takes **0.5 heart per second**. It is harmless to the bender.
Limit (Eilon)	**Fails on an earthbender whose element strength ≥ yours**: their feet stay on top and the action bar tells you "Stronger earthbender". Also fails on bosses and flying targets.
Cooldown	10 s
Mastery	Duration (above); patch 3×3 → 5×5 at M6; at M8 works on dirt and gravel too (not stone); at M10 holds **two** targets (two patches).
Sneak: Sand Cloud	A 6-block sphere of hanging sand around the aimed point for 5 s: everything inside has Blindness and takes 0.25 heart/s; projectiles entering it fall short. Cooldown 10 s.
Counters	Earth Pillar under yourself escapes it (the pillar lifts you out); an Air Jump or Fire Jet escapes; Water Shot at the patch turns it to mud, which frees everyone; a frozen patch (Freeze) is solid ground.
Edge cases	A patch never reaches under a block entity, a bed or a door. A quicksanded mob that was pathing stops pathing (we set no-AI for the duration: no, we just slow it; `Mob.getNavigation().stop()` once). Fall into it from above: counts as landing on soul sand (no extra fall damage). PvP off: no players. Nether: soul sand is the natural quicksand; works.

2.7 Lava Bending — כשפות לבה

Key	Slot; **press**: **Lava Path** (שביל לבה). Sneak: **Lava Shuriken** (שוריקן לבה 🌋)
Unlock	6 SP (needs total Earth ES ≥ 20; "a rare art": Ghazan, Bolin)
Lava Path	A line of **temporary lava** (`bending:temp_lava`: looks and burns like lava, flows nowhere, never spreads fire, never makes stone) runs from your feet in the direction you look, 1 wide, **6 + M blocks** long (16 at M10), 1 block per tick, through earth only (it stops at water, air gaps and anything not earth). Lasts **12 s**, then the earth cools back (the blocks revert; the look: a crust that darkens). Creatures on it take lava damage; **the bender walks on their own path unharmed** while it lasts (fire resistance on the path's cells). Cooldown 12 s.
Lava Shuriken	A spinning disc of lava, **3 hearts** + burning 4 s, range 24, speed 1.5, cooldown 4 s; at M8 **two** discs; at M10 the disc **bounces once** off a block. Lights no fires (the burn is on the target).
Passive (Eilon)	**Partial fire and lava resistance** from mastery: M1–4 nothing; M5–7 fire and lava damage **×0.5**; M8–9 **×0.3**; M10 **lava immunity for 10 s after touching lava**, then ×0.3 (you can swim a little).
Mastery	Path length, two discs, bounce, the passive. Also M6: Lava Path **beside real lava** costs no cooldown (it draws from the lake).
Counters	Water: a Water Shot on the path turns that cell to cooled stone at once (and the shot hurts a lavabender's path's owner nothing); an Ice Wall across the path stops it. Air: nothing. Fire: a firebender takes ×0.5 from the path.
Edge cases	The path never crosses a claim border (it stops). Never under a block entity. In the End, end stone melts fine (it is earth). In water: refused. On a boat/minecart: no. Mobs: zombies burn, fire-immune mobs (blaze, magma cube, zombified piglins) ignore it. Items dropped on it burn as on lava (**(added)** no: temp lava never destroys items, to spare loot: items float on it).

2.8 Metal Bending — כשפות מתכת (three cards)

Unlock gate: all three need Earth **ES ≥ 30** (Toph's invention came late) and a **Metal supply**. **Metal charge (added):** a resource 0–30 shown on the HUD when a metal card is in the loadout. It is filled by **Metal Pull** (below) from `bending:metal` blocks and ores within 8, 1 charge per metal block (an ore gives 1 charge and stays; a block of iron gives 3 and stays), or **from your inventory**: 1 iron ingot → 2 charge, 1 iron nugget → 1 (consumed). Knives, walls and armour spend charge; what a wall or armour used **returns** when it ends (the metal is yours), knives return 1 charge each when picked up (they drop as the mod's `steel_knife` item... no: they *return to charge* when they land within 16 of you after 2 s, else they are lost **(added)**).

2.8a Steel Knives — סכיני פלדה. Sneak: Metal Pull — שליפת מתכת

Unlock	5 SP
Steel Knives	**3 knives** (M1) → 5 (M5) → 7 (M10), fanned 6° apart, **1.5 hearts** each, pierce one creature, range 32, speed 2.0, 1 charge each. They **stick in the target** (drawn, 3 s) and a stuck knife returns to you as charge when you press Sneak-key again within 5 s (Lin's recall). Cooldown 2.5 s.
Metal Pull	Hold up to 2 s: metal blocks and ores within **8 blocks** (+1 per 2 M) shed glowing filings that stream to your hands: +1 charge per block per second held, at most 10 per pull; **mobs in metal armour** lose nothing (that is Metal Grip's job). Also pulls iron ingots/nuggets lying as items within 8 into charge. Cooldown 6 s.
Counters	Earth Wall stops knives; Air Shield absorbs them (1/4 strength each); Water Octopus deflects one knife per 2 s; Fire melts none (steel); **Lightning** on a knife-stuck target ×1.5. A Steel Armour wearer takes ×0.5 from knives.
Edge	No charge: "No metal" and no cooldown. Knives never break blocks. They stop at the Hex wall if Enhanced says so. PvP off: no players.

2.8b Steel Wall — קיר פלדה. Sneak: Steel Armour — שריון פלדה

Unlock	5 SP
Steel Wall	Like Earth Wall but of **iron plates** (temporary `bending:steel_plate` blocks: metal look, blast-resistant, 2 wide × 2 high at M1 to 6×4 at M10), rising in 3 ticks **from your charge** (1 charge per 2 plates; returned when it ends). Lasts 10 s; rocks bounce off; Lava Shuriken dents it (no break); an Explosion breaks one plate (its charge is lost). Cooldown 8 s.
Steel Armour	**On/off**: plates (12 charge) cover the whole body but the eyes (`LOOKS.md` 1.2): shell **16 hearts** +1 per M, **+12 armour**, toughness 6, knockback resistance 0.8, speed ×0.8 (×0.9 at M8), **lightning ×1.5** (conducts), fire ×1.0, **a magnet to Magneto** (Enhanced's `Metal.isMetalHero` hook if present: he can lift you). Charge returns when shed. Cooldown 20 s after a break. Cannot be worn with Earth Armour.
Mastery	Sizes, shell; M10: the wall has a **door** cell you can walk through (others cannot: it opens for your UUID).
Edge	In water: steel armour sinks you faster (×1.5 gravity) but does not crumble. Nether: fine. Claims: plates are temp blocks with the same rights.

2.8c Wreck Metal — הרס מתכת. Sneak: Metal Grip — אחיזת מתכת

Unlock	6 SP

Wreck Metal	Aim at a **metal device** within 10 blocks and press: it crumples: the block becomes its wrecked form for `wreckSeconds` 60 (temporary: a **dark, bent** version of the block, `bending:wrecked_*` states: a wrecked hopper passes nothing, a wrecked rail derails, a wrecked piston does not move, a wrecked redstone component is off, a wrecked furnace stops smelting, a wrecked anvil is a wrecked anvil; doors and trapdoors of iron jam shut or open); then it mends. Contents are never lost (block entities keep their NBT; the temp block carries it). Costs 0 charge, **gives 1**. Cooldown 4 s. Mastery: M5 wrecks a **3×3** of devices at once (all connected metal within 1 of the aim); M8 **5×5**; M10 also **minecarts** (derailed and stopped 10 s) and **iron golems** (Weakness and Slowness 10 s, 2 hearts).
The device list (Eilon's open question 5; `bending:metal_devices` tag)	**Default:** hopper, hopper minecart, dropper, dispenser, piston, sticky piston, observer, redstone repeater, redstone comparator, redstone torch (as "electrical"), lever, iron door, iron trapdoor, iron bars, chain, lantern/soul lantern, bell, anvil (all), cauldron, rail (all four), minecart (entity), furnace, blast furnace, smoker, lightning rod, heavy/light weighted pressure plate, tripwire hook, grindstone, stonecutter, iron/copper/gold blocks (crumple into a dented look, cosmetic), copper bulb, copper grate, copper doors and trapdoors. **Never:** chests, barrels, shulker boxes, beacons, enchanting tables, anything not metal. Servers edit the tag.
Metal Grip	Hold a **metal thing** in your grip (Enhanced's `Telekinesis` in a `Mode.METAL`, `TECH.md` 2): a mob or player in **iron, chainmail, gold or netherite armour** (at least chestplate or 2 pieces), an iron golem (M8+), a minecart, an armour stand with armour, or a metal block you tore out (an iron block, an anvil, rails): lift it, move it with your look, press again to fling (3 hearts at speed), Sneak-press to set down. Range 12 + M. Hold time ≤ 6 s + M (a player struggles free after that), cooldown 8 s. **Fails** on a player with no metal armour and on bosses. Players feel it as Enhanced's `Live.LIFTED` lock (no movement input).
Counters	A gripped player who sheds armour (opens inventory, removes the piece) drops free. A Fire Jet or Air Jump breaks a grip (the lifter loses them). Earth Armour (stone) is not metal: immune to Metal Grip.
Edge	Wrecking in a claim: refused (it is a block change). Wrecking another player's hopper line on a PvP-off server: allowed by default (`wreckNeedsPvp` false), since it mends in a minute; a server flips it. Redstone contraptions: a wrecked component is simply "off"; nothing is destroyed. Bosses: never gripped.

2.9 Earth loadout examples (added)

- **Beginner:** X Rock Throw / Hold, C Earth Wall / Push, Alt Earth Pillar / Launch, K empty.
- **Toph:** X Rock Throw, C Earth Wall, Alt Steel Knives / Metal Pull, K Earth Armour / Shed.
- **Bolin:** X Rock Throw, C Earth Pillar, Alt Lava Bending, K Earth Wall.

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3. Air — כשפות אוויר (release 1.1)

Theme. Ba Gua Zhang: circles, evasion, never a straight line; the airbender is light and quick. Colours: saffron and sky `#F2B134` / `#CFE8FF`. **Passives (added, free):** fall damage ×0.5 (feather-light landing, a dip), movement speed +10%, and jumping 1.5 blocks high; you may **double-tap Space to hover** 1 s (not flight: a cushion of air) with a 3 s cooldown.

3.1 Air Blast — הדף אוויר (free)

Key	Slot; **press**. Sneak: **Air Jump** (קפיצת אוויר)
What	A wave of air from both palms in a **cone** (20° half-angle at M1 → 30° at M10) to **8 + M blocks**: every creature in it is thrown **4 + 0.6M blocks** away (knockback velocity 1.2 + 0.1M), takes **0.5 heart** (the fall does the rest), arrows and fireballs in the cone are blown off course, fires in it go out, and items are scattered. Pushes players even with PvP off (`pushPlayersWithoutPvp` true).
Cooldown	2 s

Mastery	Distance and cone (above); M6: a blast at the ground **while gliding** throws you up 6 blocks (Eilon's "shooting air at the ground launches you higher"; also works without a glider); M10: the blast is a **sphere** when Sneak-free pressed while airborne (pushes all around).
Sneak: Air Jump	A burst under your feet: you leap **1.0 + 0.1M** blocks/tick upward (≈ 6 → 12 blocks), no fall damage on the next landing; mid-air it works once per 2 s as a second jump. Cooldown 2 s.
Counters	Earth Armour wearers are pushed ×0.5; Steel Armour ×0.6; a bender on an Earth Pillar is pushed off (M4+ of the pillar: the pillar's parapet holds them... no, keep simple: pushed). Bosses ×0.2. A Water Spout rider is knocked off.
Edge	Under water: half range, pushes the bender back too (the water resists). In the Nether/End: normal. Pushing a mob into the void or lava: that is the game. A pushed player is `hurtMarked` so the client takes the velocity.

3.2 Air Shield — מגן אוויר

Key	Slot; **on/off** (press raises, press again lowers). Sneak: **Shield Burst** (פיצוץ המגן)
Unlock	4 SP
What	A spinning **sphere of air** (radius 1.6 → 2.4 at M10) around you: arrows, rocks, knives, fireballs and ice knives that enter it are flung away; melee blows and beams go through at ×0.75 (air is not a wall). It has **strength 6 hearts** +0.8 per M (13.2 at 10): each stopped projectile costs its damage; at 0 it **breaks and bursts** (below). Costs 0.3 exhaustion/s.
Burst (Eilon)	When it breaks, it bursts in **every direction**: everyone within **4 blocks** (+0.3 per M) is thrown 6 blocks and takes **3.5 hearts** (default; Eilon's 4 is an option: `airShieldBurstHearts`, SUGGESTIONS 2). Then cooldown 12 s.
Duration	Until lowered, broken or 20 s (`airShieldMaxSeconds`; 0 = unlimited). Lowering it by hand does **not** burst (and costs a 6 s cooldown).
Mastery	Radius, strength, burst radius (above); M7: moving at full speed is allowed (M1–6: speed ×0.8 while up); M10: it also **deflects Lightning** once (the bolt curves away; the shield breaks).
Sneak: Shield Burst	Burst it **on purpose** at full strength: the same push and damage, plus the strength left is added as +0.5 heart per 2 hearts left. Cooldown 12 s. Without a shield up: raises and bursts after 10 ticks (a quick 2-heart repel), cooldown 8 s.
Counters	Rocks eat it fastest (2 hearts a rock, 6 a boulder); Fire Whip strokes go through at ×0.75; a Tornado from another airbender cancels it (both end); Water Shot passes through at ×0.75 (it is a stream, not a projectile) **(added)**.
Edge	In water: refused (bubbles). Riding: works. Bosses: the Wither's skulls are flung; the Dragon's breath passes. Enhanced's shield rules: an Air Shield is a `ShieldStyle` dome drawn by Enhanced's `ShieldFx`-pattern renderer (`ANIMATION.md` 3.2).

3.3 Air Glider — גלשן אוויר (item)

Item	`bending:air_glider` (a staff that opens into wings): crafted from **2 sticks, 1 string... (added: 3 sticks, 2 string, 2 phantom membrane... no, too rare for a kid)**: **4 sticks, 3 wool, 1 feather** in a T shape. Colour: craft it again with **any dye** (16 colours; the wings and the cloth of your looks' glider match; default saffron). Durability: none (it is bending, not a tool).
Who	Only an airbender can open it (others hold a stick). It goes in the **off hand** or main hand.
Glide	Hold **Space in the air** with the glider in a hand (or Sneak + right-click to open it): you glide like an elytra (we reuse the elytra physics by setting `FALL_FLYING` while the glider is open, as Falcon's Glide does in Enhanced), but slower (×0.8) and never faster than 1.2 blocks/tick; no rocket boosting; **Air Blast downward launches you higher** (3.1, M6; at M1–5 of Air Blast it still lifts 3 blocks). Closing: Sneak, or landing. The pose: `ANIMATION.md` 10.2.
Duration	Unlimited (it is wind); exhaustion 0.2/s.

Edge	Under water: closes. In the End over the void: it glides, it does not save you from the void. Other dimensions: works. In rain: fine. A non-airbender in Creative: no.

3.4 Air Blades — להבי אוויר

Key	Slot; **press**. Sneak: **Air Slice** (חיתוך אוויר)
Unlock	4 SP
What	A **crescent of concentrated air** (a thin blade, drawn as a translucent arc, `ANIMATION.md` 3.4) flies where you look, range 20 + M, speed 2.0, **2.5 hearts** (Eilon) +0.1 per M (3.4 at 10), cuts through **leaves, tall grass, crops (not harvested: trampled... no, cut: drops them, as shears would, only where rights allow and `airBladesCutPlants` true), cobwebs and vines**, and passes through one creature to hit a second at ×0.5.
Cooldown	1.5 s. M4: **two** blades (a V); M8: **three**; M10: blades **curve** toward the aimed creature (homing ±10°).
Sneak: Air Slice	A horizontal **sweep** around you: every creature within 3 blocks takes 2 hearts and is knocked back 2 blocks; the sweep also cuts the plants in the ring. Cooldown 3 s.
Counters	Earth Wall, Sand Wall, Steel Wall and Ice Wall stop it dead; an Air Shield (another airbender's) flings it; Fire Whip at M7 "burns" it (cancels one blade); Water Octopus arms do not stop it. Earth Armour ×0.75.
Edge	Under water: refused. It never breaks blocks; it never cuts a claim's crops (rights). Bosses: full damage. PvP off: no players.

3.5 Spiritual Projection — השלכה רוחנית

Key	Slot; **press** (begins; press again ends). Sneak: **Deep Meditation** (מדיטציה עמוקה)
Unlock	6 SP (needs Air ES ≥ 15)
What (Eilon)	You sit in **meditation** (the body stays, a real player entity in a sitting pose, invulnerable? **no**: hittable, see below) and your **spirit** leaves it: a translucent blue copy of you (`SpiritEntity`, `TECH.md` 3) that **flies freely** (Creative-like flight, speed 0.08), **passes through blocks**, **sees through walls** (everything hostile within 32 glows, caves are drawn with x-ray-like outline of air pockets: `ANIMATION.md` 2.5), **cannot interact**, cannot be hurt, and is invisible to mobs. Range: **64 + 8M blocks** from the body (128 → 144... : 64 + 8M = 144 at M10); beyond it the tether pulls you back. Duration **30 + 6M s** (90 at M10), then you snap back.
The body	Sits where you were (the meditation pose, `ANIMATION.md` 10.5), takes damage normally; **any damage snaps you back at once** (Eilon), with Nausea 3 s. Mobs *do* target it. Friends see a glowing blue thread from the body toward the spirit (within 32).
Cooldown	60 s after return (30 at M10)
Mastery	Range, duration; M5: the spirit can **open the Bending screen** (to plan a loadout) and **mark** a spot (a waypoint arrow on the HUD for 5 min); M8: you see **ores**' faint glow within 8 blocks of the spirit (Jinora's sense; `spiritSees0res` true... default **false**, SUGGESTIONS); M10: the body is **shielded by a thin Air Shield** (6 hearts) while you are out.
Sneak: Deep Meditation	Sit in meditation without leaving: heal **0.5 heart per 2 s** and restore 1 hunger per 10 s for up to 20 s; any damage or movement ends it; cooldown 30 s. (Also the pose for photos.)
Edge	Never across dimensions (the spirit stops at a portal). In water: the body drowns normally (snaps you back at 2 bubbles). In the void: the spirit cannot fly below y = min + 1. Spectators cannot. PvP: a player who hits the body sends you back; the body counts as you for every rule. Logout: the spirit ends. Death of the body: you die where the body is. Server: the spirit is a server entity the client "possesses" (camera entity, `TECH.md` 3), so other players see the spirit and the body.

3.6 Air Bubble — בועת אוויר

Key	Slot; **on/off**. Sneak: **Air Cushion** (כרית אוויר, on/off)
Unlock	2 SP
What (Eilon)	In water, a bubble of air closes around your head: you **breathe** (air refills to full and stays full), see clearly (no water fog), and the bubble is drawn around the head (`ANIMATION.md` 3.6). Costs 0.2 exhaustion/s. Lasts until turned off or you leave water; it also protects against the Drowned's trident? no, just breathing and clarity. At M6 the bubble covers **one friend** within 2 blocks too; at M10 it also gives Dolphin's Grace (swim speed).
Sneak: Air Cushion	**On/off** (default on once unlocked): a cushion of air breaks every fall: **no fall damage at all**, a landing puff. (The passive halves fall damage; this removes it.) Costs 0.5 exhaustion per saved fall.
Edge	Lava: no. The bubble pops under a Water Spout's push. Bosses: nothing. The Nether: no water, so nothing; the cushion works.

3.7 Air Scooter — קטנוע אוויר (the air ball)

Key	Slot; **on/off** (press to mount, press or Sneak-jump to dismount). Sneak: **Air Spout** (עמוד אוויר)
Unlock	4 SP
What (Eilon)	A spinning ball of air under you (an `AirScooterEntity` you ride, 0.9 blocks high, `TECH.md` 3): **great speed**: 0.7 blocks/tick (= 14 blocks/s, a sprint is 5.6) +0.03 per M (0.97 at M10), climbs 1-block steps, **crosses water** at the surface, and runs along walls for 1 s if you ride into one at M6+. You steer with W A S D and the look; jumping hops. Exhaustion 0.3/s. Duration: until dismounted or 60 s (`scooterMaxSeconds`, 0 unlimited); cooldown 5 s after dismount.
Mastery	Speed; M4: 2-block steps; M6: wall ride 1 s; M8: a passenger can **sit behind you** (a friend right-clicks the ball); M10: the ball **bursts** on dismount (2 hearts, push 4 blocks, 4 s cooldown added).
Sneak: Air Spout	A column of spinning air lifts you straight up **4 + M blocks** and holds you there while held (hold ability, 0.4 exhaustion/s, up to 10 s): the airbender's "pillar". Let go: you drift down with the cushion. Cooldown 6 s.
Counters	Quicksand under the ball: it keeps going (air). A Water Shot at the rider ends the ride (knocks you off). Lightning hits normally. Earth Wall: you stop.
Edge	Under water: the ball surfaces you. Riding it in a 1-block-high gap: no (needs 2). Boats/minecarts: no stacking. Bosses: a ball hits nothing. In the End: it does not float over the void.

3.8 Tornado — טורנדו

Key	Slot; **press**. Sneak: **Air Wheel** (גלגל אוויר)
Unlock	5 SP (needs Air Blast M4)
What (Eilon)	A **small tornado** (a `TornadoEntity`: a spinning funnel 2 blocks wide at the base, **3 + 0.5M** high → 8 at M10) rolls where you look at 0.5 blocks/tick for **4 + 0.3M s**, then fades. Everything it touches is **lifted** 1–3 blocks and spun (held in orbit for up to 2 s), takes **1 heart** on entering and **0.5 heart/s** inside, and is flung out 6 blocks when it ends or they escape. Projectiles in it are flung too. Range (how far you can send it): 16 + M.
Cooldown	8 s. At most **2** of yours at once (M8+; one before).
Mastery	Height, duration, count; M5: you may **steer** it while holding the key (it follows your look); M10: a **big** tornado (3 wide, 10 high, 2 hearts entering) when double-tapped. "Stronger bending makes the tornadoes, the shield and the other air abilities bigger, with longer range" (Eilon): yes, every size above is the mastery line.

Sneak: Air Wheel	A horizontal **wheel of wind** around your legs for 4 s: you **run at ×1.6** speed and every creature you run into is thrown 4 blocks (1 heart); Air Blades fired while wheeling cost no cooldown for the first two. Cooldown 10 s.
Counters	Earth Wall and Steel Wall stop it; a boulder dispels it; Fire Circle inside it becomes a **firenado** (**(suggestion)**: a tornado over a Fire Circle burns: +1 heart/s to everyone inside, 4 s); Water Spout vs Tornado: both end. Bosses are never lifted.
Edge	Under water: a whirlpool: half damage, pulls instead of lifts (it is still a tornado entity, in water). Claims: the tornado lifts no blocks (never; it is entities only). Items: lifted and dropped, never destroyed. A tornado never crosses a Hex wall (Enhanced hook).

3.9 Zaheer's Path — דרכו של זהיר (two cards, gated)

The gate (Eilon): when **your tamed pet dies** (any `TamableAnimal` or `OwnableEntity` you own: a wolf, a cat, a parrot, a horse you tamed... **(added: horses count; a sky bison of 8.7 counts)**; Eilon's open question 3, default: **any death of any tamed pet counts**, config `zaheerAnyPetDeath` true, else only a pet that died *while you were within 16 blocks*), you "cut your earthly tethers": the two cards below **unlock for free** (they appear on the tree with a red thread icon), and from then on **every Air ability pushes less and hits harder**: knockback ×0.7, damage ×1.3 (`zaheerPushFactor`, `zaheerDamageFactor`). This is permanent (saved), unless `/bending zaheer reset`. The HUD shows a small broken-thread icon. A server can turn the whole path off (`zaheerPath` true).

3.9a True Flight — תעופה אמיתית

Key	Slot; **on/off**; also **double-tap Space** like Enhanced's fliers. Sneak: **Hover** (hold: stand still in the air)
What	Creative-style flight (Enhanced's `applyPassives` flight attribute pattern), speed 0.06 (0.09 at M10), the flight profile `ANIMATION.md` 10.6 (Zaheer's cross-legged float at rest, arms back in flight). Exhaustion 0.4/s; at hunger 0 you glide down.
Mastery	Speed; M10: no exhaustion while hovering still.
Edge	Not in water (you swim). Not through blocks. In the End: fine. The Nether roof: no.

3.9b Suffocation — חניקה

Key	Slot; **hold** (aim at a creature within 8 + M blocks, hold the key). Sneak: **Air Vacuum** (ואקום אוויר)
What (Eilon)	You pull the air out of the target's lungs: they take **1 heart per second**, Slowness II and cannot sprint or jump, their air meter drains as if drowning (the bubbles show), **unless they get far enough from you**: the hold breaks when they are **more than 8 + M blocks** away, or out of your sight for 10 ticks, or you take damage of 1+ heart. Max hold **6 + 0.4M s**, then cooldown **20 s**. One target at a time. Players only with PvP on; never bosses; never Creative.
Mastery	Range, hold time; M10: Slowness III and the target is **lifted** 0.5 block (Zaheer and the Earth Queen).
Sneak: Air Vacuum	A 3-block sphere where you aim is emptied of air for 3 s: everyone inside takes 1 heart/s and cannot jump; fires and Fire Circles inside go out at once; a Flamethrower aimed into it dies. Cooldown 15 s.
Counters	An Earth Pillar launch or Air Jump or Fire Jet breaks line-of-sight/distance; an Air Bubble (another airbender's) makes you immune; Earth Armour and Steel Armour do not help (the air is inside). A Water Shot at the suffocator breaks the hold (damage).
Edge	Under water: the target is already drowning: it just hurries it (0.5 heart/s). Mobs that do not breathe (undead, iron golems, blazes, endermen? — the tag `bending:no_breath`: undead, golems, blazes, magma cubes, slimes, wardens, shulkers, vexes, allays) are immune.

3.10 Air loadout examples (added)

- **Beginner:** X Air Blast / Jump, C Air Blades / Slice, Alt Air Scooter / Spout, K Air Shield / Burst.
- **Tenzin:** Air Blast, Air Shield, Tornado, Spiritual Projection.
- **Zaheer:** Air Blast, Air Blades, True Flight, Suffocation.

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4. Water — כשפות מים, with ice and blood (release 1.2)

Theme. Tai Chi: flowing, continuous, turning the enemy's force. Colours: sea blue and white `#3C8DD6` / `#E8F6FF`; ice `#BFE9FF`; blood `#7A0E1E`.

4.0 The water source (Eilon)

A waterbender needs water. Every Water ability (not the ice-only shaping of existing ice, not blood bending) **draws from one of three sources**, tried in this order; the HUD's **water gauge** shows which one you have (`UI.md` 3):

1. **Open water nearby**: any water block (source or flowing), rain, snow layers or a cauldron of water within `waterReach` **8 blocks** (+1 per 2 M of the ability used), or you standing in water. Unlimited.
2. **The Water Skin** (item `bending:water_skin`): crafted from **3 leather and 1 string** (Eilon: "cow leather and a little string"). It holds **8 charges** (`waterSkinCharges`; shown as a bar on the item and on the HUD). A Water Shot costs 1 charge, a Whip 1, a Healing second 1 per 4 s, an Octopus 3 to form, Arms 2, an Ice Knife 1, an Ice Wall 3, a Spout 4; the Skin **refills** by right-clicking water (full at once), standing in rain (1 charge / 10 s), or at a cauldron (1 charge per level). You may carry several skins; the fullest is used. The Skin also **catches** water your abilities release (a shot that lands within 8 returns 0.5 charge at M6+).
3. **Plants and flowers** (Eilon; needs **mastery 7 in Water Shot or Healing**): with no other source, the ability draws from the nearest plant within 8 (`bending:plants`: crops, flowers, grass, ferns, leaves, saplings, bushes, cactus, kelp...): the plant **dries** (crops reset to age 0 — no: become **dead bush**/withered: a flower becomes a dead bush, tall grass vanishes, leaves turn to brown "dry leaves" temp state for 60 s, crops lose 2 growth stages; a cactus is a water source without drying, Hama's trick). One plant = 1 charge. Never in a claim (rights).

With **no source**: the action bar says "No water" (אין מים), no cooldown, and the gesture plays a dry reach. Water that an ability leaves in the world (a shot's splash, a spout's pool) is **temporary water** that evaporates in 2–5 s (never a permanent water block, never flows).

Passives (added, free): swim speed +30%, no slowness in water, Water Breathing for 10 s after any Water ability under water (not infinite: that is Air's). **Full moon:** every Water ability deals ×1.25 on a full-moon night (`fullMoonFactor`), the HUD shows the moon (`UI.md` 3.6). **Day of the sun (suggestion):** a solar eclipse does not exist in Minecraft; nothing.

4.1 Water Shot — יריית מים (free, "very easy to unlock")

Key	Slot; **press**. Sneak: **Water Whip** (שוט מים, hold)
What	A stream of water (a `WaterShotEntity`, drawn as a ribbon) flies where you look: **1.5 hearts** +0.15 per M (2.85 at 10), knockback 1.0, range 20 + M, speed 1.8; it **extinguishes** a burning target and any fire it passes (1.7), **×1.5 vs firebenders**, and **wets** the target (3 s; lightning ×1.5). Cost 1 charge.

Cooldown	1 s. M4: two streams (one per hand) per press; M8: three; M10: the stream **arcs** over a wall (a lobbed shot when you aim above 30°).
Sneak: Water Whip	**Hold**: a whip of water hangs from your hand (length **4 + 0.6M** → 10 at M10) and lashes at what you look at every 10 ticks: **1 heart** per lash, pulls the target 1 block toward you per lash (Katara's whip), reach = length. Costs 1 charge to start, 0.4 exhaustion/s. Max hold 8 s, cooldown 4 s.
Counters	Fire **evaporates** the whip's tip (a Fire Whip crossing it ends both for 1 s); an Earth Wall stops a shot; an Air Shield lets it through at ×0.75; a Fireball hit by a shot bursts harmlessly (1.7). Freeze on your own shot in flight = an ice knife (**(suggestion)** combo at M10).
Edge	Under water: the shot is a pressure jet, range 8, no cost. Lava: a shot into lava hisses and does nothing to the lava (no obsidian: temporary rule). Nether: Skin only. Bosses: full damage (a Blaze takes ×2 as vanilla water would... no, vanilla does not; we give **×1.5 vs fire-type mobs**: blaze, magma cube, strider: `waterVsFireMobs`). Rain: half cost.

4.2 Healing — ריפוי

Key	Slot; **hold**. Sneak: **Cleanse** (טיהור, press)
Unlock	3 SP
What (Eilon)	Hold: glowing water gathers on your hands and heals **yourself** (no target in sight within 4 blocks) or **the friend you look at** within 4 + 0.3M blocks: **1 heart per 2 s** at M1 (1 heart per 1.5 s at M5, **1 heart per second at M10**); also slowly removes Wither and Poison (halves their time every 2 s); costs 1 charge per 4 s and 0.5 exhaustion/s; the target stands in a glowing water veil (`ANIMATION.md` 4.2). Mobs: heals your pets and villagers; never hostile mobs. Players: anyone, PvP or not. Both of you must be mostly still (movement speed over 0.2 blocks/tick breaks it).
Cooldown	None (a hold); 20 s max hold then 2 s rest.
Mastery	Speed and range (above); M4: heals **two** friends in your look's cone (2 blocks apart); M7: plant-drawn source (4.0); M10: **revives a downed Enhanced healer**? no — M10: heals **1 heart/s** and removes **Weakness, Slowness, Blindness, Nausea** too.
Sneak: Cleanse	A splash on the target (or yourself): removes **Poison, Wither, Nausea, Blindness, Hunger** at once and 1 heart; cost 2 charges; cooldown 10 s. Also puts out a burning target.
Counters	None needed; a healer in water is hard to interrupt (the source never runs dry). Fire Healing (5.6) is the weaker twin.
Edge	Healing a boss: never. Healing yourself at full health: nothing happens, no cost (no mastery XP either). Nether: Skin only; the Skin empties fast (1.9). Spectators: no.

4.3 Water Octopus — תמנון מים

Key	Slot; **on/off**. Sneak: **Octopus Lash** (הצלפת תמנון)
Unlock	5 SP (needs Water Shot M3)
What (Eilon)	Water wraps your body and **eight arms** (6 at M1, 8 at M5) wave around you (`ANIMATION.md` 4.3). **Defence:** each arm **slaps away one projectile** (an arrow, a rock, a knife, a fireball; not beams or blades) per 2 s within 2.5 blocks, and melee blows against you are **×0.75**. **Offence:** when you hit someone with your hand or a weapon within 3 blocks, **2 arms strike them too**: **+1 heart** (+0.1 per M) and a push of 1 block; also any creature that comes within 2 blocks and is hostile to you is lashed once per 2 s for 0.5 heart. Cost **3 charges** to form, 0.4 exhaustion/s; **speed ×0.85** while up. Lasts until turned off, until you leave water reach for 10 s (the arms need a source to keep their shape; the Skin counts), or 40 s (`octopusMaxSeconds`, 0 unlimited). Cooldown 10 s after it drops.
Mastery	Arms, damage; M4: reach 3 → 4; M7: arms also **catch a thrown rock and throw it back** (1 heart) once per 4 s; M10: the arms **grab**: a creature that hits you is **held** for 1 s (no movement) once per 5 s.

Sneak: Octopus Lash	All arms lash forward in a cone of 4 blocks: **2 hearts**, pull 2 blocks toward you; cooldown 4 s. Without an octopus up: forms it and lashes (5 charges).
Counters	Fire: a Flamethrower on you evaporates one arm per second (the octopus shrinks); Fire Circle removes 3 arms; Freeze from another waterbender freezes your arms solid for 2 s (you cannot act); Lightning ×1.5 on you (you are wet). Earth: rocks are slapped; a boulder breaks 2 arms. Air: Air Blast pushes you ×1 (the arms don't root you).
Edge	Under water: the octopus is free and the arms are 8 at any mastery (you are in your element). Nether: the Skin keeps it 10 s per 3 charges. Riding: the octopus drops. Bosses: the arms do not grab them.

4.4 Water Spout — עמוד מים (the vortex)

Key	Slot; **hold** (ride while held). Sneak: **Wave Ride** (רכיבה על גל)
Unlock	6 SP (needs Water ES ≥ 15; "needs high mastery; otherwise very small")
What (Eilon)	A **vortex of water** rises under you from a water source **you stand in or beside** (within 3 blocks; the Skin makes a small one: height ×0.5): a spinning column (a `WaterSpoutEntity` you ride) that lifts you to a height of **2 + M blocks** (3 at M1 — "very small" — 12 at M10) and **moves with you**: W A S D steers at 0.3 blocks/tick over water, 0.15 over land (the spout drags its base; over land it loses 1 block of height per 2 s away from water). While riding you can use every other ability. Let go: the column collapses into a splash and you fall (the water cushions: no fall damage within 12 blocks). Cost 4 charges to start, 0.5 exhaustion/s, max ride 15 s +1 per M, cooldown 10 s.
Attack (Eilon: "ride onto an enemy and push them, dealing heavy damage")	**Steering the spout into a creature** rams it: **4 hearts** +0.3M (7 at 10), knockback 3 blocks, once per 2 s per creature; the spout's spin also pulls creatures within 2 blocks of the base in and lifts them 1 block, 0.5 heart/s.
Mastery	Height, ride time, ram damage (above); M5: works from the Skin at full height; M8: the spout **freezes** on command (Sneak-press while riding: it becomes a temporary ice pillar for 6 s and you stand on it); M10: two riders (a friend may jump on).
Sneak: Wave Ride	A **wave** rises from the water and carries you forward 12 + M blocks at 0.8 blocks/tick along the surface (Korra's surf), pushing creatures ahead of it 3 blocks (1 heart); works only from water (not the Skin). Cost 2 charges, cooldown 8 s.
Counters	Lightning on the spout collapses it (1.7); Fire does little (steam: ×0.5 to the rider); Freeze from another waterbender freezes the column (you are stuck on top 2 s); an Air Blast knocks you off; a Tornado meeting a Spout ends both; Earth: a Rock Throw at the base does nothing, at the rider full damage.
Edge	Lava: never (no source). The Nether: Skin-only small spout. The End: no water unless you brought it. Under a ceiling: the spout stops below it. Bosses: a ram deals full damage, no lift.

4.5 Water Arms — זרועות מים

Key	Slot; **on/off**. Sneak: **Arm Grab** (אחיזת זרוע)
Unlock	4 SP

What (Eilon)	Your two arms are wrapped in long arms of water (Ming-Hua): **reach** for melee becomes **4 + 0.5M blocks** (4.5 → 9 at M10; vanilla is 3) against players, animals and monsters; **attack damage +1 heart** (+0.1 per M) on every hand blow (an empty hand deals 1 heart base + the bonus: a water fist); attack speed +20%. You may **not hold items** in your hands while the arms are on (your hands are water: the hotbar still works for selection, items are shown on your back... simpler: items in hand are not drawn and do nothing; `ANIMATION.md` 4.5). Cost 2 charges, 0.2 exhaustion/s; lasts until off or 60 s (`armsMaxSeconds`, 0 unlimited); needs a source within reach every 10 s. Cooldown 6 s.
Mastery	Reach, damage; M4: the arms **climb**: hold Space at a wall within reach and you are pulled up 4 blocks (Ming-Hua's climbing), 2 s cooldown; M7: the arms **swing**: Sneak + Space at a ceiling pulls you up to it; M10: arms **freeze on hit** for +0.5 heart (ice tips).
Sneak: Arm Grab	Grab the creature you look at within reach and **pull it to you** (1 heart, it lands 1 block in front of you); players only with PvP; bosses never. Cooldown 4 s.
Counters	Fire evaporates the arms if you are held in a Flamethrower 2 s (they drop; 6 s cooldown); Freeze from another waterbender freezes them (you cannot attack 2 s); Lightning ×1.5 on you. Earth Armour takes ×0.5 from water fists.
Edge	Under water: the arms are free and reach +2. Nether: the Skin feeds them (1 charge / 10 s). Bosses: full damage, no grab. A wrapped hand cannot use a bow, shield, or eat (open the Bending screen or toggle off first); the HUD says so.

4.6 Ice — קרח (three cards)

4.6a Ice Knives — סכיני קרח. Sneak: Ice Spikes — חודי קרח

Unlock	3 SP (needs Water Shot M2)
Ice Knives	Water freezes in your hand into **3 knives** (M1) → 5 (M6) → 7 (M10), **1.5 hearts** each, range 24, speed 2.0, pierce 0; a hit **wets and chills**: Slowness I 3 s (II at M8); the knives shatter into ice particles. Cost 1 charge per 3 knives, cooldown 2 s. **Fire ×1.5 vs the chilled target** (1.7).
Ice Spikes	A line of **temporary ice spikes** (the mod's `ice_spike` block, 1 block, pointed) erupts from the ground where you aim, 5 + M long, 1 block per tick, through any ground (not only water: the spikes are made of your water): **2 hearts** to anything they rise under + knockback up; the spikes stand 5 s then melt. Cost 3 charges, cooldown 6 s.
Counters	Fireball melts knives in flight (1.7); Earth Wall stops them; Air Shield flings them; a Flamethrower melts spikes in 1 s. Steel Armour ×0.5 from knives.
Edge	Nether: knives melt after 8 blocks (half range), spikes last 2 s. Lava: a spike into lava does nothing. Bosses: full.

4.6b Freeze — הקפאה. Sneak: Snow Throw — זריקת שלג

Unlock	3 SP
Freeze (Eilon: "freeze water")	Aim at **water**: the surface within a **3 + M radius** disc (4 → 13 at M10) turns to **temporary ice** (frosted ice look, 20 s +2 per M, then it melts back; you may walk across — a bridge; boats stop). Aim at a **creature standing in or on water, or wet** (Water Shot in the last 3 s, rain): its **feet freeze**: it cannot move for **2 + 0.2M s** (4 at M10), takes 0.5 heart; players with PvP only; bosses never. Aim at **another waterbender's water construct** (octopus, arms, spout, whip): it **freezes solid** 2 s (4.3–4.5). Aim at a Fire Circle or lava path: a hiss, nothing (Fire melts Ice). Cost 1 charge per 10 blocks of ice (free on open water), cooldown 4 s.
Snow Throw	A **snowball volley**: 4 + M snowballs (the mod's `snow_burst` projectiles, 0.5 heart each, Slowness I 2 s, extinguish fire, ×3 vs blazes like vanilla snowballs), range 16, cooldown 2 s. From M6 the snow **settles**: a 3×3 of temporary snow layers where the volley lands (slows walkers). Cost 1 charge.

Counters	Fire melts the ice disc (a fireball melts a 3×3 hole), a firebender frees frozen feet in 1 s by any fire ability (the self-cast Fire Circle does it); Earth Pillar under frozen feet frees you; Air Jump frees you.
Edge	Deep ocean: the disc freezes the top layer only. Nether: nothing to freeze unless you brought water; the frozen-feet use works on wet targets (rain does not exist there: only Water Shot wets). Freezing a creature mid-air: no (needs ground). Frozen players keep looking around and can bend.

4.6c Ice Wall — קיר קרח. Sneak: Ice Dome — כיפת קרח

Unlock	4 SP (needs Freeze M2)
Ice Wall (Eilon: "between you and an enemy")	A wall of **temporary ice** rises in 3 ticks **between you and the creature you look at** (at the midpoint, perpendicular to the line; with no creature, 3 blocks ahead facing you): 3×2 at M1 → 7×4 at M10, lasts **10 s**, see-through (ice), stops projectiles and creatures; **fire melts it** (1 fireball per 4 blocks, a Flamethrower 1 block/s: 1.7); rocks break 1 block each. Cost 3 charges (free on open water), cooldown 8 s.
Ice Dome	A dome of ice (radius 2 → 3 at M10) closes over you (and friends within it) for 8 s: nothing gets in or out but Fire melts a hole in 2 s; you may bend from inside (Ice Knives through your own dome: the dome lets your projectiles out). Cost 5 charges, cooldown 15 s.
Edge	Nether: ice lasts half as long. Lava: an ice wall touching lava hisses away in 1 s. Claims: temp blocks, same rights. Water: a wall in water is an ice wall in water (fine).

4.7 Blood Bending — כשפות דם (two cards, gated)

The limits (Eilon): 1. It **never works on a waterbender whose element strength ≥ yours** (they stand firm, their eyes glow blue at you). 2. **Until you fully master it** (Blood Bending at **mastery 10**), it works **only on a full-moon night** (moon phase 0 in `level.getMoonPhase()`, between 13000 and 23000 of the day; the HUD shows the moon and a red tint grows in the corners while you blood-bend, `ANIMATION.md` 2.2). In the Nether and the End there is no moon: only M10 works there. Server options: `bloodMoonOnly` true, `bloodWorksOnPlayers` true (with PvP), `bloodBending` true.

Unlock: 8 SP and Water **ES ≥ 40** (Hama taught it to a master). Mastery rises by use **only on full moons** until M10 (so reaching M10 takes about 66 full-moon nights of practice... too slow: mastery XP is **×4** for blood bending uses, so ~165 successful puppet-moves, roughly 15 full moons; **and** it may be bought with SP up to 7 as any ability).

4.7a Blood Bending — כשפות דם. Sneak: Blood Fling — הטלת דם

What (Eilon)	**Hold** the key on a creature or player within **10 + M blocks**: they are **lifted** 1 block and **move like a puppet**: your look moves them (they float where you aim, up to 6 blocks from where they stood per second), they cannot move, attack, bend or use items; their limbs twist (`ANIMATION.md` 12.4, the Live bit `BLOODBENT` their client reads to lock input, as Enhanced's `PinLock`). Damage **0.5 heart per second** held. Max hold **4 + 0.6M s** (10 at M10), then cooldown **25 s**. One target (two at M10: the second within 4 blocks of the first). Cost 0.6 exhaustion/s, no water.
Fails on	a waterbender with ES ≥ yours; bosses; Creative/Spectator; players with PvP off; the undead, golems and anything in `bending:no_blood` (undead, iron/snow golems, blazes, magma cubes, slimes, wardens, shulkers, vexes, allays, endermen? — endermen have blood: yes they are bent).
Sneak: Blood Fling	While holding: **fling** the puppet where you look at speed 1.6 (**3 hearts** on landing + fall damage), ending the hold (cooldown as above). Without a hold: grab and fling at once (2 s of hold then fling: a quick 2.5-heart throw).
Mastery	Range, hold time, two targets; the M10 full-moon freedom.

Counters	**Another waterbender's Blood Bending** on you: the stronger wins (ES), a tie favours the defender; a **firebender's Lightning** hitting the blood bender breaks every hold; **damage of 2+ hearts** to the blood bender breaks the hold; Earth Armour wearers are bent (flesh under stone) but Steel Armour at M10 holds 50% of the time (Toph's... no: no exception; removed). An Air Bubble: no help. Spiritual Projection: a spirit cannot be bent; a meditating body can.
Edge	Under water: works. Through walls: needs line of sight to start, keeps the hold through 1 block. A bent player's client locks movement input (`MovementInputUpdateEvent`), their camera is free. Riding: dismounted. The Nether/End: M10 only. A mob bent over lava: its death is yours (SP counts).

4.7b Bending Severance — שלילת כשפות (full mastery)

Unlock	Appears when Blood Bending is **M10** (free then).
What (Eilon)	**Up close** (within **2 blocks**, Amon's touch: you must be bending the target with 4.7a, then press this card's key, or press it with the target within 2 and not held: it does both: a 1 s hold then the touch), you take the target's bending away for a time **by their strength**: `severanceSeconds = 600 × 1 / (1 + ES/10)` → **ES 1 ≈ 9 minutes** ("about 10 minutes for a weak bender"), **ES 40–60 ≈ 1.5–2 minutes** ("1–2 for a fairly skilled one"), ES 100 ≈ 55 s. The target's HUD greys out, every ability refuses ("Your bending is gone"), their looks stay, their points stay; the time runs down on their HUD; **a Healing (4.2) of 20 s at M10 by another waterbender, or Fire Healing M10 for 40 s, restores it early (suggestion; `severanceHealable` true)**. Cooldown **120 s**. Players only with PvP; also works on **(suggestion)** Enhanced heroes if present (their powers via `Powers.setPower`? **no**: never; only bending).
Fails on	the same list as 4.7a, plus anyone with no element.
Edge	The severed may still fly with a glider? **No**: the glider needs bending. They can still use the Water Skin to drink? There is no drinking. They keep passives? **No**: passives are bending too (the airbender falls like anyone). Death does not restore it early (the timer is saved on the player).

4.8 Water loadout examples (added)

- **Beginner:** X Water Shot / Whip, C Ice Knives / Spikes, Alt Healing / Cleanse, K Freeze / Snow.
- **Katara:** Water Shot, Healing, Water Octopus, Ice Wall.
- **Ming-Hua:** Water Arms, Ice Knives, Water Shot, Freeze.
- **Amon:** Blood Bending, Bending Severance, Water Shot, Ice Knives.

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5. Fire — כשפות אש, with lightning (release 1.3)

Theme. Northern Shaolin: sharp, linear, fast, driven by the breath. Colours: flame orange and gold `#FF6A1F` / `#FFD23F`; **Azula's blue fire** `#3FA4FF` / `#CFF2FF` (the look changes every fire colour, `LOOKS.md` 4.3); lightning white-blue `#DDF4FF`.

Passive (Eilon): fire hurts firebenders less: fire, lava and burning damage **×0.5** (`fireResistFactor`; ×0.3 at Fire ES ≥ 50), burning time ×0.5; you can **extinguish yourself** by sneaking (1 s). **(added)** The sun: by day fire abilities deal ×1.1, at night ×1.0 (Sozin's Comet is a **(suggestion)** rare event).

5.1 Fireball — כדור אש (free)

Key	Slot; **press**. Sneak: **Small Fireball** (כדור אש קטן)

What	A ball of fire from the fist, range 24 + M, speed 1.6, **2 hearts** +0.2 per M (3.8 at 10) + burning **3 s** (4 at M6); it bursts on impact in a 1.5-block flash (0.5 heart to others nearby) and lights **no fires** unless `[fire] setsFires` and `mobGriefing` (then up to 3 fire blocks, rights).
Size (Eilon)	The ball grows with mastery: radius 0.3 → 0.9 blocks (hitbox and look), the burst 1.5 → 3 blocks at M10 (1 heart to bystanders).
Cooldown	1.2 s (24 ticks). M5: two balls (one per fist) per press; M8: three; M10: a **great fireball** on double-tap (radius 1.2, 5 hearts, burst 4, cooldown 4 s).
Sneak: Small Fireball (Eilon: "a very skilled bender can choose to keep it small so it deals little damage")	From **M6**: a marble of fire, **0.5 heart**, no burning, no burst, lights a campfire/candle/TNT? — lights **campfires and candles** (not TNT, not blocks), range 16, cooldown 0.5 s: for lighting things and for play. Before M6 the Sneak use is a normal fireball with half damage.
Counters	Water Shot bursts it harmlessly (1.7); Air Blast deflects it; an Earth Wall or Ice Wall stops it (ice melts: 1 ball per 4 blocks); Air Shield flings it (eats 2 hearts). A waterbender takes ×1.0; a frozen target ×1.5.
Edge	Under water: refused. Rain: ×0.8. Nether: ×1.25. Bosses: full; the Wither, blazes, magma cubes are fire-immune (vanilla): 0 damage, the HUD says "Immune". Items on the ground: never burned.

5.2 Fire Punch — מכת אש

Key	Slot; **on/off** (fists blaze). Sneak: **Fire Kick** (בעיטת אש, press)
Unlock	2 SP
What	While on, every **empty-hand melee hit** is a fiery fist: **+1.5 hearts** (+0.1 per M) and burning 2 s, a flash of flame, and the hit **ignores 2 armour points per M**... no: **ignores 20% of armour** (fire goes through the gaps). Reach 3 (vanilla). Costs 0.1 exhaustion per hit. Lasts until off or 60 s (0 unlimited). Cannot hold items while on (the fists are the weapon).
Mastery	Damage; M4: the third hit in a row **knocks back 2 blocks** (a combo); M7: every hit also throws a **small fire jab** 3 blocks ahead (0.5 heart) if the fist misses; M10: hits do **+2.5 hearts**.
Sneak: Fire Kick	A spinning kick with a crescent of flame: **2 hearts** to everyone in a 2.5-block arc in front, knockback 2, burning 2 s; cooldown 3 s; usable mid-air (a flip).
Counters	Water Octopus arms take the hit (×0.75) and put the fire out on you? no: on the target. A frozen target ×1.5. Steel Armour ×1.0, Earth Armour ×0.5.
Edge	Under water: ×0.5 and no burning. Bosses: full. PvP off: no players.

5.3 Flamethrower — להבור

Key	Slot; **hold**. Sneak: **Fire Breath** (נשיפת אש, hold; from M7)
Unlock	4 SP (needs Fireball M3)
What (Eilon: "a continuous jet of fire from both hands")	Hold: two jets of fire from the palms (Enhanced's `ContinuousBeam` in a flame style, `TECH.md` 2): range **8 + 0.6M** (14 at M10), a cone 1.5 blocks wide at the end, **1 heart per 0.5 s** per creature in it (damage type with no cooldown, as Enhanced's beams) + burning 3 s; melts ice walls 1 block/s and snow; lights no fires unless `setsFires`. Cost 0.8 exhaustion/s; max hold **6 s** then 4 s rest (the breath); you move at ×0.6 while firing.
Mastery	Range; M4: 1 heart per 0.4 s; M7: **Fire Breath** unlocked; M10: the jet **pushes** targets back 0.5 block/s.

Sneak: Fire Breath (Eilon: "breathe fire from the mouth, bigger and stronger")	From **M7**: a single wider jet from the mouth: range 10 + 0.6M, cone 2.5 wide, **1.5 hearts per 0.5 s**, max hold 4 s, rest 6 s, you cannot move while breathing (head back, the dragon's breath pose). Before M7, Sneak is the normal jet from one hand at ×0.5 (a "warm-up").
Counters	A Water Shot across the jet puts it out for 1 s and the waterbender takes nothing extra; an Air Blast bends the jet aside; a Tornado snuffs it; Earth Wall stops it; an **Air Vacuum** kills it. Firebenders take ×0.5.
Edge	Under water: refused. Rain: ×0.8. Nether: ×1.25. Bosses: full. Lighting TNT: never. A jet into a Lava Path: nothing.

5.4 Fire Whip — שוט אש

Key	Slot; **hold** (the whip is in your hand while held). Sneak: **Whip Lash** (הצלפת שוט, press)
Unlock	4 SP
What (Eilon)	A whip of fire hangs from your hand (length **5 + 0.7M** → 12 at M10) and **strikes what you look at** every 12 ticks at full reach: **1.5 hearts** per stroke (+0.1 per M) and **everything it hits catches fire** (burning 4 s; **blocks** too: 1 fire block at the hit point if `setsFires` and rights; a Fire Whip on an Ice Wall melts 2 blocks per stroke). Cost 0.5 exhaustion/s, max hold 10 s, cooldown 4 s. Movement ×0.8 while held.
Mastery	Length; M4: two whips (one per hand) when double-tapped: strokes alternate every 6 ticks; M7: a stroke **cuts a rock in flight** (2.1) or burns an Air Blade; M10: the whip **wraps**: a struck creature is pulled 2 blocks toward you (Zuko's catch).
Sneak: Whip Lash	One **long crack** at full length +3: **3 hearts**, burning 4 s, knockback 2; cooldown 3 s.
Counters	Water Whip crossing it: both end for 1 s; Water Shot on the wielder: the whip is out for 2 s; Air Shield: strokes at ×0.75; an Earth Wall blocks it (the whip strikes stone and scorches it: a smoke puff).
Edge	Under water: refused. Rain: strokes ×0.8 and no block fires. Nether: ×1.25. PvP off: no players. Flammable blocks around you: the whip's own body lights nothing; only the strike point does, and never in a claim.

5.5 Fire Circle — מעגל אש

Key	Slot; **press**. Sneak: **Fire Jet** (סילון אש, hold)
Unlock	5 SP (needs Fire Punch M3)
What (Eilon: "spin with your hands to send a ring of fire into everyone around you; a skilled bender also uses the legs, for stronger fire")	You spin (the gesture: 360° in 12 ticks, `ANIMATION.md` 5.5) and a **ring of fire** expands from you to **4 + 0.4M blocks** (8 at M10) over 10 ticks: everyone it crosses takes **2 hearts** (M1–5, hands only) or **3 hearts** (**M6+**: the leg sweep joins: a second, lower ring 4 ticks later) + burning 3 s + knockback 2; the ring **snuffs** incoming Water Shots... no: water beats fire: a Water Shot cuts a gap (1.7). Melts ice within the radius, puts nothing out. Fires: up to 12 fire blocks on the ring's edge if `setsFires` (never in claims).
Cooldown	8 s. M10: a **double circle** (press twice within 10 ticks: the second ring at ×0.5 damage).
Sneak: Fire Jet	**Hold**: jets from your feet and palms **lift you**: 0.9 blocks/tick upward at first then hover-climb 0.2/tick, steerable (W A S D at 0.3), for up to **3 + 0.3M s** (6 at M10): Ozai's and Azula's jet-flight, short bursts, not true flight; fall damage from the end of a jet ×0.5. Cost 1 exhaustion/s, cooldown 6 s. Also breaks a grip (Metal Grip, Quicksand, frozen feet).

Counters	Water: a Water Shot into the ring cuts it; a Water Spout rider is above it; Air Blast pushes the ring? no: snuffs the part of the ring inside its cone (1.7); Air Vacuum kills it; a Tornado over a Fire Circle makes a **firenado** (3.8). Earth Armour ×0.5. Ice Wall: melts a 2-block hole.
Edge	Under water: refused. In rain: ×0.8. Nether: ×1.25. A ring never crosses a Hex wall or a claim's edge (the fire blocks). Bosses: full damage, knockback ×0.2.

5.6 Fire Healing — ריפוי באש

Key	Slot; **hold**. Sneak: **Burn Out** (צריבת רעל, press)
Unlock	3 SP
What (Eilon: "heal others and help against poison; a bit weaker than water healing")	Hold on a **friend** you look at within 4 blocks (or yourself with no friend in sight): warm golden fire-light flows into them: **1 heart per 3 s** at M1 (1 per 2 s at M5, 1 per 1.5 s at M10: always a step behind Water's), and **Poison and Wither tick slower** (halved) while held; costs 0.5 exhaustion/s; max hold 20 s, rest 2 s; both must be still. (Chi-based healing from the dragons' teaching in *Korra*.)
Sneak: Burn Out	Burns the **Poison and Wither** out of the target at once (and 0.5 heart of damage to them: it hurts), plus Hunger; cooldown 15 s.
Mastery	Speed; M10: removes Weakness and Slowness too, and restores 1 hunger per 10 s.
Edge	Never hostile mobs or bosses. In water: works (it is chi, not flame). Nether: works. PvP: heals anyone.

5.7 Lightning Generation — יצירת ברק

Key	Slot; **hold to charge, release to fire** (min charge 10 ticks). Sneak: **Lightning Arc** (קשת ברק, press)
Unlock	8 SP (needs Fire ES ≥ 30, "cold-blooded fire")
What (Eilon)	Hold: the two-finger form circles (`ANIMATION.md` 5.7, Iroh's path: both arms in slow circles that meet at the stomach then extend), sparks crawl on your arms, and on release a **bolt** (a `LightningBoltFx` line with 2–4 branches, instantaneous raycast to **range 32 + M**, stopped by blocks) hits the first creature on the line: **4.5 hearts** (Eilon) +0.15 per M (5.85 at M9), **paralysis 2 s** (Eilon; a `bending:paralysed` effect: no movement, no attack, no bending; players' input locked; bosses 0.5 s), **×1.5 vs wet/water/metal**, ×0.5 vs Earth Armour (1.7); a bolt that hits water spreads to everyone in that water within 4 blocks (×0.5). Charging longer (up to 30 ticks) adds nothing but the look; you cannot move while charging (Slowness... locked).
Cooldown	**20 s** (Eilon: "a long cooldown"; `lightningCooldown` 400 ticks).
Mastery	Damage, range; M5: paralysis 2.5 s; **M10: both hands** (Eilon: "fully mastered: shoot it with both hands for 6 hearts"): the bolt is two bolts that join, **6 hearts**, 3 s paralysis, cooldown 25 s; the dual-hand form is the default at M10 and a tap (release before 10 ticks) gives the single bolt.
Sneak: Lightning Arc	A short crackling arc from your fingers to **one creature within 6 blocks**: **1.5 hearts**, paralysis 0.5 s, cooldown 5 s (a jab, Mako-style).
Lightning Redirection (5.8)	**A bolt aimed at a firebender who redirects** goes back at you (5.8).
Counters	An Earth Wall or Steel Wall stops it (the steel wall **conducts**: anyone touching the wall takes 1 heart); an Ice Wall is shattered (4 blocks); an Air Shield at M10 deflects one bolt; a lightning rod within 8 of the aimed point **draws** the bolt (vanilla's rod: a nice touch: `lightningRodsAttract` true). Standing in water yourself: you take **1 heart** when you fire (a hazard).

Edge	**Vanilla lightning:** the bolt is the mod's own FX and damage type (`bending:lightning`, in `minecraft:is_lightning` so creepers it hits become **charged**, villagers do not turn into witches? — vanilla does that on a real `LightningBolt` entity only: we spawn **no** vanilla bolt, so no witches, no fires, no copper oxidation changes: **(added)** "pure" lightning). Thunderstorm: ×1.25. Nether: ×1.0 (not fire). Bosses: full damage, 0.5 s paralysis. PvP off: no players. A paralysed player can still open the Bending screen (cannot act).

5.8 Lightning Redirection — הטיית ברק

Key	Slot; **hold** (the stance: Iroh's redirection path). Sneak: **Ground It** (הארקה, press)
Unlock	8 SP (needs Fire ES ≥ 35; "needs high mastery")
What (Eilon)	Hold the key as a bolt comes: while the stance is held, **any lightning that hits you** (another firebender's bolt or arc, **vanilla lightning**, Thor's beam if Enhanced is present and in `is_lightning`, Love's) **enters your arm, passes through your stomach** (the red-hot path drawn on your body and in first person, `ANIMATION.md` 5.8) **and leaves the other arm** where you look, **within 10 ticks of the hit** (you must release the key, or it fires along your look automatically at 10 ticks at M8+). Success chance **by mastery**: M1 20%, M2 30% ... **M9 100%** (`redirectChance` list; Eilon: "unskilled: the lightning will most likely hit you; skilled: 100% back"); on failure you take the bolt at ×1.25 (it went the wrong way through the heart: Iroh's warning) and the stance drops. On success you take **0** and the bolt leaves at **the same damage** (6 hearts if it was 6), with your paralysis duration, **at the attacker if you look at them, or anywhere else you look** (Eilon: "100% back at the attacker, or in another direction").
Hold	The stance may be held up to **5 s** (then 2 s rest, no cooldown: it is a stance); while held you move at ×0.5 and cannot use other abilities; it costs 0.3 exhaustion/s. At M10 you may also hold it **1 s after** a bolt begins its charge in your sight (the HUD flashes the attacker's charge: "Lightning!" when a firebender within 40 blocks starts charging at you).
Sneak: Ground It	Send the lightning you hold **into the ground**: no damage anywhere, a crater of sparks (no blocks), and **you heal 1 heart** (Zuko's lesson: the energy passes safely). Also discharges a vanilla thunderstorm bolt harmlessly.
Mastery	The chance (above); M5: the window is 15 ticks; M8: auto-fire at the window's end; M10: redirect **two bolts** in one hold (Azula's dual bolt, 5.7 M10, counts as one bolt of 6).
Edge	Standing in water: the redirected bolt also shocks you 1 heart (1.7). Steel Armour on: redirection **fails always** (it conducts through the armour: the HUD says so). Vanilla lightning (a storm strike): redirectable, 2.5 hearts, sets no fire, charges a creeper you aim at. Bosses: a redirected bolt deals full damage to them.

5.9 Fire loadout examples (added)

- **Beginner:** X Fireball / Small, C Fire Punch / Kick, Alt Flamethrower / Breath, K Fire Whip / Lash.
- **Zuko:** Fireball, Fire Whip, Fire Circle / Jet, Lightning Redirection.
- **Azula:** Fireball (blue), Flamethrower, Fire Circle / Jet, Lightning.
- **Iroh:** Fire Breath (Flamethrower), Fire Healing, Lightning, Lightning Redirection.

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6. Progression

6.1 The skill tree (one per element)

Each element's tree is a **graph of cards** (`UI.md` 2.2): tiers from the root (the free starter) outward. An ability can be unlocked when **its parent is unlocked and at the required mastery** and you have the SP. Unlocking gives the ability at mastery 1 and puts it in the first empty loadout slot if

any. Mastery levels are bought on the same card.

Earth tree

```
Rock Throw (free)
├─ Earth Pillar (2)                    [Rock Throw M1]
├─ Earth Wall (3)                      [Rock Throw M2]
│   └─ Sand Bending (3)                [Earth Wall M1, Rock Throw M3]
│       └─ Quicksand (4)               [Sand Bending M3]
├─ Earth Armour (4)                    [Rock Throw M2]
└─ Lava Bending (6)                    [ES ≥ 20]
Metal Bending (needs ES ≥ 30):
    Steel Knives (5) — Steel Wall (5) — Wreck Metal (6)   [each needs the previous at M1]
```

Air tree

```
Air Blast (free)
├─ Air Bubble (2)
├─ Air Blades (4)                      [Air Blast M2]
├─ Air Shield (4)                      [Air Blast M2]
├─ Air Scooter (4)                     [Air Blast M3]
│   └─ Tornado (5)                     [Air Blast M4, Scooter M1]
└─ Spiritual Projection (6)            [ES ≥ 15]
Zaheer's Path (a pet's death): True Flight (free), Suffocation (free)
```

Water tree

```
Water Shot (free)
├─ Healing (3)
│   └─ Water Octopus (5)               [Water Shot M3, Healing M1]
├─ Water Arms (4)                      [Water Shot M2]
├─ Ice Knives (3)                      [Water Shot M2]
│   └─ Freeze (3)                      [Ice Knives M1]
│       └─ Ice Wall (4)                [Freeze M2]
├─ Water Spout (6)                     [ES ≥ 15]
└─ Blood Bending (8)                   [ES ≥ 40]
    └─ Bending Severance (free)        [Blood Bending M10]
```

Fire tree

```
Fireball (free)
├─ Fire Punch (2)
│   └─ Fire Circle (5)                 [Fire Punch M3]
├─ Flamethrower (4)                    [Fireball M3]
├─ Fire Whip (4)                       [Fireball M2]
├─ Fire Healing (3)
├─ Lightning Generation (8)            [ES ≥ 30]
└─ Lightning Redirection (8)           [ES ≥ 35]
```

6.2 The mastery curve

Level	XP needed (cumulative uses)	SP to buy (optional, ≤ level 7)	Typical time for an active player
1	0	—	at unlock
2	10	1	first evening
3	25	2	

Level	XP needed (cumulative uses)	SP to buy (optional, ≤ level 7)	Typical time for an active player
4	50	2	a week of play
5	90	3	
6	150	3	
7	230	4	a month
8	340	— (use only)	
9	480	—	
10	660	—	the long game

Mastery XP is also shown as a thin bar under the ability's icon on the HUD. A level-up plays the element's chime and a small particle ring, and says "Rock Throw: mastery 5" in the action bar.

6.3 Full-tree cost (for tuning)

Element	Unlock SP (all cards)	Buying every mastery to 7	Total
Earth	1+2+3+3+4+4+6+5+5+6 = 39 (the first 3 free)	10 cards × 15 = 150	~186
Air	2+4+4+4+5+6 = 25 (Zaheer's two free)	9 × 15 = 135	~157
Water	3+5+4+3+3+4+6+8 = 36	10 × 15 = 150	~183
Fire	2+5+4+4+3+8+8 = 34	8 × 15 = 120	~151

(Buying a card's levels 2–7 costs 1+2+2+3+3+4 = 15 SP.) At 1 SP per zombie and 25 per Warden, a full tree is a long but reachable goal; the numbers are Eilon's open question 6 and all in `[progression]`.

---

7. Balance tables (one row per ability)

Hearts; range in blocks; cooldown in seconds; "M" = scales with mastery (the formula is in the ability's section).

7.1 Earth

#	Ability	Key	Type	Cost SP	Dmg (M1 → M10)	CD	Range	Duration	Sneak variant (dmg, CD)	Counters
E1	Rock Throw	slot	press	0	2 → 3.8 (boulder 6)	1.5 (3)	24+M	—	Hold Rock (same, —)	Earth Wall, Air Shield, Octopus
E2	Earth Armour	slot	on/off	4	shell 12 → 25.5; +8 armour	20 after	self	∞ (0=∞)	Shed Blast (1.5, 20)	Water ×1.5, Freeze
E3	Earth Pillar	slot	press	2	aimed 1	3	self / 12	15 s	Earth Launch (0, 4)	Rock breaks it
E4	Earth Wall	slot	press	3	—	8	3 ahead	10 s	Wall Push (1.5, 10)	Rocks, Lava Shuriken, erosion

#	Ability	Key	Type	Cost SP	Dmg (M1 → M10)	CD	Range	Duration	Sneak variant (dmg, CD)	Counters
E5	Sand Bending	slot	press	3	Stream 1 + blind	2.5	10 → 16	wall 6 s	Sand Wall (—, 6)	Water, Air Blast
E6	Quicksand	slot	press	4	0.5/s	10	12	4 → 7.6 s	Sand Cloud (0.25/s, 10)	stronger ES, Pillar, Air Jump
E7	Lava Bending	slot	press	6	lava; shuriken 3 + burn	12 / 4	6+M / 24	12 s	Lava Shuriken (3, 4)	Water, Ice Wall
E8	Steel Knives	slot	press	5	1.5 × 3 → 7	2.5	32	—	Metal Pull (—, 6)	Earth Wall, Air Shield, Steel Armour
E9	Steel Wall	slot	press	5	—	8	3 ahead	10 s	Steel Armour (shell 16 → 25, 20)	Explosions, Lightning ×1.5
E10	Wreck Metal	slot	press	6	golem 2 (M10)	4	10	60 s wrecked	Metal Grip (fling 3, 8)	shed armour, Fire Jet

7.2 Air

#	Ability	Type	Cost SP	Dmg	CD	Range	Duration	Sneak variant	Counters
A1	Air Blast	press	0	0.5 + throw 4 → 10 blocks	2	8+M cone	—	Air Jump (—, 2)	Earth/Steel Armour ×0.5
A2	Air Shield	on/off	4	burst 3.5 (or 4)	12 broken / 6 lowered	r 1.6 → 2.4	20 s (0=∞)	Shield Burst (3.5, 12)	rocks eat it, Tornado
A3	Air Glider	item	craft	—	—	—	∞	—	—
A4	Air Blades	press	4	2.5 → 3.4	1.5	20+M	—	Air Slice (2, 3)	any wall, Fire Whip M7
A5	Spiritual Projection	press	6	—	60 (30)	64+8M	30+6M s	Deep Meditation (heal, 30)	hit the body
A6	Air Bubble	on/off	2	—	—	self (+1 friend M6)	∞ in water	Air Cushion (on/off)	Spout push
A7	Air Scooter	on/off	4	burst 2 (M10)	5	speed 14 → 19 b/s	60 s (0=∞)	Air Spout (—, 6)	Water Shot, walls
A8	Tornado	press	5	1 + 0.5/s	8	16+M	4 → 7 s	Air Wheel (1, 10)	walls, boulder, Spout
A9	True Flight	on/off	pet	—	—	—	∞	Hover	—
A10	Suffocation	hold	pet	1/s	20	8+M	6 → 10 s	Air Vacuum (1/s, 15)	distance, damage, Bubble

7.3 Water

#	Ability	Type	Cost SP	Dmg	CD	Range	Duration	Water cost	Sneak variant	Counters
W1	Water Shot	press	0	1.5 → 2.85 (×1.5 fire)	1	20+M	—	1	Water Whip (1/lash, 4)	Fire Whip, walls
W2	Healing	hold	3	heal 0.5/s → 1/s	—	4	20 s hold	1 / 4 s	Cleanse (—, 10)	—
W3	Water Octopus	on/off	5	+1 → +1.9 per hit	10	2.5 → 4	40 s (0=∞)	3	Octopus Lash (2, 4)	Flamethrower, Freeze, Lightning
W4	Water Spout	hold	6	ram 4 → 7	10	h 3 → 12	15+M s	4	Wave Ride (1, 8)	Lightning, Air Blast, Freeze
W5	Water Arms	on/off	4	+1 → +1.9; reach 4.5 → 9	6	reach	60 s (0=∞)	2	Arm Grab (1, 4)	Flamethrower, Freeze
W6	Ice Knives	press	3	1.5 × 3 → 7 + slow	2	24	—	1 / 3	Ice Spikes (2, 6)	Fire ×1.5 melt, walls
W7	Freeze	press	3	0.5 + root 2 → 4 s	4	16	ice 20+2M s	1 / 10	Snow Throw (0.5 × 4+M, 2)	Fire, Pillar, Air Jump
W8	Ice Wall	press	4	—	8	midpoint	10 s	3	Ice Dome (—, 15)	Fire melts, rocks
W9	Blood Bending	hold	8 (ES≥40)	0.5/s	25	10+M	4 → 10 s	0	Blood Fling (3, 25)	stronger ES, Lightning, 2 hearts
W10	Bending Severance	press	free @M10	—	120	2	9 min → 55 s	0	—	ES, Healing M10

7.4 Fire

#	Ability	Type	Cost SP	Dmg	CD	Range	Duration	Sneak variant	Counters
F1	Fireball	press	0	2 → 3.8 (great 5) + burn	1.2 (4)	24+M	—	Small Fireball (0.5, 0.5)	Water Shot, Air Blast, walls
F2	Fire Punch	on/off	2	+1.5 → +2.5 + burn	—	3	60 s (0=∞)	Fire Kick (2, 3)	Octopus, Earth Armour

#	Ability	Type	Cost SP	Dmg	CD	Range	Duration	Sneak variant	Counters
F3	Flamethrower	hold	4	1 per 0.5 s (0.4 @M4)	4 rest	8 → 14	6 s hold	Fire Breath (1.5/0.5 s, 6 rest; M7)	Water Shot, Air Blast, Vacuum
F4	Fire Whip	hold	4	1.5 → 2.4 per stroke + fire	4	5 → 12	10 s hold	Whip Lash (3, 3)	Water Whip, Water Shot
F5	Fire Circle	press	5	2 (3 from M6) + burn	8	r 4 → 8	—	Fire Jet (—, 6)	Water Shot gap, Air Vacuum
F6	Fire Healing	hold	3	heal 0.33/s → 0.67/s	—	4	20 s hold	Burn Out (−0.5 to target, 15)	—
F7	Lightning	charge/release	8 (ES≥30)	4.5 → 5.85; 6 dual @M10 + paralysis 2–3 s	20 (25)	32+M	—	Lightning Arc (1.5, 5)	walls, Redirection, Air Shield M10
F8	Lightning Redirection	hold	8 (ES≥35)	returns the bolt (20% → 100%)	—	5 s hold	—	Ground It (heal 1)	Steel Armour fails it

---

8. Fun extras (suggestions; Eilon decides)

Everything here is **(suggestion)**: not in Eilon's spec, proposed because it fits the shows. Each is a separate lane in `PLAN.md` so saying no costs nothing.

8.1 Being the Avatar (later idea, not built)

A fifth choice that bends all four elements, with the Avatar State (all abilities ×1.5 for 20 s when below 3 hearts, glowing eyes and tattoos, the pose), the past lives as advice lines, and a world rule of **one Avatar per world**. It changes the balance of everything above, so it is a 2.x idea: the data model keeps room for it (`BenderData.elements` is a set, not a single value; `TECH.md` 4).

8.2 Training dummies

A block (`bending:training_dummy`, crafted from hay, sticks and a pumpkin... a wooden post with a straw body) that takes bending hits and gives **mastery XP at half rate**, up to mastery **5** (you cannot master anything on a dummy), shows the damage number above it for 1 s, and counts hits per minute. A **metal dummy** (iron bars + the dummy) trains metal and lightning; a **water trough** next to it is a water source.

8.3 Bending scrolls in structures

Four kinds of **Bending Scroll** (Earth/Air/Water/Fire), found in the loot of strongholds, jungle temples (Earth), desert pyramids (Fire), ocean monuments and shipwrecks (Water), and the end cities and tall mountains' structures of 8.4 (Air). Reading one of **your** element gives **3 SP** (or 1 mastery level to a chosen ability, up to 7); another element's scroll can be traded to a bender of that element or **sold** to the Tutorial Master (8.4) for 1 SP.

8.4 Element temples and a tutorial master per element

Four structures, one per element, rare (one per ~2,000 blocks): the **Earth Kingdom gatehouse** (walls, a Dai Li training ground, badgermole statues) in mountains and savannas; the **Air Temple** (a mountaintop spire with spinning gates, a glider launch) on peaks; the **Water Tribe pavilion** (an ice dome on a frozen ocean, a healing hut, a spirit oasis pool) in frozen oceans; the **Fire Sages' temple** (a volcano-side pagoda with braziers and the Sun Warriors' dragon stairs) in badlands and basalt deltas. Each holds a **Tutorial Master** NPC (an element master in the look of a teacher: a Bumi-like earth master, Gyatso-like monk, a Pakku-like elder, a Jeong Jeong-like hermit; **not** named after the characters, so Eilon's looks stay the player's) who: explains the element in 6 short lines (EN/HE), gives the first free scroll (8.3) once per player, trains (a dummy field), and sells looks' cosmetics for scrolls. Enhanced's `AsgardTalkScreen`/NPC goal patterns are the model.

8.5 A boss per element

Four spirit bosses in the temples' depths (Enhanced's boss framework pattern, `TECH.md` 2): the **Badgermole King** (Earth: tunnels, rock walls, seismic slams; drops the Earth boss scroll: +30 SP and the **Lava Bending** unlock for free), the **Storm Spirit** (Air: a tornado that must be dispelled by Air Blasts; drops the Air scroll and a **sky bison egg**, 8.7), **Koh-like "the Faceless"** (Water: in the spirit oasis, moves only when you are not looking at him, steals your look for the fight; drops the Water scroll and **Spirit Water**, 1.6), and the **twin dragons Ran and Shaw** (Fire: a dance, not a fight: match their fire with Fire Circle and Fireball in rhythm; they teach the **true colours**: a rainbow fire cosmetic). Each 30 SP, once per player.

8.6 Duels and ranks

`/bending duel <player>`: a consented 1v1 with PvP on inside a 24-block ring for 3 minutes, no SP for the kill, a **rank** per element on a scoreboard (Elo-like: `BenderRank`), shown as a small badge on the HUD. Pro-bending ring (three zones, push the other team back) as a 2.x idea.

8.7 Companions

Sky bison (a flying mount from the Storm Spirit's egg or found at the Air Temple; six legs, a saddle for 3, `ModelPart` model, flies at 0.6 blocks/tick, lands to graze; Appa's "yip yip" is a synthesised call), **winged lemur** (a shoulder pet like a parrot, fetches dropped items), **polar bear dog** (Naga: a ride and a guard, from the Water pavilion), **fire ferret** (Pabu: a shoulder pet that warms you, puts out your burning). Each is a `TamableAnimal`, so its death counts for Zaheer's path (3.9), which is a story beat on purpose.

8.8 Small things that fit

Sozin's Comet (a 1-day event every ~40 days: fire ×2, the sky red); the **Day of Black Sun**... no eclipse exists; **Pai Sho** table block (decor); **Cabbage merchant** (a wandering trader variant whose cart you must not break); **Tea** (Iroh's jasmine tea: a drink that restores 2 hunger and gives

10 s of ×1.1 bending); **White Lotus tile** item as the badge of a player who unlocked every card.

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9. What is deliberately not in 1.x

- The Avatar (8.1), chi-blocking, combustion bending (Sparky Sparky Boom Man), seismic sense as a full x-ray (only Earth Armour's M8 glow), platinum, spirit portals, the Spirit World as a dimension, Vaatu/Raava, mecha tanks, bending for mobs (no bending zombies). Each is a clean later lane.

ANIMATION.md — Animation and effects bible

Bending — Animation and Effects Bible

This is the build reference for every movement and every effect of the mod. It is written against **Enhanced's animation system**, which Bending copies into `com.kolodkingal.bending.client.render` (see `TECH.md` 2 for the exact class list): gestures (`GestureId` + `Gestures.define` keyframes), stances (`Stance` + `Stances` rows), flight and run profiles, `FirstPersonArms.Offset`, `HeroAnimator` blending, `GestureEffects` flourishes, `HandTracker`, and the FX toolkit (`EnergyGeometry`, `FxShapes`, `SandShapes`, `EnhancedRenderTypes`, `ShieldFx`, `BeamRenderer`, `EnergyParticle`). Where a building block does not exist in Enhanced, section 14 lists it as **NEW** and names the lane that builds it.

0. How to read this document

0.1 Units and signs (Enhanced's conventions, unchanged)

- **Ticks.** 20 per second. A keyframe's time is the tick it is reached.
- **Angles are degrees**, written for a **right-handed** player; the pipeline mirrors a left-handed one. `M` is the **main** arm (right), `O` the **off** arm (left). Arm angles are `(x, y, z)`:
  - `x`: negative raises the arm **forward and up** (−90 is level forward, −176 is straight overhead); positive swings it **back**. 0 hangs at the side.
  - `y`: turns the arm about its own axis / inward-outward in the horizontal plane (for `M`, +y swings the hand across the body).
  - `z`: for `M`, **+z spreads the arm outward** (away from the body); for `O`, **−z** spreads outward. So a symmetric pose is written `m(x, y, z)` and `o(x, −y, −z)`.
- **Legs** in gesture keys drive **x only** (`legs(right, left)`: negative lifts the leg forward, positive back). Leg **z** (spread) is available in **stance rows** (`c.leg(right, x, z, w)`), and Bending adds it to gesture keys (**NEW**, 14.1), so a horse stance and a kick can be keyed.
- **Head** `(x, y, z)` is **additive** on top of the look (x down positive).
- **Whole body:** `lean` (degrees, forward positive; a pose-stack pitch about a pivot 0.40 blocks up on the ground), `roll` (bank), `yaw` (a visual turn; must start at 0 and end on a multiple of 360), `drop` (blocks the body sinks; positive is down), `twist` (shoulder twist in degrees: becomes `body.yRot`).
- **Ease codes** (the curve of the segment *arriving* at the key): `S` smoothstep (default), `O` ease-out cubic (snap then settle), `I` ease-in cubic (accelerate into an impact), `L` linear, `E` sine in-out, `B` ease-out-back (10% overshoot). The gesture's **weight envelope** is `envelope(t, attack, dur, release)`: cubic rise over `attack`, hold, sine fall over `release`.
- **Modes:** a plain `m(x,y,z)` is ABS (lerped toward by the weight); `mAdd` is ADD (added to what is below); `mWorld(phi, y, z)` holds the arm at a **world elevation** (0 hangs, 90 level, 180 up) regardless of body lean; `mAim(Target, dx, dy, dz)` points the arm at a target (`LOOK`, `HELD`, `PROJECTILE`, `ANCHOR`, `STRIKE`…) plus an offset.
- **Tremble:** `mTremble(amplitude°, speed rad/tick)` on a key; off under *Reduced motion*.
- **Groups:** at most one gesture per group plays; a new one fades the old over 3 ticks. Bending's groups: `CAST` (every ability), `TAKEOFF`, `LAND`, `BRAKE`, `RUN`, `RIDE` (**NEW**: mount/dismount of scooter and spout), `HIT` (**NEW**: hit reactions, so they never cancel a cast).

- `force`: the gesture replaces the vanilla punch on the arms it holds. Every attack gesture below is `force = true`; stances, rides and reactions are not.

0.2 Keyframe tables

Each ability has a keyframe table in this shape:

| t | ease | M (x,y,z) | O (x,y,z) | legs R/L x (z) | head | body (lean / twist / drop / yaw) | notes |

An empty cell means the channel is **not keyed** at that time (the layer below shows through: walking legs keep walking). A dash in a whole column means the channel is never driven by this gesture. `REST` means the arm's rest pose of the current stance (`EARTH_REST_M` etc., 1.2). After the table: **timing** (`dur / attack / release`, group, force), **lock** (does it stop movement? how: a `MovementInputUpdateEvent` lock bound to a `Live` bit, never the gesture itself), **blend** (what it does over walking, flying, swimming, sneaking), **started by** (the `GestureRoutes` arm: the ability's cooldown edge, a `Live` bit, a toggle flip, or an FX payload kind), **prediction** (the client's own key press starts it at once when the client can tell it will work).

0.3 First person

Hand motion is `FirstPersonArms.Offset` per hand: `(dx, dy, dz, rx, ry, rz)` in view space (blocks: +x right, +y up, +z away from the camera; degrees) in right-hand signs; the left hand mirrors `dx`, `ry`, `rz`. The empty off hand slides in when used (`used > 0.01`). Offsets are capped at 0.8 blocks. **Camera** effects go through `ViewportEvent.ComputeCameraAngles` (shake: pitch/roll sine with distance falloff and the reduced-motion gate) and `ComputeFovModifierEvent` (FOV). **Overlays** are `LayeredDraw.Layer`s above `CAMERA_OVERLAYS`. All of this exists in Enhanced and is reused as is.

0.4 Effects

Particles are Enhanced's `EnergyParticle` (`bending:energy`, colour + scale, fullbright, 10–21 ticks, friction 0.9) plus vanilla types (`BLOCK` dust of a state, `DUST`, `SPLASH`, `FLAME`, `SMALL_FLAME`, `SOUL_FIRE_FLAME` for blue fire, `ELECTRIC_SPARK`, `CLOUD`, `SNOWFLAKE`, `ITEM_SNOWBALL`, `FALLING_WATER`, `DRIPPING_WATER`, `BUBBLE`, `POOF`, `CAMPFIRE_COSY_SMOKE`, `LAVA`, `DRIPPING_LAVA`, `ASH`, `WHITE_ASH`, `GLOW`, `END_ROD`, `SONIC_BOOM`, `GUST`, `SMALL_GUST`, `TRIAL_SPAWNER_DETECTION`), and **two new particle types** (**NEW**, 14.5): `bending:grain` (a tinted sand/earth grain with gravity, 20–40 ticks) and `bending:droplet` (a water bead with gravity and a splash on landing). Meshes are drawn in `RenderLevelStageEvent.AFTER_PARTICLES` into the copied render types: `ENERGY_BEAM` (additive, for fire, lightning, glows), `ENERGY_VEIL` (translucent, for air shapes and water bodies), `MATTER_FLAT` and `MATTER_TEXTURED` (Enhanced's `SandRenderTypes.FLAT` / `textured`: normal alpha, for earth, sand, water and ice surfaces), `energyField(tex)` (textured additive lattice: the air shield's lines). Rule from Enhanced: **matter is not light**: earth, sand, water and ice use normal alpha; fire, lightning and spirit glows use additive.

Budgets. Every effect row gives *particles per tick (or per event)* at density 1.0. The client option *Effect density* (0.25–1.0, default 1.0; `bending-client.toml [effects] density`) multiplies every count; vanilla's Particles option (All / Decreased / Minimal = 1.0 / 0.5 / 0.15) multiplies again, as Enhanced's `SandFx.density()` does. Nothing is spawned beyond **48 blocks** from the camera, meshes are not drawn beyond **96**, and each player's effects are capped (per-player caps in the rows). The design target: **4 benders fighting at once stay under ~600 live particles and ~40 mesh draws per frame.**

Lighting. No dynamic world light exists in Enhanced and none is added (no light-block tricks: they would be temp blocks with light updates, the one thing that lags). Fire, lightning and spirit forms are drawn **fullbright and additive**, which reads as emissive; fire casts no light on the ground. **(suggestion, 14.7):** a client-only "fake light" pass is possible later (a translucent warm disc under a flame, `MATTER_FLAT`), cheap and loader-free.

0.5 Sound cues

Each ability lists **wind-up / release / impact (or loop)** cues by event name; the synthesis recipe and both subtitles are in `AUDIO.md`. Loops follow the player through the copied `PowerLoopSounds` (a `Kind` per loop with a predicate on `BenderData`).

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1. The four styles, as poses

1.1 What the shows do, in Minecraft's joints

Element	Martial art	Body	Arms	Legs	Rhythm
Earth	Hung Gar	upright, heavy, **low**: horse stance (`drop` 0.18–0.32), lean forward 4–8°, shoulders square	fists and **open palms pushed from the hip**, elbows bent, slow wind-up then a hard stop (`I` into the hit, `0` out)	**wide**: legs spread z ±14, stomps (a leg lifts to x −35 then slams, `drop` dips 0.1)	wait, root, strike: long attack ticks, short release
Air	Ba Gua Zhang	light, upright, often **turning** (`yaw` 90–360), lean with the circle, `drop` 0	**open palms**, arms drawing **circles** (y and z sweep through 60–120° over 10–16 ticks), one arm leads, the other follows a quarter turn behind	**stepping**, never planted: legs x alternate ±18 as the body turns (walk cycle keeps running under most gestures)	continuous: `S` and `E` easings, weights overlap, no hard stops except the blast's release
Water	Tai Chi	upright, **soft knees** (`drop` 0.06–0.10), slow weight shifts (`lean` ±4 alternating)	**flowing**: both arms move together in **S-curves**, wrists lead (y leads x by 2 ticks), palms turn over (y sweeps), pulls and pushes	shoulder-width, one leg forward (x −12 / +12), weight shifts between them	slow in, quick out: a 12-tick gather, a 4-tick release; `E` everywhere, `0` on the release
Fire	Northern Shaolin	upright, **long** bow stance (front leg x −30, back leg x +25, `lean` 10), chest forward	**straight punches** and chops: arms snap to level forward (x −90) in 2–3 ticks with `0`, pulled back with `I`; fists	**kicks** (leg x −80 to −110, z ±20) and stamps	explosive: attack 1–2 ticks, hold 2, release 4–6; the **breath**: a 6-tick inhale (chest up: head −6, lean −3) before big moves

Element	Martial art	Body	Arms	Legs	Rhythm
Blood	(Hama / Amon)	hunched, lean 12–18°, head down 6, shoulders in (twist ±6 alternating)	**claw hands**: arms forward and up at −70, y ±25 inward, z toward the centre, **fingers** implied by a 3° tremble at 2.2 rad/tick and small y pulses; one hand pulls, the other pushes	feet planted, knees bent (`drop` 0.12)	slow, trembling, the puppet's limbs follow the hands by 2 ticks
Lightning	Iroh's form	upright and still, lean 0, breath in	the **two-finger** path: both arms draw **slow circles** (14 ticks) that **meet at the stomach** (x −40, y ±40, z toward the centre), then the leading arm **extends** to level (x −90, y 0) with the other braced at the hip; redirection runs the path **in reverse** through the stomach and out the other arm	bow stance, planted	24-tick wind-up, 2-tick release

1.2 Pose library (the `StancePoses` of Bending)

These are the constants every table below refers to (degrees, right-handed; `_0` is the mirror unless given).

Name	M (x,y,z)	O (x,y,z)	legs R/L x (z)	head	lean / drop	Meaning
`NEUTRAL`	0,0,8	0,0,−8	0/0 (±2)	0	0 / 0	vanilla rest
`EARTH_REST`	8,0,14	8,0,−14	0/0 (±6)	0	2 / 0.04	heavy shoulders
`EARTH_HORSE`	−38,12,40	−38,−12,−40	0/0 (±16)	2	4 / 0.24	horse stance, fists at the hips' height, forearms level
`EARTH_PUSH`	−88,−4,6	−60,10,−30	0/0 (±14)	0	8 / 0.18	palm thrust, rear arm chambered
`EARTH_LIFT`	−120,0,30	−120,0,−30	0/0 (±12)	−6	−6 / 0.10	both palms up as the rock rises
`EARTH_CLAW`	−72,18,10	20,0,−20	0/0 (±10)	0	4 / 0.12	the holding hand, fingers down
`EARTH_STOMP_UP`	−30,0,30	−30,0,−30	−36/0	−4	−4 / 0	the right leg raised
`EARTH_STOMP_DOWN`	−20,0,34	−20,0,−34	0/0 (±16)	6	6 / 0.30	the slam
`EARTH_GUARD`	−100,30,20	−100,−30,−20	0/0 (±14)	0	2 / 0.20	forearms crossed before the chest (the wall rises)
`AIR_REST`	−10,0,10	−10,0,−10	0/0 (±3)	−2	0 / 0	light, palms open
`AIR_PALMS`	−70,−30,−10	−70,30,10	—	0	0 / 0	both palms forward, close together

Name	M (x,y,z)	O (x,y,z)	legs R/L x (z)	head	lean / drop	Meaning
AIR_CIRCLE_A	−120,−40,40	−30,20,−10	−10/10	0,−10,0	−2 / 0	a quarter of the circle: leading hand high
AIR_CIRCLE_B	−60,40,−30	−100,−40,30	10/−10	0,10,0	2 / 0	the opposite quarter
AIR_GUARD	−60,−20,−24	−60,20,24	0/0 (±6)	0	0 / 0.02	palms up and out: the shield
AIR_SIT	−20,30,−40	−20,−30,40	−90/−90 (±55)	4	0 / 0.72	meditation: cross-legged, hands on the knees (**NEW** leg z in keys)
WATER_REST	4,6,10	4,−6,−10	−6/6	0	0 / 0.06	soft knees, one foot forward
WATER_GATHER	−40,30,−20	−40,−30,20	−12/12	2	−3 / 0.10	both hands drawn in at the belly, palms up
WATER_PUSH	−86,10,0	−86,−10,0	−14/16	0	6 / 0.08	both palms out, level
WATER_WAVE	−110,−30,30	−50,30,−20	−10/10	0	2 / 0.08	"wave hands like clouds": high hand leads
WATER_WHIP	−70,40,−10	−30,−20,−20	−10/10	0	4 / 0.08	one arm cocked, wrist leading
WATER_HEAL	−60,20,−10	−60,−20,10	−6/6	6	2 / 0.08	palms down, hovering
FIRE_REST	6,0,12	6,0,−12	−8/8 (±6)	0	2 / 0.02	ready, fists loose
FIRE_BOW	−30,0,34	−30,0,−34	−30/25 (±8)	0	10 / 0.14	bow stance, fists chambered at the hips
FIRE_PUNCH	−90,0,0	−10,0,−30	−30/25 (±8)	0	12 / 0.12	straight punch, rear fist chambered
FIRE_DOUBLE	−90,−6,4	−90,6,−4	−30/25 (±8)	0	12 / 0.12	both fists forward (flamethrower)
FIRE_BREATH	−40,0,50	−40,0,−50	−20/20 (±10)	−14	−8 / 0.04	arms back and wide, head back, chest open
FIRE_KICK	−60,0,40	40,0,−20	−105/15 (+22/0)	0	−10 / 0.02	right leg at head height (**NEW** leg z)
LIGHTNING_MEET	−40,38,−24	−40,−38,24	−24/22 (±8)	4	4 / 0.10	fingers meeting at the stomach
LIGHTNING_POINT	−92,0,2	−20,10,−16	−30/25 (±8)	0	6 / 0.10	the two fingers extended
REDIRECT_IN	−92,0,2	−30,0,−30	−24/22 (±8)	0	2 / 0.12	one arm receives (level), the other low
REDIRECT_OUT	−30,0,30	−92,0,−2	−24/22 (±8)	0	2 / 0.12	the bolt leaves the other arm
BLOOD_CLAW	−72,26,−6	−66,−22,8	0/0 (±8)	6	14 / 0.12	both claws forward, hunched
BLOOD_HIGH	−120,10,−10	−60,−24,10	0/0 (±8)	2	10 / 0.10	one claw raised (the puppet lifts)

Name	M (x,y,z)	O (x,y,z)	legs R/L x (z)	head	lean / drop	Meaning
`PUPPET`	−100,−40,−60	−100,40,60	−20/30 (+25/−25)	−10,20,0	−10 / −0.4 (lifted)	the victim: arms wrenched up and back, legs apart, head back
`PARALYSED`	−20,0,46	−20,0,−46	0/0 (±14)	−8	−6 / 0	rigid, arms out, 2° tremble at 4.0

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2. Shared first-person and screen rules

2.1 Hands

Every bender's first person follows Enhanced's `FirstPersonArms`: the gesture's hand motion in view space, the ambient stance motion, and the element's **idle hands** when a slot key with a hold or ready stance is held (section 9). Blue fire, water ribbons, rock chips and sparks are spawned at the drawn hand position (`FirstPersonArms.mainHand(player, out)`), as Enhanced's beams are, so a Water Whip really hangs from your hand.

2.2 Overlays (`LayeredDraw.Layer`s above `CAMERA_OVERLAYS`)

Overlay	When	What is drawn	Fade
Earth Armour eye slit	Earth or Steel Armour on, first person	a stone (or steel) frame covering the screen except a horizontal slit 22% of the height centred at 42% from the top, with rough inner edges (a 32-segment jagged polyline, `MATTER_FLAT`, dark ochre `#3A2E1C` at α 0.92; steel: `#2B2F36`); cracks (white lines) grow as the shell's health falls below 50%, 30%, 15%; the slit widens by 6% at Earth Armour M8+	forms over 20 ticks from the edges inward (the armour's forming time); crumbles out in 6 ticks with chips
Spiritual projection	the spirit is out	a cool blue vignette (28 nested outlines as Enhanced's `PhoenixOverlay`, `#7FD0FF` α from 0.35 at the edge to 0), a faint waving ribbon border (sine 0.6 Hz), saturation ×0.7 and a blue tint via a post shader `bending:shaders/post/spirit.json` (vanilla `color_convolve`: Saturation 0.7, ColorScale 0.85/0.95/1.2), a thin **tether line** at the bottom centre with the distance to the body ("64 m") and a compass tick toward it; through-wall glows (`EntityGlow`: hostile mobs light blue, players white, villagers green)	in over 10 ticks at departure, out over 6 at return (with a white flash 0.3 at the snap-back)
Blood-bending tint	a Blood Bending hold is active (caster)	a red vignette (`#7A0E1E` α 0.30 at the edges), pulsing with a slow heartbeat (0.9 Hz: two beats), the moon drawn large and pale at the top-left corner behind the HUD	in 8 ticks, out 12
Being blood-bent (victim)	`Live.BLOODBENT` on you	a stronger red vignette (α 0.5), the view **tilts** ±3° with the puppet's sway (camera roll), a heartbeat at 1.4 Hz	out over 20 ticks when freed
Full-moon glow	a waterbender outdoors on a full-moon night	a soft silver sheen (`fillGradient` top third `#E8F0FF` α 0.08 → 0) and the HUD moon icon glows (`UI.md` 3.6)	—
Paralysed	`bending:paralysed` on you	a white-blue flash 0.4 then a static-noise tint: 6 horizontal jittering bars (random y each 2 ticks, α 0.15) for the duration	out at once

Overlay	When	What is drawn	Fade
Quicksand	you stand in quicksand	the bottom 30% of the screen is a sand gradient (`#D8B76A` α 0.5 → 0) that rises as you sink	—
Underwater bubble	Air Bubble on, under water	the water fog is removed (`ViewportEvent.RenderFog` set far plane 128, cancel) and a thin circular bubble rim is drawn (a ring at the screen's edge, α 0.25, with 6 slow highlights)	—
Severed	bending severed	every HUD icon greys (`UI.md` 3.5); no vignette (the loss is quiet)	—
Fire Breath	breathing	a heat ripple: the bottom half of the screen's hue shifts +3° every 2 ticks (a cheap wobble via two alternating `fillGradient`s at α 0.06)	—

2.3 Camera

- **Shake** (never with *Reduced motion*): Earth stomps and landings (`strength 0.6`, 8 ticks, within 12 blocks), boulder impacts (0.8, 10, 16), Earth/Steel Wall rising (0.3, 6, 8), Shed Blast (0.7, 8, 12), Lava Path (0.2, 4, 6), Air Shield burst (0.5, 8, 10), a Tornado within 6 blocks (0.15 continuous), Water Spout ram (0.4, 6, 8), Lightning hit (0.9, 6, 20: a sharp high-frequency jolt: pitch `sin(5.1·age)`), Fire Circle (0.3, 6, 8), Shatter of your own armour (0.5, 8). Formula as Enhanced's `ArenaFx.shake`: `pitch += s·fade·sin(2.3·age)`, `roll += 0.6·s·fade·sin(1.77·age+1)`.
- **FOV**: Air Scooter and Air Wheel ×1.12 (eases in 6 ticks); True Flight at full speed ×1.08; Water Spout ride ×1.05; Fire Jet ×1.1; Lightning charge − ×0.96 (a slight pull-in over the 24 ticks, then snaps to 1.0 on release); Spiritual projection ×1.05 constant.
- **Roll**: Air Scooter banks the camera 4° into turns (lerp with `turn`), glider 6°; True Flight 5°.

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3. Earth

3.0 Earth idle: the ready stance (held slot key, or the element's `IDLE`)

Stance	tauIn / tauOut	Rows
`EARTH_IDLE` (ambient, on while a bending key is held with nothing to fire, or 40 ticks after any Earth gesture)	5 / 6	`EARTH_HORSE` at weight w: arms lerp to the horse, legs spread z ±16·w, `drop` 0.24·w, lean 4·w; a slow **breath**: arms x ±2.5 at 0.09 rad/tick; every 60 ticks a tiny weight shift (lean alternates ±1.5 over 20 ticks). Head looks along the camera (the shared head rule).
`EARTH_GRIP` (aimed, while `Live.EARTH_HOLDING`)	2 / 4	`EARTH_CLAW`: M aims at the held rock (`aim(HELD, −20, −6, 0)`) with tremble 2.4 at 1.9; O braced back; lean 4; `occupy(M)`.
`EARTH_ARMOURED` (ambient, while Earth/Steel Armour is on)	6 / 6	arms +10° x heavier swing, z ±4 wider, `drop` 0.03; the run character `EARTH_ARMOUR` (0.9 arms, 0.8 legs, lean 6, bob 0.035: a stomping walk).

Ambient particles of a master (Earth ES ≥ 60): 1 `bending:grain` (the ground's colour) every 10 ticks rising 0.3 blocks around the feet and settling; at ES 100 a faint ochre dust ring (radius 1.2, `MATTER_FLAT`, α 0.08) under the feet while standing still. Budget: ≤ 2 particles/s per master.

Earth first-person idle (key held): both hands low and wide, palms down (`M dx +0.10 dy −0.14 dz +0.05 rx +20`, O mirrored), breathing ±0.01 in dy.

Earth sound bed: none while idle; a master's ambient grain has no sound.

3.1 Rock Throw — זריקת סלע

Third person, `EARTH_ROCK_THROW` (dur 16 / attack 2 / release 6, group CAST, force):

t	ease	M	O	legs R/L (z)	head	body	notes
0	S	EARTH_REST	EARTH_REST		0	lean 2, drop 0.04	
3	O	−40,0,36	−40,0,−36	−34 / 0	−6	lean −6, drop 0	the right leg lifts: the stomp begins
5	I	−118,0,28	−118,0,−28	0 / 0 (±16)	−4	lean 2, **drop 0.30**	**stomp**: the foot slams, the rock jumps out of the ground (particles 3.1e), both palms rise under it
9	S	−120,−6,24	−30,10,−26	0 / 0 (±16)	0	lean 0, drop 0.22	the rock floats at chest height (tick 5–9: 4 ticks of float, the "hold" of Hung Gar)
11	I	**−92,−8,4**	−10,0,−30	0 / 0 (±14)	2	**lean 10**, drop 0.16, twist 12	**the push**: the palm drives forward from the hip, the body turns in; the rock leaves at tick 11 (the server's launch tick)
16	S	EARTH_REST	EARTH_REST	(±6)	0	lean 2, drop 0.04, twist 0	recovery

Timing notes: the server lifts the rock at the press (tick 0) and holds it 11 ticks before launching, so the look matches: the rock entity is spawned "rising" at tick 0 with a 5-tick rise (0.0 → 1.2 blocks) and a 6-tick float, then flies. **Lock:** none (you may walk; the legs show the walk except for the stomp key at tick 3–5, which drives the right leg only). **Blend:** over walking the stomp becomes a step; over flying (Creative) the stomp is skipped (legs not keyed when `wAir > 0.5`: the `legs` channel is masked by `1 − wAir`, **NEW** 14.3 channel masks); never under water (refused). **Started by:** the slot's cooldown edge (`cooldownGesture`: Rock Throw's configured 30 vs Hold Rock's 0: a hold starts no cooldown, so the Sneak use is told apart by `Live.EARTH_HOLDING` rising). **Prediction:** on the press when earth is in reach (the client raycasts the ground within 6) and the slot is ready.

Mastery variants: M4–6 (two rocks): the O arm mirrors the M push 2 ticks later (`0` at tick 13: −92,8,−4) and the twist goes −12; M7–9: a third rock from the first palm again at tick 15 (the gesture is `EARTH_ROCK_THROW_3`, dur 20). M10 boulder (`EARTH_BOULDER`, dur 26 / 3 / 8): the stomp at 5 is deeper (drop 0.36), both arms rise to −150 by tick 12 (the boulder floats over the head, `EARTH_LIFT` with x −150), then **both palms thrust** at tick 18 (`FIRE_DOUBLE`-like −92,±6,±4) with lean 14 and a 0.1 step forward (`drop` 0.20 → 0.10); recovery to 26.

Sneak: Hold Rock, `EARTH_ROCK_LIFT` (dur 12 / 2 / 5, not force): the stomp keys 0–5 as above, then at tick 8 the M settles into `EARTH_CLAW` aimed at the rock and hands over to the `EARTH_GRIP` stance (the gesture ends on the stance's pose: no pop). Throwing from the hold plays `EARTH_ROCK_THROW` from tick 9 (a `param` of 9 skips the stomp). Setting down: `EARTH_ROCK_DOWN` (dur 10): the claw opens (y 18 → 0) and lowers to x −60 with lean 6, as the rock sinks back.

First person:

t	M hand (dx, dy, dz, rx, ry, rz)	O hand
0–3	0, −0.06, 0, +10, 0, 0 (hands drop as the leg lifts)	mirrored
5	+0.02, **+0.22**, +0.06, −30, 0, +8 (both hands fly up with the stomp, palms up)	mirrored
9	0, +0.16, +0.10, −25, 0, +4 (float)	−0.02, +0.04, 0, −10, 0, −4 (the off hand drifts back)
11	**+0.04, +0.02, +0.30, −8, −6, 0** (the thrust)	−0.08, −0.04, −0.08, 0, 0, 0
16	rest	rest

The camera dips 0.04 blocks at the stomp (the `drop` of the body is applied to the first-person camera too: Enhanced applies drop to the pose stack only; **NEW** 14.4: a small camera-drop hook, reduced-motion gated).

Effects:

Where	What	Count	Colour
Stomp (t 5)	`bending:grain` burst at the foot + a flat dust ring (`MATTER_FLAT`, radius 0.3 → 1.4 over 6 ticks, α 0.5 → 0)	10 grains, 1 ring	the ground block's tint (`BlockParticleOption` of the state)
Rock rising (t 0–5)	the **rock entity** (`RockEntity`, a block display: `renderSingleBlock` of the torn state, scale 0.9, rising with `easeOutBack`; the hole fills with a dark `MATTER_FLAT` disc that fades as the temp block reverts) + 6 grains falling off it per tick for 5 ticks	6/tick	ground tint
Float (t 5–11)	1 grain per 2 ticks dropping from the rock; the rock turns 4°/tick	0.5/tick	
Flight	the rock tumbles (Y 11°, X 7° per tick), a trail of 1 grain per tick; no glow (matter)	1/tick	
Impact (creature)	14 grains + `BLOCK` dust ×8 of the rock's state + a 0.6-block `MATTER_FLAT` puff disc; the rock vanishes	22	
Impact (block)	the same, plus a 0.2 shake within 8 blocks	22	
Boulder (M10)	the rock is 2×2×2 (8 block displays as one mesh, `renderSingleBlock` ×8 or one scaled cube with the block texture), trail 3 grains/tick, impact 40 grains + shake 0.8		
Per-player cap	6 rock entities, 60 grains alive		

Sounds: wind-up `earth_stomp` (t 3–5: a deep thud with a gravel crunch) → release `earth_rock_throw` (t 11: a low whoosh with grit) → impact `earth_rock_hit` (a cracking thump; on a creature a duller `earth_rock_hit_flesh`); boulder: `earth_boulder_throw` / `earth_boulder_hit` (bigger, a sub-bass drop). Hold: `earth_rock_lift` on lift, a quiet rumble loop `earth_hold_loop` while held (volume 0.3), `earth_rock_down` on set-down.

3.2 Earth Armour — שריון אדמה

Forming, `EARTH_ARMOUR_FORM` (dur 20 / 2 / 6, CAST, force; **locks movement** for its 20 ticks via `Live.EARTH_FORMING`):

t	ease	M	O	legs (z)	head	body	notes
0	S	`EARTH_REST`	`EARTH_REST`		0		
4	O	−30,0,40	−30,0,−40	−36 / 0	−4	lean −6	the right leg lifts
6	I	−20,0,38	−20,0,−38	0 / 0 (±18)	6	lean 6, **drop 0.32**	the **stomp** (horse stance): earth answers
12	S	−60,30,30	−60,−30,−30	(±18)	0	lean 2, drop 0.26	both arms rise, elbows out: the stone climbs the body (the model layer shows from the feet up: 3.2e)
16	E	`EARTH_GUARD`	`EARTH_GUARD`	(±16)	0	lean 2, drop 0.22	forearms cross before the chest: the chest plate closes
20	O	−40,0,50	−40,0,−50	(±16)	0	lean 2, drop 0.20	arms **snap** out as the helmet seals: hand over to `EARTH_ARMOURED`

Shedding / breaking, `EARTH_ARMOUR_SHED` (dur 10 / 1 / 4): arms to −60,0,60 / −60,0,−60 at tick 3 with `B` (a shrug), lean −6, then rest; the layer crumbles (3.2e). **Shed Blast,** `EARTH_ARMOUR_BLAST` (dur 14 / 1 / 5, force): the same shrug at tick 3, then at tick 5 both arms **slam down** to 20,0,30 / 20,0,−30 with drop 0.28 and lean 12 (`I`), a stomp of both feet; recovery to 14.

Lock: forming locks input (20 ticks); shedding does not. **Blend:** forming is refused while airborne, so no flight blend; while armoured the walk is the `EARTH_ARMOUR` run character (heavier) and jumping is 0.9 height (server: jump boost −; **(added)** the shell weighs). Under water the shell crumbles (`EARTH_ARMOUR_SHED` plays). **Started by:** `Toggle.EARTH_ARMOUR` flip (on → FORM, off → SHED) and the FX payload kind `ARMOUR_BREAK` (→ SHED with a bigger crumble); the blast by the slot's cooldown edge (20 s) while the toggle goes off in the same tick. **Prediction:** on the press when standing on earth and not airborne.

First person (forming): both hands rise with the stone: t 6 (stomp) `dy −0.10, rx +15`; t 12 `dx ±0.12, dy +0.10, dz +0.05, rx −20, rz ±15` (elbows out); t 16 hands cross in front of the face `M dx −0.14, dy +0.12, dz +0.12, ry −35`, O mirrored; t 20 hands snap outward `dx ±0.25`. Then the **eye slit overlay** (2.2) closes over 20 ticks, from tick 8 to 28 (it lags the body by 8 ticks so you see the stone climb). The hands are drawn **in stone** while armoured: the first-person arm uses the armour layer's texture via a `RenderArmEvent` override (Enhanced's `DeadpoolGearLayer` pattern).

Effects:

Where	What	Count	Colour
Stomp (t 6)	14 grains + dust ring r 0.3 → 1.6 + shake 0.6	15	ground tint
Climb (t 6–20)	the **armour layer** (`LOOKS.md` 1.1: a `ModelPart` suit of plates over the inflated player model, `CubeDeformation(0.35)`) is revealed **bottom-up** by a clip plane: `y = feet + 2.1·(t−6)/14` (an alpha mask in the texture coordinates is not available on the cutout type; use a scissor in model space: render the layer in 6 vertical slices as separate parts and show each when the plane passes it)	—	the ground's stone: the layer's texture is tinted by the torn block's map colour (grass → brown-green dirt plates; stone → grey; sandstone → yellow)
Climb particles	4 grains/tick spiralling up the body; 2 `BLOCK` dust/tick	6/tick	ground tint

Where	What	Count	Colour
Seal (t 20)	a puff of 10 grains at the head; the slit glows for 4 ticks	10	
Worn	every 40 ticks 1 grain falls from a plate; walking leaves 1 grain per step; the model's footstep particles are block dust	0.1/tick	
Hit on the shell	6 grains + `BLOCK` dust ×4 at the hit, a crack appears on the layer's texture overlay (3 crack stages as overlay textures `armour_cracks_1..3.png`, `entityTranslucent`)	10	
Break / shed	the slices fall off in 6 ticks top-down (each slice becomes 2 `FallingBlock`-style client particles: 12 grain bursts) + dust ring + shake 0.5; Shed Blast adds a stone-shard ring: 24 `bending:grain` at speed 0.6 outward + a `MATTER_FLAT` expanding ring r 0.5 → 4 over 8 ticks (α 0.6 → 0)	40–60	
Seismic Sense (M8)	hostile mobs within 16 get `EntityGlow` ochre `#C9B47A` while you stand still armoured	—	
Per-player cap	1 armour layer, 40 grains		

Sounds: `earth_armour_form` (t 6–20: a stomp, then stone sliding and grinding up, ending in a heavy "clunk" seal), `earth_armour_step` (replaces the footstep: a stony tread, volume 0.4), `earth_armour_hit` (a chip and a dull knock), `earth_armour_crack` (a sharp split at each crack stage), `earth_armour_break` (rubble collapsing with a rising cloud), `earth_shed_blast` (the same with a crack and a wide hiss of shards).

3.3 Earth Pillar — עמוד אדמה

`EARTH_PILLAR` (dur 14 / 2 / 5, CAST, force):

t	ease	M	O	legs (z)	head	body	notes
0	S	`EARTH_REST`	`EARTH_REST`				
3	O	−20,0,40	−20,0,−40	0/0 (±18)	4	lean 4, **drop 0.26**	both knees bend (the gather)
6	I	**−150,0,20**	**−150,0,−20**	0/0 (±16)	−10	**lean −8**, drop 0.06	both palms **drive upward** past the shoulders: the ground rises under you (the rise starts at tick 5)
10	S	−120,0,26	−120,0,−26	(±14)	−6	lean −4, drop 0.08	riding up, arms held high
14	S	`EARTH_REST`	`EARTH_REST`	(±8)	0	lean 2, drop 0.04	settle on top (the pillar reaches its height at tick 5 + height/0.5 ≈ 11–21; the gesture ends before, the `EARTH_RIDE_UP` stance holds the arms at −60,0,30 while the pillar still rises, `tauIn 2 / tauOut 4`, on while `Live.EARTH_PILLAR_RIDING`)

Lowering (press again): `EARTH_PILLAR_DOWN` (dur 10): both palms **press down** from −60 to 10,0,30 with lean 6 at tick 4 (`I`), the pillar sinks; rest at 10. **Earth Launch,** `EARTH_LAUNCH` (dur 12 / 1 / 4, force): a deeper crouch at tick 3 (drop 0.34, legs ±20, arms back 20,0,30 / 20,0,−30), then at tick 5 the arms **throw up** to −176,0,10 / −176,0,−10 with lean −14 and drop 0 as the stub fires you; the air pose hands over to the shared `LAUNCHED` stance (11.3: arms wide, legs trailing) until landing, which plays `LAND_SOFT` with a 0.12 drop kick.

Lock: none, but while riding the pillar the server moves you with it (`EARTH_PILLAR_RIDING`: your position is set to the top each tick, as a passenger of a `PillarEntity`... see `TECH.md` 3: the pillar is temp blocks, the player is moved by `setPos` with `hurtMarked` each tick: smooth because the client interpolates). **Blend:** the gesture's legs are keyed only at 3 (bend) so a walking player bends mid-step; flying: refused. **Started by:** the slot's cooldown edge (3 s vs the Launch's 4 s, told apart by length as Thanos's are), and `Live.EARTH_PILLAR_RIDING` for the ride stance. **Prediction:** standing on earth (client check) and ready.

First person: t 3 `dy −0.12, rx +20` (crouch); t 6 both `dx ±0.06, dy +0.34, dz +0.04, rx −45, rz ±10` (palms up, driving); t 10 hold `dy +0.22`; t 14 rest. The camera rides up smoothly (interpolated position). Launch: t 5 both hands `dy +0.40` then they trail down `dy −0.10, rx −10` in the air (the `LAUNCHED` first-person: hands slightly out and back, wobbling ±0.01 at 0.3 rad/tick).

Effects:

Where	What	Count
Rise	the **pillar** is temp blocks of the ground's state placed 1 per 2 ticks (0.5 block/tick), each new block "pops up" with 4 grains and `BLOCK` dust ×2; a `MATTER_FLAT` dust ring at the base r 0.6 → 1.5 at the start	6 per block
Top	the player rides; 1 grain falls off the top edge per 2 ticks while rising	0.5/tick
Standing	nothing	
Sink	blocks removed top-down 1 per 2 ticks with 3 grains each; the rider descends	3 per block
Aimed pillar (M5)	a dust ring at the aimed spot for 6 ticks before it rises (the tell: `MATTER_FLAT` ring r 1.0 pulsing α 0.3 → 0.6)	1 ring
Launch	a 3-block **column of dust** (`FxShapes.cylinder` r 0.5 → 0.2, h 0 → 3 over 6 ticks, α 0.5 → 0, `MATTER_FLAT`) + 20 grains upward + shake 0.4	21
Cap	1 pillar (8–16 temp blocks), 24 grains	

Sounds: `earth_pillar_rise` (a rising rumble with stone clunks every 2 ticks for the duration; built from a loop `earth_rumble_loop` + one-shot `earth_clunk` per block), `earth_pillar_sink` (the same descending), `earth_launch` (a stomp and a sharp stone "crack-whoosh"), the landing `earth_land_soft`.

3.4 Earth Wall — קיר אדמה

`EARTH_WALL` (dur 14 / 1 / 5, CAST, force):

t	ease	M	O	legs (z)	head	body	notes
0	S	`EARTH_REST`	`EARTH_REST`				
2	O	10,0,36	10,0,−36	−34 / 0	−4	lean −4	right leg lifts (a quick stomp)

t	ease	M	O	legs (z)	head	body	notes
4	I	−40,10,30	−40,−10,−30	0/0 (±18)	4	lean 6, **drop 0.28**	**stomp**: the wall jumps up in 4 ticks (server: rows placed at ticks 4, 5, 6, 7)
8	O	**`EARTH_GUARD`**	**`EARTH_GUARD`**	(±16)	0	lean 2, drop 0.22	forearms **cross and rise** before the chest: the wall tops out
14	S	`EARTH_HORSE`	`EARTH_HORSE`	(±16)	2	lean 4, drop 0.24	hand over to `EARTH_IDLE` (the bender stays in the horse behind their wall)

Wall Push, **`EARTH_WALL_PUSH`** (dur 18 / 1 / 6, force): keys 0–8 as above (if a wall stands, the gesture starts at tick 8 via `param`), then at tick 10 (`I`) **both palms thrust** to −88,±6,±4 with lean 14, drop 0.16, twist 0, a step: right leg x −20 / left +20; hold to 14; recovery at 18. The wall slides from tick 10 over 12 ticks.

Lock: none. **Blend:** walking continues (the legs are keyed only at the stomp); flying: allowed in Creative (the wall rises where aimed; the stomp is masked, 14.3). **Started by:** the slot's cooldown edge (8 s vs the push's 10 s). **Prediction:** earth under the footprint (client raycast) and ready.

First person: t 2 `dy −0.06`; t 4 both `dx ±0.04, dy −0.14, rx +25` (the stomp pulls the hands down and in); t 8 both hands **rise crossed** before the face `M dx −0.16, dy +0.14, dz +0.12, ry −30, rz +10`, O mirrored (the wall rising fills the lower half of the view); t 14 lower to the idle hands. Push: t 10 both `dz +0.32, dy +0.02, rx −10` (a two-palm shove), kick back `dz −0.06` at 12.

Effects:

Where	What	Count
Tell (t 0–4)	a `MATTER_FLAT` dust line along the wall's footprint, α 0.3 pulsing	1
Rise (t 4–7)	each row of temp blocks appears bottom-up (row n at tick 4+n) with `BLOCK` dust ×2 and 3 grains per block, and a dust curtain: a `SandShapes.wall`-like translucent plane in front of the wall, α 0.5 → 0 over 8 ticks, `MATTER_TEXTURED(dust.png)`	5 per block (≈ 30–140)
Standing	1 grain falls from the top every 20 ticks per 3 blocks of width; when a projectile hits: 6 grains + `BLOCK` dust ×3 at the hit	low
Sink (t 200)	rows vanish top-down over 8 ticks with 2 grains per block	
Push	the wall's blocks move 1 cell per 2 ticks (temp blocks re-placed: at most 28 blocks × 6 steps); a dust plume behind the wall (`FxShapes.cylinder` lying along the wall, α 0.4) and 8 grains per step; things hit get 6 grains	8/step
Cap	1 wall (≤ 28 blocks, 56 at M10), 80 grains	

Sounds: `earth_wall_rise` (a stomp into a fast rising stone slide, 0.4 s, four clunks), `earth_wall_hit` (a knock with grit, pitched by the projectile's damage), `earth_wall_sink` (a slow grinding fall), `earth_wall_push` (a heavy scrape loop `earth_scrape_loop` for the 12 ticks + a thud at the end).

3.5 Sand Bending — כשפות חול

Sand Stream, **`EARTH_SAND_STREAM`** (dur 16 / 2 / 6, CAST, force): a lighter Hung Gar, no stomp:

t	ease	M	O	legs (z)	head	body	notes
0	S	`EARTH_REST`	`EARTH_REST`				

t	ease	M	O	legs (z)	head	body	notes
4	E	**−130,20,20**	**−130,−20,−20**	0/0 (±14)	−8	lean −6, drop 0.12	both arms **scoop up** from the sides: the sand lifts in a sheet
8	S	−100,−30,40	−100,30,−40	(±14)	−4	lean −2, drop 0.14, twist −10	the arms **circle out** (a wide gather)
11	I	**−86,−10,−6**	**−86,10,6**	(±14)	2	**lean 10**, drop 0.10, twist 8	both palms **push** together: the stream flies
16	S	`EARTH_REST`	`EARTH_REST`	(±8)	0	lean 2, drop 0.04, twist 0	

Sand Wall, `EARTH_SAND_WALL` (dur 12 / 1 / 5): like `EARTH_WALL` without the stomp: arms **sweep up from the knees** (t 2: −10,0,36 / −10,0,−36 with drop 0.20; t 6: `EARTH_GUARD` with `E` and lean −2: the sand pours upward), hand-over to `EARTH_IDLE` at 12.

Lock: none. **Blend:** walking continues. **Started by:** the slot's cooldown edges (2.5 s stream vs 6 s wall). **Prediction:** sand in reach (or 4 sand in the inventory) and ready.

First person: stream: t 4 both `dx ±0.20, dy +0.22, dz +0.02, rx −30, rz ±20` (the scoop), t 8 both `dx ±0.28, dy +0.12, ry ±20` (circle out), t 11 both `dx ±0.02, dz +0.30, rx −6` (push together), rest. Wall: t 2 `dy −0.16, rx +20` (hands at the knees), t 6 `dy +0.18, dz +0.10, rz ±12` (rising palms), idle.

Effects:

Where	What	Count
Stream lift (t 0–11)	20 `bending:grain` (sand tint `#D8B76A`, red sand `#A95B2E`, soul sand `#4E3A2C`) rise from the sand within 3 blocks into a sheet 1.5 wide before the bender (a hanging `MATTER_TEXTURED(sand_swirl)` sheet, α 0 → 0.5), 3 grains/tick	3/tick
Stream flight	one `SandStreamEntity`: a cone drawn as 2 `SandShapes.funnel`-like frustums lying forward (r 0.3 → 1.2 over the length), α 0.6, scrolling, plus 6 grains/tick along it; blinded targets get 4 grains at the eyes	6/tick
Sand Wall	rows of temp **sand** blocks poured in from below: a `SandShapes.ridge` crest runs up the front face as it rises; 4 grains per block; standing: 1 grain/20 ticks per 2 blocks slides down; collapse: a `MATTER_FLAT` sand slump at the foot	4 per block
Cap	1 stream entity, 1 wall, 60 grains	

Sounds: `sand_lift` (a dry hiss rising, 0.5 s), `sand_stream` (a sustained grainy rush 0.6 s + a soft "pelt" loop), `sand_hit` (a sandy pelt on a creature), `sand_wall_rise` (pouring sand reversed: a swelling hiss), `sand_wall_fall` (the slump).

3.6 Quicksand — חול טובעני

`EARTH_QUICKSAND` (dur 20 / 2 / 6, CAST, force):

t	ease	M	O	legs (z)	head	body	notes
0	S	`EARTH_REST`	`EARTH_REST`				
4	O	**−70,−10,20**	−30,10,−30	0/0 (±16)	0	lean 4, drop 0.20	the M palm **points down at the target's feet** (`mAim(LOOK, 30, 0, 0)`: 30° below the look)
8	E	aim(LOOK, 20, −20, 0)	aim(LOOK, 20, 20, 0)	(±16)	4	lean 8, drop 0.22	both palms, fingers spread (y out), **tremble 2.0 at 2.4** (the pull)

t	ease	M	O	legs (z)	head	body	notes
14	I	aim(LOOK, 40, −10, 0)	aim(LOOK, 40, 10, 0)	(±16)	8	**lean 14**, **drop 0.30**	both hands **press downward** slowly (`I`): the target sinks
20	S	`EARTH_HORSE`	`EARTH_HORSE`	(±16)	2	lean 4, drop 0.24	hand over to the idle

Sand Cloud, `EARTH_SAND_CLOUD` (dur 14 / 2 / 5): arms **swirl up**: t 4 `AIR_CIRCLE_A`-like −120,−40,40 / −30,20,−10 with twist 10; t 8 the mirror with twist −10 (a figure-eight); t 11 both palms **open outward** at −80,±40,∓40 (the cloud blooms); rest.

Lock: none. **Blend:** walking continues. **Started by:** cooldown edges (10 s each: the two are told apart by the FX payload kind `QUICKSAND` vs `SAND_CLOUD`, `startedElsewhere`). **Prediction:** a creature in sight within 12 on sand.

First person: t 4 `M dz +0.22, dy −0.08, rx +30` (point down); t 8 both `dx ±0.10, dy −0.04, dz +0.22, rx +25, rz ±10` with tremble 0.012; t 14 both `dy −0.18, dz +0.18, rx +40` (press down); idle. Cloud: hands swirl (a circle of r 0.08 over 8 ticks) then open `dx ±0.26`.

Effects (target side): the 3×3 patch's temp blocks swap to `bending:quicksand` (a sand texture with a slow **ripple** animated texture, 4 frames); 6 grains rise and fall at the patch per tick; the sinking creature gets a `MATTER_FLAT` sand collar (a ring r 0.6 at its chest height, α 0.6) and 2 grains/tick sliding down it; the victim's first-person overlay (2.2). Victim pose (12.1). **Sand Cloud:** a `SandShapes.funnel`-like **sphere** of swirl texture r 6, rotating, α 0.45, with 10 grains/tick drifting inside; blinded targets get 4 grains at the eyes. **Cap:** 2 patches, 1 cloud, 60 grains.

Sounds: `quicksand_cast` (a low sandy groan that drops in pitch, 0.6 s), `quicksand_sink_loop` (a wet, grainy churning loop at the victim, volume 0.5), `quicksand_free` (a sucking pop), `sand_cloud` (a broad hiss swell 0.8 s, then a loop `sand_cloud_loop` at 0.3).

3.7 Lava Bending — כשפות לבה

Lava Path, `EARTH_LAVA_PATH` (dur 22 / 2 / 7, CAST, force): a slow, heavy Hung Gar with heat in it:

t	ease	M	O	legs (z)	head	body	notes
0	S	`EARTH_REST`	`EARTH_REST`				
4	O	10,0,40	10,0,−40	−36 / 0	−4	lean −6	the right leg lifts high
6	I	−30,0,36	−30,0,−36	0/0 (±20)	6	lean 8, **drop 0.34**	the **deep stomp**: the ground under the foot glows (the first path cell melts at tick 6)
12	E	**−80,0,30**	**−80,0,−30**	(±18)	2	lean 6, drop 0.26	both palms **press forward and down** slowly (`E`), fingers spread, **tremble 1.6 at 1.4**: the path runs out ahead 1 cell/tick
18	S	−70,0,34	−70,0,−34	(±18)	2	lean 6, drop 0.26	hold (the path keeps running: the gesture hands over to `EARTH_LAVA_HOLD` stance for the remaining ticks, aimed, tremble 1.6)

t	ease	M	O	legs (z)	head	body	notes
22	S	`EARTH_HORSE`	`EARTH_HORSE`	(±16)	2	lean 4, drop 0.24	

Lava Shuriken, `EARTH_LAVA_SHURIKEN` (dur 14 / 1 / 5, force): t 3 the M arm **winds across the body** (−60,50,−30, twist −18, lean 2), the O arm forward low (−40,−10,−20); t 5 the disc forms at the M hand (effects); t 8 (`I`) a **horizontal throw**: M to −80,−40,30 with twist +20 and lean 8, a step (right leg −16 / left +16); recovery at 14. M8 (two discs): the O arm throws 3 ticks later mirrored.

Lock: none (walking continues; the path follows your look as you walk). **Blend:** flying: refused. **Started by:** cooldown edges (12 s vs 4 s). **Prediction:** earth ahead and ready (path); ready (shuriken).

First person: path: t 4 `dy −0.06`; t 6 both `dy −0.16, rx +25` (stomp); t 12 both `dx ±0.14, dy −0.06, dz +0.26, rx +30, ry ∓10` (palms pressing down and out) with tremble 0.012 (the heat shimmer below the hands: 2.2 Fire Breath style ripple in the lower 20% of the screen, warm); hold; idle. Shuriken: t 3 `M dx −0.28, dz +0.02, ry −40` (wound across), t 8 `M dx +0.30, dz +0.24, ry +30` (the flat throw), rest.

Effects:

Where	What	Count
Stomp	14 grains + ring + shake 0.2; the first cell cracks: `BLOCK` dust ×6 then **glows** (a `MATTER_FLAT` disc `#FF7A1F` α 0 → 0.8 over 4 ticks) before becoming temp lava	20
Path running	each cell: 4 ticks of crack glow, then the temp **lava** state (vanilla lava's animated texture; `bending:temp_lava` reuses `minecraft:lava` textures); 2 `LAVA` + 1 `SMOKE`… (`CAMPFIRE_COSY_SMOKE` ×1) per cell per 2 ticks while it burns; heat shimmer: a `MATTER_TEXTURED(heat.png)` translucent sheet 0.5 above the path, scrolling, α 0.25	≈ 1.5/tick per cell, capped at 20/tick for the path
Cooling	the cell darkens over 20 ticks (an overlay disc `#2A1A10` α 0 → 0.9 drawn on top, `MATTER_FLAT`) then the original block returns with 3 `WHITE_ASH`	3 per cell
Shuriken	a **disc** entity: `FxShapes.disc` r 0.5 (0.65 at M8) `#FF6A1F` core + `annulus` rim `#8C2410`, spinning 30°/tick, additive glow halo (`FxShapes.glow` r 0.7 α 0.3), trail 2 `LAVA` + 1 `FLAME`/tick; impact: 16 `LAVA` + 8 `FLAME` + a 1-block `MATTER_FLAT` scorch disc for 60 ticks	3/tick
Lava passive	at M10 while immune: 2 `FLAME` rising from the feet per tick	
Cap	1 path (≤ 16 cells), 2 discs, 40 lava particles	

Sounds: `lava_path_cast` (stomp + a deep molten rumble swelling 0.8 s), `lava_path_loop` (a bubbling hiss while cells burn, follows the path's head, volume 0.5), `lava_cool` (a sizzling tick as cells cool), `lava_shuriken_throw` (a hot whoosh with a spinning flutter), `lava_shuriken_hit` (a splat-sizzle), `lava_shuriken_spin_loop` (a thin flutter while it flies, volume 0.3).

3.8 Metal Bending — כשפות מתכת

The metal gestures are **Lin Beifong's**: sharper and smaller than earth, wrists snapping, little stomp.

Steel Knives, `EARTH_STEEL_KNIVES` (dur 12 / 1 / 4, CAST, force):

t	ease	M	O	legs (z)	head	body	notes
0	S	EARTH_REST	EARTH_RES T				
3	O	**−50,20,−10**	−30,0,−26	0/0 (±12)	0	lean 2, drop 0.10	the M forearm **snaps up** across the chest, wrist cocked: the knives form on the forearm (effects)
5	I	**−88,−20,20**	−30,0,−26	−12 / 12 (±12)	2	**lean 8**, twist 10	the **flick**: wrist snaps forward and out, knives fly (t 5, 6, 7 one each)
8	S	−84,−10,16	−30,0,−26	(±12)	2	lean 6, twist 8	held out
12	S	EARTH_REST	EARTH_RES T	(±8)	0	lean 2, twist 0	

M5+ (5 knives): the O arm flicks too, 2 ticks after (t 7: −88,20,−20), knives alternate hands. **Recall** (Sneak-key within 5 s of stuck knives): `EARTH_KNIFE_RECALL` (dur 8): the M hand **opens toward the target** (aim(LOOK, −10, 0, 0), `B`) and **closes** (y −20 → 0) as the knives streak back into charge.

Metal Pull, **`EARTH_METAL_PULL`** (hold; `EARTH_METAL_PULL_START` dur 8 / 1 / 3 then the stance `EARTH_PULLING` (aimed, tauIn 2 / out 4) while `Live.METAL_PULLING`): the start: both arms **reach out and up** to −70,∓30,±20 with fingers spread (y) by tick 4 (`E`), lean 4, drop 0.12; the stance: arms at −60,∓20,±16 with **tremble 2.2 at 2.8** (the strain), head 4, lean 6; every 10 ticks the hands **pull back** 8° in x (a tug, `Stances.drift` with a sawtooth). Releasing plays `EARTH_METAL_PULL_END` (dur 8): fists close (y 0) and drop to rest.

Steel Wall, **`EARTH_STEEL_WALL`** (dur 12 / 1 / 4, force): a Lin **chop**: t 2 both arms raised to −140,∓10, ±10 (`O`), drop 0.10; t 5 (`I`) both forearms **chop down** to −60,0,±14 with lean 10 and a short stomp (right leg −20 → 0 at 6): the plates **shoot up** from the charge at the bender's feet toward the footprint (effects); t 12 `EARTH_HORSE`-lite (drop 0.14) hand-over to idle.

Steel Armour, **`EARTH_STEEL_ARMOUR_ON`** (dur 16 / 2 / 5; **locks** 16 ticks via `Live.EARTH_FORMING`): t 3 both arms cross the chest (`EARTH_GUARD` with x −90: hands at the collar), drop 0.14; t 6 (`O`) arms **sweep down the body** to −20,∓20,±26 as plates **clap** onto the body top-down (the reverse of earth: steel comes from the hands, not the ground); t 10 arms snap out to −40,0,±50 (`B`) as the last plates lock; t 16 `EARTH_ARMOURED` hand-over. **Off,** **`EARTH_STEEL_ARMOUR_OFF`** (dur 10): a shrug at 3 (`B`), plates fly off into the hands (charge returns) as the arms gather at the belly (`WATER_GATHER`-like) at 7; rest.

Wreck Metal, **`EARTH_WRECK`** (dur 14 / 1 / 5, force): t 3 the M hand **reaches** (aim(LOOK, 0, 0, 0), fingers spread y 20, `O`); t 6 (`I`) the hand **clenches and twists**: y 20 → −30, z 0 → 24, with a wrist roll implied by `ry` in first person, twist −12, lean 6: the device crumples at tick 6; t 10 the hand **flicks away** (aim(LOOK, −20, 30, 0), `O`); t 14 rest. At M5+ (3×3) both hands reach and clench (O mirrored).

Metal Grip, **`EARTH_METAL_GRIP`** (dur 10 / 1 / 4, then the stance `EARTH_METAL_HOLD` aimed at the held entity (`aim(HELD, -16, 0, 0)` with tremble 1.8 at 2.0, O braced at −20,0,−30, lean 6, `occupy(M)`) while `Live.EARTH_HOLDING` on a metal target): the start: M **thrusts open-handed** at the target (aim(LOOK, 0, 0, 0), `O` at tick 3), **closes** (y −18) at tick 5 as the grip takes; hand-over. Fling: `EARTH_METAL_FLING` (dur 10): the arm **whips** from the hold to aim(LOOK, −30, 10, 0) with twist 14 and lean 10 (`I` at tick 4), rest. Set down: `EARTH_ROCK_DOWN` reused.

Lock: Steel Armour on locks 16 ticks; nothing else. **Blend:** metal gestures key no legs except the chop's stomp, so they ride over walking cleanly; flying: knives and wreck allowed in Creative flight (the chop is masked). **Started by:** cooldown edges (2.5 s knives, 6 s pull, 8 s wall, 20 s armour-break, 4 s wreck, 8 s grip); `Toggle.STEEL_ARMOUR` flips; `Live.METAL_PULLING`, `Live.EARTH_HOLDING`; payload kinds `KNIVES_RECALL`, `WRECK`. **Prediction:** charge > 0 (the client knows the synced charge) and ready; grip: a metal target in sight.

First person: knives: t 3 `M dx −0.10, dy +0.14, dz +0.06, ry −30, rz +20` (the forearm across, knives visible on it: 3.8e), t 5 `M dx +0.16, dy +0.04, dz +0.28, ry +20, rz −10` (flick), rest; the O hand mirrors 2 ticks later at M5+. Pull: both hands out `dx ±0.16, dy +0.02, dz +0.22, rx +10, ry ∓15`, trembling 0.012, tugging back `dz −0.05` every 10 ticks; filings stream into the palms (effects). Steel Wall: t 2 both `dy +0.30, rx −40` (raised), t 5 both `dy −0.10, dz +0.14, rx +30` (chop). Steel Armour on: hands at the collar `dx ∓0.10, dy +0.16, dz +0.06` then sweep down `dy −0.14`, snap out `dx ±0.24`; the hands turn to steel (the arm texture swap) at tick 8; the **steel eye slit** overlay (2.2) closes over 16 ticks. Wreck: `M dz +0.26, ry +10` (reach), then `ry −40, rz +25, dz +0.18` (clench-twist, tremble 0.01), then `dx +0.14, dz +0.10` (flick). Grip: reach `dz +0.26`, close `ry −20`, hold with tremble; fling `dx +0.22, dz +0.22`.

Effects:

Where	What	Count
Knives form (t 3–5)	3–7 **knife quads** (`FxShapes.box`-like thin boxes 0.5 long, `MATTER_FLAT` steel grey `#B8C0CC` with a lit edge `#E8EEF5`) appear along the forearm (`HandTracker.hand` + the arm direction) one per tick with 2 `ELECTRIC_SPARK`... no: 2 `CRIT`-style white motes (energy `#E8EEF5` scale 0.5) each	2 per knife
Knife flight	`SteelKnifeEntity`: the same box, spinning about its long axis 25°/tick, a 0.6-block faint `ENERGY_BEAM` streak behind (α 0.2); no particles	0
Knife hit	4 white motes + the knife **stays** in the target (drawn stuck, offset to the hit point, 3 s) or in the wall (a temp decoration: the entity rests)	4
Recall	each knife streaks back as a `ENERGY_BEAM` ribbon 1.2 long `#D8E0EA` α 0.5 over 6 ticks, 2 motes at the hand on arrival	2
Metal Pull	**filings**: 6 `bending:grain` (steel grey, tinted by the source block: iron `#B8C0CC`, gold `#F2D15A`, copper `#D08A5A`) per tick streaming from each metal block within 8 along a curve to the hands (`EnergyGeometry.flowingPath` sampled: spawn at parametric positions), + a faint `ENERGY_BEAM` thread per block α 0.12; the charge bar fills	6 per block per tick, capped 24/tick
Steel Wall	plates (`bending:steel_plate` temp blocks, an iron-block texture with rivets) **slide up** from the ground: each row appears with 4 white motes and a metallic flash (a `MATTER_FLAT` quad α 0.6 → 0 in 3 ticks); a hit: 6 `ELECTRIC_SPARK`-like white motes and a ring on the plate	4 per plate
Steel Armour	the steel suit layer (`LOOKS.md` 1.2) appears **top-down** in 5 slices (head, chest, arms, legs, boots) at ticks 6, 7, 8, 9, 10 with 6 white motes and a clap each; worn: a glint sweep (a `RenderType.eyes` pass over a highlight mask every 80 ticks for 6 ticks); hit: 6 sparks + a dent decal stage (3 stages); break: slices fall off top-down as 5 × 8 grey grains + a clang	30 / 6
Wreck	the device swaps to its **wrecked** temp state (a crumpled texture variant, `LOOKS.md` 6: the `bending:wrecked_*` blocks share one "bent metal" atlas with the device's base colour tinted) with a 6-tick **crunch**: 10 white motes + 4 `BLOCK` dust of iron + a 0.3 shake within 6; mending (t 60): 6 motes and a soft chime	14
Metal Grip	the held thing gets a steel-grey `ENERGY_VEIL` outline (an inflated copy of its box, α 0.25, 4 corner "clamps" as thin boxes) and 2 motes/tick at the hand; fling: a 1-block `ENERGY_BEAM` streak and 8 motes on impact	2/tick
Cap	8 knives, 1 wall, 1 armour, 1 grip, 40 motes/grains	

Sounds: `metal_knives_form` (three quick steel "shinks" 0.4 s), `metal_knife_throw` (a thin whistle), `metal_knife_hit` (a "tink" into stone / a thud into flesh), `metal_knife_stick` (a quivering twang), `metal_recall` (a reversed shink glide), `metal_pull_loop` (a scraping, grinding hum with filings hiss, volume 0.5, pitch rises with the charge), `metal_pull_end` (a clink), `metal_wall_rise` (plates sliding and locking: four clanks), `metal_wall_hit` (a resonant plate ring), `metal_armour_on` (five claps of plates and a final lock), `metal_armour_off` (plates peeling: five reversed claps), `metal_armour_hit` (a dented clang), `metal_armour_break` (a cascade of falling plates), `metal_wreck` (a crunching, groaning twist of metal 0.6 s), `metal_mend` (a soft reversed crunch with a chime), `metal_grip` (a clamp clank), `metal_fling` (a swung-metal whoosh).

3.9 Earth hit reactions (what a target shows)

See 12.1: **rock** hits stagger (a `HIT_STAGGER` reaction gesture on players: lean −14 over 2 ticks, back to 0 over 8, head +6; mobs: the shared `LivingEntityRenderer` hit tilt, 12.1), **boulders** knock down (players: `HIT_KNOCKDOWN`, lean −40 and drop 0.5 for 10 ticks, then up), **quicksand** gives the sinking pose, **wall push** staggers, **Steel Knives** flinch (head −4 twist 6 for 4 ticks), **Metal Grip** gives the `LIFTED` pose (Enhanced's `Live.LIFTED`: arms wide, legs trailing, drop −0.3).

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4. Air

4.0 Air idle: the ready stance

Stance	tauIn / tauOut	Rows
`AIR_IDLE` (ambient, key held / 40 ticks after an Air gesture)	4 / 5	`AIR_REST` plus a **slow circle**: the M hand draws a 12° circle (x −10 + 6·sin(0.12·age), z 10 + 6·cos(0.12·age)), the O hand the same a half-turn behind; weight shifts: lean 2·sin(0.06·age), legs x ±4 alternating with it (the Ba Gua "mud-wading" sway); head level.
`AIR_GUARDING` (aimed, while the Air Shield is up)	3 / 4	`AIR_GUARD` with both palms drifting out and in (z ±4 at 0.2 rad/tick), `occupy` none (you can fight inside it).
`AIR_RIDING` (ambient, on the scooter)	2 / 4	10.1.
`AIR_GLIDING`	2 / 4	10.2.
`AIR_MEDITATING`	4 / 8	10.5.

Ambient particles of a master (Air ES ≥ 60): a thin `GUST`-like breeze: 1 `CLOUD` (small, α 0.4, `EnergyParticle` white `#F4F8FF` scale 0.6) every 8 ticks drifting in a slow circle at knee height, 2 at ES 100; a soft `AIR_LOOP` breeze sound at volume 0.15 while standing still (reduced-motion: no particles).

Air first-person idle (key held): both hands up and open at shoulder height (`dx ±0.16, dy +0.06, dz +0.08, rx −10, ry ∓20`), slowly circling (r 0.02 at 0.12 rad/tick).

4.1 Air Blast — הדף אוויר

`AIR_BLAST` (dur 14 / 1 / 5, CAST, force):

t	ease	M	O	legs (z)	head	body	notes
0	S	`AIR_REST`	`AIR_REST`				
3	E	**−110,30,−30**	**−110,−30,30**	−8 / 8	−4	lean −6, twist −12	both arms **circle up and across** (a gathering sweep from the outside in: the hands pass above the head)
6	E	−60,50,−40	−60,−50,40	−10 / 10	0	lean −2, twist 12	the circle comes down the other side: hands close together at the chest, palms forward
8	**O**	**`AIR_PALMS`**	**`AIR_PALMS`**	−20 / 14	2	**lean 10**, twist 0	**the release**: both palms thrust forward together, a step into it (the front leg forward); the blast leaves at tick 8
14	S	`AIR_REST`	`AIR_REST`		0	lean 0	the hands float back down

Air Jump, **`AIR_JUMP`** (dur 12 / 1 / 4, force): t 3 both palms **press down** beside the thighs (10,∓10, ±24 with `E`), knees bend (drop 0.18, legs ±8); t 5 (`0`) palms **snap down** to 30,∓10,±30 as the body **shoots up** (drop 0 → the jump happens on the server at tick 5), lean −6; in the air the `LAUNCHED` stance (11.3) takes over until landing (`LAND_SOFT`, with a puff). Mid-air second jump: the same from the flight pose.

Lock: none. **Blend:** the step at tick 8 drives the legs only for 2 ticks; walking continues otherwise; while **gliding** (3.3), the blast aimed at the ground plays `AIR_BLAST_DOWN` (dur 10): palms sweep **down** to 20,0,±30 at tick 5 (`0`) from the glider grip, the body pitches up 10° (lean −10) as you rise; the glider pose resumes. Under water: half range, the same gesture slowed ×1.5 (`Slot.rate`). **Started by:** the slot's cooldown edge (2 s; the jump's 2 s is told apart by `Live.AIR_JUMPING` rising). **Prediction:** always (the blast needs no target).

First person: t 3 both hands sweep up and in: `M dx −0.12, dy +0.30, dz +0.06, ry −40, rz −20` (O mirrored); t 6 hands together low at the chest `dx ∓0.04, dy +0.04, dz +0.10, ry ∓10`; t 8 **thrust**: `dx ±0.06, dy +0.06, dz +0.34, rx −12` (both palms fill the lower centre); recovery. The FOV kicks ×1.03 for 4 ticks at the release (not with reduced motion). Jump: hands press down `dy −0.22, rx +35` then snap `dy −0.30`, the camera's rise is the jump.

Effects:

Where	What	Count
Gather (t 3–8)	2 white `EnergyParticle` (`#F4F8FF`, scale 0.5, α 0.5) per tick trailing the hands (`HandTracker.hand`), drawn as a wisp: a 6-point `EnergyGeometry.polyline` ribbon α 0.2 following each hand's last 6 positions	4/tick + 2 ribbons
Release	a **blast cone**: a translucent `ENERGY_VEIL` cone (`FxShapes.cylinder` r 0.3 → 2.5, length 0 → range over 8 ticks, white α 0.18 → 0) with 3 rotating **swirl rings** (`EnergyGeometry.ring` r growing with the cone, α 0.3, white) and 10 `CLOUD` + 6 `GUST`-style white motes spawned along the cone's axis; grass and leaves in the cone **sway** (client: `GUST` vanilla particles at ground level, 4)	20
On a target	6 white motes at the chest + the target's tumble (12.2)	6
Air Jump	a `MATTER_FLAT` white dust ring at the feet r 0.3 → 1.6 over 6 ticks α 0.4 + 8 `CLOUD` + a 1-block `FxShapes.cylinder` puff under the feet α 0.3	9
Blast while gliding	the cone fires downward; the glider's cloth **snaps taut** (the glider layer's wings scale 1.0 → 1.08 for 3 ticks)	

Where	What	Count
Cap	2 cones alive, 30 motes	

Sounds: `air_gather` (t 3–8: a soft inhale-like whoosh rising), `air_blast` (t 8: a hard, breathy "whumpf" with a sharp onset and a 0.4 s decaying rush), `air_hit` (a hollow thump when a creature is thrown), `air_jump` (a short down-whoosh and a puff).

4.2 Air Shield — מגן אוויר

Raise, **`AIR_SHIELD_UP`** (dur 12 / 2 / 5, CAST, not force): t 3 both arms **circle outward and down** from `AIR_REST` to −30,∓20,±50 (`E`); t 7 they **circle up** to −120,∓20,±30 (the hands trace a sphere around the body); t 12 `AIR_GUARD` hand-over to `AIR_GUARDING` (palms out at the sides, as if holding the sphere). **Lower,** **`AIR_SHIELD_DOWN`** (dur 8): palms **press inward** to the chest (−60,∓40,∓10 at tick 4) and drop to rest; the sphere thins. **Break / Burst,** **`AIR_SHIELD_BURST`** (dur 12 / 1 / 4, force): from the guard, both arms **snap wide** to −60,0,±80 at tick 3 (`B`) with lean −6 and drop 0 → −0.08 (a tiny hop); recovery.

Lock: none; the shield moves with you (speed ×0.8 at M1–6: an attribute, not a lock). **Blend:** walking, sprinting and jumping continue; the guard stance keys only the arms; the stance's weight falls to 0.4 while sprinting (so the arms can swing). Under water: refused. **Started by:** `Toggle.AIR_SHIELD` (on → UP, off → DOWN) and the payload kind `SHIELD_BURST` (→ BURST). **Prediction:** on the press when ready and not in water.

First person: UP: hands sweep out `dx ±0.30, dy −0.08` then up `dy +0.28, dx ±0.14` then settle at the guard: `dx ±0.22, dy +0.06, dz +0.12, rx −10, ry ∓25` (both palms at the screen's edges, slightly turned out), with a slow drift (±0.015 at 0.2 rad/tick) and, when a projectile is stopped, a 2-tick **flinch** of the nearer hand (`dz −0.04`). DOWN: hands come to the chest `dx ∓0.02, dz +0.06` and fall. BURST: both hands fly to the edges `dx ±0.40, dy +0.04` in 3 ticks; the camera shakes 0.5.

Effects: the shield is Enhanced's shield pipeline (`ShieldStyle` + the sphere passes): a `ShieldStyle(veil 0xDDEBFF, lines 0xFFFFFF, texture "textures/entity/air_lattice.png", radius 1.6→2.4, strength, maxTicks)`; the veil is drawn **faint** (base α 0.05, rim 0.25: air is nearly invisible) in `ENERGY_VEIL`, the lattice is a **swirl**, not a hex web: `EnergyGeometry.sphere` with the texture scrolling **around** (scroll on U at 0.05/tick, tilesU 4, tilesV 2) in `energyField(air_lattice)` at α 0.15 / 0.35; plus **3 orbiting wind rings** (`EnergyGeometry.ring` r = radius at tilted normals rotating 6°/tick, α 0.25 white, width 0.05) and 2 `CLOUD`-like white motes/tick drifting on the surface. A stopped projectile: a **ripple** (`ShieldFx.hit` pattern: `circleOnSphere` arc growing over 10 ticks) and 6 white motes; the projectile is flung with 2 motes. Strength: as it falls below 40% the rings **wobble** (their normals jitter ±8°, smooth) and the veil flickers at 2 Hz (the shield's `look.flicker`); **burst**: the sphere swells ×1.4 over 4 ticks and vanishes, an expanding `ENERGY_VEIL` **shockwave sphere** (r → burst radius over 8 ticks, α 0.3 → 0) + a `MATTER_FLAT` ground ring + 24 `CLOUD` outward + `GUST` ×6 + shake 0.5. **Cap:** 1 shield, 20 motes. (Lowering by hand: the sphere fades over 6 ticks as Enhanced's `SHIELDS_GOING` does.)

Sounds: `air_shield_up` (a rising swirl into a sustained hush), `air_shield_loop` (a soft, airy rotating hum, volume 0.35, pitch rises slightly as strength falls), `air_shield_hit` (a muffled "whup" + a flutter), `air_shield_down` (the hush falling away), `air_shield_burst` (a deep "boom-whoosh" with a wide stereo rush, 0.9 s).

4.3 Air Glider — גלשן אוויר (item)

Covered in 10.2 (the glider pose). Opening: `AIR_GLIDER_OPEN` (dur 8 / 1 / 3, group RIDE): the M hand raises the staff overhead (−160,0,10) at tick 3 (`0`); the wings **snap open** (the glider layer's wing parts rotate from folded (z ±5) to open (z ±85) over 4 ticks with `B`); hand-over to `AIR_GLIDING`. Closing: the reverse in 6 ticks.

4.4 Air Blades — להבי אוויר

`AIR_BLADES` (dur 12 / 1 / 4, CAST, force): a **cutting** arm, Ba Gua's palm change:

t	ease	M	O	legs (z)	head	body	notes
0	S	`AIR_REST`	`AIR_REST`				
3	E	**−100,50,−40**	−40,−20,20	−8 / 8	0,−12,0	lean −2, **twist −20**	the M arm winds **across the body**, palm edge leading (the hand passes the opposite shoulder), the body coils
5	**O**	**−86,−60,40**	−30,10,−20	−14 / 10	0,10,0	lean 6, **twist 24**	the **slice**: the arm sweeps horizontally across the view in 2 ticks, palm edge out; the blade leaves at tick 5
8	S	−80,−50,36	−30,10,−20		0,6,0	lean 4, twist 16	hold
12	S	`AIR_REST`	`AIR_REST`		0	lean 0, twist 0	

M4 (two blades): the O arm slices the other way at tick 7 (−86,60,−40 with twist −20): a V. M8 (three): the M arm slices again at tick 9 (dur 14). **Air Slice,** `AIR_SLICE` (dur 14 / 1 / 5, force): a **full spin**: t 2 both arms out level (−80,0,±60), t 10 `yaw(360)` with the arms held out and drop 0.06, legs alternating (−16/16 at 4, 16/−16 at 8), lean 4; t 14 `AIR_REST`, yaw 360 (one turn). The sweep hits at tick 6 (the server's hit tick).

Lock: none. **Blend:** walking continues (the slice's step is 2 ticks); the spin rides over a walk (yaw is visual, the walk cycle keeps running); flying (Creative/True Flight): fine. Under water: refused. **Started by:** cooldown edges (1.5 s vs 3 s). **Prediction:** always.

First person: t 3 `M dx −0.34, dy +0.10, dz +0.06, ry −50, rz +15` (the hand crosses to the far left edge); t 5 `M dx +0.36, dy +0.06, dz +0.20, ry +40, rz −20` (the sweep across: the hand crosses the whole lower screen in 2 ticks); hold; rest. Second blade: the O hand sweeps the other way at 7. Slice: both hands at the edges `dx ±0.36, dy +0.02, dz +0.10`, the **camera yaws 360°** over 8 ticks (the only gesture that turns the first-person camera: `ViewportEvent` yaw += the gesture's yaw track; **reduced motion: no camera yaw, the hands sweep alone**).

Effects:

Where	What	Count
Wind-up	a 4-point ribbon trailing the M hand (white α 0.25)	1 ribbon
Blade	an `AirBladeEntity` drawn as a **crescent**: a translucent `ENERGY_VEIL` arc (an `annulus` segment of 100°, outer r 0.9, inner 0.7, tilted to the throw plane) with a bright thin edge ribbon (`ENERGY_BEAM` white α 0.7, width 0.03) and a faint motion blur (the last 3 positions as fading arcs α 0.3/0.15/0.05); 1 white mote/tick	1/tick

Where	What	Count
Cut	on a creature: 6 white motes + a 0.4 s **slash line** (an `ENERGY_BEAM` ribbon across the hit point α 0.6 → 0); on plants: vanilla block-break particles of the plant (`levelEvent 2001`)	6
Slice	a horizontal **disc of wind** (`FxShapes.annulus` r 0 → 3 over 4 ticks, white α 0.3 → 0, `ENERGY_VEIL`) + 12 white motes on the ring + grass sways (`GUST` ×4)	16
Cap	3 blades, 20 motes	

Sounds: `air_blade_cast` (a short slicing whoosh with a high "shing" of air), `air_blade_fly_loop` (a thin whistle following the blade, volume 0.3), `air_blade_hit` (a crisp cut with a thump), `air_slice` (a spinning whoosh rising then falling, 0.7 s).

4.5 Spiritual Projection — השלכה רוחנית

Sit, `AIR_MEDITATE_DOWN` (dur 16 / 2 / 6, CAST, not force; **locks movement** via `Live.MEDITATING` for the whole meditation): t 4 arms gather to the chest (−70,∓40,∓10, `E`), drop 0.10, head 6; t 8 the body **sinks** (drop 0.40, legs x −40/−40 z ±30: the knees fold out), arms lower to the knees (−30,∓30,±40); t 16 `AIR_SIT` (drop 0.72, legs x −90 z ±55, hands resting palms up: −20, ±30,∓40), head 4, lean 0: hand-over to `AIR_MEDITATING` (10.5: a breathing sway ±1.5° lean at 0.05 rad/tick, the hands rising 2° on the inhale). **The spirit leaves,** `AIR_SPIRIT_OUT` (dur 10, on the **spirit entity**, 10.6): the spirit copy stands up out of the body (drop 0.72 → 0 over 10 ticks with `0`, arms spreading to −40,0,±40) while the body stays seated. **Return,** `AIR_SPIRIT_IN`: the body's head lifts (head −6 → 0 in 4 ticks) and the hands open (y ±30 → ±50) for 6 ticks; then `AIR_MEDITATE_UP` (dur 10): the reverse of sitting. **Deep Meditation** (Sneak): the sit alone, with a soft blue glow. **Snap-back on damage:** `AIR_MEDITATE_UP` plays at ×1.5 rate (`Slot.rate`) with a flinch (`HIT_STAGGER` layered, group HIT).

Lock: input locked while meditating (the body cannot move); the spirit is controlled instead (`TECH.md` 3: the client's camera entity is the `SpiritEntity`; W A S D steer it). **Blend:** sitting replaces every locomotion layer (`wAir`, `wRun`, `wSprint` forced to 0 while `MEDITATING`: **NEW** 14.3 "stance owns the body" flag). **Started by:** `Live.MEDITATING` rising/falling; `Live.PROJECTING` for the spirit. **Prediction:** on the press when on the ground and ready.

First person: on the way down the camera lowers with the body (the eye height follows `drop`: vanilla handles a sitting pose's eye height only for vehicles; **NEW** 14.4: an eye-height offset bound to `MEDITATING` = −0.9 blocks, eased over 16 ticks); the hands rest low and open at the bottom corners (`dx ±0.30, dy −0.30, dz +0.10, rx +40, ry ∓30`). When the spirit leaves, the view **lifts out of the body**: the camera entity becomes the spirit (the transition: a 10-tick camera move from the body's eyes to the spirit's, with the overlay 2.2 fading in). In the spirit the hands are the spirit's: translucent blue, drawn with `entityTranslucent` and the spirit tint (`#9FDCFF` α 0.45), floating a little higher than normal (`dy +0.06`, bob ±0.02 at 0.1 rad/tick).

Effects:

Where	What	Count
Sitting down	6 `CLOUD`-like white motes at the ground; a dust ring	6
Meditating body	a faint **blue aura**: `PowerAuraLayer`-style `energySwirl` over the inflated model, `#7FD0FF` α 0.18 slowly scrolling; 1 `END_ROD` (small, white-blue) rising from the head every 10 ticks; the **tether**: an `ENERGY_BEAM` thread from the body's forehead to the spirit (`HandTracker.forehead`), `#9FDCFF` α 0.25, drawn as a sagging `polyline` (a catenary of 12 points), visible within 32 blocks	0.1/tick + 1 thread

Where	What	Count
The spirit	the player's own model drawn **translucent blue** (Enhanced's `AstralBodyRenderer` / `AstralLayer` pattern: `entityTranslucent` with the spirit tint, no armour, no held items, name tag faint), legs folded (the spirit floats in the sit pose with arms free: a stance `SPIRIT_FLOAT` = `AIR_SIT` legs, arms at −30,∓20, ±30 drifting), a soft bob; 2 blue motes (`#9FDCFF`, scale 0.5)/tick trailing	2/tick
Through walls (spirit's client)	`EntityGlow`: hostile mobs `#8FD3FF`, players white, villagers/animals green; **caves**: a client raycast grid every 10 ticks marks air pockets within 24 blocks as faint `ENERGY_VEIL` boxes (α 0.08, blue) — "search for caves"	≤ 64 boxes
Snap-back	a white flash (overlay) + 12 blue motes at the body + 6 at the spirit's last spot, the thread **snaps** (drawn straight for 2 ticks then gone)	18
Deep Meditation	the aura only, plus 1 green-white mote (`#CFF5E0`) per 2 ticks rising when healing ticks	0.5/tick
Cap	1 spirit, 1 thread, 20 motes	

Sounds: `spirit_sit` (a soft breath out and cloth settling), `spirit_leave` (a rising shimmer with a bell-like inhale, 1.2 s), `spirit_loop` (a quiet, hollow wind with distant chimes, volume 0.3, for the spirit's client only), `spirit_return` (the shimmer reversed, a soft "whum"), `spirit_snap` (a sharp intake + a crack), `spirit_meditate_loop` (a slow breathing hush, volume 0.2).

4.6 Air Bubble — בועת אוויר

`AIR_BUBBLE_ON` (dur 10 / 1 / 4, CAST, not force): under water, both hands **circle around the head** (t 3: −150,∓30,±30 with `E` (hands up by the ears), t 6: −170,∓10,±10 (meeting over the head), t 10 rest) as the bubble forms. **Off:** the hands brush down past the face (−120 → −40 in 6 ticks) as it pops. **Air Cushion:** no gesture (a passive); the landing puff plays `LAND_SOFT` with a lighter dip (0.06).

Lock: none. **Blend:** swimming is vanilla's (Enhanced hands the body to vanilla when `getSwimAmount > 0`), so under water the gesture plays only the **arms** when not swimming (treading water); while swimming the bubble simply appears (no gesture). **Started by:** `Toggle.AIR_BUBBLE`. **Prediction:** in water and ready.

First person: hands rise beside the view `dx ±0.26, dy +0.24, dz +0.08, rx −30` then meet above `dx ±0.06, dy +0.34`; the overlay 2.2 (the fog lifts) fades in over 6 ticks.

Effects: a **bubble** around the head: an `ENERGY_VEIL` sphere r 0.55 centred on the head (follows `head`), base α 0.10 rim 0.35, white-blue `#E8F6FF`, with 2 slow highlights (`EnergyGeometry.ring` arcs α 0.3) and 1 vanilla `BUBBLE` particle per 4 ticks rising from its top; forming: 8 `BUBBLE` + a 0.1-radius pop ring; popping: 12 `BUBBLE` + `BUBBLE_POP`. Friend's bubble (M6): the same on them. Cushion landing: 6 `CLOUD` + ring. **Cap:** 2 bubbles, 10 bubbles/s.

Sounds: `air_bubble_on` (a glassy "bloop" reversed into a hush), `air_bubble_loop` (a soft underwater hum with breathing, volume 0.3), `air_bubble_off` (a pop), `air_cushion` (a soft "fwoomp").

4.7 Air Scooter — קטנוע אוויר

Mount, **`AIR_SCOOTER_MOUNT`** (dur 10 / 1 / 4, group RIDE): t 3 both arms **spin down and around** (−40,∓50,∓30 → the hands sweep low in a circle, `E`), drop 0.10; t 6 (`0`) a **hop**: drop −0.2 (up), legs fold (x −50/−50, z ±30: the ride crouch), arms out for balance (−30,0,±40); t 10 hand-over to `AIR_RIDING` (10.1). **Dismount:** `AIR_SCOOTER_DISMOUNT` (dur 8): a hop off (drop −0.1 then `LAND_SOFT`), arms down. **Air Spout,** **`AIR_SPOUT`** (hold; `AIR_SPOUT_UP` dur 10 / 1 / 4, then the stance `AIR_SPOUTING`

while `Live.AIR_SPOUTING`): both palms **press down** beside the hips (20,∓10,±30 at tick 4, `0`) as the column lifts you (the server moves you up 1 block per 2 ticks); the stance: arms held down and out (10,0,±40) with a drift ±4 at 0.3 rad/tick, legs hanging (x 10/10), head level, lean 0; release: `LAUNCHED` → `LAND_SOFT`.

Lock: riding uses a vehicle entity (`AirScooterEntity`), so vanilla's riding pose is **replaced** by the stance (Enhanced hands the body to vanilla when `isPassenger`; **NEW** 14.3: a `rideOverride` set of vehicle types the animator keeps control of). **Started by:** the mount/dismount edges of the vehicle (`isPassenger` of a scooter); `Live.AIR_SPOUTING`. **Prediction:** the mount on the press when ready and on the ground.

First person: mount: hands sweep low `dy −0.20, dx ±0.20`, then rise to the riding hands: slightly out and down, palms down (`dx ±0.22, dy −0.10, dz +0.10, rx +25`), **bobbing** with the ball (±0.03 at 0.5 rad/tick, scaled by speed) and leaning into turns (`dx` shifts ∓0.06 with `turn`). FOV ×1.12, camera roll 4° into turns, a slight forward tilt (pitch +3°) at full speed. Spout: hands down `dy −0.26, rx +30` trembling 0.008.

Effects (10.1 for the ride): the **ball**: an `ENERGY_VEIL` sphere r 0.5 under the player (white `#F4F8FF`, base α 0.12, rim 0.4) with **4 swirl bands** (`EnergyGeometry.ring` at tilted normals spinning 25°/tick, α 0.35, width 0.06) and the air lattice texture scrolling fast (`energyField(air_lattice)`, α 0.2); 3 white `CLOUD`-like motes/tick kicked up behind (more on dirt: `BLOCK` dust ×2 of the ground), a `MATTER_FLAT` dust streak on the ground (a lane 0.8 wide fading over 10 ticks); over water: `SPLASH` ×4/tick and a ripple ring; wall-ride: `BLOCK` dust from the wall. Mount: 10 `CLOUD` + ring. Dismount burst (M10): the shield-burst sphere r 2 + 16 `CLOUD`. Spout: a **column** (`SandShapes.funnel`-like frustum r 0.8 → 0.5, height to the player, swirl texture `air_swirl.png` scrolling up, α 0.35) + 4 motes/tick along it + a ground dust ring. **Cap:** 1 ball, 1 column, 30 motes.

Sounds: `air_scooter_mount` (a spin-up whoosh), `air_scooter_loop` (a continuous airy whir, pitch and volume follow speed (0.8–1.3), volume 0.5), `air_scooter_dismount` (a whoosh down), `air_scooter_burst` (the shield burst, shorter), `air_spout_loop` (a rushing updraft hum, volume 0.5), `air_spout_end` (the rush fading).

4.8 Tornado — טורנדו

`AIR_TORNADO` (dur 18 / 2 / 6, CAST, force): the Ba Gua **spin**:

t	ease	M	O	legs (z)	head	body	notes
0	S	`AIR_REST`	`AIR_REST`				
4	E	−90,40,−20	−90,−40,20	−10 / 10	−4	lean −4, drop 0.06	both arms level, hands close, palms facing: the funnel seeds between them (effects)
10	E	−100,0,30	−100,0,−30	10 / −10	0	lean 0, drop 0.08, **yaw 360**	a **full turn** with the arms opening (the hands trace the funnel's spiral)
13	**O**	**−96,−10,10**	−60,20,−30	−18 / 12	2	**lean 10**, twist 10	the **send**: the M palm pushes forward, the funnel rolls away (server: the tornado entity spawns at 13)
18	S	`AIR_REST`	`AIR_REST`		0	lean 0, twist 0, yaw 360	

Steering (M5, key held): the stance `AIR_STEERING` (aimed): the M arm points at the tornado (`aim(PROJECTILE, −8, 0, 0)`: the `PROJECTILE` target is the player's tornado entity) with a slow circling of the hand (r 6° at 0.25 rad/tick), O braced. **Air Wheel,** `AIR_WHEEL` (dur 14 / 1 / 5, then the stance `AIR_WHEELING` for 4 s): t 3 arms down and back (20,0,±40), a crouch (drop 0.16); t 6 (`0`) arms **sweep forward and up** to −70,∓20,±30 as the wheel forms at the legs; the stance: arms held forward-low (−50,∓10,±24) with a lean of 14 (a runner's lean) and legs **blurred**: the run cycle at ×1.6 speed (the `RunCharacter` `AIR_WHEEL`: legs 0.4 amplitude (hidden in the wheel), arms 0.6, lean 14, bob 0.01).

Lock: none. **Blend:** the spin is visual yaw over a walk; flying: allowed. Under water: the whirlpool version plays the same gesture at ×1.3 rate. **Started by:** cooldown edges (8 s vs 10 s), `Live.AIR_STEERING`, `Live.AIR_WHEELING`. **Prediction:** always.

First person: t 4 both hands level, close, palms facing `dx ±0.08, dy +0.10, dz +0.26, ry ∓40` (a funnel of motes spins between them); t 10 hands open `dx ±0.30` while the **camera yaws 360°** over 6 ticks (reduced motion: no yaw); t 13 `M dz +0.34, dx +0.04` push; rest. Steering: the M hand points `dz +0.28` circling 0.02. Wheel: hands forward-low `dy −0.10, dz +0.20`, FOV ×1.12, the camera **bobs** ±0.04 at 0.8 rad/tick.

Effects:

Where	What	Count
Seed (t 4–13)	a mini funnel between the hands: `SandShapes.funnel` r 0.1 → 0.3, h 0.4, white swirl texture α 0.4, spinning 40°/tick + 2 motes/tick	2/tick
Tornado	the `TornadoEntity`: **two counter-rotating funnels** (Enhanced's tornado recipe, white): outer r 0.9 → radius·1.2 at the top, h = height, 28 seg × 8 rings, 3 turns, texture `air_swirl.png` α 0.45 → 0.2, scroll 0.06, twist 0.05; inner 0.6 → 0.8r, α 0.3, counter-rotating; **6 ring bands** (`EnergyGeometry.ring` α 0.2 at 6 heights, radii following the funnel, spinning 20°/tick); 8 `CLOUD` motes/tick on a helix + 4 `BLOCK` dust of the ground per tick at the base (grass, leaves, snow it crosses) + items/entities inside spin (server)	12/tick
Lifted creatures	2 white motes/tick each; their pose: `LIFTED`-tumble (12.2)	2/tick each
End	the funnel **unravels** top-down over 8 ticks (rings fly off outward as 6 fading arcs) + 16 `CLOUD`	16
Firenado (suggestion)	the funnels tinted `#FF8A3A` with `FLAME` ×6/tick and `SMOKE` at the top	
Air Wheel	a **horizontal wheel** around the legs: 2 `EnergyGeometry.ring`s r 0.9 (normals ±Y, tilted 10° with speed) spinning 50°/tick, α 0.4 white + a translucent disc α 0.1; 4 motes/tick + ground dust streak; hits: 6 motes	4/tick
Cap	2 tornadoes, 1 wheel, 60 motes	

Sounds: `tornado_cast` (a spinning whoosh that builds over 0.6 s into a roar onset), `tornado_loop` (a deep, wide rotating roar with a whistling top layer, volume 0.7, follows the entity), `tornado_end` (the roar unwinding 0.8 s), `air_wheel_loop` (a fast whirring rush, volume 0.5), `air_wheel_hit` (a thump).

4.9 Zaheer's Path — דרכו של זהיר

The unlock moment (`AIR_TETHERS_CUT`, dur 40 / 4 / 10, played once when the path opens, not force; locks input for 40 ticks): the bender **kneels** (t 10: drop 0.48, legs x −40/−40 z ±20, head 12, arms hanging 10,0,±16: grief), holds (t 10–24: a breath), then **rises into the air** (t 32: drop

−0.6 (lifted), legs trailing x 20/20, arms spreading to −40,0,±60, head −6, lean −4: the first flight, `B`), and settles (t 40) into `ZAHEER_HOVER` (10.6). Screen: a slow white flash and the HUD thread icon appears.

True Flight (10.6 for the profile): take-off `AIR_TAKEOFF_ZAHEER` (dur 12): a small crouch (drop 0.12 at 3) then the body **lifts straight** with arms sweeping down (−20,0,±50 → 20,0,±30 at 8, `O`) and legs trailing. Flight: the profile's anchors (10.6). Hover (Sneak): `ZAHEER_HOVER`: cross-legged (`AIR_SIT` legs: x −90 z ±55), hands on the knees, a slow spin (yaw 0.4°/tick) and bob (0.06 at period 60). Landing: `LAND_SOFT` with a feather dip.

Suffocation, **`AIR_SUFFOCATE_START`** (dur 10 / 1 / 4, then the stance `AIR_SUFFOCATING` while `Live.SUFFOCATING`): t 3 the M hand **reaches** at the target (aim(LOOK, 0, 0, 0), fingers spread y 20, `O`), the O hand low and back (10,0,−30); t 6 the hand **closes slowly** (y 20 → −10, `E`) and **draws back** 6° in x: pulling the air; the stance: the M arm aimed at the target with **tremble 1.4 at 1.8**, pulsing back 4° every 20 ticks (each pulse = the victim's gasp), the O arm braced, head 4, lean 8 (leaning in), legs planted (±10). Release: `AIR_SUFFOCATE_END` (dur 8): the hand **opens** (y −10 → 30, `B`) and drops. **Air Vacuum,** **`AIR_VACUUM`** (dur 14 / 1 / 5, force): both hands reach (−80,∓20,±10 at 3), then **clench and pull in** to −60,∓50,∓20 (`I` at 6) with lean −8 (leaning back: pulling the air out of the sphere); hold to 10; rest.

Lock: the unlock moment locks 40 ticks; suffocating locks nothing (you may walk slowly: speed ×0.5 attribute). **Blend:** flying with the profile. **Started by:** the payload kind `TETHERS_CUT`; `Live.SUFFOCATING`; cooldown edges (Vacuum 15 s). **Prediction:** suffocation on the press with a breathing creature in sight within range.

First person: tethers cut: the camera lowers (eye offset −0.5 over 10 ticks) then **rises** 1.2 over 8 ticks with a white bloom; hands hang then spread wide `dx ±0.36`. Suffocation: `M dz +0.30, ry +25` (reach, fingers open) then `ry −15, dz +0.22` (closing), trembling 0.012, pulsing `dz −0.04` every 20 ticks; the O hand low `dy −0.16`. Vacuum: both `dz +0.26` then `dx ∓0.10, dz +0.14, ry ∓30` (clench in).

Effects:

Where	What	Count
Tethers cut	a column of white light (`FxShapes.cylinder` r 0.6, h 6, α 0.4 → 0 over 20 ticks, `ENERGY_BEAM`) + 30 white motes rising + a ground ring; a **red thread** (an `ENERGY_BEAM` polyline `#B33A3A` from the chest to the ground, α 0.6) that **snaps** at tick 24 (two halves fall and fade)	30
True Flight	the flight profile's effects: a faint white **wake** (two `FlightEffects` ribbons from the feet, α 0.15, white, 1.2 blocks) + 1 `CLOUD`/tick at speed; take-off: a dust ring + 8 motes; hover: 1 mote per 10 ticks drifting around	1/tick
Suffocation	a **thread of pulled air**: an `ENERGY_VEIL` ribbon from the victim's mouth to the caster's hand, white α 0.25, wavering (`flowingPath` jitter 0.1); 3 `CLOUD`-like motes/tick drawn **from the victim toward the hand** (spawned at the victim's head with velocity toward the hand); the victim's air meter bubbles (vanilla, server-set `airSupply`); the victim's pose 12.2 (clutching); every 20 ticks a `BUBBLE_POP`-like gasp mote at the victim	3/tick
Vacuum	a 3-block **sphere of absence**: an `ENERGY_VEIL` sphere r 3 with inverted rim (α 0.0 centre → 0.3 rim, dark-blue tint `#1C2A40`) and 8 motes/tick flying **inward** to the centre; fires inside go out with `SMOKE` ×2 each; creatures inside show the gasp mote	8/tick
Cap	1 thread, 1 sphere, 30 motes	

Sounds: `tethers_cut` (a slow rising chord with a snapping string, 2 s), `flight_takeoff` (a soft lift whoosh), `flight_loop` (a gentle wind rush, volume follows speed 0.2–0.5), `suffocate_start` (a sharp inhale reversed), `suffocate_loop` (a thin sucking wind with a heartbeat under it, volume 0.5; the

victim hears their own choking: `suffocate_victim_loop`), `suffocate_end` (a gasp of released air), `air_vacuum` (a deep implosive "thoomp" then silence for 0.3 s then a soft rush back), `air_vacuum_loop` (a hollow pressure hum).

4.10 Air hit reactions

Pushed targets **tumble** (12.2); blade cuts flinch; suffocated targets clutch; tornado-lifted spin. An airbender who is hit while in a spin finishes the spin (yaw gestures are never cut).

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5. Water

5.0 Water idle: the ready stance

Stance	tauIn / tauOut	Rows
`WATER_IDLE` (ambient)	6 / 6	`WATER_REST` plus Tai Chi **"wave hands like clouds"**: the M hand draws a vertical figure-eight (x −20 + 20·sin(0.07·age), y 10·sin(0.14·age), z 10 + 8·cos(0.07·age)), the O hand the mirror a half-phase behind; the weight shifts (lean 3·sin(0.035·age), legs x ∓8 following); head level; drop 0.06.
`WATER_SOURCE` (aimed, while `Live.WATER_DRAWING`: a source is being drawn, 3 ticks before any ability)	1 / 3	the M hand **reaches toward the source** (`aim(ANCHOR, 0, 0, 0)`: the anchor is the water block / the skin at the hip / the plant) for 3 ticks: a quick pull.
`WATER_OCTOPUS` (ambient while the octopus is up)	3 / 5	arms slightly out (−20,0,±30), elbows soft, a slow sway ±3; `occupy` none.
`WATER_ARMS` (ambient while arms are on)	3 / 5	5.5.
`WATER_RIDING` (spout)	2 / 4	10.3.
`WATER_HEALING` (aimed, hold)	2 / 4	5.2.
`BLOOD_HOLD` (aimed)	1 / 4	5.7.

Ambient particles of a master (Water ES ≥ 60): 1 `bending:droplet` every 10 ticks lifting from a nearby water source (within 4) and orbiting the hands once before falling; at ES 100 a thin water ring (`MATTER_FLAT` `#3C8DD6` α 0.15, r 0.8) rotates slowly at the feet while standing still. Full moon: 2 extra `END_ROD`-like silver motes per 10 ticks.

Water first-person idle: both hands at the belly, palms up, slowly circling (`dx ±0.10, dy −0.10, dz +0.14, rx −20, ry ∓20`, r 0.03 at 0.07 rad/tick); a droplet orbits them.

The source draw (every Water ability starts with it): 3 ticks before the ability's own keyframes, the M hand reaches toward the source and a **water ribbon** flies from it to the hand (an `ENERGY_VEIL` ribbon `#3C8DD6` α 0.5, width 0.08, `flowingPath` jitter 0.15 over 3 ticks, `MATTER` alpha (not additive), with 3 `bending:droplet`s along it). From the **Skin**: the ribbon rises from the hip (the Skin item is drawn on the belt: `LOOKS.md` 6). From a **plant**: the plant greys over 3 ticks (a `BLOCK` tint change via the temp state) and 4 droplets rise from it. Every table below starts at the ability's tick

0 = after the draw; the draw is `WATER_DRAW` (dur 3 / 1 / 1, group CAST, not force) and the ability's gesture starts as it ends (the gesture is started with `param = 3` delay, or the animator chains them: **NEW** 14.6 gesture chaining).

5.1 Water Shot — יריית מים

`WATER_SHOT` (dur 14 / 2 / 5, CAST, force): Tai Chi "push hands":

t	ease	M	O	legs (z)	head	body	notes
0	E	`WATER_GATHER`	`WATER_GATHER`	−12 / 12	2	lean −3, drop 0.10	the water gathers at the belly between the palms (effects)
5	E	**−100,−20,20**	−60,30,−10	−14 / 10	0	lean 0, drop 0.08	the M hand **rises in an arc** (wrist leading: y leads x), the O hand stays low: the stream rises with the hand
8	**O**	**−86,0,4**	−40,10,−20	**−20 / 16**	2	**lean 8**, drop 0.06	the **push**: the palm shoots forward, the weight shifts to the front leg; the shot leaves at tick 8
11	S	−80,0,6	−40,10,−20		2	lean 6	
14	E	`WATER_REST`	`WATER_REST`	−6 / 6	0	lean 0, drop 0.06	

M4 (two streams): the O hand pushes at tick 10 (−86,0,−4), lean 10. M8: a third from the M hand at 12 (dur 16). M10 lob: the push aims up (`mAim(LOOK, −30, 0, 0)` at 8).

Water Whip (Sneak, hold): **`WATER_WHIP_START`** (dur 8 / 1 / 3) then the stance `WATER_WHIPPING` (aimed, tauIn 2 / out 4) while `Live.WATER_WHIPPING`: start: from the gather, the M arm **sweeps up and back** to `WATER_WHIP` (−70,40,−10: the whip hangs behind the shoulder); the stance: the arm **lashes** every 10 ticks: a 4-tick swing from `WATER_WHIP` to aim(LOOK, −10, 20, 0) (`0`) and a 6-tick return (`E`), the wrist leading (y swings 40 → −20 → 40), lean 4 → 8 → 4 with each lash, the O hand balancing low; `occupy(M)`. End: the arm lowers and the whip falls (`WATER_WHIP_END` dur 8).

Lock: none. **Blend:** walking continues; the push's step drives the legs for 3 ticks; under water: the same at ×1.2 rate (water is the element). **Started by:** the slot's cooldown edge (1 s), `Live.WATER_WHIPPING`. **Prediction:** a source in reach (the client checks: water within 8, a skin with charge, or M7+ with a plant) and ready.

First person: draw: the M hand dips toward the source `dz +0.30, dy −0.14` (3 ticks). Shot: t 0 both hands at the belly `dx ±0.08, dy −0.12, dz +0.14, rx −20` (water pooling between them); t 5 `M dx −0.10, dy +0.18, dz +0.12, ry −30` (the arc up and across); t 8 `M dz +0.34, dy +0.06, dx +0.04, rx −10` (push); recovery. Whip: the M hand cocks back and high `dx +0.24, dy +0.26, dz −0.04, ry +30`, then lashes `dz +0.34, dx −0.08, ry −25` over 4 ticks, every 10 ticks; the whip ribbon hangs from the hand (effects).

Effects:

Where	What	Count
Gather (t 0–5)	a **water blob** between the hands: an `ENERGY_VEIL`... no, a `MATTER_TEXTURED(water_still)` sphere r 0.15 → 0.3 (vanilla's `block/water_still` animated texture, tinted `#3C8DD6`, α 0.7) + 2 droplets/tick orbiting	2/tick + 1 sphere

Where	What	Count
Stream flight	the `WaterShotEntity` drawn as a **ribbon** of water: an `ENERGY_VEIL`-style ribbon chain through the last 8 positions (width 0.25 head → 0.08 tail, `MATTER_TEXTURED(water_flow)` α 0.75 with a white highlight ribbon α 0.4 on top, width 0.05) + 2 droplets/tick falling off + 1 `SPLASH`/tick	3/tick
Hit	a **splash**: 14 `SPLASH` + 6 droplets + a `MATTER_FLAT` wet disc on the ground under the target (α 0.3, 40 ticks); the target **drips** (12.3)	20
Miss (block)	10 `SPLASH` + a wet decal (`webDecal`-style quad with `wet.png`, α 0.4 → 0 over 60 ticks)	10
Vs fire	a **steam burst**: 10 `CLOUD` white + the fire's particles stop; a fireball hit: 16 `CLOUD` + a hiss	10–16
Whip	a **hanging rope of water**: a chain of 10 segments (verlet-ish: each follows the previous with gravity, client-simulated) from the hand, `MATTER_TEXTURED(water_flow)` ribbon width 0.14 → 0.08, α 0.75, highlight ribbon; droplets 1/tick from the tip; lash: the tip whips to the target (the chain's end is driven to the hit point over 4 ticks) with 8 `SPLASH` at the hit	1/tick + 8
Cap	3 shots, 1 whip, 40 droplets	

Sounds: `water_draw` (a liquid "shloop" rising, 0.3 s), `water_shot` (a liquid whoosh with a wet onset), `water_hit` (a splash and a slap), `water_splash_block` (a splash), `water_vs_fire` (a hiss and a steam puff), `water_whip_swing` (a wet whoosh), `water_whip_crack` (a liquid crack), `water_whip_loop` (a soft sloshing while it hangs, volume 0.25).

5.2 Healing — ריפוי

`WATER_HEAL_START` (dur 10 / 2 / 4, CAST, not force) then the stance `WATER_HEALING` while `Live.WATER_HEALING`: start: from the gather the hands **rise together** to `WATER_HEAL` (palms down at chest height, −60,∓20,∓10) over 8 ticks (`E`), head 6 (looking at the hands or the friend), lean 2, drop 0.08; the stance: the hands **hover** with a slow circular stir (x ±4, y ±6 at 0.15 rad/tick, both hands in the same direction: Katara's circles), aimed at the friend (`aim(LOOK, −10, 0, 0)` for the M hand when a friend is targeted, else both at the belly: self-healing hands press to the chest (−70,∓30,∓20)), a gentle sway; `occupy` both. End: the hands lower (`WATER_HEAL_END` dur 8). **Cleanse (Sneak):** `WATER_CLEANSE` (dur 10 / 1 / 4, force): a **splash**: both hands scoop from the belly (t 3: `WATER_GATHER`) and **throw upward** at the target (t 6: `aim(LOOK, −20, ∓10, 0)`, `0`), lean 6; rest.

Lock: none, but healing breaks if either moves fast (server). **Blend:** the stance rides over standing; walking slowly keeps it. **Started by:** `Live.WATER_HEALING`; cooldown edge (Cleanse 10 s). **Prediction:** a source in reach, and a target below full health in sight (or yourself below full).

First person: hands rise to the centre, palms down, `dx ±0.10, dy +0.04, dz +0.20, rx +20, ry ∓15`, stirring (a circle r 0.02 at 0.15 rad/tick); the glowing water pools between them (effects); self-healing: the hands turn inward toward the camera `rx +50, dz +0.08` (palms toward you). Cleanse: scoop `dy −0.12` then throw up `dy +0.30, dz +0.24`.

Effects:

Where	What	Count
Hands	a **glowing water veil** on the hands: `MATTER_TEXTURED(water_still)` sphere r 0.2 per hand α 0.6 tinted `#8FE3FF` + an additive glow (`FxShapes.glow` r 0.3 `#CFF5FF` α 0.3, `ENERGY_BEAM`); 2 `END_ROD`-like glow motes (`#CFF5FF`, scale 0.4)/tick drifting up	2/tick

Where	What	Count
The healed	a **water veil** over the target: `PowerAuraLayer`-style `energySwirl` with `water_flow` texture over the inflated model, `#8FE3FF` α 0.3 scrolling up; 3 glow motes/tick rising around them; a ring of 6 droplets orbiting at the waist; each heart healed: a brief white pulse of the veil (α +0.3 for 4 ticks)	3/tick
Stream	an `ENERGY_VEIL` ribbon from the healer's hands to the target, `#8FE3FF` α 0.35, gentle (`flowingPath` jitter 0.05), with 2 droplets/tick flowing along it	2/tick
Cleanse	a splash arc from the hands to the target (a ribbon over 4 ticks) and 12 `SPLASH` + 8 glow motes bursting at them; poison particles stop	20
Cap	2 targets (M4), 30 motes	

Sounds: `heal_start` (a soft liquid shimmer rising), `heal_loop` (a gentle bubbling with a warm pad underneath, volume 0.4; a soft chime every heart healed: `heal_tick`), `heal_end` (the shimmer fading), `cleanse` (a splash with a bright "pling" and a sigh).

5.3 Water Octopus — תמנון מים

Form, **`WATER_OCTOPUS_FORM`** (dur 20 / 2 / 6, CAST, not force): t 4 the gather (`WATER_GATHER`, drop 0.12); t 8 the arms **circle outward and up** (−110,∓40,±30, `E`), the water rising around the body; t 14 both arms **sweep down and out** to −30,0,±50 (`E`) as the arms of water spread; t 20 `WATER_OCTOPUS` stance hand-over. **Drop:** `WATER_OCTOPUS_DROP` (dur 10): the hands **clap together** at the chest (−70,∓50,∓20 at tick 4, `O`) and the arms collapse into a puddle; rest. **Lash (Sneak),** **`WATER_OCTOPUS_LASH`** (dur 12 / 1 / 4, force): both arms **whip forward** from the sides (t 3: −40,0,±60; t 6: `WATER_PUSH` with `O` and lean 10; rest), every tentacle lashes with them.

The attack strikes (auto, when you hit someone): `WATER_TENTACLE_STRIKE` (dur 8, group HIT... no: group `CAST` would cancel casts; a **new group** **`AUTO`** (**NEW**, 14.2) for reactive extra gestures): a small shoulder twist (twist ±10) as two tentacles strike; the arms are not keyed (your punch plays).

Lock: none; speed ×0.85 (attribute). **Blend:** the octopus stance adds only a sway; walking, sprinting, jumping as usual; under water: the arms are 8 at any mastery and the sway is larger (±6). **Started by:** `Toggle.WATER_OCTOPUS`; payload kinds `OCTOPUS_LASH`, `TENTACLE_STRIKE`, `TENTACLE_SLAP` (a projectile slapped). **Prediction:** source in reach and ready.

First person: form: gather `dy −0.12`, rise `dy +0.30, dx ±0.24, ry ∓30`, sweep down and out `dx ±0.34, dy −0.06, rx +20`; the **tentacles are visible in first person**: the two nearest arms (the ones attached at the shoulders) are drawn curling into the lower corners of the view (the octopus renderer draws all arms from the body's anchor points, which the first-person camera sees from inside: the two front-lower arms sit at the screen's bottom edges, waving). Lash: both hands sweep forward `dz +0.32`. Drop: hands clap at the centre `dx ∓0.02, dz +0.14`.

Effects:

Where	What	Count
Form	the water **climbs the body**: the octopus layer (`LOOKS.md` 1.3: a `MATTER_TEXTURED(water_flow)` body veil over the inflated model, α 0.55 `#3C8DD6`, plus **8 tentacles**: each a chain of 7 segments (`ENERGY_VEIL`-style ribbon quads with the water texture, width 0.22 root → 0.06 tip, α 0.7), rooted at the body (2 at each shoulder, 2 at each hip), **swaying** with a per-arm phase: `angle = base + 25°·sin(0.12·age + i·0.8)` on two axes, tips curling (the last 3 segments bend 30° more)) rises over 14 ticks (the veil's clip plane bottom-up; tentacles grow from root to tip 1 segment per 2 ticks); 4 droplets/tick	4/tick + layer

Where	What	Count
Standing	1 droplet per 2 ticks drips from a tip; the sway; under water the tentacles move slower and wider	0.5/tick
Slap	the nearest tentacle **snaps** to the projectile (its tip driven to the hit point over 2 ticks, then back over 6) with 6 `SPLASH`	6
Strike (on your hit)	two tentacles snap to the target with 8 `SPLASH` and a wet disc	8
Lash	all 8 tentacles whip forward in a fan (tips to the cone's edge, 4 ticks) + 16 `SPLASH`	16
Grab (M10)	two tentacles wrap the creature (their last 4 segments bend around its box) for 20 ticks	
Drop	the layer collapses in 6 ticks (clip top-down) into a **puddle**: a `MATTER_FLAT` disc r 1.2 α 0.5 fading over 40 ticks + 20 `SPLASH`	20
Fire on it	a tentacle **evaporates**: 10 `CLOUD` and it shrinks tip-first over 10 ticks	10
Freeze on it	the tentacles **stiffen** (sway → 0 over 2 ticks) and tint `#BFE9FF` with frost lines (`ENERGY_BEAM` white cracks α 0.5) for 2 s	
Cap	1 octopus (8 chains), 40 splashes	

Sounds: `octopus_form` (a surging swell of water rising around, 1 s), `octopus_loop` (a soft lapping and sloshing, volume 0.35, pitch 0.9), `octopus_slap` (a wet slap), `octopus_strike` (two quick wet thwacks), `octopus_lash` (a wide liquid whip-crack), `octopus_drop` (a collapse into a splash), `octopus_steam` (a hiss), `octopus_freeze` (a crackling freeze).

5.4 Water Spout — עמוד מים

Rise, `WATER_SPOUT_UP` (dur 16 / 2 / 5, group RIDE) then the stance `WATER_RIDING` (10.3) while `Live.WATER_SPOUTING`: t 4 the gather with a deep knee bend (drop 0.18, legs ±10); t 8 both arms **spiral upward** (−140,∓30,±20 with y sweeping 30 → −30, `E`): the column rises under the feet (server: the spout entity lifts you 0.5 block/tick); t 12 arms settle out and down (−40,0,±40) for balance, legs **slightly bent and apart** (x −10/−10, z ±12: standing on water); t 16 hand-over. The ride (10.3): arms out and low, swaying with the column's spin (twist ±6 at 0.2 rad/tick), the body leaning into steering (`lean 8·fwd`, `roll 6·side`), a bob (0.04 at period 16). **Ram:** `WATER_SPOUT_RAM` (group AUTO, dur 8): a forward lean kick (lean +14 for 3 ticks, `0`). **Collapse:** `LAUNCHED` → the fall → `LAND_SOFT` with a big splash. **Wave Ride (Sneak),** `WATER_WAVE_RIDE` (dur 12 / 1 / 4, then the stance `WATER_SURFING` for the ride): both arms **sweep back** (20,0,±40 at 3) then **forward and up** (−90,∓10,±20 at 6, `0`) as the wave rises behind; the surf stance: a surfer's crouch (drop 0.16, legs x −20/−20 z ±14: sideways: the body yaws 30° to the travel direction via `twist 30`), arms out (−40,∓20,±50), lean 6. **Freeze (M8, Sneak while riding):** the column freezes: the ride stance stops swaying (the `tauOut` 4 brings it still) and a stomp gesture (`EARTH_WALL`'s stomp keys) plays.

Lock: the spout is a vehicle (`WaterSpoutEntity`): vanilla's riding pose is overridden (14.3). **Blend:** flying no; swimming no (you ride). **Started by:** mount/dismount edges, `Live.WATER_SPOUTING`, `Live.WATER_SURFING`, payload kind `SPOUT_RAM`. **Prediction:** standing in/by water (or a skin at M5+) and ready.

First person: rise: gather `dy −0.14` then both hands spiral up `dy +0.34, dx ±0.14, ry ∓40` (the camera rises with the body: smooth interpolation of the vehicle), then settle out `dx ±0.28, dy −0.08, rx +15`; riding: the hands sway with the spin (`dx ±0.03 sin`), the camera **rolls** 3° with steering and bobs 0.04; FOV ×1.05; water spray at the bottom of the screen (droplet particles spawned in front of the camera at the column's top ring: 2/tick, visible as flying drops). Ram: a 3-

tick forward lurch (`dz +0.06` of both hands, camera pitch +2). Surf: hands out `dx ±0.30, dy −0.04`, the camera yawed 30° from travel (the body's twist: the first-person camera keeps the look; only the hands shift `dx` by +0.08).

Effects:

Where	What	Count
The column	the `WaterSpoutEntity` renderer: a **twisted frustum** (`SandShapes.funnel` with `water_flow` texture, r 1.2 at the base → 0.7 at the top, 3 turns, scroll up 0.08/tick, twist 0.06, α 0.75 `#3C8DD6`, `MATTER_TEXTURED`) with a white **foam ring** at the top (`EnergyGeometry.ring` width 0.12 α 0.6) and at the base (r 1.6, α 0.5); **spiral highlight ribbons** (2 `ENERGY_VEIL` helices α 0.4, width 0.08, white); 6 droplets/tick flung outward from the top ring + 4 `SPLASH`/tick at the base + 2 `BUBBLE` rising inside	12/tick
Over land	the base loses the splash; a wet trail (`MATTER_FLAT` disc chain α 0.3, 60 ticks)	1 disc/4 ticks
Ram	20 `SPLASH` at the target + a foam burst ring	20
Collapse	the frustum **falls** (its height → 0 over 6 ticks, α 0.75 → 0) into a 3-block **splash disc** (`FxShapes.disc` r 0 → 3 α 0.6 → 0 over 8 ticks) + 30 `SPLASH` + 10 droplets	40
Frozen (M8)	the frustum turns to **ice**: texture swap to `ice` (translucent `#BFE9FF` α 0.8), the spin stops, frost lines, 6 `SNOWFLAKE`; it is a temp block pillar for 6 s then shatters (16 `BLOCK` ice dust)	6 / 16
Wave Ride	a **wave**: a `SandShapes.ridge` lane in water (crest 1.1, width 3, length 6, the `water_flow` texture, α 0.8) running ahead of the rider with a foam crest ribbon (white α 0.6) + 8 `SPLASH`/tick + 4 droplets	12/tick
Cap	1 spout, 1 wave, 60 splashes	

Sounds: `spout_rise` (a rushing surge rising in pitch, 1 s), `spout_loop` (a roaring, churning water column with a whirl, volume 0.7, pitch with height 0.9–1.1), `spout_ram` (a heavy splash-thud), `spout_collapse` (the column dropping into a broad splash), `spout_freeze` (a crackling freeze with a deep creak), `wave_ride` (a long rolling wave with foam hiss, loop `wave_loop` volume 0.6), `wave_end` (the wave breaking).

5.5 Water Arms — זרועות מים

On, `WATER_ARMS_ON` (dur 12 / 2 / 4, CAST, not force): the gather at 3; at 6 both arms **thrust forward** (−80,∓10,±6, `O`) and the water **runs down them** from the shoulders to the hands; at 9 the arms **flex wide** (−60,∓30,±40, `B`: the new arms test their reach); at 12 the `WATER_ARMS` stance: arms slightly forward and out (−30,∓10,±24), **always moving**: a slow serpentine (x ±6, y ±8 at 0.2 rad/tick, each arm in counter-phase: Ming-Hua's arms never rest), `occupy` none (the vanilla punch plays as a lunge: the arm's punch swing is kept, and the water arm **extends** to the reach: effects). **Off:** the arms **shake off** the water (t 3: −40,∓20,±50 with tremble 3 at 3.0; t 8 rest) and it splashes down. **Grab (Sneak),** `WATER_ARM_GRAB` (dur 12 / 1 / 4, force): the M arm **shoots out** (aim(LOOK, 0, 0, 0), `O` at 3), **closes** (y −20 at 5), and **hauls back** (−40,30,20 at 8, `I`, lean −10: the body leans back with the pull); rest. **Climb (M4, Space at a wall):** `WATER_ARM_CLIMB` (dur 10): both arms reach up (−160,∓10,±10) and the body is pulled up 4 blocks (server velocity), arms sweeping down to −60 as you rise (the pull). **Swing (M7):** the same to a ceiling, then the `LAUNCHED` stance.

Lock: none. **Blend:** the stance's serpentine adds to walking arms (ADD rows, so the swing shows through); sprinting is fine; swimming: vanilla (the arms are drawn still extended on the vanilla pose). **Started by:** `Toggle.WATER_ARMS`; cooldown edges (grab 4 s); payload kinds `ARM_CLIMB`, `ARM_SWING`. **Prediction:** source in reach and ready.

First person: on: gather `dy −0.12`, thrust `dz +0.30`, flex `dx ±0.30`; the arms are **drawn as water** from the elbows down: the first-person arm render is replaced (`RenderArmEvent` override: the arm's texture becomes the water texture with the highlight ribbon, slightly longer (scale z 1.3), translucent) and the hands are water claws (a `ModelPart` of three water "fingers"); the serpentine is a slow `dx ±0.03, dy ±0.02` wander. Items in hand are hidden. Punching: the hand shoots forward (`dz +0.40` for 2 ticks, beyond the normal swing) and the water arm's extension stretches toward the target (the water arm mesh extends to the hit point). Grab: `dz +0.40, ry −20` then haul back `dz −0.10, dx +0.16`.

Effects:

Where	What	Count
On	water runs down the arms: 6 droplets/tick for 6 ticks sliding from the shoulders; the **water arm layer** (`LOOKS.md` 1.3): each arm wrapped in a `MATTER_TEXTURED(water_flow)` sleeve (an inflated arm part, `CubeDeformation(0.25)`, α 0.65 `#3C8DD6`) plus an **extension**: a tapered ribbon chain from the hand to `reach` blocks along the arm's direction (width 0.18 → 0.05, 6 segments with a lazy S-curve (sin along the length ±0.15), α 0.7) ending in a **hand of water** (3 short ribbons as fingers); highlights; 1 droplet per 2 ticks from the tips	6/tick then 0.5/tick
Punch	the extension **snaps straight** to the target over 2 ticks (its end driven to the hit point) and relaxes over 6; 8 `SPLASH` at the hit	8
Grab	the extension wraps the target (the last 2 segments bend around its box) for 4 ticks and **drags**; 6 `SPLASH`	6
Climb / swing	both extensions shoot up to the wall/ceiling (anchored: `aim(ANCHOR)`), the body is pulled; 6 droplets at the anchor	6
Off	16 `SPLASH` + a puddle disc	16
Freeze on them	the sleeves tint ice `#BFE9FF` and stop moving 2 s; frost lines	
Fire on them	the extensions shrink tip-first with 8 `CLOUD`	8
Cap	1 pair, 30 splashes	

Sounds: `water_arms_on` (a surge running down with a wet flex), `water_arms_loop` (a quiet continuous slosh, volume 0.25), `water_arm_strike` (a wet snap-thwack), `water_arm_grab` (a wet grip and a drag), `water_arm_climb` (a liquid pull with a creak), `water_arms_off` (a shake-off splash).

5.6 Ice — קרח

Ice Knives, **`WATER_ICE_KNIVES`** (dur 12 / 1 / 4, CAST, force): the gather (t 0, `WATER_GATHER`); t 3 the M hand **rises and freezes the water** (−70,30,−10 with a 2-tick **hold**: the knives crystallise, a `tremble 1.0 at 4.0` for 2 ticks: the frost shiver); t 6 (`O`) a **flick** forward and out (−88,−30,24, lean 8, twist 12), knives fly at 6, 7, 8; t 12 rest. M6+: the O hand flicks at 8 too. **Ice Spikes (Sneak),** **`WATER_ICE_SPIKES`** (dur 14 / 1 / 5, force): the gather; t 4 both palms **press down** toward the aimed ground (`aim(LOOK, 30, ∓10, 0)`, `E`); t 7 (`I`) both hands **jerk up** 20° (−70) as the spikes **erupt** (server: spikes rise 1 block per tick from 7); lean −6, drop 0 → −0.04 (a hop); t 14 rest.

Freeze, **`WATER_FREEZE`** (dur 14 / 1 / 5, force): the gather; t 4 the M arm **sweeps out level** over the water (`WATER_WAVE`: −110,−30,30 / −50,30,−20, `E`); t 7 (`O`) the arm **drops and stiffens**: −80,0,20 with the hand flat (y 0), **tremble 1.2 at 4.5** for 3 ticks (the freezing shiver through the arm), the body **sinks** (drop 0.14) and stiffens (lean 2): the ice spreads from the aim point at tick 7 outward 1 block per tick; t 14 rest. On a creature: the same, aimed at its feet. **Snow Throw (Sneak),**

WATER_SNOW_THROW (dur 10 / 1 / 4, force): the gather at the belly (snow forms: effects); t 4 both hands **scoop up** (−100,∓20,∓10, E); t 6 (0) **fling** forward (WATER_PUSH, lean 8): the snowballs fly in a spread at 6; t 10 rest.

Ice Wall, **WATER_ICE_WALL** (dur 14 / 1 / 5, force): the gather; t 4 both arms **sweep across** low (−40,∓40,∓20 with y crossing: the hands pass each other, E), drop 0.14; t 7 (0) both arms **rise fast** to −130,∓10,±20 with the frost shiver (tremble 1.4 at 4.5, 3 ticks), lean −4: the wall **shoots up** at the midpoint; t 10 hold; t 14 WATER_REST. **Ice Dome (Sneak),** **WATER_ICE_DOME** (dur 14 / 1 / 5): t 4 arms gather; t 7 both arms **rise and arch overhead** (−170,∓20,±20 with the shiver) and **sweep down the sides** to −60,0,±60 at 10 (E: tracing the dome); rest.

Lock: none. **Blend:** walking continues (the freeze's sink drives no legs); under water: Freeze aimed up freezes the surface above you. **Started by:** cooldown edges (2 s / 6 s; 4 s / 2 s; 8 s / 15 s), each pair told apart by length as Thanos's are; Live none. **Prediction:** source in reach (ice from the Skin or water) and ready; Freeze: water or a wet creature in sight.

First person: knives: gather dy −0.12; the M hand rises dy +0.14, dz +0.12, ry −20 and **shivers** (tremble 0.015 for 2 ticks) as three ice knives appear fanned on the fingers (effects); flick dz +0.32, dx +0.12, ry +20. Spikes: hands press down dy −0.16, dz +0.22, rx +35 then jerk up dy +0.06. Freeze: the M hand sweeps across dx −0.24 → +0.20, dy +0.08 then drops flat dy −0.06, dz +0.24, rx +30 and shivers; the frost spreads from the aim point. Snow: scoop dy +0.20, dx ±0.10, ry ∓30 then fling dz +0.30. Wall: cross low dx ∓0.16, dy −0.12 then rise dy +0.34, dx ±0.12 with the shiver. Dome: rise overhead dy +0.40, dx ±0.06 then sweep down and out dx ±0.36, dy −0.10; the dome's inside is seen (ice walls around the view).

Effects:

Where	What	Count
Knives form	3–7 **ice knives**: thin boxes 0.5 long (MATTER_FLAT ice #BFE9FF α 0.85 with a white edge) appearing on the fingers with 3 SNOWFLAKE each and a frost **crackle** (an ENERGY_BEAM white bolt 0.3 long, 2 ticks)	3 each
Knife flight	the IceKnifeEntity: the box spinning end over end 30°/tick, a faint sparkle trail (1 SNOWFLAKE/2 ticks)	0.5/tick
Knife hit	the knife **shatters**: 10 BLOCK dust of ice + 4 SNOWFLAKE; the chilled target gets frost at the hit (12.3)	14
Spikes	each spike is a temp block bending:ice_spike (a pointed model, 3 states of rise: 0.33 / 0.66 / 1.0 height, cycled at ticks 0, 1, 2 of each spike) with 6 BLOCK ice dust and 2 SNOWFLAKE on eruption; a frost **crack line** runs along the ground ahead (an ENERGY_BEAM white polyline α 0.5, 2 ticks ahead of the spikes); melting: 4 DRIPPING_WATER each	8 per spike
Freeze (water)	ice spreads: temp frosted_ice-look blocks placed outward in rings 1 block/tick, each with 3 SNOWFLAKE and a frost crackle bolt; a white **frost ring** (MATTER_FLAT, r growing with the ice, α 0.4 → 0)	3 per block, capped 30/tick
Freeze (feet)	the victim's feet are **encased**: a MATTER_FLAT ice block mesh (0.9 × 0.6 × 0.9) at their feet with frost lines, 8 BLOCK ice dust; it shatters on release (12 dust)	8
Freeze (constructs)	frost lines over the construct (ENERGY_BEAM white cracks α 0.5 over its surface) and a tint shift to #BFE9FF	
Snow	4+M ITEM_SNOWBALL-trailing projectiles (vanilla snowball render), impact 6 SNOWFLAKE + ITEM_SNOWBALL ×4; the settled snow (M6): temp snow layers with 10 SNOWFLAKE	10

Where	What	Count
Ice Wall	rows of temp **ice** blocks (vanilla ice texture, translucent) rise bottom-up over 3 ticks, each block with 3 `SNOWFLAKE` + a frost crackle; a white frost **bloom** on the wall's face (an `ENERGY_BEAM` branching bolt tree across the face α 0.6 → 0 over 6 ticks); standing: 1 `SNOWFLAKE`/20 ticks; melting (fire): `DRIPPING_WATER` ×4 per block + `CLOUD` ×6 (steam); end: blocks fade to `DRIPPING_WATER` ×2	4 per block
Ice Dome	a hemisphere of temp ice blocks (r 2–3, hollow) closing **top-down**... bottom-up over 4 ticks; inside: the light is blue-tinted (the ice blocks do that)	4 per block
Cap	8 knives, 1 spike line (≤ 15), 1 ice disc (≤ 169 blocks at M10: placed over 13 ticks), 3 walls, 1 dome, 60 snowflakes	

Sounds: `ice_knives_form` (three crystalline "tinks" with a frost hiss), `ice_knife_throw` (a thin icy whistle), `ice_knife_shatter` (a glassy shatter), `ice_spikes` (a cracking eruption: a rapid series of ice cracks rising in pitch), `freeze_cast` (a sharp frost hiss sweeping outward with crackles, 0.8 s), `freeze_feet` (a crunching encasement), `ice_shatter` (glassy break), `snow_throw` (a soft fluffy whoosh ×4), `snow_hit` (a soft pat), `ice_wall_rise` (a fast crystalline growth: crackles rising into a resonant ring), `ice_wall_break` (shatter), `ice_melt_loop` (a hissing drip, volume 0.3), `ice_dome` (the wall sound, longer, with a hollow boom as it closes).

5.7 Blood Bending — כשפות דם

Grab, `BLOOD_GRAB` (dur 16 / 2 / 5, CAST, force; **locks movement** via `Live.BLOODBENDING` while held) then the stance `BLOOD_HOLD` (aimed, tauIn 1 / out 4) while `Live.BLOODBENDING`:

t	ease	M	O	legs (z)	head	body	notes
0	E	`WATER_REST`	`WATER_REST`		0		
4	E	−50,20,10	−50,−20,−10	0/0 (±8)	4	**lean 8**, drop 0.08	the hands rise slowly, fingers curling (y in), the body **hunches**
8	E	**`BLOOD_CLAW`**	**`BLOOD_CLAW`**	(±8)	6	**lean 14**, drop 0.12	both claws forward, **tremble 3.0 at 2.2** begins (Hama's hands)
12	**I**	aim(LOOK, −10, 15, 0) with tremble 3	aim(LOOK, −10, −15, 0) with tremble 3	(±8)	8	lean 16, drop 0.14	the claws **clench** toward the target (y pulls in 12°): the victim is seized at tick 12 and lifted
16	S	`BLOOD_HIGH`-M aimed	`BLOOD_CLAW`-O aimed	(±8)	6	lean 14, drop 0.12	hand over: one claw high (the puppet hangs from it), the other low

The stance **`BLOOD_HOLD`**: the M claw aims at the victim (`aim(HELD, −20, 10, 0)`: the held entity), raised when the victim is high; the O claw at the victim's feet (`aim(HELD, 30, −10, 0)`); both tremble 3.0 at 2.2; **the hands move the puppet**: the bender's look drives the victim's position (server), so the claws follow it; a slow alternating twist (±6 at 0.1 rad/tick: the puppeteer's shoulders); head 6, lean 14, drop 0.12, legs planted ±8; `occupy` both. **Release:** `BLOOD_RELEASE` (dur 10): the claws **open** (y → ±40, `B`) and drop, the body straightens (lean 14 → 0 with `E`). **Fling,** **`BLOOD_FLING`** (dur 12 / 1 / 4, force): from the hold, both arms **whip** from the claws to aim(LOOK, −20, ∓30, 0) with twist 20 and lean 4 → 18 (`I` at 4), a half-step (legs −14/14); rest at 12. **Severance,** **`BLOOD_SEVER`** (dur 30 / 2 / 8, force; locks 30 ticks): the victim is drawn to within 1 block (the hold); t 6 the M claw **rises** to the victim's forehead (`aim(HELD, −45, 0, 0)` → the forehead), the O claw to its chest (`aim(HELD, 0, 0, 0)`), both trembling; t 14 (`I`) the M **thumb presses** (x −6, a 2-

tick push forward) and **holds** (t 14–24: the hands go **still**: tremble 0 — the one moment a blood bender is still), head 8 looking into the victim's eyes, lean 18; t 26 the M hand **draws back** (`0`) as the victim's limbs **sag**; t 30 release into `WATER_REST`. The victim's pose: 12.4.

Lock: the bender cannot move while holding (`BLOODBENDING` input lock); the victim cannot move (`BLOODBENT` lock). **Blend:** none needed (standing). Under water: the same. **Started by:** `Live.BLOODBENDING` (grab), its fall (release), payload kinds `BLOOD_FLING`, `BLOOD_SEVER`. **Prediction:** a target in sight within range on a full moon (or M10): the client knows the moon phase and its own mastery.

First person: the hands rise into the view as **claws**: `dx ±0.16, dy +0.08, dz +0.22, rx −20, ry ∓25, rz ∓15`, then clench (`ry` pulls ∓10 more, `dz +0.02`), trembling 0.016 (strong), the whole view **tilts** 2° forward (camera pitch +2: the hunch); holding: the hands follow the puppet's height (the M hand `dy` from +0.00 to +0.30 as the victim rises; the O hand `dy −0.10`), the overlay 2.2 (red vignette, heartbeat). Fling: both hands whip `dx ±0.20, dz +0.30` in 3 ticks. Sever: the M hand rises to the centre-top of the view `dx −0.02, dy +0.26, dz +0.30, rx −30` (the thumb on the forehead, the victim's face fills the view since they are 1 block away), the O hand at the centre `dy +0.02, dz +0.30`; the tremble stops for 10 ticks; a red pulse; then the hands fall. Release: hands open and fall `dy −0.20`.

Effects:

Where	What	Count
Wind-up	2 **dark-red motes** (`#7A0E1E`, scale 0.5, α 0.6)/tick rising from the claws; the moon's light: a faint silver glow on the hands (`FxShapes.glow` r 0.2 `#E8F0FF` α 0.2)	2/tick
Seize	the victim is outlined (`EntityGlow` dark red `#7A0E1E`); **4 red threads** (`ENERGY_BEAM` `#9A1E2E` α 0.45, width 0.03, `flowingPath` jitter 0.08) run from the claws to the victim's wrists and ankles (`HandTracker`-like points on mobs: the model's limb pivots if humanoid, else box corners); 3 red motes/tick at the victim	3/tick + 4 threads
Hold	the threads **pulse** with the heartbeat (α 0.45 ↔ 0.7 at 0.9 Hz); the victim's pose 12.4 (the puppet, with a 2-tick lag behind the claws); 1 mote/tick	1/tick
Fling	the threads **snap** (fade in 3 ticks), 12 red motes burst at the victim, a motion streak (`ENERGY_BEAM` red α 0.3)	12
Severance	the threads **all gather at the forehead** (converging over 8 ticks); at the press, a **dark pulse**: an `ENERGY_VEIL` sphere r 0 → 2 over 6 ticks, `#2A0A10` α 0.5 → 0, and the victim's **element colour** (their aura) **drains**: 16 motes of their element's colour flow from them into the blood bender's hand over 10 ticks, then **grey out** (fade to `#777777`); a silver flash from the moon; the victim's HUD greys (`UI.md`)	20
Release	the threads fade in 6 ticks, 6 red motes	6
Refused (stronger waterbender)	the threads form and **break at once** (2 ticks), the defender's eyes flash blue (`GlowingEyesLayer` `#8FE3FF` for 10 ticks) and 8 blue motes burst at them; the caster staggers (`HIT_STAGGER`)	8
Cap	2 victims (M10), 8 threads, 30 motes	

Sounds: `blood_grab` (a low, wet, straining groan with a heartbeat entering, 0.8 s), `blood_loop` (a slow heartbeat with a tense string drone and a faint liquid creak, volume 0.6; the victim hears a louder, faster heartbeat: `blood_victim_loop`), `blood_move` (a creaking strain with each large puppet move), `blood_fling` (a wet snap and a whoosh), `blood_release` (a breath out, the heartbeat

stopping), `blood_sever` (a long inhale, silence, then a deep soft thud and a sigh of released air, 1.5 s; the victim hears a high ringing fade `severed_victim`), `blood_refused` (a glassy chime and a snapping string: the defender's strength).

5.8 Water hit reactions

Soaked targets drip (12.3); frozen feet give the stuck pose; a creature hit by a spout tumbles; the bloodbent puppet is 12.4; a severed bender's **loss** is 12.5.

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6. Fire

6.0 Fire idle: the ready stance

Stance	tauIn / tauOut	Rows
`FIRE_IDLE` (ambient)	3 / 5	`FIRE_BOW`-lite: fists chambered at the hips (−30,0,±34), the front leg forward (x −14/+12, z ±8), lean 6, drop 0.08; the **breath**: head −2 ↔ +1 and lean 6 ↔ 4 at 0.1 rad/tick (a visible inhale every 3 s); small fist pulses (x ±2 at 0.3 rad/tick: tension).
`FIRE_PUNCHING` (ambient while Fire Punch is on)	3 / 5	fists raised to a guard (−70,∓20,∓10 → the hands before the chin), elbows in; the breath; the punch swing is **kept** (the vanilla swing plays through the ADD rows).
`FIRE_BREATHING` (aimed, while Fire Breath)	2 / 4	6.3.
`FIRE_WHIPPING` (aimed)	2 / 4	6.4.
`FIRE_JETTING`	1 / 3	6.5.
`LIGHTNING_CHARGING` (aimed)	1 / 4	6.7.
`LIGHTNING_REDIRECTING` (aimed)	1 / 3	6.8.

Ambient particles of a master (Fire ES ≥ 60): 1 `SMALL_FLAME` (blue: `SOUL_FIRE_FLAME` for Azula) every 8 ticks from a fist, rising 0.3 and dying; at ES 100 a heat shimmer sheet over the shoulders (`MATTER_TEXTURED(heat.png)` α 0.15, 1 × 0.6 blocks) while standing still; a faint crackle sound every 4 s at volume 0.1.

Fire first-person idle: fists chambered low at the sides (`dx ±0.22, dy −0.18, dz +0.04, rx +10, ry ∓10`), rising 0.01 on the inhale; a flame flickers on the knuckles (1 `SMALL_FLAME` per 10 ticks at each hand).

6.1 Fireball — כדור אש

`FIRE_FIREBALL` (dur 12 / 1 / 4, CAST, force): Northern Shaolin straight punch:

t	ease	M	O	legs (z)	head	body	notes
0	S	`FIRE_REST`	`FIRE_REST`				

t	ease	M	O	legs (z)	head	body	notes
2	O	**−20,0,30**	−70,−20,−10	−30 / 25 (±8)	−3	lean 4, drop 0.10	the **chamber**: the M fist pulls back to the hip, the O fist guards; the bow stance drops in; the **inhale** (head up)
4	**O**	**`FIRE_PUNCH`-M (−90,0,0)**	−10,0,−30	−32 / 26 (±8)	2	**lean 12**, twist 14	the **punch**: the fist snaps level forward in 2 ticks, the rear fist pulls back, the hips turn; the fireball leaves at tick 4
7	S	−88,0,2	−10,0,−30		2	lean 12, twist 12	held out (the fist smokes)
12	S	`FIRE_REST`	`FIRE_REST`	−8 / 8	0	lean 2, twist 0	

M5 (two): the O fist punches at tick 6 (−90,0,0 mirror, twist −14). M8 (three): the M fist again at 8 (dur 14). **Great fireball (M10, double-tap),** `FIRE_GREAT_FIREBALL` (dur 20 / 2 / 6): t 4 both fists chamber at the hips (−20,0,±30), a deep bow stance (drop 0.16), a 4-tick **inhale** (head −8, lean −4); t 10 both arms **circle out and back in** (−100,∓50,±40 → −80,∓30,∓10 at 10: the hands gather the fire before the chest); t 13 (`0`) **both palms thrust** (`FIRE_DOUBLE`, lean 14); recovery. **Small Fireball (Sneak),** `FIRE_FLICK` (dur 8 / 1 / 3): a wrist flick: the M hand rises to −60,10,10 at 2 and flicks to −80,−10,14 at 4 (`0`); rest.

Lock: none. **Blend:** the bow stance's legs are keyed 2–4 only, so walking shows otherwise; while flying (Creative, Fire Jet): the legs are masked (14.3); under water: refused. **Started by:** the slot's cooldown edge (1.2 s; the great one 4 s; the flick 0.5 s). **Prediction:** always (not in water).

First person: t 2 `M dx +0.10, dy −0.16, dz −0.10, rx +10` (the fist pulls back past the edge), O up as a guard `dx −0.12, dy +0.10, dz +0.14, ry +20`; t 4 **punch**: `M dx +0.02, dy +0.06, dz +0.40, rx −5` (the fist shoots to the centre; the fireball bursts from it); t 7 hold; recovery. Camera: a 2-tick FOV kick ×1.02 at the punch (not with reduced motion). Great fireball: both fists chamber `dy −0.16`, circle out `dx ±0.30, dy +0.14`, gather `dx ±0.06, dy +0.04, dz +0.14`, thrust `dz +0.40` both. Flick: `M dy +0.06, dz +0.22` quick.

Effects:

Where	What	Count
Chamber (t 2–4)	the fist **ignites**: 4 `FLAME` (Azula: `SOUL_FIRE_FLAME`; every "FLAME" below follows the look's palette) + a `FxShapes.glow` orb r 0.12 `#FFD23F` (blue: `#CFF2FF`) at the fist	4
Flight	the `FireballEntity`: a **core sphere** (`FxShapes.glow` r = size, core `#FFF4C2`, rim `#FF6A1F`; blue: core `#E8FBFF` rim `#3FA4FF`) + a **flame tail**: 5 trailing glow discs shrinking (α 0.6 → 0) + 3 `FLAME` + 1 `SMOKE`/tick; the ball **tumbles** (its texture-less: no need)	4/tick
Burst	an **expanding flash sphere** (`ENERGY_VEIL` … `ENERGY_BEAM` sphere r 0 → burst radius over 4 ticks, α 0.6 → 0) + 12 `FLAME` + 6 `LAVA`-like embers (`EnergyParticle` `#FF8A1F` scale 0.4 with gravity: use `bending:grain` tinted orange with the ember flag) + 4 `SMOKE` + a `MATTER_FLAT` **scorch disc** on the ground (α 0.4, 100 ticks)	22
Great fireball	r 1.2, tail 8 discs, 6 `FLAME`/tick, burst r 4: 30 `FLAME` + 12 embers + 8 `SMOKE`, shake 0.5	8/tick
Small fireball	r 0.1, 1 `SMALL_FLAME`/tick, no burst (a 0.3 flash)	1/tick
Vs water	the ball **snuffs**: 16 `CLOUD` steam + a hiss, no burst	16
Cap	6 fireballs, 60 flames	

Sounds: `fire_ignite` (t 2: a sharp fire catch with a breath, 0.2 s), `fireball_throw` (a punchy whoosh with a crackling body), `fireball_fly_loop` (a soft roaring flutter, volume 0.3, follows the ball), `fireball_burst` (a fiery "boom-crackle" 0.6 s; the great one adds a sub-bass drop), `fire_flick` (a tiny puff), `fire_vs_water` (a hiss and a steam sigh).

6.2 Fire Punch — מכת אש

On, `FIRE_PUNCH_ON` (dur 10 / 1 / 4, CAST, not force): the fists **clench and ignite**: t 3 both fists rise to the guard (−70,∓20,∓10, `0`) with a sharp **inhale** (head −6 for 2 ticks); t 5 the fists **flare** (a 2-tick tremble 2 at 5.0: the ignition); t 10 `FIRE_PUNCHING` hand-over. **Off:** the fists open and shake out (tremble 3 for 3 ticks) and drop; the flames die. **Each hit:** the vanilla swing with an ADD row: the punching arm gets +10° x snap (via `FIRE_PUNCH_HIT`, group AUTO, dur 6: at tick 1 the arm +8 x (ADD), lean +4, twist +8 toward the punch; the third hit of a combo (M4) adds a step (legs −20/16) and lean +10). **Fire Kick (Sneak),** `FIRE_KICK` (dur 14 / 1 / 5, force; **NEW** leg z keys): t 3 the body **coils**: twist −20, lean 4, the M arm back (20,0,30), O forward (−60,−10,−20), weight on the left leg (legs 10/−6); t 6 (`0`) **the kick**: right leg to `FIRE_KICK` (x −105, z +22), the body leans back (lean −10), twist +25, arms swing (M −60,0,40 / O 40,0,−20) — the crescent of flame leaves the foot at tick 6; t 9 the leg sweeps across (z +22 → −10 with x −90, `E`); t 14 `FIRE_REST`, twist 0. Mid-air: the same with a flip: `yaw` 0 (no), but `lean` runs −10 → −60 → 0 (a back-flip feel via lean only, since full flips need pitch > 90: **lean is clamped to ±112 in the profile; a flip is a later idea**).

Lock: none. **Blend:** the kick's legs are keyed 3–9 (walking pauses on the legs for 6 ticks); flying: the kick plays without the stance legs. Under water: ×0.5 damage, the same gesture at 0.8 rate. **Started by:** `Toggle.FIRE_PUNCH`; payload kinds `FIRE_PUNCH_HIT` (with the combo count), cooldown edge (kick 3 s). **Prediction:** toggle always; kick always.

First person: on: fists rise to the guard `dx ±0.14, dy +0.08, dz +0.16, ry ∓15` and ignite (the knuckles burn: 2 `FLAME` per hand per tick at the fist, drawn in front of the camera); the fists glow (a `glow` orb at each). Hit: the vanilla swing plus `dz +0.08` snap. Kick: the camera **leans back** 6° (pitch −6, roll +4 with the twist) for 4 ticks (reduced motion: none), a flame crescent sweeps across the lower screen (the kick's leg is not drawn in first person: a **flame arc effect** at the bottom of the view stands for it: an `ENERGY_BEAM` arc ribbon sweeping left to right over 3 ticks); hands swing `dx +0.20 / −0.20`.

Effects:

Where	What	Count
Fists on	2 `FLAME`/tick per fist (blue for Azula) + a glow orb r 0.12 + 1 `SMOKE`/4 ticks; the flames **flare** (4 extra) on each punch	4/tick
Hit	a **flash** at the target (`ENERGY_BEAM` sphere r 0.6 α 0.6 → 0 over 3 ticks) + 8 `FLAME` + 4 embers; the target burns (vanilla fire render)	12
Combo (M4)	the third hit adds a `FxShapes.cylinder` jet 1 block behind the fist (α 0.4, 3 ticks) and 6 extra `FLAME` + knockback dust	6
Jab (M7)	a small fireball (6.1 small) from the fist	
Kick	a **crescent of flame** (`ENERGY_BEAM` annulus segment of 120°, r 1.2 → 2.5 over 4 ticks, `#FF6A1F` → `#FFD23F` edge, α 0.7 → 0) sweeping with the foot + 12 `FLAME` along it + 4 embers; targets flash	16
Cap	1 pair of fists, 40 flames	

Sounds: `fire_fists_on` (an inhale and two fire catches), `fire_fists_loop` (a low crackling flicker, volume 0.3), `fire_punch_hit` (a punch thud with a fiery "fwoosh"), `fire_combo` (a bigger whoomp), `fire_fists_off` (a flame dying out), `fire_kick` (a sweeping fire whoosh with a sharp crack), `fire_kick_hit` (a flaming thud).

6.3 Flamethrower — להביור

`FIRE_FLAMETHROWER_START` (dur 8 / 1 / 3, CAST, force) then the stance `FIRE_FLAMETHROWING` (aimed, tauln 2 / out 4) while `Live.FIRE_STREAMING`: start: t 2 both fists chamber at the hips (`FIRE_BOW`), a 2-tick inhale (head −6); t 5 (`0`) both arms **shoot forward** to `FIRE_DOUBLE` (−90,∓6,±4), palms open (y 0), lean 12, the bow stance (−30/25); the stance: both arms aimed along the look (`aim(LOOK, 0, ∓6, 0)`) with **tremble 1.0 at 3.5** (the recoil hum), the body braced (lean 12, drop 0.12, legs −30/25 z ±8), head level; `occupy` both; every 20 ticks a small **recoil** (both arms +4 x for 2 ticks, lean +2). End: the arms **drop** (`FIRE_FLAMETHROWER_END` dur 8: `E` to `FIRE_REST`) and the jets die. **Fire Breath (Sneak, M7+),** **`FIRE_BREATH_START`** (dur 10 / 2 / 4) then the stance `FIRE_BREATHING` (aimed) while `Live.FIRE_BREATHING`: start: both fists chamber and pull back and **down** (10,0,±40 at 4), the chest **opens**: head **−14** (back), lean **−8**, a 6-tick inhale (the ribs: the body scale is not animatable; the head tilt and lean back read as the breath); at 8 the head **snaps forward** (head +6) and the breath begins; the stance: `FIRE_BREATH` (arms back and wide −40,0,±50, head aimed: the head drives the jet: `aim` on the head via the look (the shared head rule already looks along the camera) with head +6 additive, lean −2 → +6 pulsing at 0.3 rad/tick (the exhale push), drop 0.04, legs −20/20 z ±10; **locks movement** (`Live.FIRE_BREATHING` input lock: you cannot walk while breathing). End: the head drops (head +12 for 3 ticks: a cough) and the arms fall. Before M7, Sneak plays `FIRE_FLAMETHROWER_START` one-handed (the M arm only).

Lock: flamethrower none (speed ×0.6 attribute); breath locks. **Blend:** the flamethrower stance rides over slow walking (the legs keep walking at 0.6 speed; the bow stance legs are **masked** by the walk amount: when `limbAmount > 0.2` the stance's leg rows weight → 0: 14.3); flying (Fire Jet): the arms still aim. Under water: refused. **Started by:** `Live.FIRE_STREAMING`, `Live.FIRE_BREATHING`. **Prediction:** on the press when not in water (the client sets a local "predicted stream" that ends when the server's bit arrives or 12 ticks pass).

First person: start: fists chamber `dx ±0.20, dy −0.16` then both **shoot forward** `dx ±0.10, dy +0.04, dz +0.38, rx −6, ry ∓6` (both palms at the lower centre, the jets leaving them: the jets' origins are `FirstPersonArms.hand`); trembling 0.010 while firing; the recoil `dz −0.03` every 20 ticks; the heat ripple overlay (2.2) on the lower half; FOV ×0.98 (a slight tunnel). Breath: the hands pull back and down to the bottom corners `dx ±0.30, dy −0.26, dz −0.04`, the camera **tilts up** 8° over 6 ticks (the inhale: pitch −8) then **snaps down** 10° (pitch +2 relative) as the breath starts; the jet comes from the **camera** (origin = eye − 0.2 y, forward): the lower-middle of the view is fire; the heat ripple covers the whole screen at α 0.1; the camera shakes 0.15 continuously (reduced motion: none).

Effects:

Where	What	Count
The jets	Enhanced's `ContinuousBeam/BeamRenderer` with a **flame style** (`BeamStyle FLAME`: core `#FFF8D0`, inner `#FFE070`, main `#FFA030`, body `#E85A10`, fringe `#8C2A08`; blue: `#F4FDFF`, `#CFF2FF`, `#6FC4FF`, `#3FA4FF`, `#1B4FCC`), **two beams** (one per hand: `BeamPayload` per hand, `offHand` true for the second), drawn **wider and softer** than an energy beam: the 5 layers at ×2.2 width (fringe 0.66, body 0.37, main 0.2, inner 0.11, core 0.05), taper **inverted** (narrow at the hand 0.25 → full width at 2 blocks → 1.4× at the end: a cone), `flowingPath` jitter 0.08 (fire wavers), the strands replaced by **flame tongues**: 4 `crossfade`d short ribbons branching off the last third, upward-biased (gravity −: fire rises); + 6 `FLAME` + 2 `SMOKE` per tick spawned along each jet + 1 `LAVA`-like ember/tick; at the end point: a `glow` splash r 0.6 and a scorch disc	18/tick for both jets
Hit	the target burns; 6 `FLAME` + 3 embers/hit	9 per hit
Melting ice	`DRIPPING_WATER` ×4 + `CLOUD` ×4 per block	8
Fire Breath	one beam from the mouth (`HandTracker.forehead` − 0.25 y), style FLAME at ×3 width, cone taper to 2.5 wide, 10 `FLAME` + 4 `SMOKE`/tick + heat shimmer sheet along it; the bender's head glows (`GlowingEyesLayer`-style mouth mask `fire_mouth.png`, `#FFD23F`)	14/tick
Cap	1 stream (2 beams) per player, 60 flames	

Sounds: `flamethrower_start` (an inhale and a double ignition "fwoosh"), `flamethrower_loop` (a roaring jet with a crackling layer, volume 0.7, pitch 1.0; blue fire: pitch 1.15 with a hissing top), `flamethrower_end` (the roar cutting out with a puff), `fire_breath_inhale` (a long dragon's inhale 0.6 s), `fire_breath_loop` (a deeper, wider roar with a vocal rasp, volume 0.8), `fire_breath_end` (a cough-puff and embers), `fire_melt` (hiss + drips).

6.4 Fire Whip — שוט אש

`FIRE_WHIP_START` (dur 8 / 1 / 3, CAST, force) then the stance `FIRE_WHIPPING` (aimed, tauIn 2 / out 4) while `Live.FIRE_WHIPPING`: start: t 2 the M fist chambers low and back (20,0,30, `0`); t 5 the arm **sweeps up and out** in an arc to −60,−30,50 (`E`) as the whip **ignites along the arm** and uncoils from the fist (effects); the stance: the M arm held up and back (`FIRE_WHIP`-like: −70,−20,40), **strikes every 12 ticks**: a 3-tick lash (`0`) to aim(LOOK, −10, 10, 0) with lean 4 → 12, twist 0 → 16, the wrist leading (y −20 → 20), the O fist guarding at the chest (−70,−20,−10); then a 7-tick return (`E`); legs in the bow stance (masked while walking); `occupy(M)`. M4 (two whips, double-tap): the O arm mirrors with lashes offset by 6 ticks. End: the arm **snaps the whip out** (a flick to −90,−30,30 for 2 ticks, `B`) and it dies (`FIRE_WHIP_END` dur 8). **Whip Lash (Sneak),** **`FIRE_WHIP_LASH`** (dur 14 / 1 / 5, force): the M arm **winds back far** (40,20,40 at 3, twist −20, lean −4); t 6 (`I`) the **crack**: aim(LOOK, −15, 20, 0) with lean 14, twist 24, a long step (legs −34/28); hold 2; t 14 rest.

Lock: none (speed ×0.8). **Blend:** walking at 0.8; the bow legs masked while walking; flying: fine; under water: refused. **Started by:** `Live.FIRE_WHIPPING`; cooldown edge (lash 3 s). **Prediction:** not in water.

First person: start: the M fist chambers `dx +0.18, dy −0.18, dz −0.06` then sweeps up and out to the right edge `dx +0.34, dy +0.22, dz +0.04, ry +35, rz −20` (the whip uncoils from the hand down out of view and back in as it hangs); lashes: the hand snaps to the centre `dx +0.06, dy +0.10, dz +0.34, ry −20` over 3 ticks and returns over 7; the whip's coil is drawn from the hand (effects). Lash: wind back `dx +0.38, dy +0.10, dz −0.10` then crack `dx 0, dz +0.40`. The O fist guards `dx −0.14, dy +0.08, dz +0.14`.

Effects:

Where	What	Count
The whip	a **rope of fire**: a chain of 12 segments from the hand (verlet-like: gravity **upward** 0.3 (fire rises) and drag, so it hangs up and back like a flame, not down like water), drawn as a `BeamRenderer`-style 3-layer ribbon along the chain (widths 0.22 / 0.12 / 0.04, FLAME palette, additive) with 2 `FLAME`/tick along it and 1 `SMOKE`/2 ticks from the tip; the tip has a small glow orb	2/tick
Lash	the tip is **driven to the target** over 3 ticks (the chain's end point set, the rest follows) with a **crack flash** (`ENERGY_BEAM` sphere r 0.5 α 0.7 → 0 in 3 ticks) + 8 `FLAME` + 3 embers at the hit; the ground/block hit gets a **fire block** (if allowed) or a scorch disc	11
Cutting a rock (M7)	6 grains + 8 `FLAME` at the cut, the rock halves	14
Wrap (M10)	the chain's last 4 segments **wrap** the target's box for 6 ticks while it is pulled	
Lash (Sneak)	the chain stretches to length +3 straight (a 2-tick snap), a bigger flash r 0.9, 12 `FLAME`, knockback dust	16
End	the chain **burns out** tip-first over 4 ticks (α → 0) with 6 `SMOKE`	6
Cap	2 whips, 40 flames	

Sounds: `fire_whip_ignite` (a rising fiery hiss as the whip uncoils, 0.4 s), `fire_whip_loop` (a crackling, fluttering flame with a soft wind, volume 0.4), `fire_whip_strike` (a whip-crack with a fiery whoosh), `fire_whip_hit` (a crack-thud and a flare), `fire_whip_lash` (a big crack with a long whoosh), `fire_whip_end` (a flame snapping out).

6.5 Fire Circle — מעגל אש

`FIRE_CIRCLE` (dur 16 / 2 / 5, CAST, force):

t	ease	M	O	legs (z)	head	body	notes
0	S	`FIRE_REST`	`FIRE_REST`				
3	O	−60,40,−30	−60,−40,30	0/0 (±10)	−4	lean 2, drop 0.10, **twist −30**	both arms **wind across the body**, fists lit, the shoulders coil hard
4	L	−80,0,60	−80,0,−60		0	**yaw 90**	the spin begins: arms fly out level
12	L	−80,0,60	−80,0,−60		0	**yaw 360**, lean 0, drop 0.12, twist 0	a **full spin** over 8 ticks with the arms out: the ring leaves the fists at 4 and runs out (server: the ring expands from tick 4 over 10 ticks)
16	S	`FIRE_BOW`	`FIRE_BOW`	−30/25 (±8)	0	yaw 360, lean 10	lands in the bow stance

M6+ (the leg sweep): `FIRE_CIRCLE_LEGS` (dur 20 / 2 / 6): the same to tick 12, then at 13 the body **drops** (drop 0.42, the left leg folds x −60 z −20: a crouch) and the **right leg sweeps** (`FIRE_KICK` legs: x −30 z +60 → the leg level, sweeping: **yaw 360 again** from 13 to 19 with the leg out, arms braced on the ground side (M −40,0,60 / O 20,0,−30), lean 10) — the second, lower ring leaves at 14; t 20 up into `FIRE_BOW`. The whole gesture turns 720° (two yaw turns: allowed, "a whole number of turns"). **Fire Jet (Sneak, hold),** `FIRE_JET_START` (dur 6 / 1 / 2) then the stance `FIRE_JETTING` (tauIn 1 / out 3) while `Live.FIRE_JETTING`: start: fists **point down** (30,0,±24 at 3, `0`), knees bend (drop 0.14); the stance: fists and feet jetting: arms **down and back** (30,∓10,±30) with tremble

1.5 at 4.0, legs **straight and slightly back** (x 12/12 z ±8), head level, **lean follows the steering**: `lean = 20·fwd` (leaning into the direction of travel), roll `10·side`; a bob 0.02 at period 6 (the thrust). End: `LAUNCHED` → `LAND_SOFT` with a fiery dip (6 `FLAME`).

Lock: none. **Blend:** the spin is visual yaw; the walk keeps running under it (the legs are not keyed 4–12), so you can run and spin; flying: the circle plays (the ring at your height); under water: refused. Jet: the body is in the air (`wAir` 1 with the `FIRE_JET` flight profile: 10.7). **Started by:** cooldown edges (8 s; M10 double at 10 ticks apart), `Live.FIRE_JETTING`. **Prediction:** not in water; jet always.

First person: t 3 both hands cross `dx ∓0.16, dy +0.06, dz +0.12, ry ∓40` (fists lit); t 4–12 both hands fly to the edges `dx ±0.38, dy +0.08, dz +0.06` and the **camera yaws 360°** over 8 ticks (reduced motion: no yaw; the ring is still seen expanding); fire sweeps across the view (the ring passes through the camera's position: the ring's ribbon is drawn around you, so you see it leave); t 16 the bow stance hands. Leg sweep: the camera **drops** 0.4 (eye offset) for 6 ticks and yaws again; the lower screen is flame. Jet: hands point down at the bottom corners `dx ±0.28, dy −0.30, dz +0.06, rx +40`, trembling 0.012, jets of flame from them (the first-person jet effects: two `FLAME` streams downward from the hands, 6/tick each, drawn in view space); FOV ×1.1; the camera pitches +4 into travel; a soft shake 0.1.

Effects:

Where	What	Count
Wind-up	the fists ignite (4 `FLAME` each) + a crossing of two flame ribbons before the chest (the hands' trails)	8
The ring	an **expanding ring of fire**: `EnergyGeometry.ring` r 0 → radius over 10 ticks at waist height (normal Y), width 0.45, FLAME palette (3 rings: fringe/main/core widths 0.45/0.25/0.08), additive; **flame tongues**: 24 short upward ribbons (`crossfade`d, 0.3–0.6 high) riding the ring; 16 `FLAME`/tick along the ring (capped) + 6 `SMOKE`/tick; the ground under the ring gets a **scorch ring** (`MATTER_FLAT` annulus α 0.35, 100 ticks); fire blocks where allowed; targets crossed flash	22/tick for 10 ticks
Leg ring (M6)	a second ring 0.3 above the ground, 4 ticks later, FLAME, width 0.3, with 12 `FLAME`/tick	12/tick
Snuffed by water	a gap opens in the ring (the ring's arc is drawn with a missing segment where water cut it) + 10 `CLOUD`	10
Fire Jet	**jets from the hands and feet**: 4 `BeamRenderer` short beams (0.8 long, FLAME, width ×1.5, pointing down and back from `HandTracker.hand` and `.foot`) + 6 `FLAME` + 2 `SMOKE`/tick per jet (capped 20/tick total) + a **heat trail**: `FlightEffects` ribbons from the feet (FLAME body colour α 0.3, 1.5 blocks); landing: a 1.2-block `MATTER_FLAT` scorch + 8 `FLAME`	20/tick
Cap	2 rings, 1 jet set, 80 flames	

Sounds: `fire_circle_wind` (two fire catches and a sharp inhale), `fire_circle` (a spinning whoosh that erupts into a wide ring roar, 1 s, with a crackle tail), `fire_circle_legs` (the same with a lower second roar), `fire_circle_snuff` (a steam hiss), `fire_jet_start` (two jet ignitions), `fire_jet_loop` (a roaring thrust with a flutter, volume 0.6, pitch with speed), `fire_jet_end` (the jets cutting, embers falling), `fire_land` (a fiery thud).

6.6 Fire Healing — ריפוי באש

`FIRE_HEAL_START` (dur 10 / 2 / 4, CAST, not force) then the stance `FIRE_HEALING` (aimed, tauIn 2 / out 4) while `Live.FIRE_HEALING`: start: both palms **rise to the chest** (−70,∓30,∓14, `E`), a 4-tick **inhale** (head −6, lean −4); at 8 the palms **turn out** toward the target (aim(LOOK, −10, ∓12, 0)), the exhale (head +2, lean +2); the stance: the palms held out and slightly apart, a slow **breathing** cycle (x ±4, lean ±3 at 0.1 rad/tick: the healer breathes with the patient), fingers spread (y ±12 pulsing); self-healing: the palms press to the own chest (−80,∓40,∓20). `occupy` both. End: the palms lower (`FIRE_HEAL_END` dur 8). **Burn Out (Sneak),** `FIRE_BURN_OUT` (dur 12 / 1 / 4, force): both palms rise (−70,∓30,∓14) and at 5 **clap toward the target** (aim(LOOK, 0, ∓4, 0), `0`, lean 6): a sharp flash; t 12 rest.

Lock: none (both must stand still: server). **Started by:** `Live.FIRE_HEALING`; cooldown edge (Burn Out 15 s). **Prediction:** a target below full health in sight (or self).

First person: the palms rise to the centre `dx ±0.10, dy +0.06, dz +0.20, rx −10, ry ∓20`, warm light between them, breathing ±0.01; self: palms toward the camera `rx +50, dz +0.08`. Burn Out: a clap `dx ∓0.02, dz +0.30`.

Effects:

Where	What	Count
Hands	a **warm glow** between the palms: `FxShapes.glow` r 0.25 `#FFD98A` (core `#FFF4C2`) α 0.5, pulsing with the breath; 2 **ember-glow motes** (`EnergyParticle` `#FFC46A` scale 0.4, slow, rising)/tick	2/tick
The healed	a **golden fire aura**: `energySwirl` over the model `#FFD98A` α 0.25 scrolling slowly up (the dragons' fire: warm, not burning); 3 glow motes/tick rising; each heart: a pulse	3/tick
Stream	an `ENERGY_BEAM` soft ribbon (fringe only, width 0.3, `#FFC46A` α 0.25) from the palms to the target, gently wavering	1
Burn Out	a **flash** at the target (`ENERGY_BEAM` sphere r 1 α 0.7 → 0 in 4 ticks, gold) + 10 glow motes + the poison particles **burn away** (8 `SMOKE` tinted green-grey)	18
Cap	1 target, 30 motes	

Sounds: `fire_heal_start` (a soft inhale and a warm crackle rising into a hum), `fire_heal_loop` (a low warm pad with a soft crackle and a breathing cycle, volume 0.4; a soft chime per heart `heal_tick` at pitch 0.9), `fire_heal_end` (the hum fading), `fire_burn_out` (a bright flash-sizzle and a sigh).

6.7 Lightning Generation — יצירת ברק

Charge, `LIGHTNING_CHARGE` (dur 24 / 3 / 1, CAST, force; **the gesture is held** at its last key while the key is down (`HeroAnimator.hold`), **locks movement** via `Live.LIGHTNING_CHARGING`): Iroh's form:

t	ease	M	O	legs (z)	head	body	notes
0	E	`FIRE_BOW`	`FIRE_BOW`	−24/22 (±8)	0	lean 4, drop 0.10	the bow stance, a **breath in** (head −4 over 4 ticks)
6	E	**−120,−30,50**	**−40,−40,−30**		−2	lean 2	the M arm **circles up and out**, the O arm circles **down and in**: the two-finger hands trace opposite circles (y and z sweep)

t	ease	M	O	legs (z)	head	body	notes
12	E	−60,50,−40	−120,30,30		0	lean 4, twist −10	the circles continue (the arms have swapped high/low): sparks follow the fingertips
18	E	**`LIGHTNING_MEET`**	**`LIGHTNING_MEET`**		4	lean 4, drop 0.12, twist 0	the fingers **meet at the stomach** (both hands at the belly, fingertips touching): the charge gathers
24	**O**	**`LIGHTNING_POINT-M (−92,0,2)`**	−20,10,−16	−30/25 (±8)	0	**lean 6**, twist 12	the M arm **extends**, two fingers out, the O hand braced at the hip: **held here** until release

On **release** (`LIGHTNING_RELEASE`, dur 10 / 1 / 4, force): a sharp **recoil**: the pointing arm kicks back 10° (x −82) and the body leans back 4 (lean 2) for 2 ticks (`0`), the head snaps −3; then the arm lowers to `FIRE_BOW` over 8 ticks. The bolt fires at release tick 0. **Dual (M10),** **`LIGHTNING_CHARGE_DUAL`**: the same to 18, then **both** arms extend (`FIRE_DOUBLE` with two fingers: −92,∓4,±2) at 24, both braced; release recoils both. **Lightning Arc (Sneak),** **`LIGHTNING_ARC`** (dur 10 / 1 / 4): a quick jab: the M hand to `LIGHTNING_MEET` at 3 (`E`) and **jabs** out (−90,−10,10, `O`) at 5 with twist 10; rest.

Lock: charging locks movement; released, free. **Blend:** standing only (locked); if the player is airborne when the key is pressed the charge still plays (a falling bender charges: legs trail). Under water: works (you take 1 heart). **Started by:** `Live.LIGHTNING_CHARGING` (rise → CHARGE held; fall → RELEASE if the cooldown started in the same tick, else a cancel: `LIGHTNING_CANCEL` dur 8: the arms drop and the sparks die); the Arc by its cooldown edge (5 s). **Prediction:** on the press when ready: the charge is local at once; the release is predicted on key-up after 10 ticks.

First person: t 0 fists chamber `dx ±0.20, dy −0.16`; t 6–18 the hands **circle**: the M hand traces a circle of r 0.22 clockwise (centre `dx +0.12, dy +0.04, dz +0.24`), the O hand counter-clockwise (centre `dx −0.12`), two fingers out (the hand model is the vanilla hand: the "two fingers" are read from the **spark effect** at the fingertips and the hand's `ry` turn toward the camera); t 18 the hands **meet at the bottom centre** `dx ±0.03, dy −0.16, dz +0.22, rx +30, ry ∓30` with blue-white sparks crawling between them; t 24 the M hand **extends to the centre** `dx +0.04, dy +0.06, dz +0.44, rx −8` with an **arc of lightning** crackling between the two fingers and the O hand (at the hip `dx −0.22, dy −0.20`); the **FOV pulls in** to ×0.96 over the 24 ticks; the hum rises; **release**: a white flash (overlay α 0.6 → 0 in 4 ticks), the hand kicks back `dz +0.36`, the camera jolts (shake 0.3, high frequency) and FOV snaps to 1.0; reduced motion: no flash beyond α 0.2, no jolt.

Effects:

Where	What	Count
Circles (t 6–18)	**sparks crawl on the arms**: 2 `ELECTRIC_SPARK` + 1 white-blue `EnergyParticle` (`#DDF4FF`, scale 0.4)/tick per hand at `HandTracker.hand`, plus a thin **arc** between the hands when they are within 1 block: `EnergyGeometry.bolt(a, b, 0.02, jitter 0.3, #DDF4FF, seed, phase)` crossfading every 1.5 ticks	6/tick + 1 arc
Meet (t 18–24)	a **crackling ball** at the stomach: `FxShapes.glow` r 0.15 → 0.3 `#FFFFFF` core / `#9FD8FF` rim + 4 short random bolts off it per tick (`crossfade`), 4 `ELECTRIC_SPARK`/tick; the body's **eyes glow** white-blue (`GlowingEyesLayer` mask `#DDF4FF`) while charging	8/tick
Held (24+)	the arc from the fingers to the O hand persists (0.4 long); 3 sparks/tick at the fingertips; a faint blue glow on the extended arm (a `ENERGY_BEAM` sleeve ribbon α 0.15)	3/tick

Where	What	Count
The bolt	an **instantaneous bolt**: `EnergyGeometry.lightningPath(hand, hit, jitter 0.35)` drawn in **3 layers** (`ThorEffects.beam` recipe: halo `#4DB8FF` 0.30 wide, blue `#9FD8FF` 0.15, core `#FFFFFF` 0.055) for **8 ticks** fading (α 1 → 0 with 2 **reshapes** at ticks 2 and 4: a flicker), **2–4 branches** (`lightningPath` from random path points, reach 1–3 blocks, widths halved), a **flash sphere** at the origin r 0.6 and at the hit r 1.0 (`ENERGY_BEAM`, α 0.8 → 0 in 3 ticks), 20 `ELECTRIC_SPARK` at the hit + 8 along the path, a scorch disc; the sky **flashes** faintly for clients within 24 blocks (a white overlay α 0.15 for 2 ticks); dual (M10): two bolts from the two hands **joining** 1 block ahead into one 1.5× bolt	28 + 1 bolt
The hit	the target is **lit white** for 4 ticks (`EntityGlow` white, or an overlay tint via `LivingEntityRenderer` colour: 12.5 paralysed), 12 sparks, the paralysis pose; in water: 4 radiating bolts along the surface (`lightningPath` 4 blocks)	12
Arc (Sneak)	a 6-block bolt at half widths, 4 ticks, 8 sparks	8
Cancel	the ball **pops**: 10 sparks fly off, the eyes dim	10
Cap	1 charge, 2 bolts alive, 60 sparks	

Sounds: `lightning_charge` (a rising, crackling electric whine from 0 to 24 ticks, 1.2 s: a sine sweep 180 → 2400 Hz with crackle bursts every 3 ticks; the two-finger circles have soft "zip" sparks), `lightning_hold_loop` (a tense crackling hum, volume 0.5, pitch slowly rising), `lightning_strike` (a deafening crack: a 2 ms click, a broadband burst, a sub thump and a 0.8 s rumble tail; the dual bolt adds a second crack 40 ms later), `lightning_hit` (a sizzling impact with a "zap" and a short ring), `lightning_arc` (a short crackling zap), `lightning_cancel` (sparks fizzling out), `thunder_distant` (a rumble heard 24–64 blocks away, 0.6 s later).

6.8 Lightning Redirection — הטיית ברק

Stance, **`LIGHTNING_REDIRECT_START`** (dur 8 / 1 / 3, CAST, not force) then the stance `LIGHTNING_REDIRECTING` (aimed, tauIn 1 / out 3) while `Live.REDIRECTING`: start: the bow stance; the M arm **rises level** toward the threat (aim(LOOK, 0, 0, 0): the receiving arm points at where the bolt will come from), two fingers out; the O arm **low and back** at the hip (−30,0,−30): `REDIRECT_IN`; the body **sideways** (twist 30: the stomach turned away from the line, Iroh's lesson that the bolt must not cross the heart), head looking at the threat, lean 2, drop 0.12; the stance holds with a calm **breath** (lean ±1.5, no tremble: the redirector is calm). **The bolt arrives (success),** **`LIGHTNING_REDIRECT_PASS`** (dur 12 / 1 / 4, force; plays at the hit tick, group CAST replaces the stance's gesture): t 0 the bolt enters the receiving (M) arm; t 3 (`E`) the M arm **sweeps down** to the stomach (−40,30,−20) and the body **bends** forward (lean 14, drop 0.18: the energy passes through the belly: the overlay shows it); t 5 the **O arm rises** from the hip (`E`) to level (−92,0,−2) as the M arm settles at the hip: `REDIRECT_OUT`; t 6 the bolt **leaves** the O arm (the server fires the redirected bolt at tick 6 of the pass, i.e. 6 ticks after the hit: the window); recoil (O −82 for 2 ticks); t 12 `FIRE_BOW`. If the player **looks elsewhere** before tick 6, the O arm aims there (aim(LOOK)). **Failure (hit),** **`LIGHTNING_REDIRECT_FAIL`** (dur 14): the M arm receives (t 0), the body **convulses**: `PARALYSED` with a violent tremble (4 at 6.0) and lean −12 at t 3 (`O`), a stagger back (drop 0.2, legs 10/−20), then `HIT_KNOCKDOWN` (12.1). **Ground It (Sneak),** **`LIGHTNING_GROUND`** (dur 12): from the pass's tick 5, instead of rising, the O arm **drives down** to the ground (60,0,−20 with lean 18 and drop 0.26, `I` at 7: a kneel); the bolt goes into the earth; t 12 up to `FIRE_BOW`, and 1 heart heals (a soft gold pulse).

Lock: the stance slows (speed ×0.5 attribute), no lock; the pass locks for 12 ticks. **Blend:** the stance's legs masked while walking; flying: allowed (airborne redirection, Zuko on the airship). **Started by:** `Live.REDIRECTING`; payload kinds `REDIRECT_PASS`, `REDIRECT_FAIL`, `REDIRECT_GROUND` (sent

at the hit). **Prediction:** the stance on the press when ready.

First person: the stance: the M hand **level at the centre-right** `dx +0.14, dy +0.08, dz +0.40, rx −6` (two fingers toward the threat), the O hand low left `dx −0.26, dy −0.22, dz +0.02`, the camera **yawed 20°** toward the threat side relative to the body (the body's twist is third-person only; in first person the hands shift `dx` +0.08 to show the sideways stance), breathing; **the pass**: a **white-blue bolt enters the right hand** (the bolt effect ends at the hand), the hand **sweeps down to the bottom centre** `dx +0.02, dy −0.26, dz +0.18, rx +40` over 3 ticks while a **bright line runs down the right edge of the screen, across the bottom (the stomach) and up the left edge** (the overlay: an `ENERGY_BEAM`-style bolt drawn in screen space as a 3-layer polyline along that U path, white core, 6 ticks, with the red-hot "path" afterglow `#FF6A4A` α 0.4 fading over 20 ticks), the camera **dips** 0.15 and pitches +6 (the bend), then the O hand **rises to the centre** `dx −0.04, dy +0.06, dz +0.42` and the bolt **leaves it** (the bolt effect from that hand along the look); a flash; recovery. Failure: a white flash α 0.8, the paralysis overlay (2.2), the camera jolts and falls back (pitch −20 over 6 ticks, the knockdown). Ground It: the O hand drives to the bottom `dy −0.34, dz +0.14` and the bolt goes down out of view, a gold pulse (`#FFD98A` α 0.3 fade) as the heart heals.

Effects:

Where	What	Count
Stance	1 spark/2 ticks at the receiving fingertips (anticipation); the eyes faintly lit (`#DDF4FF` α 0.5)	0.5/tick
The pass	the incoming bolt's path **continues onto the body**: a 3-layer bolt ribbon (`ThorEffects.beam` recipe at half widths) drawn **along the arm to the stomach and up the other arm** using `HandTracker.hand(M)` → `HandTracker` belly (a new tracked point: the body's centre, **NEW** 14.8) → `HandTracker.hand(O)`, alive 6 ticks (the bolt is *inside* the bender for 6 ticks: visible on the body as the glowing path, additive, with 6 sparks/tick along it); then the **outgoing bolt** (6.7's bolt recipe) from the O hand; the body's **eyes blaze** white for 8 ticks; a `MATTER_FLAT` scorch under the feet (the earth took a little)	6/tick + 2 bolts
Failure	the bolt hits (6.7 hit effects ×1.25), the body convulses (12.5), 20 sparks over 8 ticks, smoke 4	24
Ground It	the body path ends at the O hand pointing **down**: a bolt into the ground (`lightningPath` 1.5 blocks down, branching along the ground surface 4 ways 3 blocks each, 4 ticks), a 2-block `MATTER_FLAT` scorch star, 16 sparks; a **gold pulse** on the bender (`energySwirl` `#FFD98A` α 0.4 for 10 ticks)	16
Cap	1 body path, the bolts of 6.7	

Sounds: `redirect_stance` (a slow calm inhale with a faint electric tension, 0.6 s), `redirect_loop` (a very quiet high hum, volume 0.15), `redirect_pass` (the crack of the hit at once, then a **rising electric whoosh through the body** 0.3 s: a filtered noise sweep up, a short pause, then the strike's crack again as it leaves: three events, 0.6 s total), `redirect_fail` (the strike crack + a sizzling convulsion + a body fall), `redirect_ground` (the pass's whoosh then a deep earth thud with a crackle spreading and a soft gold chime).

6.9 Fire hit reactions

Burning targets flail (12.3 burn); a fireball or whip knock staggers; Fire Kick knocks down; lightning gives the **paralysis** pose (12.5); a Fire Circle crossing pushes.

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7. Movement, rides and forms

7.1 Walking and running per element (`RunCharacter`)

Element	walkArms	sprintArms	sprintLegs	lean	twist	bob	Character
Earth	0.85	1.05	1.0	8	4	0.030	heavy, planted, a little slower-looking
Earth, armoured	0.9	0.8	0.8	6	2	0.035	a stomping walk; the armour's footsteps are stone
Air	1.1	1.25	1.1	4	7	0.012	light and bouncy; the arms swing wide; airbenders **land softly** (the shared `LAND_SOFT` with half the dip)
Air Wheel	0.6	0.6	0.4	14	3	0.010	legs hidden in the wheel
Water	1.0	1.1	1.0	5	6	0.018	smooth, the arms flowing (an extra 4° y sway on the arm swing)
Fire	0.95	1.2	1.05	7	8	0.022	sharp, forward, the fists closed (y −10)
Blood (while holding)	—	—	—	—	—	—	cannot move

Sprinting leans by `lean`, as Enhanced does. **Idle at rest** (not holding a key): the element's `IDLE` with weight 0.4 (a hint of the stance: a bender never stands quite like a villager).

7.2 Jumping and landing

Every bender uses Enhanced's `LAND_SOFT` (a dip) on landings over 3 blocks; airbenders at half dip with a puff; earthbenders in armour with `LAND_HERO`'s deeper dip, dust and a 0.3 shake. Earth Launch, Air Jump, Fire Jet and the Spout collapse all end in the shared `LAUNCHED` stance (11.3) while airborne.

7.3 Swimming

Vanilla owns the body while swimming (Enhanced's rule). Waterbenders get **no** special swim pose in 1.x; the passives are speed. Air Bubble's sphere follows the head in the swim pose. **(suggestion)** a waterbender swim stroke later.

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8. (Reserved: see 10 for rides and 11 for shared stances.)

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9. Idle stances summary

Element	Ready stance (key held)	Resting hint (weight 0.4)	Ambient master effect
Earth	`EARTH_IDLE` horse stance, drop 0.24, breathing fists	shoulders heavy, feet apart	grains rising at the feet

Element	Ready stance (key held)	Resting hint (weight 0.4)	Ambient master effect
Air	`AIR_IDLE` circling palms, swaying weight	palms open, light sway	a circling breeze
Water	`WATER_IDLE` wave-hands figure-eights, soft knees	one foot forward	a droplet orbiting the hands, full-moon motes
Fire	`FIRE_IDLE` bow stance, fists chambered, visible breath	fists loose, forward lean	a knuckle flame, heat shimmer

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10. Rides, flight and forms

10.1 The Air Scooter ride (`AIR_RIDING` stance; the vehicle `AirScooterEntity`)

The body **crouches on the ball**: legs folded (x −55/−55, z ±28: knees out), drop 0.45 (the ball is 0.9 high and the rider's seat is at its top, so the entity's passenger offset places the hips; the stance's drop is relative to the seat), lean `6 + 10·fwd`, roll `12·side` (banking into turns), arms **out for balance** (−30,∓10,±45) with a wobble (±3 at 0.5 rad/tick scaled by speed), head level; a bob 0.03 at period 7 (the ball's spin). Hop: a `drop` kick −0.15. Wall-ride: roll 60° toward the wall (the body leans into it; the ball is on the wall). Passenger (M8): the same pose behind, arms holding the rider's waist (aim(ANCHOR) at the rider). Dismount: `AIR_SCOOTER_DISMOUNT`.

10.2 The glider pose (`AIR_GLIDING` stance; `FALL_FLYING` with the glider open)

Vanilla's elytra owns the body when `isFallFlying` (Enhanced hands over). Bending **overrides** it for the glider (14.3 `rideOverride` includes "fall-flying with the glider"): the body **horizontal** (lean 85–100 by speed via a `FlightProfile GLIDER`: p1 88, p2 100, pBack 70, kClimb 0.3, diveAdd 10, maxRoll 35, kTurn 0.9, kStrafe 0.4, tauRoll 5, no spring, bob 0.02 period 40, armDrift 0, legSway 2, legFlutter 3, lazyRoll 2, noBrake false), **both arms up gripping the staff** overhead (−150,∓12, ±8: the hands at the glider's crossbar, `mWorld` not used: the arms follow the body), legs **trailing straight and together** (x 0/0 z ±2, a slow flutter ±3), head looking along the flight (the shared head rule keeps the look). The **glider layer** (`LOOKS.md` 5): a staff with two wings rendered from the hands' midpoint, wings angled 12° up, flexing with speed (the wing parts' z rotation ±5°·speed at 0.4 rad/tick). Launch: `AIR_GLIDER_OPEN` (4.3). Air Blast downward: `AIR_BLAST_DOWN` (4.1): the body pitches up 10° (lean 80) and the hands stay on the staff (the blast comes from the feet in this case: effects from `HandTracker.foot`). Landing: the wings fold (`B` over 4 ticks) and `LAND_SOFT`. First person: the hands are **up out of view** (both hands `dy +0.46, dz +0.10`: only the staff's lower end and the wings' shadow edge are drawn: a glider underside quad above the camera, translucent cloth); the wind streaks (Enhanced's first-person wind) at speed; FOV ×1.06.

10.3 The Water Spout ride (`WATER_RIDING`; `WaterSpoutEntity`)

Standing on the column: legs **apart and bent** (x −12/−12, z ±14), drop 0.10, arms **out and low** (−30,∓10,±48) swaying with the spin (twist 6·sin(0.2·age), arms z ±4 counter), lean `8·fwd`, roll `6·side`, a bob 0.04 at period 16; the head level. Ram: `WATER_SPOUT_RAM` (a lean kick). Freeze (M8): still. Surf (`WATER_SURFING`): the surfer's crouch (5.4). First person: 5.4.

10.4 The Earth Pillar rise (`EARTH_RIDE_UP`)

While the pillar lifts you: arms held high (−60,0,±30) with the palms up, legs together (±4), drop 0.04, head −4: a rising hero; the server moves the player (3.3). At the top the stance fades (tauOut 4) into `EARTH_IDLE`.

10.5 Meditation (`AIR_MEDITATING`)

`AIR_SIT`: cross-legged (legs x −90 z ±55: **NEW** leg z in rows is Enhanced's; in keys it is 14.1), drop 0.72, hands on the knees palms up (−20,±30,∓40), head 4 (slightly bowed), a **breathing sway**: lean 1.5·sin(0.05·age) and the hands rising 2° with the inhale, the head ±1. Eyes **closed**: a `GlowingEyesLayer`-style mask that draws **closed lids** (a skin-coloured strip over the eyes: `eyes_closed.png`, cutout, tinted by the skin's face colour sampled at runtime from the skin texture: **NEW** 14.9) while meditating. Deep Meditation: the same with a soft glow. The sitting is also what the **body** shows while the spirit is out.

10.6 The spirit form (`SpiritEntity`, `SPIRIT_FLOAT` stance) and Zaheer's flight

Spirit: the player's model drawn translucent blue (`AstralBodyRenderer` pattern: `entityTranslucent`, tint `#9FDCFF` α 0.45, no armour or items, a faint `energySwirl` aura α 0.15, the name tag at α 0.4), floating in `AIR_SIT` legs with the arms **free** (−30,∓20,±30 drifting ±6 at 0.1 rad/tick), drop 0 (it floats at its own position), a bob 0.08 at period 50, a slow lean into movement (`lean 15·fwd`), roll `8·side`; **the flight profile** `SPIRIT`: p1 20, p2 35, pBack −10, kClimb 0.2, diveAdd 15, maxRoll 20, kTurn 0.6, kStrafe 0.3, tauRoll 6, no spring, bob 0.08/50, armDrift 6, legSway 0 (the legs stay folded), legFlutter 0, lazyRoll 3, noBrake true (a spirit stops dead), defaultSpeed 0.08. Passing through blocks: the renderer draws it anyway (no culling by blocks; it is translucent). Effects: 2 blue motes/tick trailing, the tether thread (4.5).

Zaheer's True Flight (`FlightProfile ZAHEER`): p1 35, p2 70, pBack 0, kClimb 0.5, diveAdd 20, maxRoll 30, kTurn 0.8, kStrafe 0.5, tauRoll 5, no spring, bob 0.06/60, armDrift 4, legSway 3, legFlutter 2, lazyRoll 2, noBrake false, defaultSpeed 0.06; anchors: **HOVER** = `ZAHEER_HOVER` (cross-legged `AIR_SIT` legs, hands on knees), **CRUISE** = arms **swept back** (30,∓10,±30), legs together trailing (x 10/10), the body leaning 35; **SPRINT** = arms back and tight (40,0,±14), legs straight (x 20/20), lean 70: Zaheer's arrow. Take-off `AIR_TAKEOFF_ZAHEER` (4.9) ends on the hover anchor (the shared test rule). Brake: `BRAKE` (Enhanced's) with the arms flaring forward. Landing: `LAND_SOFT` with a feather dip.

10.7 Fire Jet flight (`FlightProfile FIRE_JET`)

Short bursts only (3–6 s): p1 15, p2 30, pBack −5, kClimb 0.3, diveAdd 10, maxRoll 25, kTurn 1.0, kStrafe 0.6, tauRoll 3, **spring** (omega 0.35, zeta 0.5: Ozai's jerky thrust), bob 0.02/6, armDrift 0, legSway 0, legFlutter 0, lazyRoll 1, noBrake true, defaultSpeed 0.07; anchors: HOVER = `FIRE_JETTING` (fists down and back, legs straight back), CRUISE = the same with lean 15, SPRINT = lean 30, fists further back (40,∓10,±30). No take-off gesture beyond `FIRE_JET_START`; landing `LAND_SOFT` + a fiery dip.

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11. Shared stances and transitions

11.1 Armour forming and crumbling

Earth Armour (3.2): bottom-up in 6 slices over 14 ticks (feet, shins, thighs/hips, chest/arms, shoulders, head), each slice revealed with 6 grains and a stone clunk; crumbling top-down in 6 ticks with the slices' grains. **Steel Armour** (3.8): top-down in 5 slices over 5 ticks (head, chest, arms, legs, boots) with white motes and claps; peeling off top-down. **Water Octopus** (5.3): the body veil bottom-up over 14 ticks and the tentacles growing root to tip; collapsing top-down into a puddle in 6 ticks. The slice reveal is one shared helper, `LayerReveal` (**NEW** 14.10): a model-space clip that shows parts of a layer when a plane (y) passes them, with a per-part "pop" (scale 1.1 → 1.0 over 2 ticks) and a particle burst callback.

11.2 The glider opening and closing, the scooter mount, the spout rise

4.3, 4.7, 5.4: all in group `RIDE` (so they never cancel a `CAST`), each ending on its ride stance.

11.3 `LAUNCHED` (airborne after a launch)

While `Live.LAUNCHED` (set by Earth Launch, Air Jump, Fire Jet's end, Spout collapse, Air Blast's self-knockback, Blood Fling on you, Tornado): arms **wide and back** (−70,0,±70 → drifting to −40,0, ±50 as the fall continues), legs **trailing** (x −20/20 alternating slowly ±10 at 0.2 rad/tick: a kick), head −6 (looking ahead/down), lean `-10 + 30·(vy < 0)` (leaning forward once falling), a small roll wobble ±4. It ends on landing (`LAND_SOFT`) or water (`SPLASH` ×12 and the swim pose).

11.4 `LIFTED` and `BLOODBENT` (held in the air by someone else)

Enhanced's `Live.LIFTED` pose (arms wide, legs trailing, drop −0.3) for Metal Grip and Tornado; `BLOODBENT` is 12.4.

11.5 Transitions between elements' looks

Choosing an element (the awakening) plays `AWAKEN_<ELEMENT>` (dur 40 / 4 / 10, locks 40 ticks): Earth: a stomp and both arms rise (the ground ring); Air: a spin (yaw 360) with arms out and a lift (drop −0.3 then down); Water: the gather and a slow outward circle with a water ring; Fire: a deep breath, fists ignite, both arms thrust up (−176) with a column of flame. Changing a look (Sneak + O): Enhanced's `LOOK_CHANGE` (a quick turn) with the element's burst (20 motes).

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12. Hits and defeat: what a target shows

Reactions are gestures in group `HIT` on players (never cancelling a `CAST`; they layer by weight) and, for mobs, a shared `VictimPoses` hook (**NEW** 14.11): a `LivingEntityRenderer.setupRotations` mixin tail that tilts/shakes any living entity carrying one of the mod's effects or `Live`-like synced flags (the mod attaches a small synced `VictimState` to mobs it hits: `bending:victim` attachment, 1.5 s), and a `HumanoidModel.setupAnim` tail for humanoid mobs (zombies, skeletons, villagers, piglins, illagers) that keys their arms and head for the puppet and paralysis poses.

12.1 Earth: stagger, knockdown, stuck

Reaction	Players (`HIT` gesture)	Mobs (`VictimPoses`)	Effects
`HIT_STAGGER` (rock, wall push, knife, whip, fireball)	dur 10: lean −14 at 2 (`0`), head +6, arms flail +20 x ADD, legs 10/−14; back to 0 at 10 (`E`)	body pitch −12 → 0 over 8 ticks (renderer tilt), head yaw jitter ±6 for 4 ticks	6 grains / sparks / splashes by element at the hit
`HIT_KNOCKDOWN` (boulder, Fire Kick, blood fling landing, redirect fail)	dur 20: lean −40 at 3 (`0`), drop 0.5 at 6 (`I`: on the ground), arms up (−120,0,±40), legs x −40/−40; rise: lean 0 and drop 0 at 20 (`E`)	pitch −60 for 10 ticks then up (the mob lies back), dust	a dust puff
`STUCK` (quicksand, frozen feet)	stance while the effect lasts: legs **still** (the walk cycle weight → 0: 14.3), arms pulling up (−100,∓30,∓20) with a tug every 20 ticks (ADD +15 x for 4 ticks), lean −6	legs frozen (a `walkAnimation` speed 0 via the state), the body shivers (frozen) or sinks 0.4 (quicksand: the renderer translates −0.4 y over 20 ticks)	ice block at the feet / the sand collar
`LIFTED` (Metal Grip, Tornado)	Enhanced's	the renderer lifts and tumbles: pitch 20·sin, roll 15·cos at 0.3 rad/tick	motes

12.2 Air: tumble, clutch, spin

Reaction	Players	Mobs	Effects
`HIT_TUMBLE` (Air Blast, Air Slice, Tornado fling)	`LAUNCHED` plus a **roll**: roll 0 → 360 over 12 ticks (a `roll` track in the gesture: **NEW** 14.1 roll keys are Enhanced's `roll(v)`: exists), arms wide; ends on landing	the renderer rolls the body 360° over 12 ticks and pitches −30	6 white motes trailing
`CLUTCHING` (Suffocation)	stance: both hands at the throat (−130,∓50,∓30), head +14 (bowed), lean 12, drop 0.14, a shudder (tremble 2 at 3.0), knees buckling (drop 0.14 → 0.26 over the hold); every 20 ticks a gasp (head −10 for 3 ticks)	head down, body pitch 15, shivering; humanoids: arms to the neck	gasp motes
`SPINNING` (inside a tornado)	`LIFTED` with a yaw spin 30°/tick	the renderer yaws 30°/tick and lifts	

12.3 Water: soaked, burning, chilled

Reaction	Players	Mobs	Effects
`SOAKED` (Water Shot, whip, spout)	no pose; a flinch (`HIT_STAGGER` at half weight)	a flinch	**drips**: 2 `DRIPPING_WATER`/tick from the model's lowest points for 3 s + a darker tint (`LivingEntityRenderer` colour ×0.9 for 3 s: the wet look: **NEW** 14.11 colour hook)
`BURNING` (any fire)	vanilla's fire render plus `HIT_FLAIL` (dur 12: arms shake tremble 4 at 4.0, head −8, lean −6 → 6) once per ignition	the vanilla fire; a shake	
`CHILLED` (ice knives)	a shiver: tremble 2 at 5.0 on both arms for 20 ticks (ADD rows via `CHILLED` state), head +4	the renderer shivers (x/z jitter 0.01)	frost crackles on the body (`ENERGY_BEAM` white lines α 0.5), 1 `SNOWFLAKE`/2 ticks, a blue tint ×0.9

12.4 Blood bending: the puppet

Players (`BLOODBENT` stance, tauIn 1 / out 6): `PUPPET`: arms **wrenched up and back** (−100,∓40,∓60: the elbows out, the hands high behind), the legs **apart and bent** (x −20/30, z ±25), the head **back** (−10) and turned (y 20), lean −10, **lifted** (drop −0.4: the server lifts the entity; the stance adds the body's hang), a **tremble 3 at 2.6** on everything (the strain), and **the limbs follow the claws**: the caster's M claw's elevation drives the victim's arm x (ADD −20·clawUp), the O claw drives the legs (the victim's `aim(ANCHOR)` on the caster's hands, 2 ticks behind): when the caster raises a hand, the puppet's arms rise; when they lower it, the puppet sags. The victim's own input is locked (`MovementInputUpdateEvent`), the camera free (so they see the caster), the view tilting with the body (2.2). On release: the puppet **drops** (`LAUNCHED` → `LAND_SOFT` / `HIT_KNOCKDOWN` if from height). **Mobs (`VictimPoses`):** the renderer lifts the entity (the server does), pitches the body −10 → +10 following the caster's O claw, rolls ±8 with the tremble, and humanoid mobs get the arms up and back (the `HumanoidModel` tail: arms x −100, y ∓40, z ∓60) and the head back; animals (quadrupeds) get legs splayed (a `QuadrupedModel` tail: legs x ±40 outward, **NEW** 14.11) and the body rolled 15. Effects: the red threads (5.7), 1 red mote/tick.

12.5 A bender losing their bending (Bending Severance) and paralysis

Severed (`SEVERED`, a one-shot `BLOOD_SEVERED` dur 60 / 4 / 16, locks 60 ticks): at the touch, the body **goes rigid** (t 0–8: `PARALYSED`-like, arms out, tremble 1 at 3.0), then **sags**: t 16 the arms fall (10,0,±14), the head drops (+18), the knees give (drop 0.36, legs x −30/−30 z ±12: a kneel), lean 10; t 16–48 kneeling, swaying slightly, the head slowly lifting (+18 → +8); t 60 rising slowly (`E`) to `NEUTRAL` with the head still down (+6: the `SEVERED` stance keeps head +6 and arms 2° heavier for the whole severed time, and the element's idle hint is **gone**: weight 0). Effects: the element's colour drains from the body (an `energySwirl` in the element's colour α 0.4 → 0 over 20 ticks flowing **upward and out** of the body: 16 motes of the colour rising from the chest), then a grey dust ring; the HUD greys. The victim's first person: the hands fall limp to the bottom corners for 60 ticks (`dy −0.34, rx +50`), the camera lowers 0.3 (the kneel) and a desaturation post effect (saturation 0.6 for 60 ticks, then 0.85 for the severed duration via `spirit.json`'s sibling `severed.json`: Saturation 0.85, ColorScale 1/1/1). Sound: `severed_victim` (the high ring fading into muffled quiet: a low-pass on the music for 3 s is **not** possible on vanilla's sound engine; the ring alone).

Paralysed (`PARALYSED`, while `bending:paralysed`): players: the stance `PARALYSED` (arms out rigid, legs locked apart, head −8, a **fast tremble 2 at 6.0** on arms and head, lean −6), input locked; the walk cycle weight → 0; mobs: the renderer shakes (x/z jitter 0.02 at 6 rad/tick) and the humanoid tail puts the arms out (−20,0,±46), the head back; effects: 2 `ELECTRIC_SPARK`/tick crawling on the body, a white-blue flicker tint (`#DDF4FF`, colour hook, 3 Hz), the eyes white; sound `paralysed_loop` (a buzzing crackle, volume 0.4) at the victim; the first-person overlay 2.2. Release: the body **slumps** (`HIT_STAGGER` at full weight: lean −14 then up) and 6 sparks fall off.

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13. Budgets: the four-bender fight

The worst case the design is tuned for: an earthbender in armour behind a wall throwing rocks, an airbender on a scooter with a shield up shooting blades, a waterbender with the octopus whipping, a firebender flamethrowing, all within 32 blocks of one camera, at *Effect density* 1.0 and vanilla Particles "All":

Source	Particles/tick	Meshes/frame	Entities
Earth: wall (standing) + armour + 3 rocks	2 + 0.1 + 3 + impacts (22 per hit, ≈ 1 hit/s) ≈ **6/tick avg**	wall dust 0, armour layer 1, rocks 3, rings ≤ 2	3 rocks
Air: scooter + shield + 3 blades	3 + 2 + 3 + hits 6 ≈ **9/tick**	ball 1 (+4 rings), shield 1 (+3 rings), blades 3, cones ≤ 2	1 ball, 3 blades
Water: octopus + whip	0.5 + 1 + slaps 6/2 s ≈ **5/tick**	octopus veil 1 + 8 chains, whip 1 chain	0
Fire: flamethrower (2 jets)	18/tick + hits 9 ≈ **22/tick**	2 beams (5 layers each) + tongues ≈ 10 draws	0
Total	**≈ 42/tick → with 10–21-tick lifetimes ≈ 600 live particles**, under vanilla's 16,384 cap and comparable to a vanilla rainstorm	**≈ 35 batched draws** in 4 render types (one `endBatch` each)	7

Rules that keep it there: counts above are **per tick caps**, not "per effect spawn" guesses; `density()` scales everything; nothing beyond 48 blocks spawns; meshes beyond 96 are skipped; `MAX_EFFECTS 160` as `SandFx`; each `*Fx` class keeps a bounded list and drops the oldest; one `RenderLevelStageEvent` subscriber (`BendingWorldFx`) draws every element's meshes in **four batches** (`MATTER_FLAT`, `MATTER_TEXTURED`, `ENERGY_VEIL`, `ENERGY_BEAM`), never one batch per effect. Server side, a 4-bender fight runs ≤ 7 projectile entities, ≤ 2 ride entities, ≤ 2 walls (≤ 112 temp blocks), 1 pillar, and the hazards (quicksand patches, lava paths, tornadoes) are **ticked as server-side objects**, not entities, except the tornado (an entity so it can carry riders). Payloads: one FX payload per event (not per tick) except keep-alives (beams every 8 ticks, tornado position every 4 ticks), as Enhanced does.

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14. Reuse: what comes from Enhanced, what is new

14.0 Reused as is (copied into `bending.client.render`, package name changed, nothing else)

`GestureId` (shape; Bending's own constants), `Gestures` (+ `Def`, `Builder`, `ArmKey`, `Track`, `Mode`, `Target`), `GestureTable`, `GestureRoutes` (Bending's own arms), `GestureEffects`, `HeroAnimator` (renamed `BenderAnimator`), `HeroAnimState`, `HeroPoses`, `PoseContext`, `Stance`, `Stances`, `StancePoses`, `FlightProfile`, `FlightPoses`, `FlightGestures` (takeoff/land/brake keys), `RunCharacter`, `RunPoses` (minus the speedster styles), `FirstPersonArms` (+ `Offset`), `FirstPersonMoves` (the 14 rows), `AnimMath`, `BodyLean`, `HandTracker` + `HandTrackerLayer`, `AimMath`, `HeroClient` interface (renamed `ElementClient`), `HeroClients`, `AnimDebugCommands` (`/bending_anim`), `AnimStateOverlay`, the mixins `PlayerModelMixin`, `PlayerRendererMixin`, `AbstractClientPlayerMixin` (skins), `ItemInHandRendererAccessor`, `CapeLayerMixin`, `HumanoidArmorLayerMixin` (hide armour under suits), the FX toolkit (`EnergyGeometry`, `FxShapes`, `SandShapes` → `MatterShapes`, `EnhancedRenderTypes` → `BendingRenderTypes` with `MATTER_FLAT`/`MATTER_TEXTURED` = `SandRenderTypes`), `BeamStyle`/`BeamStyles`/`BeamSpec`/`ContinuousBeam`/`BeamPayload`/`BeamRenderer` (+ a `Drawer` for the flame cone), `ShieldStyle`/`ShieldStyles`/`ShieldFx` + the dome passes of `WorldEffectsRenderer`, `FlightEffects` (trails, rings, wind, `lay`, `polyline`), `EnergyParticle`/`EnergyParticleOptions`, `PowerAuraLayer` (the swirl aura), `GlowingEyesLayer`, `AstralBodyRenderer`/`AstralLayer` (the spirit), `Afterimages` (not used in 1.x), `EntityGlow` + its two mixins, the overlay recipes (`PhoenixOverlay` vignette, `ApocalypseOverlay` gradient), `SlowMotionEffect` (post-shader loading, for `spirit.json`/`severed.json`), `ArenaFx.shake` (camera shake), `PowerLoopSounds`, `PinLock`/`EntombedLock`

(input locks, generalised to a `Live`-bit set), `TelekineticBlockEntity` + renderer (the Rock entity's base: `renderSingleBlock` of a state) and `Telekinesis.hold` (Hold Rock, Metal Grip), `LegIk` (not needed in 1.x), `PreviewPuppet` (screens), `HudDraw.visual` (RTL).

14.1 Building blocks that are NEW (and which lane builds them, `PLAN.md`)

#	Block	What	Lane
14.1	**Leg z in gesture keys**	`Gestures.Builder.legs(rx, rz, lx, rz)` and a `Track legZ` (width 2), sampled like `legX`; `PoseContext.leg` already takes z. Needed by the horse stance's stomps in keys, Fire Kick, the sweep, meditation, the scooter crouch in gestures	1.0 ANIM (core)
14.2	**Groups `RIDE`, `HIT`, `AUTO`**	three more `GestureId.Group`s and `HeroAnimState.SLOTS` 3 → **4** (so a cast, a ride, a hit reaction and an auto strike can overlap); `AUTO` and `HIT` gestures never take a slot from `CAST` (the slot picker prefers freeing the oldest non-CAST)	1.0 ANIM
14.3	**Channel masks and body ownership**	per-gesture masks: `legsMaskedByAir` (legs keys × (1 − wAir)), `legsMaskedByWalk` (stance leg rows × (1 − limbAmount/0.2)); a `Live`-driven **`bodyOwner`** flag that forces `wAir/wRun/wSprint` to 0 and ignores `vanillaOwnsBody` for `isPassenger` of the mod's rides and `isFallFlying` with the glider (`rideOverride`); needed by meditation, the scooter, the spout, the glider	1.0 ANIM (meditation and glider in 1.1 use it)
14.4	**Camera drop and eye-height offsets**	a `ViewportEvent.ComputeCameraAngles`... position hook (`CalculateDetachedCameraDistance` is for third person; for first person use `ViewportEvent.ComputeCameraAngles` plus a `Camera` position nudge via a small `CameraMixin.setup` tail (**one new mixin**)): the gesture's `drop` applied to the first-person camera (×0.5, reduced-motion gated) and a per-`Live` eye offset (meditation −0.9, the sever kneel −0.3)	1.0 ANIM
14.5	**Particles `bending:grain` and `bending:droplet`**	two `TextureSheetParticle`s with gravity (grain: 0.06, bounces once; droplet: 0.08, a `SPLASH` on landing), tint from options (grain: a `BlockState` tint or RGB; droplet: RGB), 20–40 ticks, `PARTICLE_SHEET_TRANSLUCENT`, shadow-lit (not fullbright: matter)	1.0 FX
14.6	**Gesture chaining**	`HeroAnimator.startAfter(s, id, afterId)`: a gesture queued to start when another ends (the water source draw, the great fireball's inhale)	1.2 ANIM (needed from Water; a `param` delay stands in before)
14.7	(suggestion) fake light discs	a warm `MATTER_FLAT` disc under flames	later
14.8	**Tracked body centre**	`HandTracker.belly(player)`: the body part's centre in world space (for the redirection path, the blood threads' anchor)	1.3 ANIM (FIRE)
14.9	**Closed eyes**	an eye-lid mask layer tinted by the skin's sampled face colour (`NativeImage` read once per skin, cached)	1.1 LOOKS

#	Block	What	Lane
14.10	**`LayerReveal`**	model-space bottom-up/top-down reveal of a `ModelPart` layer in slices with pop and burst callbacks	1.0 LOOKS/FX
14.11	**`VictimPoses`** (mobs)	`LivingEntityRendererMixin` tail (tilt, lift, shake, colour tint) and `HumanoidModelMixin` + `QuadrupedModelMixin` `setupAnim` tails reading a synced `bending:victim` attachment (puppet, paralysed, stuck, chilled, soaked), 1.5 s TTL	1.0 ANIM (stagger/knockdown/stuck), 1.2 (puppet), 1.3 (paralysed)
14.12	**Verlet chains**	a tiny client rope sim (`ChainSim`: N points, gravity vector, damping, pinned root, a driven end) for the water whip, octopus tentacles, water arms, fire whip; drawn as ribbon chains	1.2 FX (Water), reused 1.3
14.13	**Mesh entities**	renderers for `RockEntity` (block display, tumbling), `AirBladeEntity` (crescent), `AirScooterEntity` (sphere + rings), `TornadoEntity` (funnels), `WaterShotEntity` (ribbon), `WaterSpoutEntity` (funnel), `IceKnifeEntity`, `SteelKnifeEntity`, `FireballEntity`, `LavaShurikenEntity`, `SpiritEntity`, `SandStreamEntity`; one `BendingWorldFx` subscriber batching hazards' meshes (walls' dust, quicksand collars, lava shimmer, rings, bolts, threads)	per element lane
14.14	**Screen overlays**	`ArmourSlitOverlay`, `SpiritOverlay`, `BloodOverlay`, `ParalysisOverlay`, `QuicksandOverlay`, `BubbleOverlay`, `HeatOverlay`, `RedirectPathOverlay` (screen-space bolt)	per element lane
14.15	**Flame beam drawer**	a `BeamRenderer.Drawer` for the FLAME style: inverted taper, tongues instead of strands, two beams per player (`offHand`)	1.3 FX
14.16	**Lightning bolt FX**	`LightningFx`: the 3-layer branching bolt with reshapes and flash spheres, body-path variant	1.3 FX
14.17	**Water surfaces**	`MATTER_TEXTURED(water_still/water_flow)` ribbons and spheres using vanilla's animated water sprites (`TextureAtlas.LOCATION_BLOCKS` sprite lookup) with the biome water tint	1.2 FX

14.2 Tests that pin all this (the animation lane's `*AnimationTests`, server-side, pure)

As Enhanced's `AnimationMathTests`: every `GestureId` has well-formed keys (`problems()` empty, attack ≥ 1, release ≥ 1, no ADD/ABS mixing), every yaw gesture ends on a whole turn, every gesture that hands over to a stance ends on the stance's `StancePoses` constant (no pop), every ability/toggle/Live/flag is acted or explained (`GestureRoutes.acting`), no gesture is orphaned, every flight profile's take-off ends on its hover anchor, the new leg-z track obeys the same rules, the `VictimPoses` tables are within joint limits, the first-person offsets stay under 0.8, and each element's `*AnimationTests` checks the timing of each gesture (the tables above are the expected numbers: the test reads the `Def` and asserts the key at tick t equals the table's pose within 0.5°), which arm each drives, the prediction rules and the cooldown-length routing. What the poses look like is checked by eye in a real client with `/bending_anim play <gesture> <tick>` and `/bending_anim target <player>` from a second client, in both languages, three render rounds per lane (`TECH.md` 8).

LOOKS.md — Character looks

Bending — Looks

Every look is **drawn for the mod**: a 64×64 skin plus the mod's own `ModelPart` layers (hair, robes, armour plates, crowns, hoods), the way Enhanced draws Agatha's hat, Heimdall's helm or Hela's headdress (`AgathaHatLayer`-style `RenderLayer<AbstractClientPlayer, PlayerModel>` classes, a `LayerDefinition` in head-pixel space, attached per skin model through `ClientRegistrar.addLayers`). No skin is copied from a skin site; the references below are **descriptions from the shows** for the artist (the art lanes draw from them with the same `gen_<lane>.py`-style pixel scripts Enhanced used). Nothing here changes rules: a look changes how you look and the colour of your element's effects (Azula). Every element also has **Your own skin** (`own`): your skin, the element's effects.

Status of the research: the ATLA and Korra characters are well documented and the descriptions below are from the shows. *Avatar: Seven Havens* (Paramount+, first episodes released October 2025) is new: only text sources were reachable from this session, so **Jae's and Kartik's entries list exactly what is known and what needs a picture from Eilon** (section 2.5, 2.6). Eilon's spellings ג'יי and קארתיק are kept in Hebrew; in English the show spells them **Jae** and **Karthik** (the plan uses those in ids: `jae`, `karthik`; the Hebrew display names are Eilon's).

0. The skin layout plan (every look)

The base is the vanilla **64×64** skin (wide or slim arms per look). The mod's looks use the overlay regions for what sits close to the body and separate layers for anything that sticks out more than the 0.25-pixel hat layer allows.

Region (vanilla UV)	Base layer	Overlay layer (second skin)	Used for
Head `(0,0)–(32,16)`	face, skin tone, eyes, scars, tattoos (Air arrows on the brow), hairline	`(32,0)` hat: short hair volume, headbands, bangs, beards close to the face, masks (Amon), hoods' inner rim	all
Body `(16,16)–(40,32)`	tunic, sash, belt, necklace (Katara), chest emblem (Jae)	`(16,32)` jacket: open vests (Bolin's collar), robe folds, shoulder wraps (Tenzin's cape at the shoulders), armour plates close to the body (Lin)	all
Arms `(40,16)` / `(32,48)`	sleeves, bracers, wristbands, bare skin (Toph, Ghazan)	`(40,32)` / `(48,48)`: puffed sleeves, bracers that stand off (Azula's cuffs), the Air Nomad arm arrows	all
Legs `(0,16)` / `(16,48)`	trousers, boots, **bare feet (Toph)**	`(0,32)` / `(0,48)`: boot tops, leg armour (Jae's cuisses), robe hems, knee wraps	all

Layers (`ModelPart`, drawn by the mod): each is a small `LayerDefinition` in head space (top of the head y = −8, front z = −4) or body space, textured on its own PNG (64×32 or 64×64), attached to `head` / `body` / legs with `translateAndRotate`, hidden when the player wears a helmet in that slot (hair/crowns) or is invisible, exactly as `AgathaHatLayer.wears` does. Shared layer kinds, drawn once and textured per look:

Layer kind	Attached to	Geometry	Looks that use it (texture each)
`TopknotLayer`	head	a 3×3×3 bun 1.5 px above the crown + an optional 1×4×1 pin and a 2×2×2 tie; variants `bun_high`, `bun_back` (Katara: at the back), `bun_double` (Toph: a wide bun with the ornament loops: two 1×3×1 arcs)	Toph, Bumi, Katara, Pakku, Iroh, Jeong Jeong, Ozai, Azula, Ghazan, Zaheer
`CrownLayer`	head (over a topknot)	the Fire Nation **flame crown**: a 1 px thick, 5 px tall flame silhouette (3 cubes stepped) in gold, in front of the bun; Ozai's has five prongs (a wider 7 px flame), Azula's three; Bumi's **Earth crown**: a green 7×2×7 band with a gold 2×2 gem	Ozai, Azula, Bumi
`HairLayer`	head	long hair falling behind the head: a 8×6×1 slab at the back (−4..4, −2..4, 4..5) + two 2×4×1 side locks; variants `shaggy` (Zuko: slabs at the sides and back 1 px out, no slab below the ears), `bob` (Jinora: a 9×5×9 shell 0.6 px out), `braid` (Katara: a 1×6×1 braid at the back plus two loops `(2×3×1)` beside the face)	Zuko, Jinora, Katara, Ming-Hua (ponytail: a 2×7×2 tail), Suyin
`BeardLayer`	head	a 6×3×1 slab under the chin (−3..3, 0..3, −4.5..−3.5) + a 4×2×1 moustache; variants `goatee` (Ozai, Ghazan: a 2×3×1 point), `full` (Zaheer, Iroh), `moustache_long` (Gyatso: two 1×5×1 drooping strands), `pointed` (Tenzin: a 4×4×1 tapering slab)	Tenzin, Zaheer, Gyatso, Iroh, Pakku, Jeong Jeong, Ozai, Ghazan, Karthik
`RobeLayer`	body (hips)	a **skirt of robes**: four 1-px plates (front/back 8×8×1, sides 4×8×1) hanging from the hips (y 12..20 in body space), swinging ±8° with the legs' walk (driven by `rightLeg.xRot`), split at the front; variants `long` (to the ankles, 12 px), `short` (to the knees, 6 px), `cape_wrap` (Tenzin: a back plate only, 8×14×1, from the shoulders)	Tenzin, Gyatso, Jinora, Bumi, Suyin, Pakku, Iroh, Jeong Jeong, Ozai, Amon, Azula (short)
`ShoulderLayer`	body (shoulders)	two 5×3×5 pads above the shoulders (pointed: an extra 1 px spike)	Azula, Ozai, Lin (flat plates), Jae (brigade pads)
`BackLayer`	body (back)	a cable spool (Lin: a 6×6×3 drum with a 1 px rim), a glider staff when stowed (airbenders with a glider in the inventory: a 1×14×1 staff diagonal across the back), the Water Skin (a 4×6×2 pouch at the right hip)	Lin, airbenders, waterbenders
`HoodLayer`	head	Amon's hood: a 10×10×10 shell open at the face, drooping to the shoulders (two 3×6×1 flaps)	Amon
`MaskLayer`	head	a 8×8×1 plate on the face 0.5 px out with the painted features (Amon)	Amon
`ArmBandLayer`	arms	4×2×4 bands on the upper arms (Air Nomad arm wraps, Ghazan's bands, Bolin's bracers)	several
`EarthArmourLayer`	whole body	the stone suit: an inflated `PlayerModel` mesh (`CubeDeformation(0.35)`, built as `LayerDefinition.create(PlayerModel.createMesh(…))` like Enhanced's `ENERGY_WIDE/SLIM`) with the **eye slit** cut from the head cube (the head is 4 cubes: brow 8×3×8, cheeks 8×2×8, jaw 8×3×8, and two 2×2×8 side pieces leaving a 8×2 slit), textured `earth_armour.png` (stone plates with cracks; tinted by the ground's map colour at forming time: `argb` tint); split into **6 slice parts** for `LayerReveal` (`ANIMATION.md` 14.10); worn over the look, hiding the skin	Earth Armour

Layer kind	Attached to	Geometry	Looks that use it (texture each)
`SteelArmourLayer`	whole body	the same mesh, `steel_armour.png` (riveted iron plates, Lin's police armour's colour), 5 slice parts	Steel Armour
`OctopusLayer`	whole body + 8 chains	the water body veil (`MATTER_TEXTURED(water_flow)` over the inflated mesh at α 0.55) and the tentacles (ribbon chains, `ANIMATION.md` 5.3): not a `ModelPart`, a renderer	Water Octopus
`WaterArmsLayer`	arms	inflated arm parts (`CubeDeformation(0.25)`) textured by the water sprite + the ribbon extensions	Water Arms
`GliderLayer`	hands (open) / back (stowed)	a staff 1×16×1 with two wings: each a 10×1×6 plate hinged at the staff's top, folding from z ±5° (closed) to ±85° (open), textured `glider_<dye>.png` (16 colours: one texture tinted by the dye's `MapColor`), a cloth seam line	Air Glider
`WaterSkinLayer`	right hip	a 4×6×2 leather pouch with a cork, its fullness shown by the pouch's darkness (3 texture stages)	Water Skin

Slim vs wide: slim (3-px arms): Toph, Lin, Suyin, Jinora, Katara, Ming-Hua, Azula. Wide: everyone else. (Bumi, Ghazan, Bolin, Zaheer, Iroh, Ozai are broad: the wide model; Tenzin, Gyatso, Zuko, Jeong Jeong, Pakku, Amon, Jae, Karthik: wide.)

Faces: every face is drawn in the shows' flat style: almond eyes 2 px wide with a 1 px iris of the character's eye colour (Earth Kingdom green, Water Tribe blue, Fire Nation gold, Air Nomad grey), a 1 px mouth, eyebrows 1 px. Skin tones per nation: Earth Kingdom light-medium `#E8C39E` / `#D9A878`; Water Tribe medium-dark `#A8744F` / `#8B5A3C`; Fire Nation light `#F0D2B4`; Air Nomads light `#EBCDA8`; Seven Havens' South-Asian-inspired Ellora: medium `#B98156` / `#A06A45`.

Approval line: each look ends with one or two sentences Eilon can say yes or no to (Hebrew in `SUGGESTIONS.md`).

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1. Earth — the looks and the Earth layers

1.1 Toph Beifong — טוף בייפונג (default Earth look)

- **Reference.** *ATLA*, Books 2–3: a short girl of twelve; **black hair in a large, high bun** held by a **green headband** with the family's **hair ornament**: a pale green band with two loops over the bun, and thick bangs hanging over her **pale, unfocused grey-green eyes** (blind). Outfit: a **yellow-green sleeveless tunic** (`#C8C06A`) with **dark green trim** (`#3F6B3A`), a **cream long-sleeved undershirt**... (in Book 2 her arms are covered to the wrist in pale green; Book 3 Fire Nation disguise is not used), **dark green trousers** to the shin, a **dark green belt**, pale green wristbands, and **bare feet** (she sees through her feet). Earth Kingdom light skin.
- **Skin.** Slim. Base: the tunic and undershirt, the belt, the trousers ending at the shin, **bare feet** drawn as skin with toes (the legs' lower 4 rows), the bangs on the face's top 3 rows with the eyes beneath them pale (`#B6C9A6` irises, no pupils: the blind look), small mouth. Overlay: the hat layer carries the bangs' volume and the headband; the jacket layer the tunic's loose edges.

- **Layers.** `TopknotLayer bun_double` (`toph_bun.png`: black bun, the green band, the two pale loops).
- **Effects.** Element colours as Earth's default (ochre). **(suggestion)** Toph's rocks are a shade darker (`#6B5B3E` dust): no, keep one Earth palette; **her ambient master effect** is the grain ring at bare feet (no change).
- **Approval:** Toph as in Book 2: green and yellow tunic, big bun with the headband loops, bangs over pale eyes, **bare feet**. The default look for a new earthbender.

1.2 King Bumi — המלך בומי

- **Reference.** *ATLA*: the ancient king of Omashu, 112, wiry and then hugely muscular under the robes: **white hair in a tall, messy topknot**, **bushy white eyebrows**, a **white goatee** with a thin moustache, **one eye wider than the other**, a mad grin with a few crooked teeth; **green royal robes** (`#4F7A3F`) with **gold trim** (`#D4AF37`), a wide gold-trimmed collar, and a small **gold Earth Kingdom crown** (a green band with a gold leaf). Light skin, lined.
- **Skin.** Wide. Base: the green robe with the gold collar and trim down the front, gold cuffs, green boots; the face with the heavy brows (2 rows), the uneven eyes (one 2×2 open, one 2×1 squint), the grin (3 px with a dark gap). Overlay: the robe's folds; the jacket layer carries the collar's standing edge.
- **Layers.** `TopknotLayer bun_high` (white, tall: the bun is 3×4×3), `CrownLayer earth` (the green band with the gold gem), `BeardLayer goatee` (white), `RobeLayer long` (`bumi_robe.png`).
- **Effects.** Default Earth. **(suggestion)** his idle hint includes a slow head tilt (the madness): a `head(0, 0, 6)` ambient at weight 0.3: client-only, cheap, flagged for Eilon.
- **Approval:** old King Bumi in green and gold robes with his tall white topknot, crown and crazy grin.

1.3 Lin Beifong — לין בייפונג

- **Reference.** *Korra*: Chief of Republic City's metalbending police, in her fifties: **short grey hair** swept back and up, **two parallel scars on her right cheek**, **light green eyes**, a stern mouth. The **metalbending police armour**: **dark grey-green metal** (`#4A5550`) plates over a **dark olive uniform** (`#2F3A32`), a wide armoured collar, segmented shoulder and chest plates, armoured forearms and boots, and the **cable spool** on the back (two spools of metal cable at the lower back). Light skin.
- **Skin.** Slim. Base: the dark uniform with the plates painted on the body and limbs (lighter plate edges `#7C8A82`), the collar, the boots; the face with the two scars (two 1-px diagonal lines `#B88A78` on the right cheek), grey hair on the head's top and back (the hat layer adds its upswept volume). Overlay: the plates' relief on the jacket and sleeve layers.
- **Layers.** `ShoulderLayer flat` (`lin_shoulders.png`: squared plates), `BackLayer spool` (`lin_spool.png`).
- **Effects.** Default Earth; **her metal effects are the police steel grey** `#8C96A0` instead of the iron `#B8C0CC` (the metal filings and plates tint by look: a `metalColor` per look, as Enhanced's variants carry a `color`/`glow`).
- **Approval:** Chief Lin in the dark metalbender armour with the cable spool, short grey hair and the two cheek scars.

1.4 Suyin Beifong — סויין בייפונג

- **Reference.** *Korra*, Books 3–4: Zaofu's matriarch: **dark hair pulled back** into a low bun with a **silver and green metal headpiece** (a flat metal clip arcing over the hairline), **light green eyes**, a small beauty mark under the right eye (**check against a picture**), an

elegant **green and silver Zaofu robe** (`#3E7F5A` green with `#B9C4CC` silver panels and a dark green sash), long sleeves, a high collar, the **metal clan's silver accents** at the collar and cuffs. Light skin.
- **Skin.** Slim. Base: the green robe with silver panels, the sash, dark boots; the face with the mark; dark hair at the top and sides. Overlay: the robe's collar and cuffs in silver.
- **Layers.** `TopknotLayer bun_back` (dark, low) with the headpiece painted on the bun's front, `RobeLayer long` (`suyin_robe.png`, split at the sides).
- **Effects.** Metal effects in **Zaofu platinum-silver** `#C8D2DA`.
- **Approval:** Suyin in a green and silver Zaofu robe with the metal headpiece and a low bun.

1.5 Ghazan — גאזאן

- **Reference.** *Korra*, Book 3: the Red Lotus lavabender: a big, calm man with **dark brown hair tied back in a topknot/ponytail** with **shaved or short sides**, a **moustache and a pointed goatee**, a **lazy half-smile**, dark eyes, **tan Earth Kingdom skin** (`#C99667`); **a sleeveless dark green tunic** (`#2F4F3A`) open at the chest over a **brown undershirt**, **brown trousers** (`#5A4634`), **bracers** and arm bands, a red-brown sash; **(check:** whether he has a tattoo on an arm).
- **Skin.** Wide. Base: bare upper arms (tan), the tunic and bands, the trousers and boots; the face with the goatee (the hat layer gives the moustache its stand-off). Overlay: the open tunic's edges on the jacket layer.
- **Layers.** `TopknotLayer bun_high` (dark, a tail: the `tail` variant adds a 1×4×1 drop behind), `BeardLayer goatee`, `ArmBandLayer`.
- **Effects.** Default Earth; **his lava is the show's orange-red** (the one Lava palette: no change). **(suggestion)** Ghazan's look makes the Lava Path's crust a bit darker (`#1E120C`): a per-look lava tint, trivial.
- **Approval:** Ghazan: sleeveless green tunic, bare tanned arms, a topknot, a moustache and goatee.

1.6 Bolin — בולין

- **Reference.** *Korra*: a stocky, cheerful teen: **black hair swept back** with a **curl at the front**, **bright green eyes**, thick eyebrows, a big grin, fair skin; Book 1 outfit: a **dark green sleeveless vest/jacket** (`#2E5A3A`) with a **grey fur collar** (`#8E8E8A`) over a **cream shirt**, **brown wrist wraps**, a brown belt, **dark trousers** (`#3A3A3A`) and boots. (Pabu, the fire ferret, is a companion suggestion, not part of the look.)
- **Skin.** Wide. Base: the vest and shirt, fur collar painted at the neck, the belt, the trousers; the face with heavy brows and the grin; the hair's curl on the brow's top row. Overlay: the fur collar's volume on the jacket layer, the curl on the hat layer.
- **Layers.** none beyond the overlays (Bolin reads well with the second skin alone).
- **Effects.** Default Earth and Lava.
- **Approval:** Bolin in his green vest with the grey fur collar and his swept-back hair with the curl.

1.7 The Earth layers' art

- **Earth Armour** (`earth_armour.png`, 64×64 for the inflated mesh): large rough plates with dark seams, lighter chipped edges, three overlay crack textures (`earth_armour_cracks_1..3.png`, alpha only). Tinted at forming by the ground: grass → `#6E5B3D`, stone → `#8A8A86`, sandstone → `#C9B37C`, deepslate → `#4E4E52`, netherrack → `#7A3B3B`, end stone → `#D8D6A8` (the block's `MapColor`).

- **Steel Armour** (`steel_armour.png`): riveted plates, a seam down the chest, the police grey `#6E7A80` with highlights `#B8C4CC`; a dent overlay (`steel_dents_1..3.png`); a highlight mask for the glint sweep.
- **Rock** (block displays: no texture). **Quicksand block** (`quicksand.png`: 4-frame rippling sand). **Wrecked devices** (`wrecked_*.png`: one "bent metal" tile set with the device's base colour tint: 8 textures for the device families). **Steel plate block** (`steel_plate.png`). **Lava path crust** (`temp_lava` uses vanilla lava; `lava_crust.png` for the cooling overlay). **Ice spike**, **ice wall** (vanilla ice). **Sand grains** (`particle/grain.png`, 8×8), **droplets** (`particle/droplet.png`).

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2. Air — the looks and the Air layers

2.1 Tenzin — טנזין (default Air look)

- **Reference.** *Korra*: Aang's son, in his fifties: **bald** with the **blue Air Nomad arrow tattoo** (`#2E6FBF`) from the brow over the crown and down the back, arrows on the backs of the hands; a **dark brown-grey pointed beard** (greying, `#5A4A3E`) and a moustache; **grey eyes**; a serious, kind face. Korra-era airbender robes: a **yellow-orange robe** (`#E9A23B`) with **maroon-red wraps** (`#8E2F2F`) over the shoulders and a **red cape-like shawl** down the back, a **red sash** at the waist, **yellow trousers**, **brown boots**, orange arm wraps.
- **Skin.** Wide. Base: the robe and wraps, the sash, the trousers and boots; the head bald with the arrow drawn from the brow (a 2-px wide arrow point between the eyes widening to 4 px at the crown) over the top to the nape; grey eyes. Overlay: the shoulder wraps' volume, the hand arrows on the arm overlays.
- **Layers.** `BeardLayer pointed` (`tenzin_beard.png`), `RobeLayer cape_wrap` (`tenzin_cape.png`: the maroon shawl down the back, swinging), `BackLayer glider` when a glider is carried.
- **Effects.** Default Air (saffron/sky).
- **Approval:** Tenzin: bald with the blue arrow, the pointed beard, yellow robes with the red shoulder wraps and shawl. The default look for a new airbender.

2.2 Zaheer — זהיר

- **Reference.** *Korra*, Book 3: the Red Lotus leader: a lean, muscular man around forty: **dark hair tied back** low (**check:** in some episodes he is shown with shaved head and beard; the plan uses the tied-back hair of his escape), a **full dark beard** cropped short, **grey eyes**, tan-light skin, intense brows. Outfit: a **dark olive sleeveless tunic** (`#4A4A33`) with a **dark red wrap** (`#7A2A2A`) across the chest and waist, **brown trousers**, **bracers**, bare upper arms. After gaining flight: the same clothes, a wind always moving them.
- **Skin.** Wide. Base: the tunic, wrap, trousers, bracers, bare arms; the face with the beard's shape (the hat layer gives the beard volume), the tied hair at the back. Overlay: the wrap's folds.
- **Layers.** `BeardLayer full`, `TopknotLayer tail` (a low tail), `ArmBandLayer` (bracers).
- **Effects.** Default Air; with **Zaheer's Path** open, every airbender's pushes get a faint **red thread** motif in the HUD only (no world effect change). **(suggestion)** Zaheer's look makes the True Flight wake slightly darker (`#D8D8D8`): cosmetic per-look `color`.
- **Approval:** Zaheer: tied-back dark hair, a full short beard, an olive sleeveless tunic with the dark red wrap. Confirm the hair (tied back, not shaved).

2.3 Jinora — ג'ינורה

- **Reference.** *Korra*: Tenzin's daughter; Book 4 (an airbending master at 14): **short brown hair** (`#5B3F2A`) in a bob with a side part, the **blue arrow tattoo** on her brow (and hands), **brown eyes**, a calm, bookish face; the Korra-era **airbender wingsuit/robes**: **yellow-orange** (`#E9A23B`) top with a **red** (`#8E2F2F`) collar and sash, yellow trousers to the ankle, brown soft boots. Light skin.
- **Skin.** Slim. Base: the robes, the sash, boots; the face with the arrow point on the brow and the brown bob around it; brown eyes. Overlay: the bob's volume on the hat layer (the `HairLayer bob` adds the rest).
- **Layers.** `HairLayer bob` (`jinora_hair.png`), `RobeLayer short`.
- **Effects.** Default Air; **(suggestion)** her **Spiritual Projection spirit glows brighter** (`#BFE8FF` α 0.55 instead of 0.45): she is the show's spirit expert; cosmetic.
- **Approval:** Jinora as a young master: brown bob, the arrow on her forehead, yellow and red airbender robes.

2.4 Monk Gyatso — הנזיר גיאטסו

- **Reference.** *ATLA* flashbacks: Aang's elderly mentor: **bald** with the **arrow tattoo**, a **long white moustache** drooping past the chin, bushy white eyebrows, warm **grey eyes**, laugh lines; the classic **Air Nomad monk robes**: an **orange** (`#E07A1F`) over-robe with a **yellow** (`#F2C14E`) under-robe and a **saffron shawl**, a **wooden Air Nomad necklace** (a brown cord with a round pendant), bare feet or simple sandals (the show: bare feet with the arrows).
- **Skin.** Wide. Base: the robes and shawl, the necklace, the trousers in yellow, bare feet with arrows; the face with the brows and the moustache's root (the layer gives its length). Overlay: the shawl.
- **Layers.** `BeardLayer moustache_long` (`gyatso_moustache.png`), `RobeLayer long`.
- **Effects.** Default Air.
- **Approval:** Gyatso: old bald monk with the arrow, the long white moustache and orange and yellow robes.

2.5 Jae — ג'יי (*Avatar: Seven Havens*)

- **What is known (from reviews, interviews and fan wikis; the series' art book was not reachable):** an **airbender of Ellora's Bending Brigade**, 19, a platoon leader who becomes the twin Avatars' friend, mentor and airbending master, expressive, calm under pressure; the first on-screen gay male lead of the franchise (his boyfriend is Datu). Uniform: the **Bending Brigade's form-fitting, futuristic soldier's uniform in the Air Nation's colours: dark yellow, dark orange and light grey**, with a **triangular emblem in blue and dark orange on the chest plate** that echoes the Air Nomad arrow, **a running-style belt with pockets** on each side, **loose brown trousers**, **grey cuisse-style leg armour**, **knee-high boots in grey and dark orange**, and **shoulder pads** (the Brigade uniforms resemble Korra's pro-bending uniforms without helmet or arm guards). Gear: a **fold-up board with a hole in the middle that an air scooter fills** (a hover-board: a lovely glider alternative, 2.7), and he flies **gliders**.
- **What needs a picture from Eilon (open question 4):** hair colour and style, skin tone, whether he has arrow tattoos, the exact colours of the uniform panels and boots, any facial marks. **Until then the art lane draws a placeholder** (dark hair, short; Ellora medium skin tone; the uniform from the description) and marks the look "draft" in the look picker.

- **Skin (plan).** Wide. Base: the yellow/orange/grey uniform with the triangular chest emblem, the belt with pockets, brown trousers, grey leg armour on the legs' overlays, the boots. Overlay: the uniform's panel relief.
- **Layers.** `ShoulderLayer brigade` (`jae_shoulders.png`), optional `BackLayer board` (the folded board on the back when the scooter ability is carried: **(suggestion)**).
- **Effects.** Default Air; **(suggestion)** his Air Scooter is drawn **inside Jae's board** (a flat ring mesh around the ball) when his look is on: a per-look scooter style.
- **Approval:** Jae in the Bending Brigade uniform (yellow, orange, grey, the triangle emblem); **send a picture for the face and hair.**

2.6 Karthik — קארתיק (*Avatar: Seven Havens*)

- **What is known:** an **airbender of the Order of the White Lotus**, "Captain Karthik", a tall man in his forties: **brown hair with white streaks coming in underneath**, **scruff / a short beard**, **tired eyes**, gentle and calm but stern and forceful with orders; Nisha's surrogate father; nicknamed "the Human Hurricane" and "Warrior of the Wind"; voiced by Akshay Khanna; fans debate whether he is a villain.
- **What needs a picture:** his **outfit and its colours** (White Lotus of the Havens era: unknown whether white and blue), his skin tone, hair length, any emblem. Placeholder until then: a **white and pale-blue long coat** with the **white lotus tile** on the chest (the Order's symbol from *ATLA*), brown hair with white at the temples and nape, a short beard, Ellora medium skin.
- **Skin (plan).** Wide. Base: the coat, a dark under-tunic, trousers, boots; the face with the beard's root and the white-streaked hair. Overlay: the coat's lapels.
- **Layers.** `BeardLayer full` (short, brown with grey), `RobeLayer long` (the coat's skirt), `HairLayer short_back` (the white streaks at the nape).
- **Effects.** Default Air; **(suggestion)** his Tornado is 10% wider (the Human Hurricane): no, cosmetic only: a per-look `tornadoTexture` with a darker swirl.
- **Approval:** Karthik as a White Lotus airbender: brown hair with white streaks, a short beard, a white and blue coat with the lotus; **send a picture for the outfit.**

2.7 The Air layers' art

- **Glider** (`glider_base.png`, tinted by dye): a dark wooden staff, two cloth wings with a seam; 16 dyes → the dye's `MapColor` tint; the default saffron (`#E9A23B`). Stowed: the staff on the back. **(suggestion)** `jae_board.png`: Jae's fold-up board as an alternate glider skin, chosen in the look picker when Jae is the look.
- **Air textures:** `air_lattice.png` (a soft swirl lattice, white on transparent, 64×64 tileable) for the shield and scooter; `air_swirl.png` (a streaky wisp texture, 64×64) for tornadoes and spouts; `eyes_closed.png` (two 2×1 lid strips in the face's UV, tinted at runtime).
- **Spirit:** no texture: the player's skin tinted blue.

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3. Water — the looks and the Water layers

3.1 Katara — קטארה (default Water look)

- **Reference.** *ATLA*, Book 3 (her most familiar look): a girl of fourteen: **dark brown hair** (`#3A2518`) in a **high bun at the back** with two **"hair loopies"** (loops of hair framing the face from the temples to the chin) and bangs parted at the middle; **blue eyes** (`#3B7DD8`); **medium-dark Water Tribe skin** (`#A8744F`); her **mother's necklace**: a **blue choker band**

with a carved pale blue pendant. Outfit: a **blue Water Tribe tunic** (`#2F5F9E`) with **white fur trim** at the collar and hem (`#EDEDE4`), **dark blue sleeves**, a **lighter blue sash**, dark blue trousers tucked into **fur-trimmed boots**, white wristbands. (Book 1's parka with the thick fur is a **(suggestion)** second variant `katara_parka`.)

- **Skin.** Slim. Base: the tunic, sash, trousers, boots with white trim; the necklace at the neck (1 row of blue with a 2×1 pale pendant); the face with the bangs; blue eyes. Overlay: the fur trim's volume, the sleeves.
- **Layers.** `TopknotLayer bun_back` + `HairLayer braid` (the loops: `katara_hair.png`), `WaterSkinLayer` when a skin is carried.
- **Effects.** Default Water (sea blue). Her healing glow is the default `#8FE3FF`.
- **Approval:** Katara in blue with the white fur trim, her bun and hair loops, and her mother's necklace. The default look for a new waterbender.

3.2 Amon / Noatak — אמון / נואטק

- **Reference.** *Korra*, Book 1: the Equalist leader: a **white-cream face mask** (`#EDE6D6`) with a **red circle on the forehead** (`#B22222`), narrow dark eye slits, painted brows and a thin red line for the mouth, under a **dark hood** (`#1F2228`); a **long dark coat** with a **high collar**, a **dark red lining** (`#4A1A1A`) showing at the cuffs and the hem, dark gloves, boots. **Noatak** unmasked (Book 1 finale): a man around forty with **dark brown hair swept back**, **blue eyes**, a strong jaw, a faint painted "scar" washed off: **(suggestion)** a second variant `noatak` without the mask and hood, same coat.
- **Skin.** Wide. Base: the coat with the collar and red lining, gloves, boots; the head: under the mask, Noatak's face (so the `noatak` variant is the same skin with the mask and hood layers off). Overlay: the collar.
- **Layers.** `MaskLayer` (`amon_mask.png`), `HoodLayer` (`amon_hood.png`), `RobeLayer long` (the coat's skirt).
- **Effects.** Default Water; **his blood-bending threads are darker** (`#5A0A14`) and his Severance pulse is **black** (`#0A0608`): a per-look `bloodColor`. His Water Shot is the default blue (he is still a waterbender).
- **Approval:** Amon with the white mask, red circle, dark hood and coat; and, if you want, Noatak's face without the mask as a second look.

3.3 Ming-Hua — מינג-הואה

- **Reference.** *Korra*, Book 3: the Red Lotus waterbender **born without arms**; a pale woman with **long black hair** falling loose with a **thin braid** at each side (**check:** a high ponytail in some scenes), **dark eyes**, **dark lips**, a sharp face; a **dark blue sleeveless tunic** (`#1F3A66`) with **grey wraps** around the torso, dark trousers, bare shoulders where her water arms root.
- **Skin.** Slim. Base: the tunic and wraps, trousers, boots; the face with the dark lips; **the arm regions are drawn transparent above the shoulder stumps** (the top 2 rows of each arm are skin: the stumps; the rest of each arm is **transparent**), so without Water Arms the model's arms are invisible but for the stumps: Ming-Hua has no arms. This is the one look that changes the body's silhouette; it is cosmetic (the hitbox is the player's).
- **Layers.** `HairLayer long` + braids (`minghua_hair.png`). With **Water Arms on**, the `WaterArmsLayer` draws her arms as water from the shoulders (the inflated arm parts fully textured in water, as the show): the look is made for the ability. Without Water Arms she **punches with invisible arms**: **(suggestion)** her look draws **thin water stubs** (0.4-block ribbons) from the stumps whenever a source is in reach, as a hint; decide.
- **Effects.** Default Water; her ice tips are the default.

- **Approval:** Ming-Hua with long black hair, dark blue tunic, and **no arms** (water arms when the ability is on). Say whether the arms should stay invisible when Water Arms is off, or show short water stubs.

3.4 Master Pakku — מאסטר פאקו (White Lotus)

- **Reference.** *ATLA*, Book 3 finale: the Northern Water Tribe master, old and stern: **grey hair in a high topknot**, a **grey moustache and a short grey goatee**, wrinkles, **sharp blue eyes**, medium Water Tribe skin. The **Order of the White Lotus** robes of the finale: **white** (`#F2F0E8`) long robes with **pale blue** (`#9CC4E8`) trim at the collar, cuffs and hem, a **white lotus tile emblem** (a white lotus on a blue disc) on the chest, a blue sash, white boots. (**Check:** the exact trim colour; the Order's soldiers wear white with a tall white hat; the elders' robes are white with blue.)
- **Skin.** Wide. Base: the white robes with blue trim and the emblem, the sash, boots; the face with the brows and the goatee's root; grey hair. Overlay: the collar.
- **Layers.** `TopknotLayer bun_high` (grey), `BeardLayer goatee` (grey), `RobeLayer long` (`white_lotus_robe.png`: **shared by Pakku, Iroh and Jeong Jeong** with a per-look tint of the trim: Pakku blue, Iroh red-gold, Jeong Jeong red).
- **Effects.** Default Water.
- **Approval:** old Master Pakku in the white and blue White Lotus robes with the lotus emblem, grey topknot and goatee.

3.5 The Water layers' art

- **Water Skin** (`water_skin.png`: a leather pouch, 3 fullness stages), the item icon (16×16, 9 stages for the bar).
- **Water textures:** vanilla's `block/water_still` and `block/water_flow` sprites (looked up from the block atlas, tinted by the biome's water colour), `water_foam.png` (a white foam ribbon), `ice_frost.png` (frost lines overlay), `wet.png` (a dark wet decal), `quicksand`… (Earth's).
- **Octopus / Arms:** no textures beyond the water sprites; the tentacle ribbons use `water_flow` with a highlight.
- **Blood:** the thread colour per look (`bloodColor`), the moon overlay (`moon_full.png`, 32×32, for the HUD and the vignette corner).

---

4. Fire — the looks and the Fire layers

4.1 Zuko — זוקו (default Fire look; "after he joined Team Avatar")

- **Reference.** *ATLA*, Book 3 (Western Air Temple onward): a sixteen-year-old with **shaggy black hair** grown out to the jaw, falling over his brow, the **burn scar over his left eye** (`#A0564A` and `#B86E5C`, from the brow to the ear, the eye narrowed), **gold eyes** (`#D8A62A`), pale Fire Nation skin. Outfit: the **dark red Fire Nation tunic** (`#7A1F1F`) with a **darker, almost black** (`#2A1A1A`) under-layer at the shoulders and a **dark brown belt**, **gold trim** (`#C9A24A`) at the collar, **red trousers**, dark boots, red arm wraps with gold edges.
- **Skin.** Wide. Base: the tunic and under-layer, belt, trousers, boots, wraps; the face with the scar (a 3×3 area of darker tones around the left eye, the eye 1 px narrower), gold eyes, the hair's fringe. Overlay: the hair's shag on the hat layer (the `HairLayer shaggy` adds the sides).
- **Layers.** `HairLayer shaggy` (`zuko_hair.png`).
- **Effects.** Default Fire (orange/gold).

- **Approval:** Zuko with the shaggy hair and his scar, in the dark red and gold tunic of Book 3. The default look for a new firebender.

4.2 Uncle Iroh — הדוד אירו (White Lotus)

- **Reference.** *ATLA*: the Dragon of the West, heavyset and jolly: **grey hair in a topknot**, a **full grey beard** and moustache, **warm gold eyes** with laugh lines, pale skin. The **White Lotus robes** of the finale (white with the Order's trim; Iroh's **(check)** carries a red-gold trim as the Grand Lotus), the lotus emblem, wide sleeves. (His Book 2 green Earth Kingdom traveller's robe is a **(suggestion)** second variant `iroh_traveller`; his tea.)
- **Skin.** Wide. Base: the white robes with the trim and emblem, a wide sash, boots; the round face with the beard's root and the brows; grey hair. Overlay: the wide sleeves.
- **Layers.** `TopknotLayer bun_high` (grey), `BeardLayer full` (grey, wide: `iroh_beard.png`), `RobeLayer long` (`white_lotus_robe.png`, red-gold trim).
- **Effects.** Default Fire; **his Fire Breath is the default** but **(suggestion)** his look makes the breath's jet wider by the "dragon" texture: cosmetic no; keep default.
- **Approval:** Uncle Iroh in the white White Lotus robes with the lotus, his grey topknot and full beard.

4.3 Princess Azula — הנסיכה אזולה (blue fire)

- **Reference.** *ATLA*: **black hair in a high topknot** with the **gold flame crown** (three prongs) and **two long bangs** framing her face to the chin, **gold eyes** sharp and lined, **red lipstick**, pale skin, a cruel smile. The **Fire Nation royal armour**: **dark red/maroon** (`#6E1B1B`) with **black** panels, **gold trim**, **pointed gold shoulder pads**, armoured cuffs, a gold belt plate with the Fire Nation flame, dark trousers and boots.
- **Skin.** Slim. Base: the armour's panels, gold trim, the belt plate, the cuffs, boots; the face with the bangs (two 1-px columns beside the face on the hat layer), the eyes and lips; black hair. Overlay: the armour's relief.
- **Layers.** `TopknotLayer bun_high` (black), `CrownLayer flame` (gold, three prongs), `ShoulderLayer pointed` (`azula_shoulders.png`), `RobeLayer short` (the armour's skirt plates).
- **Effects. Every fire effect is blue**: the FLAME `BeamStyle` becomes `BLUE_FLAME` (core `#F4FDFF`, inner `#CFF2FF`, main `#6FC4FF`, body `#3FA4FF`, fringe `#1B4FCC`), every `FLAME` particle becomes `SOUL_FIRE_FLAME`, fireballs' cores `#E8FBFF` / rims `#3FA4FF`, scorch discs stay dark, embers `#9FD8FF`; sounds get a hissing top layer (pitch 1.15) for the jets. Lightning stays white-blue. The look's `color` is `#3FA4FF`, `glow` `#CFF2FF`.
- **Approval:** Azula in the dark red royal armour with pointed gold shoulders, the flame crown, two bangs and red lips; **all her fire is blue.**

4.4 Jeong Jeong — ג'ונג ג'ונג (White Lotus)

- **Reference.** *ATLA*: the Deserter: an old, gaunt master with **long white hair** swept back and falling to the shoulders, a **long white moustache** and a **wispy white beard**, a **scar across his right eye** (a vertical line from brow to cheek), **gold eyes**, pale weathered skin. The **White Lotus robes** (white with red trim as a firebender of the Order) over his old dark red rags **(check)**.
- **Skin.** Wide. Base: the white robes with red trim and emblem; the face with the scar line, the brows; white hair. Overlay: the robes.
- **Layers.** `HairLayer long_back` (white, `jeongjeong_hair.png`), `BeardLayer moustache_long` + a wispy beard slab, `RobeLayer long` (`white_lotus_robe.png`, red trim).
- **Effects.** Default Fire.

- **Approval:** Jeong Jeong: long white hair and moustache, the scar over his eye, white White Lotus robes with red trim.

4.5 Fire Lord Ozai — אדון האש אוזאי

- **Reference.** *ATLA*: the Fire Lord: **long black hair** in a **high topknot** with the **five-pronged golden flame crown** of the Fire Lord, a **pointed black goatee** and a thin moustache, **cold gold eyes**, pale skin, sharp cheekbones. The **royal robes**: **deep red** (`#7A1515`) with **black** and **gold** (`#D4AF37`) armour at the shoulders (large pointed pauldrons), a gold-trimmed black mantle, a wide gold belt with the flame, long red robes, black boots. (Bare-chested Phoenix King of the finale: a **(suggestion)** second variant `phoenix_king`.)
- **Skin.** Wide. Base: the robes, mantle, belt, boots; the face with the goatee's root and the moustache, the brows; black hair at the back. Overlay: the mantle.
- **Layers.** `TopknotLayer bun_high` (black, tall), `CrownLayer flame_five` (gold, five prongs: 7 px wide), `ShoulderLayer pointed` (`ozai_shoulders.png`, larger), `BeardLayer goatee` (black), `RobeLayer long` (`ozai_robe.png`).
- **Effects.** Default Fire; **(suggestion)** Ozai's Fire Jet and great fireball are **10% bigger** cosmetically (no rule change: the mesh scale only). His lightning is the default.
- **Approval:** Fire Lord Ozai: long black topknot with the five-pronged crown, pointed goatee, deep red and gold royal robes with big shoulder armour.

4.6 The Fire layers' art

- `white_lotus_robe.png` (shared, trim tinted), the flame crowns (`crown_flame_3.png`, `crown_flame_5.png`), the shoulder pads, the beards and topknots as above; `fire_mouth.png` (the breathing glow mask); `heat.png` (a heat shimmer noise texture, 64×64, tileable); `scorch.png` (a dark burn decal).

---

5. Items drawn on the body

Item	Where	Texture	Notes
Air Glider (stowed)	across the back	`glider_base.png` tinted	only for airbenders; hidden when a cape/robe layer would clip (the robe wins)
Water Skin	right hip	`water_skin.png` (3 stages)	waterbenders with a skin in the inventory; fullness from the fullest skin
Element Scroll (held)	hand	a vanilla-style item model `element_scroll.json` (a rolled scroll with the four symbols)	

6. Icons (`UI.md` 6 has the full list)

Every look has a **face icon** for the look picker: the skin's face region (8×8) blown up, as Enhanced's `PlayerFaceRenderer.draw` does from the skin texture; no separate art.

7. Checklist for the art lane (per look)

1. Draw the 64×64 skin (wide or slim as listed) from the reference text; check it with `SkinCheck` (64×64, PNG, valid).

2. Draw each layer texture and build its `LayerDefinition` (head space; test with `/bending_model look <id>` in a client).
3. Register the `Variant` (`id`, skin, `slim`, `color`, `glow`, and the look's extras: `metalColor`, `bloodColor`, `fireStyle`) in the element's `Looks` list; the lang keys `look.bending.<element>.<id>` in both languages.
4. Screenshot in a real client: front, back, side, first-person hands, with the Earth/Steel armour over it, with the octopus, with the glider; English and Hebrew HUD.
5. The `LooksTests` pin: every look has a skin PNG of 64×64, every layer texture exists, every look has both names, the face icon renders.

UI.md — Screens and HUD

Bending — Screens and HUD

Everything here is built with Enhanced's UI kit, copied into `bending.client.screen`: `CreatorUi.BigButton` (the card widget: icon or face, label, sub-line, colour, selected/lit states), `EditorTab.cards(...)` (the paged grid), `Popup`, `PreviewPuppet` (a `RemotePlayer` drawn in the GUI with `EntityRenderDispatcher`, animated by the animator's preview API), `CatalogSearch` (two-language search with Hebrew folding), `HudDraw.visual` (forced right-to-left layout with ICU `Bidi`), the generated config screens (`ConfigScreen`, `ConfigSectionScreen`, `ConfigListScreen`), and the `LayeredDraw.Layer` HUD pattern registered through `ClientRegistrar.guiLayer`. None of the screens pause the game.

Hebrew. Every screen has a right-to-left layout when `Language.getInstance().isDefaultRightToLeft()`: columns are mirrored (the preview moves to the right, lists to the left, text right-aligned), lines are laid out with `HudDraw.visual` so a line that starts with a Latin key name (`[X]`) is still right-to-left, and units live inside the translation (`hud.bending.seconds` = `%s שׁ`), as Enhanced learned. Numbers stay Latin digits. Key names from `getTranslatedKeyMessage()` are left as the game gives them.

---

1. Keys and how the screens open

Key	Default	Action
`key.bending.slot_1..4`	**X, C, Left Alt, K**	the loadout slots; Sneak + key is the variant (the client sends the Sneak flag with the press, Enhanced's `PRESS_SNEAK` capture)
`key.bending.menu`	**O**	opens the **Bending screen** (tabs); Sneak + O cycles the look at once
(none)	right-click	the Element Scroll (choice screen), the Water Skin (refill), the Air Glider (open/close)
`key.bending.menu` while a slot is held?	—	no chords beyond Sneak

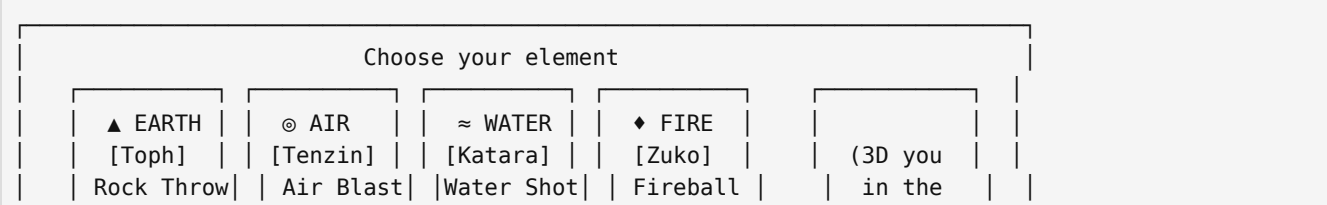
Category `key.categories.bending` = "Bending" / "כשפות". Context `IN_GAME` for all (so a clash with Enhanced's keys shows in Controls: there is none by default).

---

2. Screens

2.1 Element Choice — בחירת אלמנט (`ElementChoiceScreen`)

Opened by the Element Scroll (server → `ElementScreenPayload`), or `/bending choose`. Four cards, a preview, a confirm.



```
|   | Earth Wall| |Air Shield| | Healing  | |Fire Whip |    |  element's|  |
|   | Earth Arm.| |Air Scoot.| |Ice Knives| | Lightning|    |  default  |  |
|   |  Looks: 6 | |  Looks: 6| |  Looks: 4| |  Looks: 5|    |   look)   |  |
|   └───────────┘ └──────────┘ └──────────┘ └──────────┘    └───────────┘  |
|        Strong, rooted. Earth Kingdom.  Water is strong against fire.      |
|                  [ Become an Earthbender ]        [ Not now ]             |
└──────────────────────────────────────────────────────────────────────────┘
```

- Cards are `BigButton`s 92 × 110 (clamped to the width: `cardW = clamp((width−24−3·GAP)/4 … )`, the `SerumScreen` recipe for three cards, widened to four), each with the element's **symbol icon** (6.1), its name bold in the element colour, the default look's **face** (`PlayerFaceRenderer`), three starter abilities in grey, "Looks: n"; a selected card gets the gold double outline; hovering plays a short puff of the element's particles behind the card (`CreatorUi` has no particles: a 6-mote `GuiGraphics` sprite flurry, cheap).
- The **preview** (`PreviewPuppet`, 96 px tall) shows **you** in the selected element's default look, playing that element's idle stance (`previewHold` of `EARTH_IDLE` etc.) and, every 4 s, its starter gesture (`previewCast`).
- The **description line** (2 lines max, `font.split`) and the **matchup hint**.
- **Become** sends `ElementChoicePayload(element)`; **Not now** closes (the scroll is kept). Keys 1–4 select, Enter confirms (as `SerumScreen`).
- **Hebrew (RTL):** the cards run right to left (Earth at the right), the preview sits at the **left**, text right-aligned:

```
┌──────────────────────────────────────────────────────────────────────────┐
|                          בחרו את האלמנט שלכם                             |
|  ┌──────────┐    ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐     |
|  |          |    |  אדמה    | | ◎ אוויר  | | ≈ מים    | | ▲ אש ♦   |     |
|  | ([טוף])  |    | [טנזין]  | | [קטארה]  | | [זוקו]   | |  תצוגה   |     |
|  | זריקת סלע | | הדף אוויר| | ירית מים| | כדור אש  | | (תלת-ממד)|     |
|  |          |    | קיר אדמה | | מגן אוויר| | ריפוי    | | שוט אש   |     |
|  |          |    | שריון אדמה| |קטנוע אווי| |סכיני קרח | |ברק       |     |
|  |          |    | מראות: 6 | | מראות: 6 | | מראות: 4 | | מראות: 5 |     |
|  └──────────┘    └──────────┘ └──────────┘ └──────────┘ └──────────┘     |
|                חזקה ומושרשת. ממלכת האדמה.  מים חזקים נגד אש.             |
|              [ לא עכשיו ]        [ להפוך לכשף אדמה ]                    |
└──────────────────────────────────────────────────────────────────────────┘
```

2.2 The Bending screen — מסך הכשפות (`BendingScreen`, key O)

One screen, four tabs (Enhanced's `HeroEditorScreen` layout: tabs at y 16, a left preview column `left() = clamp(width/3, 92, 160)`, the content page to the right of it, a status line in place of the title for 6 s after an action). The **header** shows the element symbol, "Earthbender", **SP** and **ES**: `✦ 12 SP ◆ ES 23`.

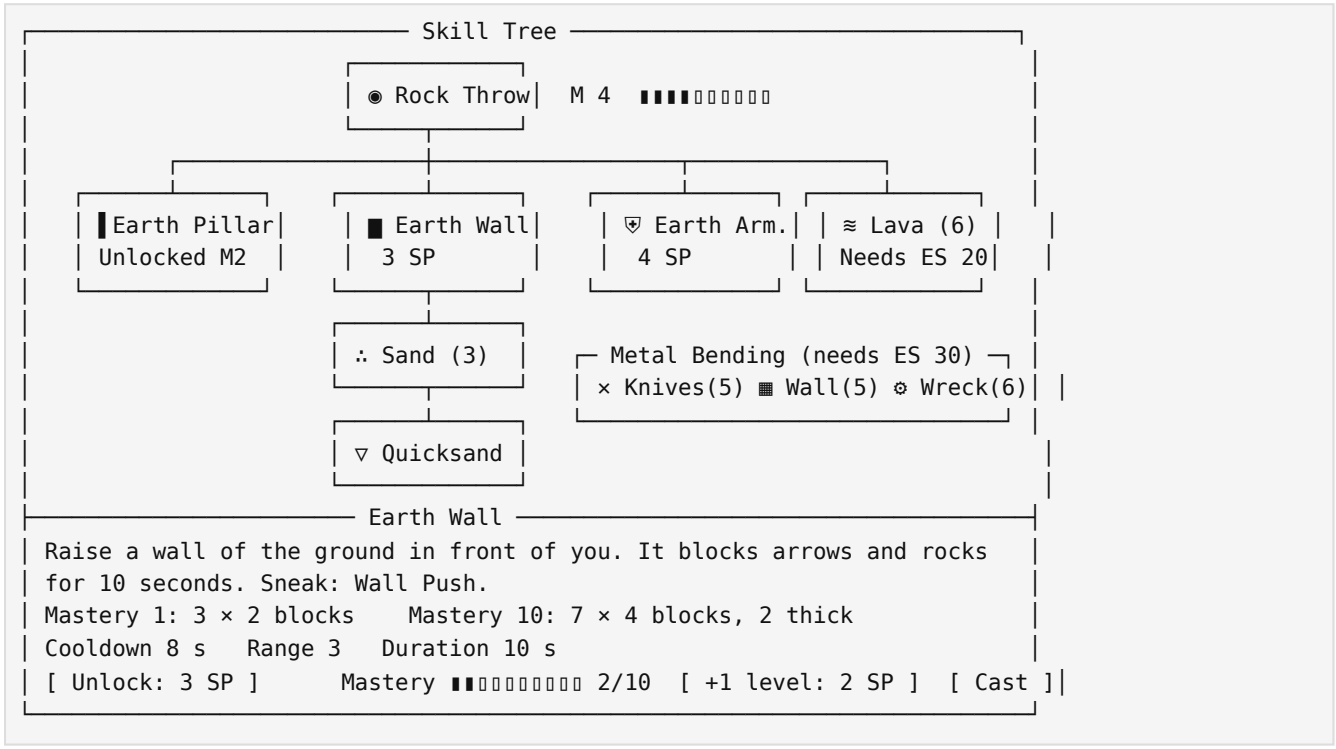
```
┌──────────────────────────────────────────────────────────────────────────┐
| ▲ Earthbender              ✦ 12 skill points   ◆ strength 23             |
| [ Skill Tree ] [ Loadout ] [ Looks ] [ Water ]            (tab buttons)  |
| ┌──────────┐ ┌──────────────────────────────────────────────────────┐     |
| |          | |                                                      |     |
| |  3D you  | |                  (tab content)                       |     |
| |  (turn by| |                                                      |     |
| |  dragging)| |                                                     |     |
| |          | |                                                      |     |
| | Stand Walk| |                                                     |     |
| | Run  Cast| |                                                      |     |
| └──────────┘ └──────────────────────────────────────────────────────┘     |
```



The preview plays the element's idle; **Cast** plays the selected ability's gesture (`previewCast`) so Eilon sees the move before unlocking it. The **Water** tab appears for waterbenders only (2.6).

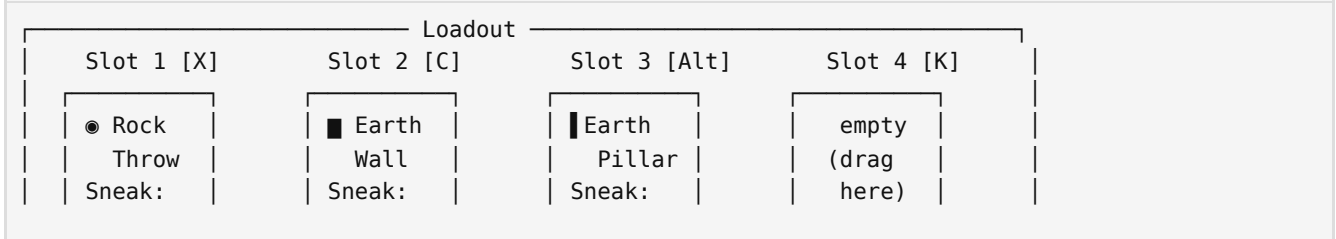
2.2a Skill Tree tab — עץ הכישורים

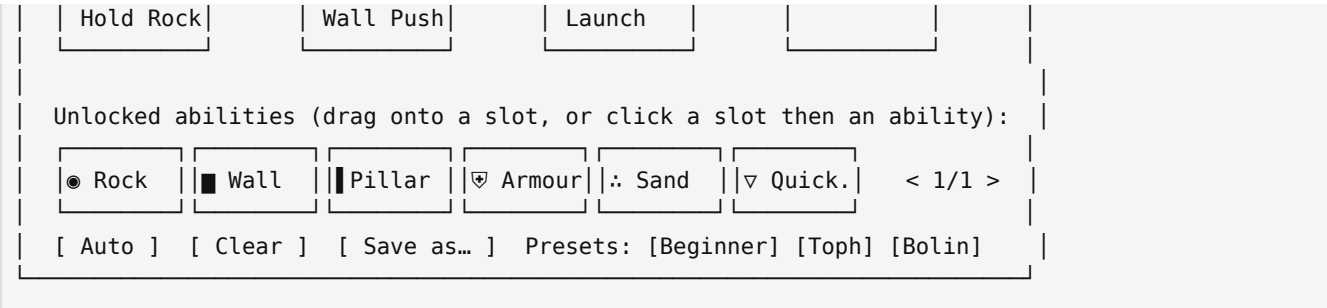
A **graph of cards** drawn on a scrollable canvas (drag to pan, wheel to zoom 0.75–1.25): the root at the top, tiers below, **connectors** as 2-px lines in the element colour (lit when the child is unlockable, grey when locked, gold when unlocked). Each node is a `BigButton` 76 × 40 with the ability's **icon** (6.2), name, and a **sub-line**: `Unlocked · M 4` or `3 SP` or `Needs Rock Throw M2`. Selecting a node fills the **detail panel** at the right (or below on narrow screens):



- **Unlock** is enabled when the parent is unlocked at the needed mastery and you have the SP (`BendingPayload.UNLOCK`). **+1 level** buys mastery up to 7 (`BUY_MASTERY`). **Cast** previews the gesture on the puppet. The mastery bar is 10 cells (`▮▯` as Enhanced's `hexBarrier()` bar; green/yellow/grey by use). A **thin XP bar** under the cells shows the progress to the next level from use.
- Zaheer's two cards appear with a **red thread** icon when the path opens; before that a greyed card with "?" and the tooltip "A loss opens this path" (no spoiler of the pet rule unless `showZaheerHint` is on: default **off**).
- **Hebrew:** the tree is mirrored left-right, text right-aligned; connectors unchanged.

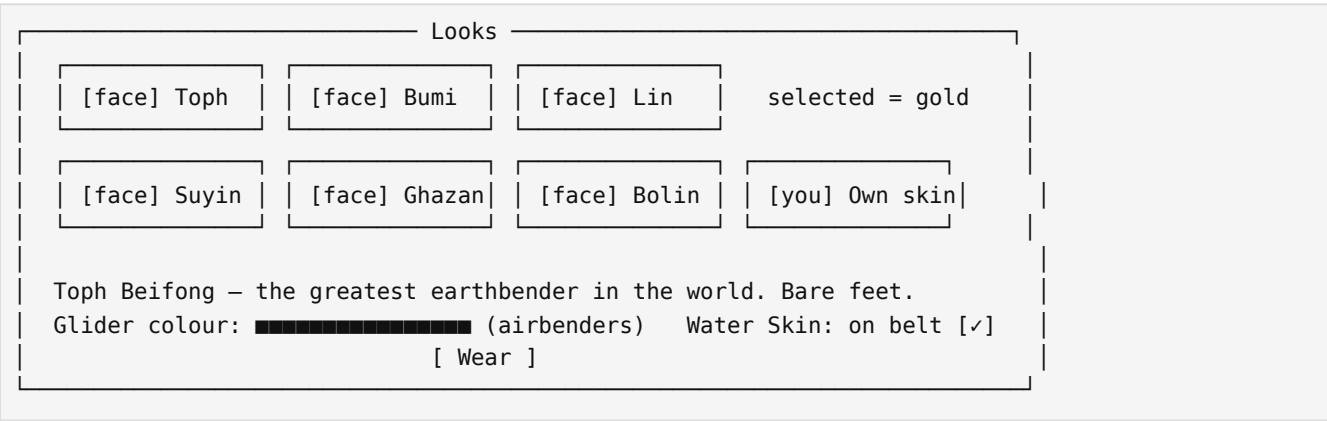
2.2b Loadout tab — סידור הקרב





- Four big `BigButton`s (the `KeysTab` cells, `stacked()`), each showing the ability's icon and name and its Sneak variant's small icon and name; a **cooldown shade** if it is cooling right now. The grid of unlocked abilities below is `EditorTab.cards("loadout", …, 62, 18)` with paging. **Drag and drop** (`mouseDragged` moving a ghost card) or click-click. **Auto** fills empty slots in tree order; **Clear** empties; **Save as…** stores up to 4 named presets per element (saved on the player, `TECH.md` 4); the element's built-in presets (`DESIGN.md` loadout examples) are one click. Changes send `BendingPayload.LOADOUT(slots)`.
- **Hebrew:** slots run right to left (slot 1 at the right); the key names stay as given.

2.2c Looks tab — מראות



- Face cards (`BigButton.face(skin)`, 20 × 20 faces at `flow`), the selected look drives the preview at once (the puppet's `PowerData`-equivalent `BenderData.look` is set locally). **Wear** sends `BendingPayload.LOOK(id)` (also Sneak + O cycles). A **draft** badge on Jae and Karthik until their pictures arrive. Airbenders see the **glider dye swatches** (16 `CreatorUi.Swatch`es; changing re-dyes the held glider: a crafting shortcut, `GLIDER_DYE` payload) and a **Jae board** toggle when Jae is worn (**suggestion**). Waterbenders: "Water Skin on belt" toggle (draw it or not).
- **Hebrew:** mirrored grid; the description right-aligned.

2.2d Water tab — מים (waterbenders)

Shows the **water sources** in reach right now (a list: "River, 4 blocks", "Water Skin 6/8", "Plants (needs M7)"), the Skin's charges as a bar, and two options: `preferSkin` (use the Skin before open water: default off) and `showSourceLines` (draw a thin line to the current source in the world: default on).

2.3 Settings — הגדרות

Enhanced's generated config screens as they are, reached from the mod list's Config button (`ClientRegistrar.configScreen`) and from a **Settings** button on the Bending screen's footer. Two files: `bending-client.toml` (sections `[hud]`, `[effects]`, `[animation]`, `[sound]`) and `bending-server.toml` (every number: `TECH.md` 5), the latter editable only in single player / LAN host (Enhanced's `not_online` tooltip).

Client options (the only ones not in the server file):

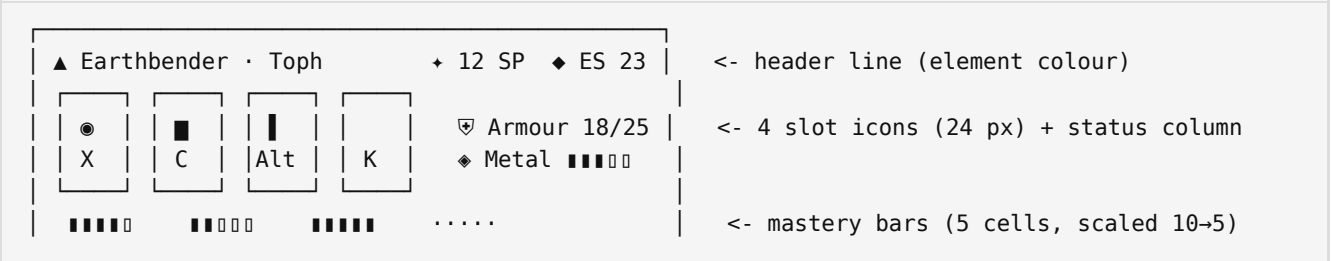
Section	Option	Default	Meaning
`[hud]`	`position`	`BOTTOM_LEFT`	`BOTTOM_LEFT`, `BOTTOM_RIGHT`, `TOP_LEFT`, `TOP_RIGHT`
	`scale`	1.0	0.75–1.5
	`showMasteryBars`	true	
	`showElementStrength`	true	
	`moonIndicator`	true	the full-moon icon for waterbenders
	`sourceLines`	true	the thin line to the water source
`[effects]`	`density`	1.0	0.25–1.0 multiplier on every particle count
	`screenEffects`	true	overlays (slit, vignettes, heat)
	`cameraShake`	true	(also off with reducedMotion)
	`fakeLight`	false	(suggestion) the warm disc under flames
`[animation]`	`animations`	true	the mod's poses (off: vanilla)
	`reducedMotion`	false	calm bob, no tremble, no shake, no camera yaw
	`firstPersonHands`	true	the hand motions
`[sound]`	`music`	true	the element themes
	`loopVolume`	1.0	the hold loops' master

---

3. The HUD (`BendingHud`, a `LayeredDraw.Layer` registered `BELOW DEBUG_OVERLAY`, like Enhanced's `PowerHud`; the

overlays of `ANIMATION.md` 2.2 are separate layers `ABOVE CAMERA_OVERLAYS`)

Placement. Default **bottom-left**, so it never collides with Enhanced's top-left text panel when both mods run: `x = 4, y = height − 4 − panelH`, above the vanilla left status bars when they would overlap (`HudHeights.left()` tells how tall they are: the panel sits above `max(hotbar+bars, 0)` only when the position is bottom-centre; bottom-left is clear of them since the bars start at the hotbar's left edge, `width/2 − 91`: on screens narrower than 2·(4+panelW+8)+182 the panel moves up by `HudHeights.left()`). Hidden when `hideGui`, the debug screen is open, the player is dead, or has no element. The panel's width eases (Enhanced's 20-tick narrowing rule).



```
 │  ◇ Hold   ◇ Push   ◇ Launch  ·····            │   <- Sneak variant mini-icon + name (hover or
Sneak held)
 └──────────────────────────────────────────────┘
```

- **Slot icon** (24 × 24): the ability icon (6.2); a **cooldown sweep**: a dark radial wedge (`GuiGraphics` triangles via a 24-segment fan, as `StoneWheelScreen.ring` draws fills) covering the remaining fraction clockwise, with the seconds (`hud.bending.seconds`, `%.1f`) in the corner; **ready flash**: a white 3-tick outline when it comes off cooldown (+ a soft `hud_ready` click, optional); **held**: an aqua outline while a hold ability runs; **on**: a green dot for toggles; **dimmed** when it cannot work now (no earth/water/metal in reach: the client runs the same reach checks as prediction) with a tiny reason glyph (✗ earth / ✗ water / ✗ metal / ✗ moon); the **key letter** under it in yellow (`getTranslatedKeyMessage()`). When **Sneak is held**, the icons **swap** to the Sneak variants (the small diamond row becomes big and vice versa) so a kid always sees what the key will do.
- **Header**: the element symbol and "Earthbender · Toph" (element name key `element.bending.<id>.bender`, the look's name after a middle dot), **SP** (`✦ 12 SP`, white; flashes gold and counts up when earned: a `+2` rises and fades for 20 ticks), **ES** (`◆ ES 23`, grey; hidden if `showElementStrength` is off).
- **Status column** (right of the icons, one line per active state, Enhanced's `hudStatus` style): the **Earth Armour / Steel Armour shell** (`⛨ 18/25` with a 5-cell bar, green/yellow/red), **Metal charge** (`◈ ▮▮▮▯▯ 12/30`), **Air Shield** strength (`◎ 9/13`), **Scooter / Spout time** (`⟲ 42 s`), **Spiritual Projection** (`☽ 48 s · body 64 m`), **Water supply** (3.3), **Blood** (`✶ 4.2 s` hold left), **Lightning** (`⚡ charging…` / `⚡ redirect ready`), **Zaheer's path** (a small broken-thread glyph in the header), **Severed** (`⊘ bending gone 1:52`, red), **Paralysed** (`⚡ 1.6 s`).
- **Mastery bars**: 5 cells per slot (level/2 rounded), tooltip-less; `showMasteryBars`.
- **Hebrew:** the panel's lines go through `HudDraw.visual`; the header reads `✦ 12 23 חוזק ◆ נק׳ כישור ▲ · טוף · כשף אדמה` right-aligned; the icons stay in slot order **right to left** when the position is a right corner, left to right in a left corner (icons are not text; the key letters under them say which is which).

3.1 Earth: nothing more. 3.2 Air: the glider's state (`⌇ glider open`) when open.

3.3 Water supply — אספקת מים (waterbenders only)

```
┌──────────────────────────────────────────────┐
│ ≈ Waterbender · Katara       ✦ 8 SP  ◆ ES 14 │
│ ┌────┐ ┌────┐ ┌────┐ ┌────┐   ~ River 4 m     │   <- the current source
│ │ ≈  │ │ ✧  │ │ ❄  │ │ ⌁  │   ◍ Skin ▮▮▮▮▮▮▯▯ │   <- the water skin's charges
│ │ X  │ │ C  │ │Alt │ │ K  │   ☾ full moon      │    <- 3.6
│ └────┘ └────┘ └────┘ └────┘                    │
└──────────────────────────────────────────────┘
```

- **Source line**: `~ River 4 m` / `~ Rain` / `◍ Water Skin` / `❀ Plants` / `✗ No water` (red). The `sourceLines` option draws a thin blue line in the world from the hand to the source block (an `ENERGY_VEIL` ribbon α 0.2).
- **Skin bar**: 8 cells (`▮▯`, blue), the fullest skin; empty: red; refilling in rain: a drip glyph.

3.4 Spiritual projection mode — מצב השלכה רוחנית

While the spirit is out, the HUD **changes**: the slot row is replaced by `☽ Spirit · 48 s left · body 64 m → ↑` (the time, the distance and an arrow toward the body; range ring: a thin arc fills as you near the limit), the overlay vignette (`ANIMATION.md` 2.2), and a hint `[X] return` (the slot key of the ability returns you). Damage to the body shows a red flash on the panel.

3.5 Severed — נשללו הכשפות

Every icon greys (`0x60000000` fill over them), the header turns grey, the status line `⊘ Your bending is gone · 1:52` counts down; nothing else changes.

3.6 The full-moon indicator — מחוון הירח המלא (waterbenders)

A **moon glyph** in the status column: `☾` grey on ordinary nights, `●` **white** on a full-moon night (phase 0, night) with "full moon" / "ירח מלא" and a faint glow; by day the phase of the coming night (`☽ tonight: full` from 10000 ticks on). For a blood bender below M10 the Blood slot icon is dimmed with the ✗ moon glyph until the full moon. The icon is `moon_full.png` (16 × 16).

3.7 Other players

Above another bender's head, no HUD. Their looks, stances and effects say everything (the design's rule, as Enhanced).

---

4. Action-bar messages (the `message.bending.*` keys, both languages)

`no_earth` "No earth here" / "אין פה אדמה"; `no_sand`, `no_metal` "No metal" / "אין מתכת"; `no_water` "No water" / "אין מים"; `protected` "Protected ground" / "קרקע מוגנת"; `stronger_earthbender` "A stronger earthbender" / "כשף אדמה חזק יותר"; `stronger_waterbender`; `no_moon` "Only under a full moon" / "רק בירח מלא"; `in_water` "Not in water" / "לא בתוך מים"; `empty_slot` "Empty slot: press O" / "משבצת ריקה: לחצו O"; `severed` "Your bending is gone" / "הכשפות שלכם נשללו"; `mastery_up` "%1*s* : *mastery*s" / "%1*s* : *רמתשליטה*s"; `points` "+%s skill points" / "+%s נקודות כישור"; `unlocked` "%s unlocked!" / "%s נפתחה!"; `zaheer_path` "You have cut your earthly tethers" / "ניתקתם את הקשרים הארציים שלכם"; `immune` "Immune" / "חסין".

---

5. Hebrew copy rules (for every lane)

- Natural Hebrew, never transliterated English words where a Hebrew word exists; the dub's terms: כשף/כשפת, כשפות אדמה / אוויר / מים / אש, כשפות דם, כשפות ברק, כשפות מתכת, כשפות לבה, כשפות קרח, כשפות חול (the last three are the mod's, in the dub's pattern).
- Second person plural for instructions ("לחצו", "בחרו"), as Enhanced's wiki; ability names are nouns (זריקת סלע).
- Sneak = התכופפות, hold = החזקה, on/off = פועל/כבוי, seconds = `%s` ש׳, mastery = רמת שליטה, skill points = נקודות כישור, element strength = חוזק האלמנט, loadout = סידור הקרב, skill tree = עץ הכישורים, look = מראה.
- Character names in Hebrew as Eilon spells them (קארתיק, ג'יי) and the common Hebrew spellings for the rest (`LOOKS.md` headings).

- `DataTests` rules as Enhanced: identical key sets, identical `%` placeholders, Hebrew letters present, no double spaces.

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6. Icons (16 × 16 for the HUD at 24 px it is drawn scaled; 32 × 32 masters, `textures/gui/icons/`)

6.1 Elements and states

Icon	Name	Drawing
`earth`	▲	a stepped mountain in ochre with a moss-green base
`air`	◎	three nested wind swirls in white on sky blue
`water`	≈	two waves, sea blue with a white crest
`fire`	♦	a three-tongued flame, orange with a gold core (Azula: a blue variant `fire_blue`)
`sp`	✦	a four-point gold star
`es`	◆	a grey diamond
`moon_full`, `moon`	● / ☾	a white disc with craters; a grey crescent
`thread_cut`	⌇	a red thread snapped
`severed`	⊘	a grey circle with a bar
`paralysed`	⚡	a white-blue bolt
`no_source` glyphs	✗	a small red cross with the source's glyph

6.2 Abilities (one per card; the Sneak variant reuses the icon with a small ◇ badge)

Earth	Air	Water	Fire
`rock_throw` a flying rock with a dust tail	`air_blast` a palm with wind lines	`water_shot` a water stream	`fireball` a fireball
`earth_armour` a stone chest plate with an eye slit	`air_shield` a wind sphere	`healing` glowing hands over a heart	`fire_punch` a burning fist
`earth_pillar` a column rising	`air_glider` the open glider (item)	`water_octopus` a figure with 8 arms	`flamethrower` two jets
`earth_wall` a wall of three blocks	`air_blades` a crescent	`water_spout` a vortex column	`fire_whip` a curling flame rope
`sand_bending` a sand swirl	`spiritual_projection` a seated figure with a blue spirit	`water_arms` two long water arms	`fire_circle` a ring of flame
`quicksand` a sinking figure	`air_bubble` a head in a bubble	`ice_knives` three shards	`fire_healing` gold hands over a heart
`lava_bending` a lava path with a shuriken	`air_scooter` a figure on a ball	`freeze` a frozen footprint	`lightning` a forked bolt
`steel_knives` three knives	`tornado` a funnel	`ice_wall` an ice wall	`lightning_redirection` a bolt bent in a U
`steel_wall` riveted plates	`true_flight` a floating figure (red thread)	`blood_bending` a red claw hand with threads	

Earth	Air	Water	Fire
`wreck_metal` a bent hopper	`suffocation` a hand over a gasping mouth (red thread)	`bending_severance` a thumb on a brow	

Sneak variants with their own art where the badge is not enough: `hold_rock`, `shed_blast`, `earth_launch`, `wall_push`, `sand_wall`, `sand_cloud`, `lava_shuriken`, `metal_pull`, `steel_armour`, `metal_grip`, `air_jump`, `shield_burst`, `air_slice`, `deep_meditation`, `air_cushion`, `air_spout`, `air_wheel`, `hover`, `air_vacuum`, `water_whip`, `cleanse`, `octopus_lash`, `wave_ride`, `arm_grab`, `ice_spikes`, `snow_throw`, `ice_dome`, `blood_fling`, `small_fireball`, `fire_kick`, `fire_breath`, `whip_lash`, `fire_jet`, `burn_out`, `lightning_arc`, `ground_it`: **38** small icons.

6.3 Items and blocks

`element_scroll`, `water_skin` (9 fill stages), `air_glider` (16 dyes, one base tinted), `bending_scroll_<element>` (suggestion), `training_dummy` (suggestion), `quicksand`, `steel_plate`, `ice_spike`, `wrecked_*` (8), `temp_lava` (vanilla lava), the advancement icons (one per element: the symbol in a frame).

6.4 Where icons are drawn

Icons are `GuiGraphics.blit` sprites from one **atlas** `textures/gui/bending_icons.png` (256 × 256: 16 × 16 cells, addressed by an `Icon` enum with u, v), so the HUD costs one texture bind.

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7. Tests for the UI (server-side, as Enhanced's client tests: no client class in a signature)

`HudTests`: the panel's lines for a given `BenderData` (keys present, cooldown formatting, the Hebrew seconds line keeps the unit next to the number through `java.text.Bidi`), the source line for each source, the moon line for each phase and time; `ScreenDataTests`: every ability has an icon cell, a name and a description in both languages, every look a face; `LoadoutTests`: slot payload round trip, presets, auto-fill order; `TreeTests`: unlock rules (parent, mastery, SP, ES gates), buying mastery caps at 7. What they look like: a real client, English and Hebrew, three render rounds per lane, screenshots per screen (`TECH.md` 8).

AUDIO.md — Sounds and music

Bending — Audio

Every sound and every piece of music is **synthesised for the mod**: no audio from the shows, no copyrighted music, no samples. The pipeline, format rules and conventions are Enhanced's (section 1); the events are listed per element with a synthesis recipe each and both subtitles (sections 2–6); the music is section 7.

1. Pipeline and rules

1.1 The toolkit (committed this time)

Enhanced's sounds were made with a numpy toolkit (`sfx.py`: oscillators, noise, filters, modal synthesis, a synthetic reverb) and per-release recipe scripts (`thanos_sounds.py`…), **none of which are in the Enhanced repository**. Bending **commits its toolkit** under `tools/sound/`:

- `tools/sound/sfx.py`: the primitives: `sine`, `saw`, `square`, `tri`, `noise(white|pink|brown)`, `glide(f0, f1, t)`, `env(a, d, s, r)` and `envexp`, `lowpass/highpass/bandpass(cutoff, q)` (biquads), `modal(freqs, decays, amps)` (struck objects: stone, metal, glass), `karplus(freq, decay)` (plucked strings), `formant(vowel)` (breath and voice-like tones), `chorus`, `reverb(size, damp, wet)` (a Schroeder/FDN synthetic hall), `limiter(−1 dBFS)`, `loop_seam(x)` (a crossfade of the last 100 ms into the first so a loop is seamless), `write_ogg(path, x)` (mono, 44.1 kHz, Vorbis q 5 through `soundfile`/`pyogg`), and `analyze(path)` (peak, LUFS-ish loudness, energy above 12 kHz, length).
- `tools/sound/recipes/<element>.py`: one function per event, named as the event, returning the signal; the tables below are these functions' specifications.
- `tools/sound/render.py`: renders every recipe into `src/main/resources/assets/bending/sounds/sfx/<event>.ogg` (music to `sounds/music/`), prints the analysis, and fails on a peak over −1 dBFS, a length under 0.5 s, a loop under 2 s or an ambience under 15 s, a non-mono or non-44.1 kHz file.
- `tools/sound/placeholder.ogg`: the 0.2 s placeholder S0 ships for every event until the sound lane renders it (its md5 is pinned by `SoundAssetTests.PLACEHOLDER_MD5`, as Enhanced's).

1.2 Format and registration (Enhanced's rules, enforced by `SoundAssetTests` and `DataTests`)

- **Ogg Vorbis, mono, 44,100 Hz**, longer than 0.5 s; loops ≥ 2 s and seamless; ambience ≥ 15 s; peak ≤ −1 dBFS.
- Registered in `registry/ModSounds` (`Registrars.create(Registries.SOUND_EVENT, "bending")`, `register(name)` → `SoundEvent.createVariableRangeEvent`), one region per lane (`// ===== earth (1.0) =====`), appended only.
- `sounds.json`: `"<event>": { "subtitle": "subtitles.bending.<event>", "sounds": [ { "name": "bending:sfx/<event>" } ] }`, `"stream": true` for music and loops over 10 s; one block per lane inserted before the next lane's block.
- Subtitles: `subtitles.bending.<event>` in `en_us.json` and `he_il.json`, Hebrew with Hebrew letters, no transliteration.

- **Takes.** Short one-shots that repeat (hits, stomps, knives) get **2–3 takes** (`sounds: [a, b, c]`, random) so they do not machine-gun; the recipe renders the takes with a different seed and ±4% pitch.
- **Loops** play through the copied `PowerLoopSounds` (`Kind(sound, volume, pitch, predicate on BenderData)`), fading in 0.2 and out 0.15 per tick, following the player's eyes; entity-bound loops (tornado, spout, fireball) through a per-entity `AbstractTickableSoundInstance` (Enhanced's `Loop` generalised to `EntitySoundLoop`).
- **Mastering**: effects normalised to about −16 LUFS-ish by `analyze`; impacts a little louder (−14), loops quieter (−20), ambience (−24); music stems 6–8 dB under the effects (Enhanced's rule).
- **Distance**: default attenuation (16 blocks) for small sounds; big ones (lightning strike, boulder impact, shield burst, tornado) are played with volume 1.5–2.0 (vanilla scales range by volume) and the lightning adds a separate `thunder_distant` played at the listener for 24–64 blocks.

1.3 Voice

No words anywhere (the shows' lines are theirs). Grunts and breaths are **synthesised** (`formant` tones with noise, no samples): the airbender's breath, the firebender's shout at a big move (a short formant "ha!" at 180 Hz with a noise burst), the blood-bending victim's gasp, the spirit's hollow wind.

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2. Shared events (release 1.0)

Event	Length	Recipe	Subtitle EN	Subtitle HE
`awaken_earth`	2.0 s	a deep stomp (`modal` 60/90 Hz, 0.3 s) → a rising rumble (brown noise, LP 200→600 Hz swell 1.2 s) → a stone "lock" (modal 320/510/780, 0.4 s) + hall reverb	You become an Earthbender	הפכתם לכשפי אדמה
`awaken_air`	2.0 s	a wide wind swell (pink noise BP 800→2400 Hz, 1 s) → three rising sine chimes (523, 659, 784 Hz, 0.3 s each) → a soft gust tail	You become an Airbender	הפכתם לכשפי אוויר
`awaken_water`	2.0 s	a liquid swell (`formant` "oo" 220 Hz with LP noise ripple 6 Hz) → a shimmer (sine cluster 1046–1568 Hz, detuned) → a slow splash tail	You become a Waterbender	הפכתם לכשפי מים
`awaken_fire`	2.0 s	an inhale (HP noise 1→2 kHz, 0.5 s) → an ignition "fwoom" (saw 110 Hz + crackle bursts, LP 3 kHz, 0.6 s) → a brass-like swell (saw stack 220/330/440, 0.8 s)	You become a Firebender	הפכתם לכשפי אש
`mastery_up`	0.8 s	a two-note chime (sine 880 → 1320 Hz with a soft attack) + a tiny sparkle (3 short sines 2–4 kHz)	Mastery rises	רמת השליטה עולה
`points`	0.5 s	a soft gold "ting" (modal 1760/2640 Hz, 0.4 s)	Skill points earned	נקודות כישור
`unlock`	1.0 s	the element's awaken chord at a third of the length, quieter	Ability unlocked	יכולת נפתחה
`scroll_open`	0.6 s	paper rustle (pink noise HP 2 kHz, 3 short envelopes)	Scroll unrolls	מגילה נפתחת
`look_change`	0.6 s	a cloth whoosh (noise BP 1 kHz sweep 0.3 s) + the element's small chime	Look changes	החלפת מראה

Event	Length	Recipe	Subtitle EN	Subtitle HE
`hud_ready`	0.15 s	a soft click (sine 1200 Hz, 20 ms) — client UI, no subtitle	—	—
`refused`	0.3 s	a dull double tap (modal 180/270, 2× 60 ms)	Cannot bend here	אי אפשר לכשף פה
`hit_stagger`	0.3 s	a body thump (sine 70 Hz + noise burst 40 ms)	Hit	פגיעה
`land_soft`	0.4 s	a padded thud (sine 55 Hz, 0.2 s) + a dust hiss (noise LP 1 kHz, 0.3 s)	Lands	נחיתה
`temp_revert`	0.5 s	a soft stone slide reversed (noise BP 300 Hz swell 0.4 s) + a click; 3 takes	Ground settles	הקרקע מתיישבת

3. Earth (release 1.0)

Event	Length	Recipe	EN	HE
`earth_stomp`	0.5 s	a deep thud (sine 50→40 Hz glide, 0.25 s) + gravel crunch (brown noise BP 400–1200 Hz, 0.15 s, granular 8 grains); 3 takes	Earthbender stomps	כשף אדמה רוקע
`earth_rock_lift`	0.6 s	a tearing crunch (noise BP 300–2 kHz, 0.2 s) → a rising rumble (sine 60→90 Hz, 0.4 s) + pebble clicks (6 modal taps 2–4 kHz)	Rock tears free	סלע נתלש
`earth_rock_throw`	0.5 s	a low whoosh (brown noise LP 800 Hz, envelope 0.35 s with grit: amplitude-modulated 40 Hz)	Rock flies	סלע עף
`earth_rock_hit`	0.5 s	a cracking thump (modal stone 180/290/450 Hz with a 5 ms broadband crack, 0.35 s) + debris clicks; 3 takes	Rock shatters	סלע מתנפץ
`earth_rock_hit_flesh`	0.4 s	the thump without the crack, duller (LP 600 Hz) + a body thud	Rock strikes	סלע פוגע
`earth_boulder_throw`	0.8 s	`earth_rock_throw` an octave down + a sub thump 35 Hz	Boulder flies	סלע ענק עף
`earth_boulder_hit`	1.0 s	a sub-bass drop (sine 45→25 Hz, 0.5 s) + a big crack + a rubble cascade (12 modal taps over 0.5 s) + reverb	Boulder smashes	סלע ענק מתרסק
`earth_hold_loop`	3.0 s loop	a quiet rumble (brown noise LP 120 Hz, slow 0.3 Hz tremolo) + occasional grit ticks	Rock hovers	סלע מרחף
`earth_rock_down`	0.5 s	a soft stone settle (modal 180/290, no crack) + a dust hiss	Rock settles	סלע מתיישב
`earth_armour_form`	1.2 s	stomp → stone sliding (noise BP 200–900 Hz rising 0.7 s, granular) → a heavy "clunk" seal (modal 140/220/360 Hz, 0.3 s)	Earth armour forms	שריון אדמה נוצר
`earth_armour_step`	0.25 s	a stony tread (modal 220/330, 60 ms + grit); 3 takes, volume 0.4	Stone footstep	צעד אבן
`earth_armour_hit`	0.3 s	a chip (modal 1200/1800 Hz, 40 ms) + a dull knock (sine 90 Hz)	Armour takes a hit	השריון נפגע
`earth_armour_crack`	0.4 s	a sharp split (broadband 3 ms + modal 800/1300 Hz decaying 0.3 s)	Armour cracks	השריון נסדק

Event	Length	Recipe	EN	HE
`earth_armour_break`	1.0 s	a rubble collapse (14 modal taps falling in pitch 600→150 Hz over 0.6 s) + a dust cloud (noise LP 1 kHz swell)	Earth armour crumbles	שריון האדמה מתפורר
`earth_shed_blast`	0.9 s	`earth_armour_break` + a crack and a wide shard hiss (noise HP 2 kHz, 0.4 s, stereo-ish via two detuned copies)	Armour bursts	השריון מתנפץ
`earth_rumble_loop`	2.0 s loop	brown noise LP 150 Hz with a 2 Hz wobble	Earth rumbles	האדמה רועמת
`earth_clunk`	0.2 s	one stone block clunk (modal 160/260/410 Hz, 0.15 s); 3 takes	Stone shifts	אבן זזה
`earth_pillar_rise`	(loop + clunks)	`earth_rumble_loop` pitched up 1.2 while rising + `earth_clunk` per block	Earth pillar rises	עמוד אדמה מתרומם
`earth_pillar_sink`	0.8 s	the rumble descending (sine 80→45 Hz) + two clunks	Earth pillar sinks	עמוד האדמה שוקע
`earth_launch`	0.6 s	a stomp + a sharp stone crack-whoosh (crack 3 ms → noise BP 400→1500 Hz 0.3 s upward)	Earth launch	זינוק אדמה
`earth_land_soft`	0.5 s	`land_soft` + a stone dip (modal 120 Hz)	Earthbender lands	כשף אדמה נוחת
`earth_wall_rise`	0.5 s	stomp → a fast rising stone slide (noise BP 300→1200 Hz 0.3 s) + four clunks 60 ms apart	Earth wall rises	קיר אדמה מתרומם
`earth_wall_hit`	0.3 s	a knock with grit (modal 200/330 Hz + noise 30 ms); pitch by the hit's damage (0.8–1.2)	Wall takes a hit	הקיר נפגע
`earth_wall_sink`	0.9 s	a slow grinding fall (noise BP 200–600 Hz, 0.7 s, falling) + a settle	Earth wall sinks	קיר האדמה שוקע
`earth_scrape_loop`	2.0 s loop	a heavy stone scrape (noise BP 150–500 Hz with 12 Hz granular stutter)	Wall scrapes	הקיר נגרר
`earth_wall_push`	0.5 s	a final thud + the scrape ending	Wall pushes	הקיר דוחף
`sand_lift`	0.5 s	a dry hiss rising (white noise HP 3 kHz → BP 1.5 kHz, 0.5 s)	Sand lifts	חול מתרומם
`sand_stream`	0.6 s	a grainy rush (pink noise BP 1–4 kHz, 0.5 s, amplitude granular 60 Hz)	Sand streams	זרם חול
`sand_hit`	0.3 s	a sandy pelt (noise HP 2 kHz, 80 ms, 4 overlapping) + a soft thud; 3 takes	Sand pelts	חול מצליף
`sand_wall_rise`	0.6 s	pouring sand reversed: a swelling hiss (noise BP 1–3 kHz, 0.5 s crescendo)	Sand wall rises	קיר חול מתרומם
`sand_wall_fall`	0.8 s	the slump: noise BP 800 Hz falling 0.6 s + a soft thud	Sand wall collapses	קיר החול קורס
`quicksand_cast`	0.6 s	a low sandy groan dropping (sine 90→50 Hz + grainy noise, 0.6 s)	Quicksand forms	חול טובעני נוצר
`quicksand_sink_loop`	2.5 s loop	a wet, grainy churn (brown noise LP 400 Hz with slow 0.5 Hz bubbling tremolo) volume 0.5	Sinking in quicksand	שוקעים בחול טובעני
`quicksand_free`	0.4 s	a sucking pop (sine glide 200→600 Hz 80 ms + a noise pop)	Freed from quicksand	שוחררו מהחול הטובעני
`sand_cloud`	0.8 s	a broad hiss swell (pink noise BP 1–3 kHz, 0.8 s)	Sand cloud	ענן חול

Event	Length	Recipe	EN	HE
`sand_cloud_loop`	3.0 s loop	a soft swirling hiss, 0.3 Hz pan-like tremolo, volume 0.3	Sand swirls	חול מסתחרר
`lava_path_cast`	0.9 s	stomp + a molten rumble swelling (brown noise LP 300 Hz + sine 55 Hz, 0.8 s) with crackling (random 2 ms clicks)	Lava path flows	שביל לבה זורם
`lava_path_loop`	3.0 s loop	bubbling hiss (noise LP 800 Hz with random 3–6 Hz bubbles: sine pops 150–300 Hz) volume 0.5	Lava bubbles	לבה מבעבעת
`lava_cool`	0.4 s	a sizzling tick (noise HP 4 kHz, 0.3 s decay) + a faint stone click	Lava cools	הלבה מתקררת
`lava_shuriken_throw`	0.5 s	a hot whoosh (noise BP 600 Hz→2 kHz, 0.4 s) with a spinning flutter (AM 25 Hz)	Lava shuriken flies	שוריקן לבה עף
`lava_shuriken_spin_loop`	2.0 s loop	a thin flutter (noise BP 2 kHz, AM 25 Hz) volume 0.3	Shuriken spins	השוריקן מסתובב
`lava_shuriken_hit`	0.5 s	a splat-sizzle (noise burst LP 1 kHz 50 ms → HP 4 kHz sizzle 0.4 s)	Lava shuriken strikes	שוריקן לבה פוגע
`metal_knives_form`	0.4 s	three quick steel "shinks" (modal 2400/3600/5200 Hz, 60 ms each, 100 ms apart)	Steel knives form	סכיני פלדה נוצרים
`metal_knife_throw`	0.3 s	a thin whistle (sine 3200→2200 Hz glide, 0.25 s) + a short air hiss; 3 takes	Steel knife flies	סכין פלדה עפה
`metal_knife_hit`	0.3 s	a "tink" (modal 3000/4500 Hz, 80 ms) into stone; a dull "thuk" (LP 800 Hz) into flesh; 3 takes each	Steel knife strikes	סכין פלדה פוגעת
`metal_knife_stick`	0.5 s	a quivering twang (sine 180 Hz with 12 Hz FM decaying, 0.45 s)	Knife quivers	הסכין רוטטת
`metal_recall`	0.5 s	a reversed shink glide (modal 5200→2400 Hz) + a catch click	Knives return	הסכינים חוזרות
`metal_pull_loop`	2.5 s loop	a scraping grinding hum (saw 70 Hz LP 400 + noise BP 300–900 Hz granular) + a filings hiss (HP 5 kHz sparkle), pitch rises with charge (0.9–1.2)	Metal pulled	מתכת נשלפת
`metal_pull_end`	0.2 s	a clink (modal 2800/4200 Hz, 0.15 s)	Metal gathered	המתכת נאספה
`metal_wall_rise`	0.6 s	plates sliding (noise BP 1–2 kHz 0.3 s) and locking: four clanks (modal 420/630/1100 Hz, 80 ms, 70 ms apart)	Steel wall rises	קיר פלדה מתרומם
`metal_wall_hit`	0.5 s	a resonant plate ring (modal 380/760/1140 Hz, 0.45 s)	Steel wall rings	קיר הפלדה מהדהד
`metal_armour_on`	1.0 s	five plate claps (modal 300/450/700 Hz, 60 ms, 120 ms apart) + a final heavier lock (modal 180/270, 0.3 s)	Steel armour locks on	שריון פלדה ננעל
`metal_armour_off`	0.9 s	five reversed claps (each 60 ms clap with a 40 ms reversed tail) descending	Steel armour peels off	שריון הפלדה מתקלף
`metal_armour_hit`	0.4 s	a dented clang (modal 500/800 Hz with a pitch dip, 0.35 s)	Steel armour dents	שריון הפלדה נמעך
`metal_armour_break`	1.0 s	a cascade of falling plates (8 claps falling in pitch over 0.7 s) + a clatter tail	Steel armour falls apart	שריון הפלדה מתפרק
`metal_wreck`	0.6 s	a crunching, groaning twist (saw 90→60 Hz with heavy FM 7 Hz + metallic noise BP 800–3 kHz, 0.5 s) + a final creak	Metal wrecked	מתכת נהרסת

Event	Length	Recipe	EN	HE
`metal_mend`	0.6 s	a soft reversed crunch (the twist reversed, LP 2 kHz, quieter) + a chime (sine 1760 Hz)	Metal mends	המתכת מתוקנת
`metal_grip`	0.3 s	a clamp clank (modal 350/520 Hz, 0.2 s)	Metal gripped	מתכת נאחזת
`metal_flin g`	0.5 s	a swung-metal whoosh (noise BP 500 Hz→1.5 kHz, 0.3 s) + a clank	Metal flung	מתכת מוטלת

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4. Air (release 1.1)

Event	Length	Recipe	EN	HE
`air_gathe r`	0.5 s	a soft inhale-like rising whoosh (pink noise BP 600→1800 Hz, 0.45 s)	Air gathers	אוויר נאסף
`air_blast`	0.6 s	a hard breathy "whumpf": a 20 ms burst (noise LP 400 Hz) → a 0.4 s decaying rush (BP 500–3 kHz falling)	Air blast	הדף אוויר
`air_hit`	0.3 s	a hollow thump (sine 110 Hz with a resonant LP, 0.25 s)	Thrown by air	נזרקו מהאוויר
`air_jump`	0.4 s	a short down-whoosh (noise BP 1.5 kHz→400 Hz, 0.25 s) + a puff	Air jump	קפיצת אוויר
`air_shiel d_up`	0.8 s	a rising swirl (noise BP sweeping 400→2000 Hz with a 1.5 Hz rotation tremolo) into a sustained hush	Air shield rises	מגן אוויר מתרומם
`air_shiel d_loop`	3.0 s loop	a soft airy rotating hum (noise BP 900 Hz, 1.5 Hz tremolo, + a sine 220 Hz very low), pitch 1.0→1.1 as strength falls, volume 0.35	Air shield hums	מגן האוויר מהמהם
`air_shiel d_hit`	0.3 s	a muffled "whup" (noise LP 600 Hz, 60 ms) + a flutter (AM 20 Hz tail 0.2 s)	Air shield deflects	מגן האוויר מסיט
`air_shiel d_down`	0.6 s	the hush falling away (noise BP 1500→300 Hz, 0.5 s fade)	Air shield lowers	מגן האוויר יורד
`air_shiel d_burst`	0.9 s	a deep "boom-whoosh": sine 50 Hz thump + a wide rush (two detuned noise copies BP 400–4 kHz, 0.7 s)	Air shield bursts	מגן האוויר מתפוצץ
`glider_op en`	0.4 s	a cloth snap (noise HP 1 kHz, 30 ms) + a flap (noise LP 800 Hz, 0.2 s)	Glider opens	הגלשן נפתח
`glider_cl ose`	0.4 s	two soft flaps and a wooden click (modal 900 Hz)	Glider closes	הגלשן נסגר
`glider_lo op`	3.0 s loop	wind over cloth (pink noise BP 300–1500 Hz with a 3 Hz flutter), volume and pitch with speed (0.3–0.7, 0.9–1.2)	Gliding	גלישה
`air_blade _cast`	0.4 s	a slicing whoosh (noise BP 2 kHz→6 kHz, 0.25 s) with a high "shing" (sine 5200 Hz, 0.1 s)	Air blade	להב אוויר
`air_blade _fly_loop`	2.0 s loop	a thin whistle (sine 2800 Hz with 6 Hz vibrato + noise HP 4 kHz), volume 0.3	Air blade whistles	להב האוויר שורק
`air_blade _hit`	0.3 s	a crisp cut (noise HP 3 kHz, 40 ms) + a thump	Air blade cuts	להב האוויר חותך
`air_slice`	0.7 s	a spinning whoosh rising then falling (noise BP 800→3000→800 Hz, 0.6 s, AM 8 Hz)	Air slice	חיתוך אוויר
`spirit_si t`	0.6 s	a soft breath out (`formant` "hh" noise LP 2 kHz, 0.4 s) + cloth settling	Sits to meditate	מתיישבים למדיטציה

Event	Length	Recipe	EN	HE
`spirit_leave`	1.2 s	a rising shimmer (sine cluster 880–1760 Hz detuned, 1 s) with a bell-like inhale (sine 1320 Hz swelling)	Spirit leaves the body	הרוח יוצאת מהגוף
`spirit_loop`	15 s loop (stream)	a quiet hollow wind (pink noise BP 200–800 Hz) with distant chimes (random sine 1–3 kHz, 1 per 3 s, reverb), volume 0.3; the spirit's client only	Spirit world	עולם הרוחות
`spirit_return`	0.9 s	the shimmer reversed + a soft "whum" (sine 110 Hz, 0.3 s)	Spirit returns	הרוח חוזרת
`spirit_snap`	0.5 s	a sharp intake (noise HP 1 kHz reversed 0.2 s) + a crack (3 ms) + a short ring	Snapped back	הרוח נמשכת בחזרה
`spirit_meditate_loop`	6.0 s loop	a slow breathing hush (noise LP 1 kHz with a 0.25 Hz envelope), volume 0.2	Meditating	מדיטציה
`air_bubble_on`	0.5 s	a glassy "bloop" reversed (sine 400→900 Hz, 0.15 s) into a hush	Air bubble forms	בועת אוויר נוצרת
`air_bubble_loop`	4.0 s loop	a soft underwater hum (sine 150 Hz + noise LP 500 Hz) with slow breathing (0.3 Hz), volume 0.3	Breathing in a bubble	נושמים בבועה
`air_bubble_off`	0.3 s	a pop (sine 600→200 Hz, 50 ms + noise click)	Air bubble pops	בועת האוויר מתפוצצת
`air_cushion`	0.4 s	a soft "fwoomp" (noise LP 300 Hz, 0.3 s)	Air cushion	כרית אוויר
`air_scooter_mount`	0.5 s	a spin-up whoosh (noise BP 400→2000 Hz with AM 6→20 Hz, 0.4 s)	Air scooter	קטנוע אוויר
`air_scooter_loop`	2.0 s loop	a continuous airy whir (noise BP 1.2 kHz, AM 18 Hz) pitch 0.8–1.3 and volume 0.3–0.5 with speed	Riding the air scooter	רוכבים על קטנוע האוויר
`air_scooter_dismount`	0.4 s	a whoosh down (noise BP 2000→400 Hz, 0.3 s)	Air scooter ends	קטנוע האוויר נגמר
`air_scooter_burst`	0.6 s	`air_shield_burst` shortened	Air ball bursts	כדור האוויר מתפוצץ
`air_spout_loop`	2.5 s loop	a rushing updraft hum (noise BP 500–2500 Hz rising tremolo 0.7 Hz), volume 0.5	Air spout	עמוד אוויר
`air_spout_end`	0.6 s	the rush fading with a soft puff	Air spout fades	עמוד האוויר נעלם
`tornado_cast`	0.7 s	a spinning whoosh building (noise BP 300→1500 Hz, AM 4→12 Hz) into a roar onset	Tornado forms	טורנדו נוצר
`tornado_loop`	4.0 s loop (stream)	a deep wide rotating roar (brown noise LP 400 Hz + pink BP 600–2400 Hz, 1.2 Hz rotation tremolo) with a whistling top (sine 1800 Hz vibrato 5 Hz, quiet), volume 0.7	Tornado roars	הטורנדו שואג
`tornado_end`	0.8 s	the roar unwinding (tremolo slowing 1.2→0.3 Hz, fade 0.8 s)	Tornado fades	הטורנדו נעלם
`air_wheel_loop`	2.0 s loop	a fast whirring rush (noise BP 1.5 kHz, AM 30 Hz), volume 0.5	Air wheel	גלגל אוויר
`air_wheel_hit`	0.3 s	a thump + a short rush	Air wheel strikes	גלגל האוויר פוגע

Event	Length	Recipe	EN	HE
`tethers_cut`	2.0 s	a slow rising chord (sines 196/294/392 Hz swelling 1.5 s) with a **snapping string** (`karplus` 110 Hz plucked hard + a 3 ms crack at 1.2 s) and a wind opening after	Earthly tethers cut	הקשרים הארציים ניתקו
`flight_takeoff`	0.5 s	a soft lift whoosh (noise BP 600→1500 Hz, 0.4 s)	Takes flight	מתרוממים לתעופה
`flight_loop`	3.0 s loop	a gentle wind rush (pink noise BP 400–1800 Hz), volume 0.2–0.5 with speed	Flying	תעופה
`suffocate_start`	0.4 s	a sharp inhale reversed (noise HP 1.5 kHz, 0.3 s reversed envelope)	Air stolen	האוויר נלקח
`suffocate_loop`	3.0 s loop	a thin sucking wind (noise BP 2–5 kHz with a slow inward swell 0.5 Hz) with a heartbeat under it (sine 55 Hz, two beats per 1.1 s), volume 0.5	Suffocating	חניקה
`suffocate_victim_loop`	3.0 s loop	a choking breath (`formant` "uh" 160 Hz with noise, gasps every 1 s) + a louder faster heartbeat (1.4 Hz) — the victim's client	Cannot breathe	אי אפשר לנשום
`suffocate_end`	0.5 s	a gasp of released air (noise BP 1 kHz burst 0.3 s with a vowel formant)	Breath returns	הנשימה חוזרת
`air_vacuum`	1.0 s	a deep implosive "thoomp" (sine 60→30 Hz, 0.15 s + noise LP 200 Hz) → 0.3 s of near-silence (a −30 dB hum) → a soft rush back	Air vacuum	ואקום אוויר
`air_vacuum_loop`	3.0 s loop	a hollow pressure hum (sine 80 Hz + filtered noise notch at 1 kHz), volume 0.3	Airless	בלי אוויר
`air_master_loop`	15 s loop (stream)	a faint breeze (pink noise BP 300–1200 Hz, slow 0.2 Hz swells), volume 0.15; ambient of a master	Breeze	רוח קלה

5. Water, ice and blood (release 1.2)

Event	Length	Recipe	EN	HE
`water_draw`	0.3 s	a liquid "shloop" rising (sine 300→700 Hz with LP noise ripple 8 Hz, 0.25 s)	Water drawn	מים נשאבים
`water_shot`	0.5 s	a liquid whoosh (noise BP 800→2500 Hz, 0.35 s) with a wet onset (sine 250 Hz click + bubble pops)	Water shot	יריית מים
`water_hit`	0.5 s	a splash (noise HP 1.5 kHz burst 0.1 s + bubbly tail LP 2 kHz 0.3 s) and a slap (sine 180 Hz, 40 ms); 3 takes	Water strikes	מים פוגעים
`water_splash_block`	0.4 s	a splash without the slap	Splash	שכשוך
`water_vs_fire`	0.6 s	a hiss (white noise HP 3 kHz, 0.4 s decaying) and a steam sigh (`formant` "hh" LP 1.5 kHz)	Water quenches fire	מים מכבים אש
`water_whip_swing`	0.4 s	a wet whoosh (noise BP 600–2 kHz with LP ripple 10 Hz, 0.3 s)	Water whip swings	שוט מים מתנופף
`water_whip_crack`	0.3 s	a liquid crack (3 ms broadband + a wet slap 180 Hz + splash tail)	Water whip cracks	שוט המים מצליף
`water_whip_loop`	2.0 s loop	a soft sloshing (noise LP 800 Hz with 2 Hz bubbling tremolo), volume 0.25	Water whip	שוט מים

Event	Length	Recipe	EN	HE
`heal_start`	0.6 s	a soft liquid shimmer rising (sine cluster 1046–2093 Hz detuned, LP ripple 6 Hz, 0.5 s)	Healing begins	הריפוי מתחיל
`heal_loop`	4.0 s loop	a gentle bubbling (sine pops 400–900 Hz, 2–4 per s) over a warm pad (saw stack 220/330 Hz LP 600 Hz), volume 0.4	Healing water	מים מרפאים
`heal_tick`	0.3 s	a soft chime (sine 1568 Hz, 0.25 s, soft attack); fire: pitch 0.9	A heart heals	לב מתרפא
`heal_end`	0.5 s	the shimmer fading	Healing ends	הריפוי נגמר
`cleanse`	0.6 s	a splash + a bright "pling" (sine 2093 Hz, 0.2 s) + a sigh (formant "ah" 200 Hz, 0.3 s)	Cleansed	טיהור
`octopus_form`	1.0 s	a surging swell of water (noise LP 1.2 kHz rising 0.7 s with 4 Hz ripple) + a sloshing settle	Water octopus forms	תמנון מים נוצר
`octopus_loop`	3.0 s loop	soft lapping and sloshing (noise LP 900 Hz, 1.5 Hz irregular tremolo), volume 0.35, pitch 0.9	Water octopus	תמנון מים
`octopus_slap`	0.3 s	a wet slap (sine 160 Hz 30 ms + splash HP 2 kHz 0.15 s); 3 takes	Tentacle slaps	זרוע מצליפה
`octopus_strike`	0.4 s	two quick wet thwacks 90 ms apart	Tentacles strike	זרועות פוגעות
`octopus_lash`	0.6 s	a wide liquid whip-crack (crack + 3 overlapping splashes)	Tentacles lash	הזרועות מצליפות
`octopus_drop`	0.8 s	a collapse into a splash (noise falling LP 2 kHz→400 Hz 0.5 s + a big splash)	Water octopus collapses	תמנון המים קורס
`octopus_steam`	0.5 s	a hiss (HP 3 kHz, 0.4 s)	Tentacle evaporates	זרוע מתאדה
`octopus_freeze`	0.5 s	a crackling freeze (random 40 clicks 2–6 kHz over 0.4 s with a sine 1200 Hz sheen)	Tentacles freeze	הזרועות קופאות
`spout_rise`	1.0 s	a rushing surge rising in pitch (noise BP 300→1500 Hz, 0.9 s) with a whirl (AM 3 Hz)	Water spout rises	עמוד מים מתרומם
`spout_loop`	3.0 s loop (stream)	a roaring churning water column (brown noise LP 600 Hz + pink BP 800–3 kHz with 2.5 Hz whirl tremolo + bubble pops), volume 0.7, pitch 0.9–1.1 by height	Riding the water spout	רוכבים על עמוד המים
`spout_ram`	0.5 s	a heavy splash-thud (sine 60 Hz thump + big splash)	Water spout rams	עמוד המים דוחף
`spout_collapse`	1.0 s	the column dropping (noise pitch falling 1500→200 Hz 0.6 s) into a broad splash	Water spout collapses	עמוד המים קורס
`spout_freeze`	0.8 s	the crackling freeze + a deep creak (saw 45 Hz FM 3 Hz, 0.5 s)	Water spout freezes	עמוד המים קופא
`wave_ride`	0.8 s	a long rolling wave rising (noise LP 1 kHz swell 0.6 s) with foam hiss (HP 3 kHz)	Wave rises	גל מתרומם
`wave_loop`	3.0 s loop	a rolling surf (noise LP 800 Hz, 0.8 Hz swell) + foam, volume 0.6	Riding the wave	רוכבים על הגל
`wave_end`	0.8 s	the wave breaking (a splash crash + foam fade)	Wave breaks	הגל נשבר
`water_arms_on`	0.7 s	a surge running down (noise BP 1 kHz→300 Hz 0.5 s) with a wet flex (two liquid slaps)	Water arms form	זרועות מים נוצרות

Event	Length	Recipe	EN	HE
`water_arms_loop`	2.5 s loop	a quiet continuous slosh (noise LP 700 Hz, 1 Hz tremolo), volume 0.25	Water arms	זרועות מים
`water_arm_strike`	0.3 s	a wet snap-thwack (crack 2 ms + slap + small splash); 3 takes	Water arm strikes	זרוע מים פוגעת
`water_arm_grab`	0.5 s	a wet grip (slap + a sucking sine 300→150 Hz) and a drag (noise LP 500 Hz 0.3 s)	Water arm grabs	זרוע מים אוחזת
`water_arm_climb`	0.5 s	a liquid pull (noise BP 600 Hz rising 0.3 s) with a creak	Water arm pulls	זרוע מים מושכת
`water_arms_off`	0.6 s	a shake-off splash (3 splashes 80 ms apart + a puddle)	Water arms fall	זרועות המים נופלות
`ice_knives_form`	0.4 s	three crystalline "tinks" (modal 3200/4800/6400 Hz, 50 ms, 90 ms apart) with a frost hiss (HP 6 kHz, 0.3 s)	Ice knives form	סכיני קרח נוצרות
`ice_knife_throw`	0.3 s	a thin icy whistle (sine 3600→2800 Hz + HP noise, 0.25 s); 3 takes	Ice knife flies	סכין קרח עפה
`ice_knife_shatter`	0.4 s	a glassy shatter (12 modal taps 2–8 kHz over 0.25 s, decaying) ; 3 takes	Ice shatters	קרח מתנפץ
`ice_spikes`	0.8 s	a cracking eruption: 10 ice cracks (broadband 2 ms + modal 1.5–4 kHz) rising in pitch over 0.6 s + a frost hiss	Ice spikes erupt	חודי קרח מתפרצים
`freeze_cast`	0.8 s	a sharp frost hiss sweeping outward (noise HP 4 kHz→BP 2 kHz, 0.7 s) with crackles (random clicks 3–7 kHz, 20 per s) and a sine sheen 1400 Hz	Water freezes	מים קופאים
`freeze_feet`	0.5 s	a crunching encasement (noise BP 800–2 kHz granular 0.3 s + two ice cracks)	Frozen in place	קפואים במקום
`ice_shatter`	0.5 s	`ice_knife_shatter` bigger (20 taps, 0.4 s)	Ice breaks	הקרח נשבר
`snow_throw`	0.3 s	a soft fluffy whoosh (noise LP 1.5 kHz, 0.25 s, soft envelope) ×4 overlapping 60 ms apart	Snow thrown	שלג נזרק
`snow_hit`	0.2 s	a soft pat (noise LP 800 Hz, 80 ms) ; 3 takes	Snow hits	שלג פוגע
`ice_wall_rise`	0.6 s	a fast crystalline growth: crackles rising (30 clicks 2–8 kHz over 0.4 s, rising density) into a resonant ring (modal 900/1350 Hz, 0.3 s)	Ice wall rises	קיר קרח מתרומם
`ice_wall_break`	0.6 s	the shatter + a low crack (sine 120 Hz)	Ice wall shatters	קיר הקרח מתנפץ
`ice_melt_loop`	3.0 s loop	a hissing drip (noise HP 3 kHz quiet + drips: sine 1200→800 Hz 60 ms, 2 per s), volume 0.3	Ice melts	קרח נמס
`ice_dome`	0.9 s	`ice_wall_rise` longer with a hollow boom at the close (sine 70 Hz + LP noise, 0.3 s)	Ice dome closes	כיפת קרח נסגרת
`blood_grab`	0.8 s	a low wet straining groan (saw 65 Hz FM 4 Hz LP 300 Hz + wet noise LP 600 Hz, 0.7 s) with a heartbeat entering (sine 50 Hz, two beats)	Blood bending takes hold	כשפות הדם אוחזות
`blood_loop`	3.3 s loop (stream)	a slow heartbeat (sine 50 Hz, 2 beats per 1.1 s) + a tense string drone (`karplus` 110 Hz bowed-like: sustained saw LP 500 Hz with 0.2 Hz swell) + a faint liquid creak (noise BP 200 Hz granular), volume 0.6	Blood bending	כשפות דם

Event	Length	Recipe	EN	HE
`blood_victim_loop`	2.2 s loop	a louder, faster heartbeat (1.4 Hz) + a strained breath — the victim's client	Blood bent	נשלטים בכשפות דם
`blood_move`	0.4 s	a creaking strain (saw 80 Hz FM 6 Hz, 0.3 s) ; 3 takes	Puppet moves	הבובה זזה
`blood_fling`	0.5 s	a wet snap (crack + liquid slap) and a whoosh (noise BP 600→2 kHz, 0.35 s)	Flung by blood bending	מוטלים בכשפות דם
`blood_release`	0.6 s	a breath out (formant "hh", 0.4 s) and the heartbeat stopping (one last beat)	Blood bending releases	כשפות הדם משתחררות
`blood_sever`	1.5 s	a long inhale (noise HP 1 kHz, 0.5 s) → 0.3 s of near-silence → a deep soft thud (sine 45 Hz, 0.2 s) and a sigh of released air (formant "ah" falling 220→160 Hz, 0.5 s)	Bending taken away	הכשפות נשללות
`severed_victim`	2.0 s	a high ring fading (sine 4186 Hz with slow 0.5 Hz amplitude wobble, 1.8 s fade) — the victim's client	Your bending is gone	הכשפות שלכם נעלמו
`blood_refused`	0.6 s	a glassy chime (modal 2093/3136 Hz, 0.4 s) and a snapping string (`karplus` 220 Hz pluck + crack)	A stronger waterbender	כשף מים חזק יותר
`water_master_loop`	15 s loop (stream)	a faint water lapping (noise LP 600 Hz, 0.4 Hz swells) with a rare drip, volume 0.15	Water murmurs	מים ממלמלים
`moon_rise`	1.5 s	a soft silver swell (sine cluster 1568/2349/3136 Hz detuned, 1.2 s, reverb) — when the full moon's night begins, for waterbenders	Full moon	ירח מלא

6. Fire and lightning (release 1.3)

Event	Length	Recipe	EN	HE
`fire_ignite`	0.2 s	a sharp fire catch (noise burst BP 1–4 kHz 30 ms + a crackle 0.15 s) with a breath (HP noise 20 ms); 3 takes	Fire ignites	אש נדלקת
`fireball_throw`	0.5 s	a punchy whoosh (noise BP 400→2500 Hz, 0.35 s) with a crackling body (random clicks 1–4 kHz, 30 per s) and a low "fwoom" (saw 90 Hz LP 300 Hz, 0.2 s); blue: pitch 1.08 + a hiss layer	Fireball	כדור אש
`fireball_fly_loop`	2.0 s loop	a soft roaring flutter (noise BP 600–2 kHz, AM 14 Hz) volume 0.3	Fireball flies	כדור האש עף
`fireball_burst`	0.6 s	a fiery "boom-crackle": sine 55 Hz thump 0.15 s + noise burst LP 2 kHz 0.1 s + crackle tail 0.4 s (clicks decaying); great: + a sub drop 40→20 Hz	Fireball bursts	כדור האש מתפוצץ
`fire_flick`	0.2 s	a tiny puff (noise LP 2 kHz, 60 ms) + a click	Small flame	להבה קטנה
`fire_vs_water`	0.5 s	a hiss (HP 3 kHz, 0.4 s) and a steam sigh	Fire quenched	האש כבתה
`fire_fists_on`	0.5 s	an inhale (HP noise 0.2 s) and two fire catches 100 ms apart	Fists ignite	האגרופים נדלקים
`fire_fists_loop`	2.0 s loop	a low crackling flicker (noise BP 800–3 kHz with random 8–20 Hz amplitude crackle), volume 0.3	Burning fists	אגרופים בוערים

Event	Length	Recipe	EN	HE
`fire_punch_hit`	0.4 s	a punch thud (sine 90 Hz 40 ms + noise LP 600 Hz) with a fiery "fwoosh" (noise BP 1–3 kHz, 0.3 s); 3 takes	Fire punch	מכת אש
`fire_combo`	0.5 s	a bigger "whoomp" (sine 60 Hz + wider noise, 0.4 s)	Fire combo	שילוב אש
`fire_fists_off`	0.4 s	a flame dying (noise BP 2 kHz fading 0.3 s + a puff)	Fists go out	האגרופים כבים
`fire_kick`	0.5 s	a sweeping fire whoosh (noise BP 500→3000→1000 Hz, 0.4 s) with a sharp crack at the peak	Fire kick	בעיטת אש
`fire_kick_hit`	0.4 s	a flaming thud (`fire_punch_hit` lower, 0.9 pitch)	Fire kick lands	בעיטת האש פוגעת
`flamethrower_start`	0.5 s	an inhale and a double ignition "fwoosh" (two fire catches 60 ms apart into a rising roar onset)	Flamethrower starts	הלהביור מתחיל
`flamethrower_loop`	3.0 s loop (stream)	a roaring jet (brown noise LP 800 Hz + pink BP 500–4 kHz, 10 Hz flutter) with a crackling layer (random clicks 40 per s); volume 0.7; blue: pitch 1.15 + hissing top (HP 5 kHz)	Flamethrower	להביור
`flamethrower_end`	0.4 s	the roar cutting out (fast fade 80 ms) with a puff and three embers (clicks)	Flamethrower stops	הלהביור נפסק
`fire_breath_inhale`	0.6 s	a long dragon's inhale (noise HP 800 Hz→LP 1.5 kHz rising, 0.55 s, with a formant "hh")	Deep breath	נשימה עמוקה
`fire_breath_loop`	3.0 s loop (stream)	a deeper wider roar (`flamethrower_loop` an octave lower in the brown layer) with a vocal rasp (formant "ah" 110 Hz distorted, quiet), volume 0.8	Fire breath	נשיפת אש
`fire_breath_end`	0.5 s	a cough-puff (noise burst LP 1 kHz 60 ms ×2) and embers	Fire breath ends	נשיפת האש נגמרת
`fire_melt`	0.5 s	hiss + drips	Fire melts ice	האש ממיסה קרח
`fire_whip_ignite`	0.4 s	a rising fiery hiss (noise BP 800→3 kHz, 0.35 s) as the whip uncoils, with crackle	Fire whip ignites	שוט אש נדלק
`fire_whip_loop`	2.5 s loop	a crackling fluttering flame (noise BP 1–3 kHz, AM 12 Hz irregular) with a soft wind, volume 0.4	Fire whip	שוט אש
`fire_whip_strike`	0.4 s	a whip-crack (3 ms broadband + a 40 ms sine 1200→300 Hz snap) with a fiery whoosh	Fire whip strikes	שוט האש מצליף
`fire_whip_hit`	0.4 s	a crack-thud and a flare (noise BP 2 kHz 0.2 s)	Fire whip lands	שוט האש פוגע
`fire_whip_lash`	0.6 s	a big crack with a long whoosh (0.5 s)	Fire whip lash	הצלפת שוט אש
`fire_whip_end`	0.3 s	a flame snapping out (a short flutter + a puff)	Fire whip goes out	שוט האש כבה

Event	Length	Recipe	EN	HE
`fire_circle_wind`	0.4 s	two fire catches and a sharp inhale	Fire gathers	אש נאספת
`fire_circle`	1.0 s	a spinning whoosh (noise BP 600–2 kHz, AM 6 Hz rising) erupting at 0.3 s into a wide ring roar (two detuned noise copies LP 2.5 kHz, 0.5 s) with a crackle tail	Fire circle	מעגל אש
`fire_circle_legs`	1.2 s	`fire_circle` with a second lower roar (LP 1 kHz) 0.2 s after	Double fire circle	מעגל אש כפול
`fire_circle_snuff`	0.4 s	a steam hiss	Fire circle cut	מעגל האש נחתך
`fire_jet_start`	0.4 s	two jet ignitions (noise bursts BP 1–3 kHz 50 ms, 40 ms apart) into a thrust onset	Fire jets	סילוני אש
`fire_jet_loop`	2.0 s loop	a roaring thrust (brown noise LP 500 Hz + BP 1–3 kHz, 25 Hz flutter), volume 0.6, pitch 0.9–1.2 with speed	Fire jet flight	תעופה בסילון אש
`fire_jet_end`	0.5 s	the jets cutting (fade 60 ms) with embers falling (6 clicks over 0.4 s)	Fire jets stop	סילוני האש נפסקים
`fire_land`	0.5 s	a fiery thud (`land_soft` + a short flare)	Firebender lands	כשף אש נוחת
`fire_heal_start`	0.6 s	a soft inhale and a warm crackle rising into a hum (saw stack 220/330/440 LP 800 Hz swelling 0.5 s)	Fire healing begins	ריפוי באש מתחיל
`fire_heal_loop`	4.0 s loop	a low warm pad (saw stack LP 700 Hz, slow 0.25 Hz breathing swell) with a soft crackle (quiet random clicks), volume 0.4	Healing fire	אש מרפאת
`fire_heal_end`	0.5 s	the hum fading	Fire healing ends	הריפוי באש נגמר
`fire_burn_out`	0.6 s	a bright flash-sizzle (noise HP 2 kHz burst 80 ms + sizzle 0.3 s) and a sigh	Poison burned out	הרעל נצרב
`lightning_charge`	1.2 s	a rising crackling electric whine: sine sweep 180→2400 Hz over 1.2 s with crackle bursts (random clicks 2–8 kHz) every 150 ms growing denser, plus soft "zip" sparks (short sine chirps 3→6 kHz, 20 ms) during the circles	Lightning charges	ברק נטען
`lightning_hold_loop`	2.0 s loop	a tense crackling hum (sine 2400 Hz + saw 120 Hz LP 500 Hz + clicks 30 per s), pitch slowly rising 1.0→1.05 over 3 s, volume 0.5	Lightning held	ברק מוחזק
`lightning_strike`	1.2 s	a deafening crack: a 2 ms click → a broadband burst (white noise, 15 ms, HP 500 Hz) → a sub thump (sine 40 Hz, 0.15 s) → a 0.8 s rumble tail (brown noise LP 200 Hz decaying) with a short ring (sine 5200 Hz, 0.1 s); dual: a second crack 40 ms later; volume 2.0	Lightning strikes	ברק מכה
`lightning_hit`	0.5 s	a sizzling impact (noise HP 3 kHz burst 60 ms → sizzle 0.3 s) with a "zap" (sine 1800→600 Hz, 50 ms) and a short ring	Struck by lightning	נפגעו מברק
`lightning_arc`	0.3 s	a short crackling zap (clicks 20 ms + sine chirp 4→1 kHz)	Lightning arc	קשת ברק
`lightning_cancel`	0.4 s	sparks fizzling (clicks thinning over 0.3 s + a falling sine 2400→400 Hz)	Lightning fizzles	הברק מתפזר

Event	Length	Recipe	EN	HE
`thunder_distant`	1.5 s	a rumble (brown noise LP 120 Hz, slow 0.6 s attack, 0.9 s decay) — played at listeners 24–64 blocks away 0.6 s after a strike	Distant thunder	רעם רחוק
`paralysed_loop`	2.0 s loop	a buzzing crackle (square 120 Hz LP 1 kHz + clicks 25 per s), volume 0.4	Paralysed	משותקים
`redirect_stance`	0.6 s	a slow calm inhale (noise HP 1 kHz LP 3 kHz, 0.5 s) with a faint electric tension (sine 2400 Hz at −30 dB)	Redirection stance	עמידת הטיה
`redirect_loop`	2.0 s loop	a very quiet high hum (sine 2400 Hz + sine 2405 Hz beating), volume 0.15	Ready to redirect	מוכנים להטות
`redirect_pass`	0.6 s	the strike's crack (short) → a rising electric whoosh through the body (noise BP 400→4000 Hz, 0.25 s, with a sine sweep 300→3000) → a 40 ms pause → the crack again	Lightning redirected	הברק הוטה
`redirect_fail`	1.0 s	the strike crack + a sizzling convulsion (clicks 60 per s, 0.5 s, with a formant "uh" grunt) + a body fall thud	Redirection fails	ההטיה נכשלה
`redirect_ground`	0.9 s	the pass's whoosh → a deep earth thud (sine 45 Hz, 0.2 s) with a crackle spreading (clicks fading over 0.4 s with LP falling) and a soft gold chime (sine 1568 Hz, 0.3 s)	Lightning grounded	הברק הוארק
`fire_master_loop`	15 s loop (stream)	a faint crackle (quiet random clicks 1–3 kHz, 4 per s) with a warm low hum, volume 0.1	Embers	גחלים

Count: shared 14, Earth 53, Air 50, Water 60, Fire 55 = **232 sound events** (plus 2–3 takes for ~40 of them: about 310 files). S0 registers every event of a release with the placeholder; the release's sound lane renders them.

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7. Music

All music is **original, written for the mod**, "in the spirit of" the shows' sound (the show's score leans on East-Asian and Inuit-inspired instruments: taiko and drum ensembles for the Earth Kingdom, flutes and wind for the Air Nomads, the tsungi horn and brass for the Fire Nation, strings and water sounds for the Water Tribes) **without any melody, bass line or chord progression of the shows**. Every instrument is **synthesised** (`sfx.py` instruments: `taiko` = modal drum (60/95 Hz) with a skin noise burst; `shakuhachi` = a breathy sine with noise and pitch wobble; `erhu` = a sawtooth with a 6 Hz vibrato and a narrow BP body resonance; `tsungi` = a brassy saw stack with a slow portamento and a formant "aw"; `bells` = modal; `koto` = `karplus` plucks; `pad` = detuned saw stack LP; `drops` = sine pops with reverb). Rendered as mono 44.1 kHz Ogg, `stream: true`, and **as stems**, following Enhanced's boss-music pattern (`HelaMusic`: several stems started together, layers fading in by phase, volume by distance, the Music source, the client option `music`).

Theme	Key / tempo / length	Instruments and shape	Stems	When it plays
Earth: "Rooted" (`music.earth`)	D minor pentatonic, 72 BPM, 64 s loop	taiko ensemble in a 4/4 with a heavy beat 1 and a rolling fill every 8 bars; a low drone (pad, D1); a koto motif of five notes rising stepwise then falling a fifth (original); a stone "clack" percussion (modal 1.2 kHz) on the off-beats	A drums, B drone + clacks, C koto motif	the Earth temple (suggestion 8.4), the Earth awakening (a 12 s sting cut from it: `music.earth.sting`), and **optionally** (`music.combatThemes`, default off) stem A while an earthbender within 24 blocks is in a fight (hit or hurt in the last 8 s)
Air: "Ba Gua" (`music.air`)	G major, 6/8 at 96 BPM, 72 s loop	a shakuhachi-like flute melody in long arcs (an original 8-bar tune, mostly stepwise with one leap of a sixth), wind chimes (random bell notes of the scale, 1 per 2 s), an airy pad swelling with the 6/8 pulse, a soft frame drum on 1 and 4	A pad + chimes, B flute, C drum	the Air temple, the Air awakening sting, combat option
Water: "Tide" (`music.water`)	A minor, 58 BPM, 80 s loop	an erhu-like lead with slow slides (an original 16-bar melody that falls, pauses, rises), water drops as percussion (sine pops with long reverb), a bowed-string pad, a slow ocean swell (filtered noise) under everything; on **full-moon nights** stem D (a high bell counter-melody) fades in	A swell + drops, B pad, C erhu, D moon bells	the Water temple, the Water awakening sting, the full moon (stem D alone, very quiet, for waterbenders outdoors: `music.moonBells`, default on), combat option
Fire: "Sozin's March"... no: **"Breath of the Dragon"** (`music.fire`)	E minor, 112 BPM, 64 s loop	a tsungi-horn-like brass lead (an original march motif of four bars: two short notes, a long, a fall), driving drums (taiko + a sharp snare-like noise burst), a low ostinato (saw bass on E and B), crackling fire as a texture; Azula's look swaps the lead to a colder, higher brass (`music.fire.blue` stem variant)	A drums + bass, B brass lead, C texture	the Fire temple, the Fire awakening sting, combat option
Bosses (suggestion, 8.5)	one per boss, 3 stems each, as Enhanced's Hela			in the boss fight
Menu / Element Choice (`music.choice`)	a 30 s quiet piece: all four motifs hinted (two bars each) over a shared pad			the Element Choice screen (fades under the choice)

Music never plays over vanilla music by accident: `SelectMusicEvent` sets the vanilla music to null while a theme plays (Enhanced's `onSelectMusic` pattern), and themes crossfade over 2 s (`RoadMusicPlayer`'s recipe).

How it is checked (Enhanced's 15.15.3): every recipe rendered and inspected as a spectrogram and waveform, peak ≤ −1 dBFS after encoding, loudness and >12 kHz energy measured and logged by `render.py`, loops played across the seam ten times, downbeats on the first sample of each loop, stems trimmed 6–8 dB under the effects; subtitles checked by `SoundAssetTests` and `DataTests`; every sound heard in a real client at 1, 8 and 24 blocks, in the four elements' fights, with the music on and off.

TECH.md — Technical architecture

Bending — Technical Architecture

Mod id **`bending`**, group `com.kolodkingal.bending`, Minecraft **1.21.1**, **one universal jar for NeoForge 21.1 and Fabric** (Fabric API allowed on Fabric only), Java 21, no runtime libraries, every model a `ModelPart`, every animation code-driven. Built on Enhanced's foundations: this document says, system by system, what is **copied** (same code, new package), what is **adapted** (same shape, element-specific content) and what is **new**.

0. The honest starting point: what Enhanced has today

Enhanced `main` (3.9.0, commit 888609c) has the **loader-neutral platform layer** of its port step P0: `platform/` (`Platform`, `Net`, `Attach`, `Registrars`, `EventHub` + `@Subscribe` processor, `ClientRegistrar`, `ClientServices`, `Keys`, `LoaderHooks`, `ItemAccess`, `Parts`, the ~80 common event classes), a complete **NeoForge implementation** (`platform/neoforge/*`, the bridge `GameBindings`/`ClientBindings`, `NeoTestHooks`), a `portLint` Gradle task that forbids NeoForge API outside `platform/neoforge/**`, the mod's own TOML config backend, and the Fabric **handshake** (`HelloPayload`). It has **no Fabric implementation, no Fabric Loom build, no universal jar**, on any branch; the "PLAN2" the code cites is not in the repository, and nothing documents how later Minecraft versions are added beyond the `r1` adapter naming and the per-`<mc>-<loader>` parity report.

So Bending's universal jar requires **building the Fabric half** (section 2.3), which either Enhanced's own P1 lane does first (and Bending copies it) or Bending's **P0 skeleton** does (and Enhanced copies it back). `PLAN.md` puts it in Bending's **release 1.0, lane L0-FABRIC**, with the NeoForge-only jar as the fallback milestone so Earth is never blocked by it. **Open question for Yuval:** share PLAN2 (and `SCRATCH/port`) if they exist; the plan below is written to match the platform layer as it stands.

1. Coexistence with Enhanced (both mods in one game)

Thing	Enhanced	Bending	Rule
Mod id / namespace	`enhanced`	`bending`	every id, texture, sound, tag, lang key, advancement, damage type is `bending:*`
Java package	`com.kolodkingal.enhanced`	`com.kolodkingal.bending`	no class name is shared on the classpath; the platform layer is a **copy**, not a dependency (two copies of `EventHub` coexist: each mod registers its own listeners on its own hubs; the NeoForge bridges each add their own native listeners)

Thing	Enhanced	Bending	Rule
Mixins	`enhanced.mixins.json`, package `…enhanced.mixin`	`bending.mixins.json`, package `…bending.mixin`	the same injection points on `PlayerModel.setupAnim` and `PlayerRenderer.setupRotations` are **additive** (`@Inject` at HEAD/TAIL, no cancels), so both animators run; Bending's `HeroPoses.apply` runs after Enhanced's and only touches a player who **has an element**, Enhanced's only a player with a power; a player with both gets Enhanced's pose then Bending's layered on (acceptable; the option `bending-client [animation] yieldToEnhanced` true makes Bending skip a player who has an active Enhanced gesture)
Payload channel	`enhanced:*`, version "14"	`bending:*`, version "1" (`ModNetwork.PROTOCOL_VERSION`), optional channel `bending:music` "1"	distinct `CustomPacketPayload.Type` ids; each mod's handshake is its own
Attachments	`enhanced:power`, …	`bending:bender`, `bending:victim`, `bending:temp_blocks`, `bending:world`	distinct keys; saved side by side on the player and the level
Config files	`enhanced-server.toml`, `enhanced-client.toml`	`bending-server.toml`, `bending-client.toml` (+ `<world>/serverconfig/bending-server.toml`)	the copied `Configs` backend writes its own files; the config screen lists its own
Key defaults	R G V B N H Y Z J (+ Ctrl chords, Alt/Del in build mode)	**X C Left-Alt K**, **O**	no default clash (1.5 of `DESIGN.md`); contexts `IN_GAME`
HUD	top-left panel	bottom-left panel (configurable corner)	no overlap by default
Creative tab	`enhanced`	`bending`	
Sound registry	`enhanced:*`	`bending:*`	
Soft hooks	—	`BendingCompat.enhanced()` (section 13)	Bending asks Enhanced's `Shelters/BossFights/Metal.isMetalHero` through reflection-free optional classes loaded only if `ModList/FabricLoader` says Enhanced is present

2. Platform and build

2.1 Copied from Enhanced as is (`com.kolodkingal.bending.platform.*`)

`Platform`, `Loader`, `LoaderHooks`, `ItemAccess`, `Parts`, `EntityAttributesCompat`, `CommonTags`; `net/Net`, `NetService`, `PacketContext`; `attach/Attach`, `AttachKey`, `AttachService`, `SyncTo`; `reg/Registrars`, `Registrar`, `RegEntry`, `ItemRegistrar/ItemEntry`, `BlockRegistrar/BlockEntry`, `ComponentRegistrar`, `RegService`; `ext/ItemExt`, `EntityExt`, `BlockEntityExt`; `event/*` (every common event, `EventHub`,

`Events`, `Subscribe`, `Subscriber`, `Priority`, `Side`, `Cancellable`, `TriState`) and `event/client/*`; `client/ClientRegistrar`, `ClientServices`, `Keys`, `KeyContext`, `KeyMod`, `HudHeights`, `BlockModels`; **the whole `platform/neoforge/*`** (`NeoPlatform`, `NeoSetup`, `NeoNet`, `NeoAttach`, `NeoPersistentData`, `NeoRegistrars`, `NeoHooks`, `NeoItemAccess`, `client/NeoClientRegistrar`, `client/NeoClientServices`, `bridge/NeoForgeBridge`, `Binding`, `GameBindings`, `ClientBindings`, `ClientConvert`, `ListenerOrderRecorder`, `ListenerOrderCommand`); the **annotation processor** (`processor/` → `SubscribeProcessor` generating `GeneratedSubscriptions`, with `tools/listener-order.txt` recorded once from Bending's own listeners); the **config backend** `config/core/*` (`ConfigSpec`, `ConfigFile`, `Configs`, `ConfigWatcher`, `Toml`, `ConfigSyncPayload`) and `config/neoforge/ConfigNeoForge`; the **GameTest adapter** `platform/test/*` (`EnhancedTest` → `BendingTest`, `TestHooks`, `TestIndex`, `TestFilter`, `ParityReporter`, `EnhancedTests` → `BendingTests`) and `platform/neoforge/test/*`; the `portLint` task (rules `neoforge`, `subscribe`, `hidden`, `impl`, plus a new rule **`fabric`**: `\bnet\.fabricmc\.` outside `platform/fabric/**`); `META-INF/services` files for the three SPIs. Only the mod id strings and the two `@Mod` entry points change.

2.2 The entry points

- NeoForge: `Bending` (`@Mod("bending")`: `NeoSetup.init(modBus)`, `NeoRegistrars.register(...)` for each registrar in order, `ModAttachments.init()`, `ModNetwork.register()`, `Configs.register(SERVER, BendingConfig.SPEC)`, `Configs.register(CLIENT, BendingClientConfig.SPEC)`, `ConfigNeoForge.install(modBus)`) and `client/BendingClient` (`@Mod(value="bending", dist=CLIENT)`: `ClientSetup.register(NeoClientRegistrar.install(container, modBus));` `ClientRegistrar.get().configScreen(ConfigScreen::new)`).
- Fabric: `platform/fabric/BendingFabric implements ModInitializer` and `platform/fabric/client/BendingFabricClient implements ClientModInitializer`, doing the same through the Fabric implementation (2.3), plus `BendingFabricGameTest` for the test adapter.
- `fabric.mod.json` (entrypoints `main`, `client`, `fabric-gametest`; `depends: fabricloader ≥ 0.16, fabric-api *, minecraft 1.21.1`; `mixins: bending.mixins.json`) and `neoforge.mods.toml` (as Enhanced's template) **both** in the jar's `META-INF`/root; each loader ignores the other's.

2.3 The Fabric half (NEW; lane L0-FABRIC of 1.0)

Class (`platform/fabric/…`)	Implements	How
`FabricPlatform`	`Platform`	`loader()` FABRIC; `isPhysicalClient` via `FabricLoader.getEnvironmentType()`; `configDir` `FabricLoader.getConfigDir()`; `modResource` via `ModContainer.findPath`; `currentServer` from a `ServerLifecycleEvents` listener; `fakePlayer` `FakePlayer.get(level, profile)` (Fabric API's `net.fabricmc.fabric.api.entity.FakePlayer`); `hooks()`, `itemAccess()`, `parts()`, `attributes()` below

Class (`platform/fabric/…`)	Implements	How
`FabricHooks`	`LoaderHooks`	`blockPlaceCancelled`: no Fabric place event exists: post the mod's own `BlockEvent.EntityPlaceEvent` on the hub **and** Fabric's `PlayerBlockBreakEvents`/claim mods' common pattern (`UseBlockCallback` is for interaction, not placement): call `BlockEvent.EntityPlaceEvent` on the hub only, and ask **Fabric Permissions API**-free claim mods through their own ("protection") events if present: documented limitation: on Fabric, claim mods that hook `PlayerBlockBreakEvents.BEFORE` are honoured for breaks (we post it); for places we post the hub event only. `blockBreakCancelled`/`blockBreakEventCancelled`: `PlayerBlockBreakEvents.BEFORE` + the hub event. `projectileImpactCancelled`: the hub event only (Fabric has none). `finalizeMobSpawn`: `mob.finalizeSpawn(...)` directly. `enderPearlTeleportCancelled`: hub only. `isShield`: `stack.getItem() instanceof ShieldItem`
`FabricItemAccess`	`ItemAccess`	Fabric Transfer API `ItemStorage.SIDED` wrapped to the item-handler-like methods
`FabricParts`	`Parts`	`EnderDragonPart` only (no generic multipart on Fabric)
`FabricAttributes`	`EntityAttributesCompat`	swim speed: none on Fabric (returns null → callers skip); creative flight: the mod's own attribute `bending:creative_flight` registered on Fabric and applied through `Abilities.mayfly` in `PowerTicker`'s equivalent (`BenderTicker.applyPassives` sets `mayfly` directly when the attribute is absent)
`FabricNet`	`NetService`	`PayloadTypeRegistry.playC2S()/playS2C().register(type, codec)` for each message and `ServerPlayNetworking.registerGlobalReceiver` / `ClientPlayNetworking.registerGlobalReceiver` with a `Context` wrapping the Fabric context; the version handshake: **`HelloPayload`** registered in the **configuration phase** (`PayloadTypeRegistry.configurationS2C/C2S`, `ServerConfigurationNetworking.registerGlobalReceiver`, a `ServerConfigurationNetworking.ConfigurationTask`): the server sends its `HelloPayload.local()`, the client answers, `HelloPayload.check` decides (the exact words NeoForge shows); optional channels: `ServerPlayNetworking.canSend(player, type)` for `hasChannel`; the senders through `ServerPlayNetworking.send` / `PlayerLookup.tracking/world/around`
`FabricAttach`	`AttachService`	Fabric API **Attachment API** (`AttachmentRegistry.create(id, b -> b.initializer(def).persistent(codec).copyOnDeath().syncWith(syncCodec, AttachmentSyncPredicate.all()/targetOnly()))`): the same semantics as NeoForge's (`SyncTo.SELF` → `targetOnly()`); `persistentData` → a `CompoundTag` attachment `bending:persistent_data`
`FabricRegistrars`	`RegService`	`Registry.register` at `onInitialize` in declaration order; `entityAttributes` via `FabricDefaultAttributeRegistry.register`; `spawnEgg` via `SpawnEggItem` (Fabric allows a deferred type through `FabricSpawnEggItem`… use vanilla `SpawnEggItem` with the type registered first: Bending registers entity types before items)
`FabricPersistentData`		the attachment mode only

Class (`platform/fabric/…`)	Implements	How
`client/FabricClientRegistrar`	`ClientRegistrar`	`EntityRendererRegistry.register`, `BlockEntityRendererFactories.register`, `EntityModelLayerRegistry.registerModelLayer`, `LivingEntityFeatureRendererRegistrationCallback.EVENT` (→ `addLayers` with a `LayerContext` over the `EntityRendererProvider.Context`), `ParticleFactoryRegistry`, `KeyBindingHelper.registerKeyBinding`, `HudRenderCallback` (→ `guiLayer`; anchors mapped to a draw order: Fabric has one callback, so the layers are drawn in registration order **after** vanilla's HUD; `BELOW DEBUG_OVERLAY` ≈ after everything but the debug screen: acceptable), `ColorProviderRegistry.ITEM`, `BuiltinItemRendererRegistry`, `DimensionRenderingRegistry.registerDimensionEffects`, `ResourceManagerHelper.registerReloadListener`, `ClientCommandRegistrationCallback`, `configScreen` → a **ModMenu** entry point if present (`modmenu` optional)
`client/FabricClientServices`	`ClientServices`	`newKey` → `KeyMapping` (Fabric has no conflict contexts or modifiers: `KeyContext` kept by the mod's own check in `ClientEvents.handleKeys`, `KeyMod` checked with `Screen.hasControlDown()` etc.); `matches` = `mapping.matches(key)` && the context's `isActive`; `hudLeftHeight` → the mod's own tally (Fabric has none: Bending reads vanilla's `Gui` fields via an accessor mixin); `renderBatched` → `BlockRenderDispatcher.renderBatched`
`bridge/FabricBridge`	the event bridge	Fabric's events are callbacks, not a bus with priorities: the bridge registers **one Fabric callback per bound event** that posts to the hub at **all priorities in order** (`EventHub.post(E)`); cancellation maps to the callback's return value (`InteractionResult.FAIL` / `false`); ~40 bindings for the events Bending uses (section 2.5), each a small adapter: `ServerTickEvents`, `ServerLivingEntityEvents.ALLOW_DAMAGE/AFTER_DEATH`, `PlayerBlockBreakEvents`, `UseBlockCallback`, `UseItemCallback`, `UseEntityCallback`, `AttackEntityCallback`, `ServerPlayerEvents.AFTER_RESPAWN/COPY_FROM`, `ServerEntityEvents.ENTITY_LOAD/UNLOAD`, `ServerPlayConnectionEvents.JOIN/DISCONNECT`, `EntityElytraEvents`, `ServerLifecycleEvents`, client: `ClientTickEvents`, `WorldRenderEvents.AFTER_PARTICLES/START`, `HudRenderCallback`, `ClientPlayConnectionEvents`, `ClientPreAttackCallback`; events with **no Fabric equivalent** (`LivingIncomingDamageEvent` amount changes → `ServerLivingEntityEvents.ALLOW_DAMAGE` + a **mixin** on `LivingEntity.hurt` for the amount; `ComputeFovModifierEvent` → a mixin on `AbstractClientPlayer.getFieldOfViewModifier`; `ViewportEvent.ComputeCameraAngles` → a mixin on `Camera.setup`; `RenderHandEvent`/`RenderArmEvent` → mixins on `ItemInHandRenderer`; `MovementInputUpdateEvent` → a mixin on `KeyboardInput.tick`; `ProjectileImpactEvent` → a mixin on `Projectile.onHit`; `LivingKnockBackEvent` → `LivingEntity.knockback` mixin; `MobEffectEvent.Applicable` → `LivingEntity.canBeAffected`) are **Fabric-only mixins** in `bending.fabric.mixins.json` (`platform/fabric/mixin/`), each posting the common event. This is the bulk of L0-FABRIC.
`FabricTestHooks`	`TestHooks`	`fabric-gametest` entry point: `FabricGameTest`-annotated generator class → `BendingTests.enhancedTests()`; `mockPlayerWithConnection` as Fabric's `GameTestHelper.makeMockServerPlayerInLevel` plus a `Connection` on an `EmbeddedChannel` (the NeoForge recipe, loader-free)

Fabric-side limitations documented in-game (a line in the Fabric config screen): claim mods are honoured for breaks and interactions, placements only through Bending's own event; swim speed passive uses an attribute Bending registers.

2.4 The universal jar (NEW; L0-FABRIC with the build owner)

Gradle **multi-project**: `common/` (compiles against vanilla via **VanillaGradle** (or NeoForge's ModDevGradle in "vanilla mode", `neoForge { version = … }` with `neoFormVersion`), Mojang mappings, `src/main/java` = everything but `platform/neoforge` and `platform/fabric`; the mixins; the processor applied here), `neoforge/` (ModDevGradle 2.0.148, depends on `common` as `compileOnly` + the `platform/neoforge` sources, `neoforge.mods.toml`, runs `runClient`, `runServer`, `runGameTestServer`), `fabric/` (**Fabric Loom** 1.8+, `mappings loom.officialMojangMappings()` (so the code is one mapping; Loom remaps the built jar to intermediary), depends on `common`, Fabric API, `fabric.mod.json`, runs `runClient`, `runServer`, `runGameTest`). **One jar**: the `universal` task **merges** the two loader jars with **Forgix**-style relocation (a build-time Gradle plugin `io.github.pacifistmc.forgix`, 1.2.x; **not** a runtime library) producing `build/libs/bending-<version>.jar` that contains both loaders' classes (the common code appears twice: once in Mojang names for NeoForge, once remapped to intermediary for Fabric, relocated to `…bending.neoforge.*` and `…bending.fabric.*`? **No**: Forgix keeps the packages and resolves the duplicate common classes by **keeping both loaders' trees under their own relocated roots and rewriting the metadata files to point at them**; the jar is about 2× a single-loader jar, acceptable). CI builds all three and tests both: `./gradlew :neoforge:runGameTestServer :fabric:runGameTest` and compares the parity reports `build/gametest/1.21.1-neoforge.json` and `1.21.1-fabric.json` (`ParityReporter` already names files per loader; a `tools/parity.py` diff fails the build on a test that passes on one loader only). **Cross-play**: both loaders speak the same `bending:*` payloads with the same codecs and the same `PROTOCOL_VERSION`; **saves move** because every attachment is the mod's own id with the mod's own `Codec` on both loaders (NeoForge's `AttachmentType` and Fabric's Attachment API both store under the mod's id in the entity's/level's NBT: NeoForge under `neoforge:attachments`, Fabric under `fabric:attachments`: **not the same NBT path**: a world moved between loaders would **lose** attachments). **Mitigation (NEW, S0 of 1.0):** `SaveBridge`: the mod also writes a mirror of each persistent attachment into the entity's/level's **persistent-data compound under its own key** (`bending:save` → the `CompoundTag` attachment `bending:persistent_data`, which is itself stored by each loader's attachment system... circular). **Correct design:** Bending stores its **player data in the level's own storage**, not in attachments: a `SavedData` (`BendingSavedData` in the Overworld's `data/bending_players.dat`, keyed by player UUID, `Codec`-encoded) for everything that must survive a loader move (element, SP, unlocks, mastery, loadouts, look, zaheer, severance timer, presets), and uses **attachments only for the synced runtime view** (`bending:bender` synced, not persisted, rebuilt from the saved data at login/respawn). `SavedData` is vanilla and identical on both loaders. The level-side data (`bending:world`: moon-bells state, severed timers of offline players, temp-block journal) is `SavedData` too (`data/bending_world.dat` per dimension). **This is the save format of section 5.**

2.5 Events Bending uses (what the bridges must bind)

Server: `ServerTickEvent.Post`, `LevelTickEvent.Post`, `PlayerTickEvent.Post`, `EntityTickEvent`, `LivingIncomingDamageEvent`, `LivingDamageEvent.Pre/Post`, `LivingDeathEvent`, `LivingKnockBackEvent`, `LivingFallEvent`, `LivingChangeTargetEvent`, `MobEffectEvent.Applicable`, `EntityJoinLevelEvent`, `EntityLeaveLevelEvent`, `PlayerEvent.PlayerLoggedIn/Out/Respawn/Changed Dimension/Clone`, `PlayerInteractEvent.RightClickItem/RightClickBlock/EntityInteract`, `AttackEntityEvent`, `BlockEvent.BreakEvent/ EntityPlaceEvent`, `ProjectileImpactEvent`, `EntityStruckByLightningEvent`, `EntityMountEvent`, `LivingEntityUseItemEvent`, `ItemTossEvent`, `RegisterCommandsEvent`, `ServerStarting/Started/Stopping/StoppedEvent`, `OnDatapackSyncEvent`, `LevelEvent.Load/Save/Unload`,

`EntityMobGriefingEvent`, `ExplosionEvent.Detonate`. Client: `ClientTickEvent.Post`, `InputEvent.Key/MouseButton.Post`, `MovementInputUpdateEvent`, `RenderLevelStageEvent`, `RenderPlayerEvent.Pre/Post`, `RenderHandEvent`, `RenderArmEvent`, `RenderGuiLayerEvent.Pre`, `ViewportEvent.ComputeCameraAngles/ComputeFogColor/RenderFog`, `ComputeFovModifierEvent`, `ClientPlayerNetworkEvent.LoggingIn/LoggingOut`, `SelectMusicEvent`, `ScreenEvent.Opening`, `FMLClientSetupEvent`-equivalent.

2.6 Later Minecraft versions

Enhanced's code names its GameTest adapter "the **r1** (1.21.1) adapter" and writes parity reports per `<mc>-<loader>`, which is the only trace of the method: **one release line per adapter**. Bending follows it explicitly: `common/` targets the lowest supported version; version-specific vanilla API differences live behind a tiny **`mc/` adapter package** (`platform/mc/McApi` with `r1` as the only implementation in 1.x: `getMoonPhase`, particle option constructors, `Camera` accessors, anything that changed name between 1.21.1 and 1.21.x), and a new Minecraft version is added as a **Gradle property matrix** (`-Pmc=1.21.4` selecting `neo_version`/`fabric_version`/`parchment`), a new `r2` adapter where something moved, and the parity reports `1.21.4-neoforge.json` / `1.21.4-fabric.json` compared against `1.21.1-*`. The universal jar is per Minecraft version (one jar per MC version, each universal across loaders), as every multi-loader mod does. **Confirm against PLAN2 if it exists.**

3. Packages

```
com.kolodkingal.bending
  Bending, BendingConfig (server spec: the element configs' define()), client/BendingClient,
client/BendingClientConfig
  platform/…                        (copied + fabric/)
  registry/   ModItems, ModBlocks, ModBlockEntities, ModEntities, ModDamageTypes, ModEffects,
ModSounds, ModParticles,
              ModTags, ModDataComponents, ModCreativeTab, ModAttachments, ModStructures
(suggestions), ModAdvancements
  bender/     Element (enum), BenderData (synced runtime view), BenderSave (SavedData record per
player), BenderProgress
              (SP, unlocks, mastery: pure), ElementStrength, Loadout, Looks (variants per
element), Look (record),
              Benders (get/set/sync helpers = Enhanced's Powers), BenderTicker (= PowerTicker),
AbilityHandler (key router),
              Ability (enum of every card: id, element, slot act, sneak act, unlock cost,
parent, gate), Mastery (curve),
              Cooldowns, Live (momentary bits), Toggle (saved bits), SkillTree (graph),
ZaheerPath, Severance
  ability/    AreaTargets, BendingRights, Raycasts, TempBlocks (+ TempBlock, TempBlockJournal),
Hazards (server-side
              objects base: Hazard, HazardTicker), Grip (= Telekinesis.hold for rocks/metal),
SourceFinder (water sources),
              MetalCharge, Victims (the bending:victim attachment writer)
  earth/      EarthServer (ElementServer plug-in), EarthAbilities, RockThrow, EarthArmour,
EarthPillar, EarthWall,
              SandBending, Quicksand, LavaBending, MetalBending (knives, wall, armour, wreck,
grip), EarthEvents, EarthFx
              (server-side payload sender)
  air/        AirServer, AirAbilities, AirBlast, AirShield, AirBlades, SpiritualProjection,
AirBubble, AirScooter, Tornado,
              ZaheerAbilities (flight, suffocation), Glider (item logic), AirEvents, AirFx
  water/      WaterServer, WaterAbilities, WaterSource, WaterShot, Healing, WaterOctopus,
WaterSpout, WaterArms, Ice
              (knives, freeze, wall), BloodBending, WaterSkinItem, WaterEvents, WaterFx
```

```
  fire/       FireServer, FireAbilities, Fireball, FirePunch, Flamethrower, FireWhip,
FireCircle, FireHealing, Lightning,
              LightningRedirection, FireEvents, FireFx
  entity/     RockEntity, SandStreamEntity, SteelKnifeEntity, LavaShurikenEntity,
AirBladeEntity, AirScooterEntity,
              TornadoEntity, SpiritEntity, WaterShotEntity, WaterSpoutEntity, IceKnifeEntity,
SnowBurstEntity, FireballEntity
  block/      QuicksandBlock, SteelPlateBlock, IceSpikeBlock, TempLavaBlock, WreckedBlock (+ 8
variants), TrainingDummyBlock (sugg.)
  item/       ElementScrollItem, WaterSkinItem, AirGliderItem, BendingScrollItem (sugg.)
  network/    ModNetwork, AbilityKeyPayload, BendingActionPayload
(unlock/buy/loadout/look/preset), ElementScreenPayload,
              ElementChoicePayload, EarthFxPayload, AirFxPayload, WaterFxPayload, FireFxPayload,
BeamPayload, SpiritPayload,
              GliderDyePayload, HelloPayload, ConfigSyncPayload, PayloadCodecs
  config/     EarthConfig, AirConfig, WaterConfig, FireConfig, ProgressionConfig,
MatchupsConfig, LimitsConfig, ExtrasConfig
  command/    BendingCommand (/bending element|points|mastery|unlock|look|zaheer|duel|debug)
  client/     ClientSetup, ClientEvents, KeyBindings, BendingHud, overlays (ArmourSlitOverlay,
SpiritOverlay, BloodOverlay,
              ParalysisOverlay, QuicksandOverlay, BubbleOverlay, HeatOverlay,
RedirectPathOverlay), SpiritCamera,
              InputLocks (= PinLock generalised), PowerLoopSounds, music/ (ElementMusic,
MusicSources…), screen/
              (ElementChoiceScreen, BendingScreen + tabs SkillTreeTab, LoadoutTab, LooksTab,
WaterTab, CreatorUi, EditorTab,
              Popup, PreviewPuppet), render/ (the copied animation + FX toolkit, ElementClient
plug-ins EarthClient/
              AirClient/WaterClient/FireClient with *Gestures/*Acting/*Fx, entity renderers,
layers, BendingWorldFx,
              VictimPoses, LayerReveal, ChainSim, LightningFx, FlameBeamDrawer), particle/
(EnergyParticle, GrainParticle,
              DropletParticle)
  mixin/      the copied client mixins (PlayerModelMixin, PlayerRendererMixin,
AbstractClientPlayerMixin, ItemInHandRendererAccessor,
              CapeLayerMixin, HumanoidArmorLayerMixin, LivingEntityRendererMixin [VictimPoses],
MinecraftMixin+LevelRendererMixin
              [EntityGlow], ParticleAccessor), new: HumanoidModelMixin, QuadrupedModelMixin,
CameraMixin, GuiAccessor;
              server: PlayerMixin (if needed), MobInvoker
  compat/     BendingCompat, EnhancedHooks (optional)
```

4. Registries (what S0 registers; the lanes fill behaviour)

Items: `element_scroll`, `water_skin` (durability-style bar via a `WATER` data component 0–8), `air_glider` (a `DYE` component: `DyeColor`), suggestions: `bending_scroll_earth|air|water|fire`, `training_dummy`.

Blocks (temporary states are real blocks with a `TempBlocks` ledger): `quicksand` (a `SoulSand`-like block: `getBlockSupportShape`, `entityInside` slows and sinks), `steel_plate`, `ice_spike` (3 `RISE` states), `temp_lava` (a `LiquidBlock`-look non-flowing block: it extends `Block` with lava's render and `entityInside` lava damage via `DamageSources.lava`, `lightLevel 15`, not a fluid: no spreading), `wrecked_hopper|rail|piston|redstone|iron_door| furnace|anvil|metal_block` (8 `WreckedBlock`s carrying the original state in their block entity `WreckedBlockEntity`: `original` + the original NBT), suggestions: `training_dummy`, `metal_dummy`. **Block entities:** `wrecked`, `training_dummy`.

Entities (all `MobCategory.MISC`, `noSummon` where internal):

Entity	Size	Tracking / interval	Saved	Notes
`rock` (`RockEntity` extends `TelekineticBlockEntity`-pattern)	0.9 (boulder 1.9)	10 / 1	no	carries a `BlockState`, held/launched, damage by speed
`sand_stream`	1.0	8 / 2	no	a cone hazard with a position; 10 ticks
`steel_knife`	0.25	6 / 5	no	sticks in targets (follows them for 3 s)
`lava_shuriken`	0.6	8 / 2	no	bounces once at M10
`air_blade`	0.9 wide 0.3 tall	8 / 2	no	passes one creature
`air_scooter`	1.0 × 0.9	10 / 1	no	a vehicle (`canAddPassenger`), steered by the rider's input (the client sends `Input` through vanilla's vehicle movement)
`tornado`	2.0 × height	12 / 2	no	carries riders (orbit), drifts, re-sent every 4 ticks
`spirit`	0.6 × 1.8	12 / 1	no	`noPhysics`, the camera entity of its owner
`water_shot`	0.4	8 / 2	no	
`water_spout`	2.0 × height	10 / 1	no	a vehicle
`ice_knife`	0.25	6 / 5	no	
`snow_burst`	0.25	6 / 5	no	a snowball variant
`fireball`	0.3–1.2	8 / 2	no	

Hazards that are **not entities** (server-side `Hazard` objects ticked by `HazardTicker` from `LevelTickEvent.Post`, Enhanced's `SandHazards` pattern): walls' lifetimes, pillars, quicksand patches, lava paths, ice discs, the Air Blast cone (one tick), Air Slice, Fire Circle rings, the Air Vacuum sphere, the Suffocation hold, the Blood hold, the Lightning charge/bolt, the redirection window, beams (`ContinuousBeam`: flamethrower, fire breath).

Damage types (`data/bending/damage_type/*.json`, all in the tag `bending:bending_damage`; the lightning ones also in `minecraft:is_lightning`; the fire ones in `minecraft:is_fire`; beams in `bypasses_cooldown` + `no_knockback`): `rock`, `boulder`, `earth_wall_push`, `shed_blast`, `pillar`, `sand`, `quicksand`, `lava_path`, `lava_shuriken`, `steel_knife`, `metal_fling`, `wreck`; `air_blast`, `air_shield_burst`, `air_blade`, `air_slice`, `tornado`, `scooter_burst`, `suffocation`, `air_vacuum`; `water_shot`, `water_whip`, `octopus`, `spout_ram`, `water_arm`, `ice_knife`, `ice_spike`, `snow`, `blood`, `blood_fling`; `fireball`, `fire_punch`, `fire_kick`, `flamethrower`, `fire_breath`, `fire_whip`, `fire_circle`, `lightning`, `lightning_arc`, `redirected_lightning`. Each has `death.attack.bending.<id>` and `.player` lines in both languages.

Effects: `paralysed` (no movement/attack/bending; the client locks input), `severed` (bending refused; duration = the timer), `chilled` (slowness + ×1.5 fire), `wet` (3 s; ×1.5 lightning), `zaheer_tether_cut`? (no: saved flag), `entombed`-like `stuck` (quicksand/frozen feet: no jump, speed ×0.2).

Particles: `energy` (copied), `grain`, `droplet`.

Tags (`data/bending/tags/`): block `earth`, `sand`, `metal`, `metal_devices`, `plants`, `unbendable`, `water_sources` (water, cauldron, snow layers); entity_type `bosses`, `no_breath`, `no_blood`, `fire_mobs`; damage_type `bending_damage`, `water_damage`, `fire_damage`, `lightning_damage`; item `metal_charge`

(ingots/nuggets → charge value in a data map? no data maps on Fabric: a `metal_charge_2` and `metal_charge_1` tag pair).

Data components: `water` (int 0–8), `glider_dye` (`DyeColor`).

Advancements (`bending:` tab): become each element, first unlock, mastery 10, full tree, Zaheer's path, a severance, redirect a bolt, the extras.

Creative tab `bending` with the three items.

5. Data and save format

5.1 `BenderSave` (vanilla `SavedData`, `data/bending_players.dat` in the Overworld; one record per player UUID)

```
Codec<BenderSave> = RecordCodecBuilder:
  element: Element (string; "none")
  points: int                              // unspent SP
  pointsEarned: int                        // lifetime SP (for stats)
  unlocked: List<String>                   // ability ids (bending ability id
"earth.rock_throw")
  mastery: Map<String, MasteryEntry{level:int, xp:int, bought:int}>
  loadout: List<String> (4 entries, "" for empty)
  presets: List<Preset{name:String, slots:List<String>}> (≤ 4)
  look: String ("" = default)
  gliderDye: String (DyeColor name)
  zaheer: boolean, zaheerAt: long
  severedUntil: long (game time, 0 = none), severedBy: UUID?
  elementChosenAt: long
  version: int (1)
```

Written on change (dirty) and on world save; read at login; **`BenderData`** (the synced runtime attachment, Enhanced's `PowerData` shape: `element`, `look`, `loadout` ids as ordinals of `Ability`, `toggles` (int bits), `live`/`liveHigh` (two longs of `Live` bits), `cooldownUntil[4]` + `sneakCooldownUntil[4]`, `armourHealth`, `metalCharge`, `shieldStrength`, `waterCharge` (the fullest skin, for the HUD), `spiritId`, `heldEntityId`, `points`, `elementStrength`, `severedUntil`, `zaheer`, the mastery levels as a `byte[]` by ability ordinal (for the HUD bars), `dirty`) is **not saved**: it is rebuilt from `BenderSave` at login/respawn/dimension change by `Benders.load(player)` and synced to everyone who sees the player (`SyncTo.ALL`) through the copied `Attach.builder("bender", BenderData::new).sync(STREAM_CODEC).build()`. Because the stream codec sends ordinals, **every ability of one version must be in the same build** (Enhanced's rule for `Power`); the saved ids are strings, so the order never matters on disk. The `/bending` commands edit `BenderSave` and call `Benders.load`.

5.2 `WorldSave` (`SavedData` per dimension, `data/bending_world.dat`)

`tempBlocks: List<TempBlock{pos, original:BlockState(+nbt), placed:BlockState, revertAt:long, owner:UUID, kind:byte}>` (the journal: reverted at load), `hazardsForRecovery` (none: hazards die with the server), `moonBellsHeard: Set<UUID>` (cosmetic), `offlineSevered: Map<UUID, long>` (so a severed player who logs out stays severed: also in `BenderSave`).

5.3 `bending:victim` (attachment on any `LivingEntity`, synced, not saved)

`kind: byte` (NONE, STAGGER, KNOCKDOWN, STUCK_SAND, STUCK_ICE, LIFTED, TUMBLE, CLUTCH, SPIN, SOAKED, CHILLED, PUPPET, PARALYSED, SEVERED), `until: long`, `caster: int` (entity id, for the puppet's claw-follow), `param: float`. Written by `Victims.mark(entity, kind, ticks, caster)`; cleared by TTL in `BenderTicker`'s entity tick (only entities with the attachment are ticked: a small set kept in `Victims`).

5.4 Config files

`config/bending-server.toml` (overridden by `<world>/serverconfig/bending-server.toml`; synced to clients in the configuration phase by the copied `ConfigSyncPayload`) with sections: `[bending]` (general: `allowWithEnhancedPower`, `pushPlayersWithoutPvp`, `pressExhaustion`, `holdExhaustionPerSecond`, `effectRadius` 64), `[progression]` (points per mob list `pointsPerMob = ["minecraft:zombie=1", …]`, `pointsFormulaDivisor` 20, `pointsCap` 50, `assistWindowTicks`, `freePoints` 3, `masteryXp` list, `masteryBuyCap` 7, `masteryBuyCost` list, unlock costs per ability `unlockCost.<id>`, gates `gate.<id>`), `[matchups]` (every factor of 1.7), `[limits]` (every cap of 1.8 and `ANIMATION.md` 13), `[earth]`, `[air]`, `[water]`, `[blood]`, `[fire]`, `[lightning]` (every number of sections 2–5 of `DESIGN.md`, keys lowerCamelCase as Enhanced: `rockDamage`, `rockCooldown`, `wallSeconds`, `armourShellHealth`, …), `[extras]` (the suggestions' switches, all `false` until built), `[music]`. `config/bending-client.toml`: `UI.md` 2.3.

6. Temporary blocks: `TempBlocks`

```
TempBlocks.place(ServerLevel, ServerPlayer owner, BlockPos, BlockState placed, int ticks, Kind
kind) -> boolean
TempBlocks.placeAll(level, owner, Map<BlockPos, BlockState>, ticks, kind) -> int placed   //
walls: all-or-nothing by the half rule
TempBlocks.revert(level, pos) / revertAll(level, owner, kind) / revertOldest(level)
TempBlocks.isTemp(level, pos) -> Optional<TempBlock>
TempBlocks.tick(level)   // LevelTickEvent.Post: reverts due entries; waits up to 200 ticks
while a player stands inside, then pushes them up
TempBlocks.onBroken(level, pos)  // BlockEvent.BreakEvent / a changed state: the ledger entry is
dropped (the original is NOT restored: whoever broke it gets nothing: temp blocks have no drops:
`WreckedBlock`/`SteelPlate`/`IceSpike`/`Quicksand`/`TempLava` loot tables are empty, and a temp
**earth** block (a real stone/dirt state) is placed with the ledger flag `noDrops`, which
`LivingDropsEvent`… no: `BlockDropsEvent` cancels drops for ledgered positions)
```

Rules: a cell is placed only if `BendingRights.mayChange(owner, level, pos)` (section 7); never over a block entity, bedrock, portal frames, end portal, reinforced deepslate, command blocks, `bending:unbendable`, or another mod's block unless it is in the tags; the **original state and block-entity NBT** are recorded (`original`, `originalNbt`) and restored exactly (`setBlock(pos, original, UPDATE_ALL | UPDATE_SUPPRESS_DROPS)`, NBT reloaded); **flags** on placement `UPDATE_CLIENTS | UPDATE_KNOWN_SHAPE | UPDATE_SUPPRESS_DROPS` (Enhanced's `ArenaLoans.FLAGS`) so pistons, water and redstone around are not triggered by the temp block beyond the visual update (no neighbour updates: a temp wall does not break a redstone line; it also means a temp wall does not update observers: fine). **Fluids:** a temp block placed in water records the water and restores it; a temp lava cell adjacent to water never makes stone (temp lava is not a fluid). **Budgets**: per player and global caps (`[limits]`); exceeding reverts the oldest of that player. **Journal**: every entry is mirrored into `WorldSave.tempBlocks` on placement and removed on revert (dirty flag; vanilla saves it with the level), and `ServerStarted`/first level tick reverts every entry still in the journal (crash recovery). **A temp block hit by an Enhanced power**: Enhanced breaks blocks through the loader's break

event or `destroyBlock`; both reach `onBroken` (`BlockEvent.BreakEvent` for the former; a `LevelTickEvent` check of `level.getBlockState(pos) != placed` for the latter, every 20 ticks per entry), which drops the ledger entry: the hole stays as Enhanced left it (a wall destroyed by Thor is destroyed; what was under it is gone too: accepted, documented).

7. `BendingRights` (Enhanced's `BlockRights`, adapted)

```
mayChange(ServerPlayer, ServerLevel, BlockPos)      // for temp blocks: vanillaRights &&
!protectedByOthers && postBreak-equivalent (BlockEvent.BreakEvent of the current state) &&
postPlace-equivalent (EntityPlaceEvent)
mayHurtBlock(...)                                    // for fires: + mobGriefing + [fire]
setsFires
vanillaRights(player, level, pos)                    // mayBuild, mayInteract (spawn protection,
border), blockActionRestricted
protectedByOthers(player, level, pos)                // EnhancedHooks (Hex wall, boss arenas,
Asgard, Mirror Dimension) if present
dimensionAllows(level, kind)                         // the 1.9 rules
```

Every ability that touches a block calls `TempBlocks.place` (which asks `mayChange`); fires call `mayHurtBlock`.

8. The element plug-in shape (Enhanced's `HeroServer`/`HeroClient`, adapted)

Server `ElementServer` (one per element, registered in `ElementServers`): `element()`, `onKey(player, data, slot, down, sneak)` (the router: `Loadout.abilityIn(slot)` → the ability's `press/hold/toggle` handler; Sneak selects the variant), `tick(player, data)`, `everyPlayerTick`, `applyBody` (passives, attributes: speed, armour, flight), `endAll` (element removed/severed), `stopHold(data)`, `onLogin/Logout/Death/Respawn/ChangeDimension/ServerStopped`, plus `onHit(attacker, target, event)` (for Fire Punch, Octopus, Water Arms: `LivingDamageEvent.Pre`), `onHurt(victim, event)` (armour shells, shields, redirection, passives). Abilities are **objects**: `interface AbilityLogic { void press(ctx); void release(ctx); void tick(ctx); void end(ctx); }` keyed by the `Ability` enum, with `AbilityContext(player, data, level, now, sneak, mastery, config)`; every number read from config through `ctx.config().get(...)` at use (never cached: Enhanced's rule). **Cooldowns** per slot in `BenderData.cooldownUntil[slot]` and `sneakCooldownUntil[slot]` (the Sneak variant shares the slot but has its own timer so the HUD can show both); `Cooldowns.start(data, slot, sneak, ticks)`; an ability that did nothing starts no cooldown (so no gesture lies: Enhanced's `AnimationSignalTests` rule).

Client `ElementClient` = Enhanced's `HeroClient` methods (`hudStatus`, `shieldStyle`, `eyes`, `stanceTargets`, `edges`, `takeoffGesture`, `lean`, `drop`, `ambient`, `aimed`, `predictedGesture(s, data, slot, now, sneaking)`, `registerGestures`, `cooldownGesture(slot, sneak, left)`, `flagGesture`, `liveGesture`, `grabGesture`, `flingGesture`, `startedElsewhere`, `gestureStarted`, `firstPerson`, `flightParticles`, `flightRibbons`, `tick`, `clear`) plus `onFx(payload)`; registered in `ElementClients`.

`AbilityHandler.onKey`: as Enhanced's: drops presses of a player who cannot act (dead, severed, paralysed, bloodbent, meditating body), always passes releases; calls `ElementServers.of(data.element()).onKey`.

9. Server/client split and sync, per effect kind

Effect kind	Server	Wire	Client
Player body state (toggles, holds, cooldowns, charges, held ids)	`BenderData` fields set by abilities	the synced attachment (every change batched into one sync per tick by `Benders.syncIfDirty`)	gestures from edges (`GestureRoutes`), stances from `Live`/`Toggle`, HUD
One-shot world FX (impacts, bursts, rings, stomps, forming, tells)	`ElementFx.send(level, near, kind, id, a, b, c, d)`	`<Element>FxPayload(byte kind, int id, Vec3 a, Vec3 b, int c, int d)` to players within `effectRadius` 64 (`Net.sendToPlayersNear`)	`<Element>Fx.handle` → a deterministic client simulation from `age` (Enhanced's `SandFx` pattern); gestures via `startedElsewhere` kinds
Continuing world objects with position (walls, pillars, quicksand, lava path, ice disc)	temp blocks (the world itself syncs them) + one FX payload at start for the dust/tell	blocks	the `BendingWorldFx` draws decorations (dust, collars, shimmer) from a client-side ledger fed by the payload and pruned when the block changes
Moving entities (rocks, blades, knives, balls, spouts, tornado, spirit, shots)	entities with `SynchedEntityData` for what the renderer needs (state, size, phase, rider)	vanilla entity sync	entity renderers; the tornado's funnel position is interpolated from 4-tick payloads (as Enhanced), the rest from entity positions
Beams (flamethrower, breath)	`ContinuousBeam` (copied)	`BeamPayload(shooter, style, range, on)` on start/stop and every 8 ticks	`BeamRenderer` with the flame drawer; the client raycasts the end itself

Effect kind	Server	Wire	Client
Holds with a victim (blood, suffocation, metal grip, puppet)	`Live` bit on the caster + `bending:victim` on the target	both synced attachments	threads drawn between caster and victim; `VictimPoses`; the victim's input lock from its own attachment
Shields (air shield, armour shell health)	`BenderData.shieldStrength` / `armourHealth`	the attachment	`ShieldStyle` dome / the armour layer's cracks; hit ripples via FX payload kinds `SHIELD_HIT`/`ARMOUR_HIT`
Lightning	the bolt is instantaneous on the server (raycast, damage)	`FireFxPayload BOLT(from, to, c=branches seed, d=flags)` to 64 blocks; `THUNDER` to 24–64	`LightningFx` 8-tick bolt, flash, thunder delayed by distance
Spirit	`SpiritEntity` owned by the player; the server tells the client to possess it	`SpiritPayload(POSSESS/RELEASE, entityId)`	`SpiritCamera` sets `mc.setCameraEntity(spirit)`; input steers the spirit (the client sends vanilla movement for the spirit as a "vehicle-like" controlled entity: the spirit is a `PlayerRideableJumping`-free `Entity` whose motion is sent through a `SpiritMovePayload(Vec3 wanted)` 20/s, the server moves it within range and validates speed)

Effect kind	Server	Wire	Client
Rides (scooter, spout)	vehicles	vanilla passenger sync + input via `Serverbound PlayerInputPacket` (vanilla sends the rider's input for vehicles)	vehicle controlled client-side as a boat is (the entity's `controlVehicle`)
Screens	`ElementScreenPayload`, `BendingActionPayload` (serverbound: UNLOCK/BUY_MASTERY/LOADOUT/PRESET/LOOK/CHOOSE), results through the attachment		
Sounds	`level.playSound` (vanilla sync) for one-shots; loops client-side from `Live/Toggle` predicates (`PowerLoopSounds`) and entity-bound loops	vanilla	
Config	the server's TOML text in the configuration phase (`ConfigSyncPayload`)		the client reads the same values (prediction uses them)

Prediction: the client plays a gesture at once on a key press when its own `BenderData` and the synced config say the ability will work (`predictedGesture`), swallowing the server's edge within 12 ticks (copied `GestureTable` logic); refused uses play out harmlessly.

10. Network: `ModNetwork`

One required channel `PROTOCOL_VERSION = "1"` (bumped once per release: "2" in 1.1, "3" in 1.2, "4" in 1.3: a 1.0 client cannot join a 1.1 server), one optional music channel. Messages: serverbound `ability_key(int slot, boolean down, boolean sneak)`, `bending_action(byte action, String id, List<String> slots, String name)`, `element_choice(Element)`, `spirit_move(Vec3)`, `glider_dye(DyeColor)`; clientbound `element_screen()`, `earth_fx`, `air_fx`, `water_fx`, `fire_fx` (the `(byte kind, int id, Vec3 a, Vec3 b, int c, int d)` record; kinds per element in ranges: Earth 1–99, Air 100–199, Water 200–299, Fire 300–399, shared 400+), `beam(int shooter, String style, float range, boolean on)`, `spirit(byte op, int id)`, `config_file(String name, String text)` (configuration phase), `hello` (configuration phase, Fabric only). Every handler is **one static call into a client class** (Enhanced's rule, so the dedicated server never loads it).

11. Key bindings (`client/KeyBindings`)

`SLOTS[4]` = `Keys.newKey("key.bending.slot_" + n, KeyContext.IN_GAME, KEYSYM, {X, C, LEFT_ALT, K}[n], "key.categories.bending")`, `MENU` = O. `ClientEvents.handleKeys` is Enhanced's (press-time Sneak capture, `consumeClick`, `SENT_DOWN`, prediction), with `MENU` opening `BendingScreen` and Sneak + `MENU` sending `bending_action(LOOK_NEXT)`.

12. Commands

`/bending element get|set <players> <element|none>`, `points get|give|set <players> <n>`, `mastery get|set <player> <ability> <level>`, `unlock <player> <ability|all>`, `loadout <player> <slot> <ability>`, `look <players> <look>`, `zaheer open|reset <player>`, `sever <player> <seconds>|restore <player>`, `temp revert_all [radius]`, `debug listener_order|anim|fx`, suggestions: `duel`, `dummy`. Operators only (level 2) except `/bending help`.

13. Enhanced compatibility (`compat/`)

`BendingCompat.enhanced()` is true when the mod list has `enhanced`; `EnhancedHooks` (loaded only then, in its own class so the JVM never resolves Enhanced's types otherwise) asks: `Shelters.keepsOut(level, actor, from, to)` before an ability's line crosses a Hex/ward/barrier (the ability stops there, as Enhanced's beams do), `BossFights.protects(level, pos)` in `BendingRights`, `Metal.isMetalHero(player)` ← Bending **answers** it for Steel Armour wearers through a registered predicate if Enhanced exposes one (today it is a seam inside Enhanced: a one-line `request(I)` to Enhanced to add `Metal.addMetalRule(Predicate<Player>)`; until then, no interop), `Powers.power(player) != NONE` for `allowWithEnhancedPower`. Enhanced never needs Bending.

14. Test strategy

14.1 GameTests (the copied adapter; `@BendingTest(template, batch, timeoutTicks)`; `TestKit` copied with `Fixture`,

`player(x,y,z, Element)`, `press(player, slot, sneak)` **added** as a kit helper (`AbilityHandler.onKey` down+up with the Sneak flag), `hold(player, slot, ticks)`, `tickBender(player)` (`player.doTick()` + `BenderTicker.tick`), `config`, `pvp`, `everyTick`, `heardSounds`, `sentPackets`, `particlesIn`, `tempBlocksOf(player)`, `moon(level, phase, night)`)

Shared classes: `SkeletonTests` (the ability list and order, `PROTOCOL_VERSION`, the plug-ins, every ability has an icon/lang/damage type, no lang sentinels, `Live` bits fit), `DataTests` (both lang files equal keys and placeholders, every sound has a file and subtitle, every look a 64×64 skin, every item/block/effect/damage type named), `SoundAssetTests`, `AnimationMathTests` (+ the new leg-z/roll rules), `AnimationSignalTests` (a move with nothing to act on starts no cooldown; a working move starts it and raises its `Live`), `NetworkTests` (hello, caps), `PlatformSeamTests`, `ConfigFileTests`, `TempBlockTests` (place/revert/journal recovery/caps/rights/fluids/block entities refused/no drops/standing player pushed up), `RightsTests` (spawn protection, border, Adventure, a listener cancelling the break event, dimension rules), `ProgressionTests` (SP per mob, assist window, no SP from players, formula for unknown mobs, unlock gates, mastery XP curve, buying caps, ES), `LoadoutTests`, `MatchupTests` (each factor), `AreaTargetsTests`, `VictimTests` (attachment TTL).

Per ability (one class per element lane; each ability at least these, in the lane's batch `earth`, `earth_blocks`, `earth_pvp`…; state-heavy ones in a batch of their own with `bossFixture`-style `fixture` that clears hazards):

Ability	Tests (names say what they check)
Rock Throw	`aRockLiftsFromEarthAndHurtsAZombie`, `noEarthMeansNoRockAndNoCooldown`, `masteryFourThrowsTwoRocks`, `masteryTenBoulderOnDoubleTap`, `theHoleRevertsAfterThreeSeconds`, `holdRockFollowsTheLookAndThrows`, `holdRockSetsDownWithoutDrops`, `aRockShattersOnAnEarthWall`, `aRockNeverBreaksGlass`, `pvpOffSparesPlayers`, `aClaimRefusesTheLift` (an `EventHub` `BreakEvent` listener cancels), `underWaterIsRefused`, `aBossTakesDamageAndLittleKnockback`
Earth Armour	`formsOnEarthInTwentyTicksAndLocksInput`, `theShellTakesHitsFirst`, `speedIsReduced`, `crumblesUnderWater`, `shedBlastHurtsAndPushes`, `lightningHalvedWaterTimesOnePointFive`, `masteryEightSeismicSenseGlows` (the glow predicate), `masteryTenReformsOnce`, `cooldownAfterBreak`, `wardenSonicBoomBypasses`
Earth Pillar	`risesUnderTheFeetAndCarriesThePlayer`, `stopsUnderACeiling`, `sinksAfterFifteenSeconds`, `aimedPillarAtMasteryFive`, `launchThrowsUpAndSparesTheFall`, `notOnEarthRefused`, `claimRefused`
Earth Wall	`risesInFourTicksThreeByTwo`, `sevenByFourAtMasteryTen`, `stopsArrowsAndRocks`, `sinksAfterTenSeconds`, `aPlayerInTheFootprintIsPushedOut`, `halfRefusedMeansNoWall`, `wallPushSlidesAndHurts`, `aNewWallReplacesTheOld`, `blockEntitiesAreNeverReplaced`, `waterUnderItIsRestored`
Sand Bending	`sandStreamBlindsAndHurts`, `needsSandOrInventorySand`, `sandWallLastsSixSeconds`, `waterEndsASandWall`, `soulSandCountsInTheNether`
Quicksand	`sinksAZombieForFourSeconds`, `failsOnAStrongerEarthbender`, `tieFavoursTheDefender`, `neverABoss`, `patchRevertsAndFrees`, `pillarEscapes`, `sandCloudBlinds`
Lava Bending	`pathRunsThroughEarthOnly`, `stopsAtWaterAndClaims`, `theBenderWalksTheirPathUnharmed`, `pathCoolsAndReverts`, `shurikenHurtsAndBurns`, `twoDiscsAtMasteryEight`, `passiveResistanceByMastery`, `lavaImmunityTenSecondsAtTen`, `itemsNeverBurn`, `netherFreeBesideLava`
Steel Knives / Metal Pull	`threeKnivesAtOneSevenAtTen`, `knivesCostChargeAndReturn`, `pullFillsChargeFromOresWithoutRemovingThem`, `pullTakesIngots`, `noChargeNoKnivesNoCooldown`, `knivesStickAndRecall`, `earthWallStopsKnives`
Steel Wall / Steel Armour	`plateWallFromCharge`, `chargeReturnsWhenItEnds`, `explosionBreaksAPlate`, `armourConductsLightning`, `armourAndEarthArmourExclude`, `masteryTenDoorOpensForTheOwner`
Wreck Metal / Metal Grip	`aHopperIsWreckedAndPassesNothing`, `aRailDerails`, `contentsSurviveTheWreckAndMend`, `mendsAfterSixtySeconds`, `theTagDecidesTheList`, `threeByThreeAtFive`, `gripLiftsAnArmouredZombie`, `gripRefusesUnarmouredPlayers`, `gripRefusesBosses`, `sheddingArmourFrees`, `flingHurts`
Air Blast	`pushesInAConeByMastery`, `pushesPlayersWithoutPvpWhenAllowed`, `deflectsArrowsAndFireballs`, `extinguishesFires`, `airJumpLaunchesAndSparesTheFall`, `blastWhileGlidingLiftsAtMasterySix`, `halfRangeUnderWater`, `bossesBarelyMove`
Air Shield	`flingsProjectilesAndPaysStrength`, `meleeGoesThroughAtThreeQuarters`, `breaksAndBurstsForThreeAndAHalfHearts`, `theBurstOptionIsFour`, `loweringNeverBursts`, `manualBurstAddsLeftover`, `refusedInWater`, `tornadoCancelsIt`, `masteryTenDeflectsLightningOnce`
Air Glider	`onlyAnAirbenderOpensIt`, `glidesSlowerThanElytra`, `dyeRecolours`, `closesInWater`, `blastDownLaunches`
Air Blades	`twoAndAHalfHearts`, `passesOneCreature`, `cutsPlantsWhereAllowed`, `twoBladesAtFourThreeAtEight`, `wallsStopIt`, `airSliceSweeps`
Spiritual Projection	`theBodySitsAndTheSpiritFlies`, `rangeByMastery`, `damageToTheBodySnapsBack`, `durationEnds`, `cannotCrossDimensions`, `theSpiritCannotInteract`, `logoutEnds`, `deepMeditationHeals`, `masteryTenShieldsTheBody`

Ability	Tests (names say what they check)
Air Bubble	`breathesUnderWater`, `coversAFriendAtSix`, `cushionRemovesFallDamage`, `popsUnderASpout`
Air Scooter	`mountsAndMoves`, `speedByMastery`, `climbsOneBlockSteps`, `crossesWater`, `wallRideAtSix`, `passengerAtEight`, `burstAtTenOnDismount`, `waterShotKnocksOff`
Tornado	`liftsAndHurts`, `twoAtEight`, `steeringAtFive`, `wallsStopIt`, `bossesNeverLifted`, `liftsNoBlocks`, `itemsSurvive`, `whirlpoolUnderWater`
Zaheer's Path	`aPetDeathOpensIt`, `onlyNearbyWithTheOption`, `flightWorks`, `suffocationHurtsUntilDistance`, `breathlessMobsImmune`, `damageBreaksTheHold`, `airVacuumSnuffsFire`, `pushLessHitHarder`, `resetCommand`
Water Shot	`needsASource`, `theSkinIsUsedWhenNoWater`, `plantsAtMasterySevenDry`, `oneAndAHalfHearts`, `timesOnePointFiveOnFirebenders`, `extinguishesAndWets`, `whipLashesAndPulls`, `twoStreamsAtFour`, `lobAtTen`, `rainIsASource`, `netherSkinOnly`
Healing	`healsAFriendByMastery`, `healsSelfWithNoTarget`, `movementBreaksIt`, `removesPoisonSlowly`, `cleanseAtOnce`, `neverHostiles`, `fullHealthCostsNothing`
Water Octopus	`formsWithThreeCharges`, `slapsProjectiles`, `extraArmsOnHits`, `lashPulls`, `flamethrowerEvaporatesArms`, `freezeStiffens`, `lightningTimesOnePointFive`, `dropsWithoutASource`
Water Spout	`tinyAtOneTallAtTen`, `ramHurts`, `overLandItShrinks`, `lightningCollapsesIt`, `freezeAtEight`, `waveRideCarries`, `needsWaterNotSkinBelowFive`
Water Arms	`reachAndDamageByMastery`, `noItemsInHand`, `grabPulls`, `climbAtFour`, `flamethrowerDropsThem`, `freezeStiffens`
Ice	`knivesSlowAndShatter`, `fireMeltsKnivesInFlight`, `spikesEruptAndMelt`, `freezeMakesABridgeThatMelts`, `freezeRootsWetFeet`, `freezeStiffensConstructs`, `snowVolley`, `iceWallAtTheMidpoint`, `fireMeltsFourBlocksPerBall`, `iceDomeCloses`, `netherHalfLife`
Blood Bending	`onlyOnAFullMoonBelowTen`, `atTenAnyNight`, `noMoonInTheNetherBelowTen`, `failsOnAStrongerWaterbender`, `puppetMovesWithTheLook`, `holdTimeByMastery`, `flingHurts`, `lightningBreaksTheHold`, `twoHeartsBreakTheHold`, `undeadImmune`, `bossesImmune`, `twoTargetsAtTen`, `masteryXpTimesFour`
Bending Severance	`needsMasteryTen`, `upClose`, `durationByStrength`, `theSeveredCannotBend`, `survivesLogout`, `healingRestoresEarlyWhenAllowed`, `expires`
Fireball	`twoHeartsAndBurning`, `sizeByMastery`, `twoAtFiveThreeAtEight`, `greatFireballAtTen`, `smallFireballFromSix`, `waterBurstsIt`, `refusedInWater`, `rainReduces`, `netherBoosts`, `fireMobsImmune`, `noFiresWithoutTheOption`
Fire Punch	`fistsAddDamageAndBurn`, `ignoresSomeArmour`, `comboAtFour`, `jabAtSeven`, `kickArcs`, `noItemsWhileOn`
Flamethrower	`twoJetsHurtEveryHalfSecond`, `rangeByMastery`, `meltsIce`, `breathAtSevenLocks`, `waterShotPutsItOut`, `vacuumKills`, `exhaustion`
Fire Whip	`strokesEveryTwelveTicksAndIgnite`, `lengthByMastery`, `blockFiresOnlyWhereAllowed`, `cutsARockAtSeven`, `wrapsAtTen`, `lashCracks`, `waterWhipCancels`
Fire Circle	`ringExpandsAndHurts`, `legsFromSix`, `waterCutsAGap`, `meltsIce`, `jetLiftsAndSteers`, `jetBreaksAGrip`, `doubleAtTen`
Fire Healing	`slowerThanWater`, `poisonTicksSlower`, `burnOutCleansesAndHurts`
Lightning	`chargeLocksAndFires`, `fourAndAHalfHeartsAndParalysis`, `dualAtTen`, `wetAndMetalTimesOnePointFive`, `earthArmourHalves`, `rodsAttract`, `inWaterShocksTheCaster`, `chargedCreeper`, `noVanillaBoltNoFireNoWitch`, `arcJab`, `cancelOnRelease`

Ability	Tests (names say what they check)
Lightning Redirection	`chanceByMastery` (seeded RNG), `successReturnsFullDamageAtTheLook`, `failureHurtsAndDrops`, `windowTenTicks`, `autoFireAtEight`, `groundItHeals`, `steelArmourAlwaysFails`, `vanillaBoltRedirectable`, `twoBoltsAtTen`

Cross-element: `MatchupTests` (every row of 1.7), `CrossElementTests` (a rock vs an air shield, a tornado vs a spout, a fire circle under a tornado, water on a lava path, lightning on a steel wall, blood vs blood), `CoexistenceTests` (two benders' caps are independent; four benders' entity and temp-block counts stay under the limits in a scripted fight).

Enhanced's rules carry over: one batch per lane; a test that changes a server setting (`f.config`, `f.pvp`, the moon) in a batch of its own; keep flying things inside the 21-block arena; `fixture` clears every static (`Hazards.clear`, `TempBlocks.clear`, `Victims.clear`, `ContinuousBeam.clear`, `ElementServers` statics); delete `run/world` and `run/config/bending-server.toml` before a full run; the success line `All N required tests passed`; CI 45 min.

14.2 The real-client check list (manual, per lane, Enhanced's procedure)

A dedicated server + **two clients** (one English, one Hebrew), the second client **watching** what others see, **3+ render rounds** per lane, screenshots named `<lane>/<ability>-<view>-<lang>.png` kept in `docs/screenshots/qa/` (not shipped), with F2:

1. Each ability, third person from the second client: wind-up, release, recovery frozen with `/bending_anim target <p>`
   - `play <gesture> <tick>` at 3 ticks (start, peak, end); walking during the cast; the Sneak variant.
2. First person: the hands at the same three ticks; the overlay (slit, spirit, blood, paralysis, bubble, heat, redirect path); camera shake and FOV with and without *Reduced motion*.
3. Effects at 1, 8, 24 and 48 blocks; with *Effect density* 0.25; vanilla Particles "Minimal"; at night and in rain.
4. Idle stance (key held), the ambient master effect (set ES with `/bending mastery`), the run character, landing.
5. Looks: every look front/back/side, with the armour/octopus/glider over it, the look picker's faces.
6. Screens: every tab in both languages, narrow window (854×480) and wide; the Element Choice with 1–4 keys; Hebrew right-to-left checks (key names, numbers, units).
7. HUD: every status line, the Sneak swap, cooldown sweeps, the water gauge in each source, the moon on a full-moon night (`/time set 18000` on phase 0: `/bending debug moon 0`), spirit mode, severed, paralysed; with Enhanced installed too (a player with a power and an element: both panels, no overlap).
8. Sounds: every event audible at 1, 8, 24 blocks; loops start/stop/follow; the music themes and stems; subtitles in both languages.
9. Cross-loader: the same checks with the **Fabric client on the NeoForge server and the NeoForge client on the Fabric server**; a world saved on one loader opened on the other keeps every bender (the `SavedData`).
10. Four-bender fight for 2 minutes: F3 frame time and entity/particle counts (the budget of `ANIMATION.md` 13).

14.3 CI

`.github/workflows/build.yml` as Enhanced's: `./gradlew build` (runs `portLint` for both loaders), `:neoforge:runGameTestServer`, `:fabric:runGameTest`, `tools/parity.py build/gametest/*.json` (fails on a loader-only failure), upload the universal jar, release on `main`. `wiki.yml` as Enhanced's for `docs/wiki/`.

PLAN.md — Build plan with lane briefs

Bending — Build Plan

The method is Enhanced's (3.0 to 3.8): a **skeleton step S0** does every shared edit once (registries, payloads, attachments, config options, lang keys in both languages, sound placeholders, tags, the plug-in stubs, the seams with neutral stubs, a protocol bump pinned by a test), then **lanes** work in parallel, each in its own **git worktree and branch** from the skeleton, each **owning its packages** and writing into shared files **only in its own region** (`// ===== <lane> (1.x) =====`), never editing a lang file, a config class, `ModNetwork`, a payload record, `ModAttachments`, `ModTags`, `ClientSetup`, `TestKit` or `DataTests` outside its region (a needed change is a `request(I)`: a ten-line commit for the seam keeper, merged within the day). A **seam keeper** (one agent, the skeleton's owner) answers requests and keeps the skeleton branch green. An **integrator** merges the lanes into the release branch in a fixed order, does the cross-lane joins, runs the lanes' batches and then the **full suite**, and ships. The **gate** for a lane to merge: the build is green (`portLint` both loaders), its tests pass twice in a row on both loaders, its lang block is complete in both languages, its real-client checklist is done by someone other than the author (the integrator or a sibling lane), it edited no file outside its ownership, no existing test changed. The gate for a release: the full suite green on both loaders with matching parity reports, the real-client check of every lane, the universal jar built and run on both loaders, and the per-release checklist below.

Conventions in every brief: the design is `DESIGN.md` (D), the animation bible `ANIMATION.md` (A), looks `LOOKS.md` (L), screens `UI.md` (U), sounds `AUDIO.md` (S), architecture `TECH.md` (T). Numbers in a brief come from those files; a lane never invents a number: it reads the config option the skeleton defined.

Estimates are in **agent-days** (one autonomous build agent, one day ≈ a long session with tests) and calendar days with the lanes in parallel. Enhanced's releases (8 lanes, ~450 lane tests, ~2,500 total) took about two to three weeks each in calendar time with the lanes parallel; the estimates below assume that pace.

Release 1.0 — Core + Earth (כשפות אדמה)

Contents: the repository, the platform (both loaders, the universal jar), the bender data model, progression, the loadout and keys, the Element Choice, the Bending screen, the HUD, the animation and FX toolkits, temp blocks and rights, the audio pipeline, **every Earth ability** with its gestures, effects, sounds and looks (Toph default), GameTests and the real-client check. Network protocol **"1"**.

S0-1.0 — the skeleton (seam keeper; 3–4 agent-days; branch `v10-skel`)

1. **Repository**: `bending` with `common/`, `neoforge/`, `fabric/`, `processor/`, `tools/`, `docs/` (this plan's eight documents under `docs/plan/`, `docs/DESIGN.md` = D rewritten as the living design, `docs/wiki/` Home EN/HE), the CI workflows, `.gitattributes`, the license line as Enhanced's.
2. **Platform copy**: every class of T 2.1 copied from Enhanced with the package and mod id changed; the processor; the config backend; the GameTest adapter; `portLint` with the `fabric` rule; `META-INF/services`; the NeoForge entry points and a **stub** `platform/fabric/`

(`FabricPlatform` etc. as compiling no-op shells throwing `UnsupportedOperationException` with "L0-FABRIC fills this") so the Fabric subproject compiles from day one.
3. **Data model**: `Element`, `Ability` (every card of all four elements, all releases: ids, elements, acts, costs, parents, gates: so later releases never reorder), `BenderData` + `BenderSave` + `WorldSave` + `bending:victim` (T 5), `Benders`, `BenderTicker`, `AbilityHandler`, `Loadout`, `Mastery`, `ElementStrength`, `SkillTree`, `Cooldowns`, `Live` (every bit of all releases, in element regions), `Toggle`, `Severance` and `ZaheerPath` as data holders with no behaviour.
4. **Registries**: every item, block, block entity, entity type (**all releases**, with placeholder entity classes that do nothing: so ordinals never move), damage type JSON, effect, sound event (**232**, with `tools/sound/placeholder.ogg`), particle type, tag file, data component, advancement JSON, creative tab; `ModAttachments`; `ModNetwork` with **every payload of 1.0** (handlers calling lane stubs) and protocol "1".
5. **Config**: `BendingConfig` with every section and **every option of all four elements** (D's numbers as defaults, keys named in T 5.4), `BendingClientConfig` (U 2.3); every option's comment names its unit.
6. **Lang**: every key of 1.0 in `en_us.json` and `he_il.json` (elements, abilities with Sneak variants, HUD, screens, messages, subtitles for all 232 sounds, items, blocks, effects, death messages, config translations, key names), with a region per lane marked by sentinel keys `"_lane.earth": ""` removed at release (Enhanced's 3.0 rule) and `tools/add_lang.py` to merge blocks by union.
7. **Plug-in stubs**: `EarthServer`/`EarthClient`, `AirServer`/`AirClient`, `WaterServer`/`WaterClient`, `FireServer`/`FireClient` registered in `ElementServers`/`ElementClients`, every method a no-op; `ElementFx` senders; `TempBlocks`, `BendingRights`, `Hazards`, `Grip`, `AreaTargets`, `Raycasts`, `SourceFinder`, `MetalCharge`, `Victims` as **interfaces with neutral implementations** where another lane fills them (the seams), each pinned by `SkeletonTests.theSeamsAreNeutralUntilTheLanes`.
8. **Client copy**: the animation system, the FX toolkit, the UI kit, `PowerLoopSounds`, overlays' base, `ClientSetup` with a region per lane, `KeyBindings`, `ClientEvents`, `BendingHud` skeleton (header + 4 empty slots), `BendingScreen` with four empty tabs, `ElementChoiceScreen` shell; `GestureId`/`Stance`/`RunCharacter` enums with **every constant of all releases** in regions (Enhanced appends only; here they are all present from S0 so the lanes never collide); `GestureRoutes` arms per lane.
9. **Tests**: `TestKit` with the new helpers (T 14.1), `SkeletonTests`, `DataTests`, `SoundAssetTests` (with the 1.0 placeholder list), `NetworkTests`, `PlatformSeamTests`, `ConfigFileTests`, `AnimationMathTests`, `AnimationSignalTests` (empty regions per lane), `tools/parity.py`. **The full skeleton run passes on NeoForge** (`All N required tests passed`, N ≈ 120) and compiles on Fabric.
10. **Docs**: `docs/HEROES-AND-BOSSES.md`'s equivalent `docs/LANES.md`: the regions, the seams, the rules, this release's lanes and merge order.

Lanes of 1.0 (in parallel from `v10-skel`)

Lane	Owns	Agent-days	Depends on
L0-FABRIC	`platform/fabric/**`, `fabric/` build, `bending.fabric.mixins.json`, the `universal` task, CI's Fabric job	6–8	none (the biggest risk: start first)

Lane	Owns	Agent-days	Depends on
L1-CORE	`bender/**` behaviour (progression, SP on kills, mastery XP, unlock/buy, loadout, ES, severance timer, Zaheer flag), `ability/AreaTargets`, `Raycasts`, `BendingRights`, `TempBlocks` (+ journal, recovery), `Hazards`, `Grip`, `Victims`, `command/`, `item/ElementScrollItem`, `compat/`	4–5	none
L2-EARTH	`earth/**`, `entity/RockEntity`, `SandStreamEntity`, `SteelKnifeEntity`, `LavaShurikenEntity`, `block/Quicksand`	SteelPlate	TempLava
L3-ANIM	the animation core additions (A 14.1, 14.2, 14.3, 14.4: leg z keys, groups RIDE/HIT/AUTO and 4 slots, channel masks and body ownership, camera drop/eye offsets), `VictimPoses` (stagger, knockdown, stuck, lifted, soaked, chilled: A 12.1–12.3) with `HumanoidModelMixin/QuadrupedModelMixin/LivingEntityRendererMixin`, shared stances (`LAUNCHED`, `LIFTED`), run characters, `LAND_SOFT` variants, the awakening gestures (A 11.5), `AnimDebugCommands/AnimStateOverlay` wiring, `AnimationMathTests` extensions	4–5	none
L4-FX	`GrainParticle`, `DropletParticle` (A 14.5), `BendingRenderTypes` (`MATTER_*`), `BendingWorldFx` (the four-batch world renderer with the client ledger), `MatterShapes`, `ChainSim` (A 14.12, used from 1.2 but built now: cheap), camera shake service, overlay base classes, `EntityGlow` wiring, the fake-light option stub, `EffectDensity`	3–4	none
L5-UI	`ElementChoiceScreen`, `BendingScreen` + `SkillTreeTab/LoadoutTab/LooksTab/WaterTab` (the Water tab hidden until 1.2), `BendingHud` (every status line, cooldown sweeps, Sneak swap, SP/ES, placement option), the icon atlas (**all** icons of U 6, drawn now), action-bar messages, RTL layout, `HudTests/ScreenDataTests/LoadoutTests/TreeTests`	5–6	L1's progression API (interfaces from S0)
L6-LOOKS	the six Earth looks (L 1.1–1.6: skins, layers, face icons), the shared layer kinds (`TopknotLayer`, `CrownLayer`, `HairLayer`, `BeardLayer`, `RobeLayer`, `ShoulderLayer`, `BackLayer`, `HoodLayer`, `MaskLayer`, `ArmBandLayer`: built now for all releases), `Looks/Look`, skin application (`SuitSkins`-pattern mixin), `LooksTests`, the Element Scroll / Water Skin / Glider item art	4–5	none
L7-SOUND	`tools/sound/*` (the committed toolkit), the **shared and Earth** sounds of S 2–3 (67 events, ~100 files), `PowerLoopSounds` kinds for Earth, `EntitySoundLoop`, the Earth theme and the choice music (S 7), `SoundAssetTests` lists	4–5	none
L8-DOCS	`docs/wiki/` pages EN + HE for 1.0 (Home, Becoming a Bender, Earth, Controls, Looks, Items, Commands, Server Configuration, FAQ, Building), README EN/HE, `docs/DESIGN.md` 1.0 section, the Hebrew `docs/DESIGN.he.md`	2–3	the others' final numbers (last to merge)

Merge order (integrator; branch `v10`): S0 → L0-FABRIC (so every later merge runs on both loaders) → L1-CORE → L3-ANIM → L4-FX → L2-EARTH → L6-LOOKS → L5-UI → L7-SOUND → L8-DOCS; then the joins: `TempBlocks/Grip` neutral stubs replaced by L1's (L2 compiled against the interface), the HUD reading Earth's status lines, the looks in the picker, the sound events wired to the FX kinds, the icon atlas cells checked against `Ability`, one comma convention for the enum regions. **Integration: 2–3 agent-days.** Then the **gate**: full suite both loaders (target ≈ 450 tests), parity diff empty, the real-client checklist (T 14.2) on both loaders and cross-loader, a world moved between loaders keeps the bender, a 4-bender Earth fight within budget (four earthbenders is the

worst case for temp blocks), Enhanced installed alongside (no crash, both HUDs, keys distinct), `mod_version 1.0.0`, tag `v1.0.0`. **Calendar: ~3 weeks.** Fallback milestone if L0-FABRIC slips: **1.0-neoforge** (the NeoForge jar alone) ships with Earth; L0 lands in 1.0.1 without touching gameplay.

Lane briefs (paste into the agent's prompt; each begins with the shared preamble)

Preamble (every lane): "You are lane `<LANE>` of Bending 1.0, a Minecraft 1.21.1 mod (NeoForge + Fabric universal jar, no runtime libraries, every model a `ModelPart`, every animation code-driven, English and Hebrew everywhere). Work in your own git worktree on branch `v10-<lane>` from `v10-skel`. You own only the paths listed; in shared files write only inside your `// ===== <lane> (1.0) =====` region; never edit lang files, config classes, `ModNetwork`, payload records, `ModAttachments`, `ModTags`, `ClientSetup`, `TestKit` or `DataTests` outside your region: a needed change is a ten-line `request(I)` commit on a branch `v10-skel-req-<lane>` for the seam keeper. Read `docs/plan/DESIGN.md`, `ANIMATION.md`, `LOOKS.md`, `UI.md`, `AUDIO.md`, `TECH.md` and `docs/LANES.md` first; every number comes from the config option S0 defined (never a literal). Tests go in your own classes in your own batch (`@BendingTest(template = ARENA, batch = "<lane>")`); a test that changes a server setting gets its own batch; run `./gradlew :neoforge:runGameTestServer -Pbending.tests=<Your*>` and `:fabric:runGameTest -Pbending.tests=<Your*>` after deleting `run/world` and `run/config/bending-server.toml`; the full suite twice before you ask to merge. Check every visual in a real client (dedicated server, English and Hebrew clients, a second client watching, three render rounds) and put screenshots in `docs/screenshots/qa/<lane>/`. Every payload handler is one static call into a client class. Commit often with clear messages; when done, write `docs/lanes/<lane>.md`: what you built, what you left, what the integrator must join."

L0-FABRIC: "Implement the Fabric half of the platform layer per `TECH.md` 2.3 and the universal jar per 2.4: `FabricPlatform`, `FabricHooks`, `FabricItemAccess`, `FabricParts`, `FabricAttributes` (with the mod's `creative_flight` attribute), `FabricNet` (play channels + the configuration-phase `HelloPayload` handshake using `HelloPayload.check`), `FabricAttach` (Fabric Attachment API, `SyncTo.SELF` → `targetOnly`), `FabricRegistrars`, `client/FabricClientRegistrar`, `client/FabricClientServices`, `bridge/FabricBridge` with one adapter per event in `TECH.md` 2.5 (Fabric callbacks where they exist, Fabric-only mixins in `bending.fabric.mixins.json` where they do not: `LivingEntity.hurt` amount, `AbstractClientPlayer.getFieldOfViewModifier`, `Camera.setup`, `ItemInHandRenderer.renderArmWithItem/renderPlayerArm`, `KeyboardInput.tick`, `Projectile.onHit`, `LivingEntity.knockback`, `LivingEntity.canBeAffected`), `FabricTestHooks` and the `fabric-gametest` entry point. Build: `fabric/` with Fabric Loom and `officialMojangMappings()`, `common/` compiled once, the `universal` merge task, CI running both loaders' GameTests and `tools/parity.py`. Prove it: `SkeletonTests`, `PlatformSeamTests`, `NetworkTests` and `ConfigFileTests` pass on Fabric with the same results as NeoForge; a NeoForge client joins a Fabric server and the reverse; a version mismatch is refused with NeoForge's words; `BridgeTests` for Fabric (the pattern of Enhanced's `BridgeLivingTests`: post a native callback, assert the common event's getters and cancellation at the right priorities). Document Fabric's limitations in `docs/LANES.md`. Do not touch gameplay packages."

L1-CORE: "Build the bender core per `DESIGN.md` 1 and `TECH.md` 5–8, 12–13: `BenderSave` (SavedData), `Benders.load/ save/sync`, SP on kills (bending damage types and the assist window, the per-mob table and formula, no SP from players), mastery XP and levels (the curve, the buy cap 7 and costs), unlocks with gates (parent, mastery, SP, ES), ES, the loadout and presets, `AbilityHandler` routing with Sneak and the refusal rules (dead, severed, paralysed, bloodbent, meditating), `Cooldowns`, hunger costs, the Element Scroll and `ElementChoicePayload`, the awakening, `/bending` commands, `AreaTargets` (copied), `BendingRights`, `TempBlocks` with the journal, caps, fluids, block-entity refusal, no drops, the standing-player rule and crash recovery, `Hazards`/`HazardTicker`,

`Grip` (Enhanced's `Telekinesis.hold` adapted for `RockEntity`/metal targets), `Victims`, `compat/EnhancedHooks`. Tests: `ProgressionTests`, `LoadoutTests`, `TempBlockTests`, `RightsTests`, `AreaTargetsTests`, `VictimTests`, `CommandTests`, `SaveTests` (a save written and reloaded byte-equal; a player data record surviving a simulated loader move: the same `SavedData` bytes read on both loaders). No ability behaviour: that is L2's."

L2-EARTH: "Build every Earth ability per `DESIGN.md` 2 and `ANIMATION.md` 3 (with 1, 2, 11, 12, 14 for the shared rules): server logic in `earth/` (`RockThrow`, `EarthArmour`, `EarthPillar`, `EarthWall`, `SandBending`, `Quicksand`, `LavaBending`, `MetalBending` with knives/pull/wall/armour/wreck/grip, the Rooted passive, the lava passive), the entities and blocks listed in `TECH.md` 4 for Earth, `EarthFx` senders with the payload kinds 1–99, the client `EarthClient` (stances `EARTH_IDLE/GRIP/ARMOURED/RIDE_UP/LAVA_HOLD/PULLING/METAL_HOLD`, every gesture of A 3 with the exact keyframes, first-person offsets, prediction, cooldown routing), `EarthFx` client (every effect row of A 3 with its counts, the armour `LayerReveal` slices, overlays), the renderers, the wrecked blocks' textures (L 1.7). Use L1's `TempBlocks`/`Grip`/`Hazards` through the S0 interfaces (the neutral stubs make your tests pass only after L1 merges: the integrator runs them together; until then test against a local fake in your batch). Tests: `EarthTests`, `EarthBlockTests`, `EarthMetalTests`, `EarthAnimationTests` (A 14.2), one per ability per `TECH.md` 14.1's Earth rows, in batches `earth`, `earth_blocks`, `earth_metal`, `earth_pvp`, `earth_anim`."

L3-ANIM: "Extend the copied animation system per `ANIMATION.md` 14.1–14.4 and 11–12: leg-z keys in `Gestures` (`legs(rx, rz, lx, lz)`, a `legZ` track), groups `RIDE`/`HIT`/`AUTO` and 4 slots with the slot rules, channel masks (`legsMaskedByAir`, `legsMaskedByWalk`) and `bodyOwner`/`rideOverride`, the first-person camera drop and eye-height offsets (a `CameraMixin.setup` tail, reduced-motion gated), `VictimPoses` with the three mixins and the `bending:victim` reader (stagger, knockdown, stuck, lifted, tumble, soaked, chilled, the puppet and paralysed poses as data so 1.2/1.3 only switch them on), the shared stances `LAUNCHED`/`LIFTED`, the run characters of A 7.1, `LAND_SOFT` variants, the four awakening gestures and `LOOK_CHANGE`, `InputLocks` (a `Live`-bit set), `/bending_anim` and the overlay, the pose library constants of A 1.2 as `StancePoses`. Tests: `AnimationMathTests` extended (leg-z rules, yaw whole turns, hand-overs, 4 slots, masks), `VictimPoseTests` (angles within limits, TTL), `AnimationSignalTests` regions. Real client: every pose of A 1.2 frozen and photographed on wide and slim models."

L4-FX: "Build the FX foundation per `ANIMATION.md` 0.4, 13, 14.5, 14.12 and `TECH.md` 9: `GrainParticle` and `DropletParticle` with their options and sprites, `BendingRenderTypes` (the copied energy types + `MATTER_FLAT`/`MATTER_TEXTURED`), `MatterShapes` (Enhanced's `SandShapes` with the water/ice additions), `BendingWorldFx` (one `RenderLevelStageEvent` subscriber drawing every element's decorations in four batches from a bounded client ledger fed by the FX payloads and pruned by block changes), `ChainSim`, the camera-shake service (Enhanced's `ArenaFx.shake`), `EffectDensity` (vanilla Particles × the client option × the 48/96-block rules), the overlay base class (vignette, gradient, screen-space bolt), `EntityGlow` wiring, `LightningFx` and `FlameBeamDrawer` **skeletons** (the geometry, tested, used in 1.3). Tests: pure geometry tests (`FxMathTests`: ring/funnel/ribbon vertex counts, chain sim stability, density arithmetic), `ParticleTests` (options codecs). Real client: a debug command `/bending_fx <effect>` playing each primitive."

L5-UI: "Build every screen and the HUD per `UI.md`: `ElementChoiceScreen`, `BendingScreen` with `SkillTreeTab` (the graph canvas with pan/zoom, connectors, the detail panel, unlock/buy/cast), `LoadoutTab` (drag and drop, presets, auto), `LooksTab` (faces, glider dyes, the draft badge), `WaterTab` (hidden until the Water element exists), `BendingHud` (every line of U 3, the cooldown sweep, the Sneak swap, dimmed icons with reason glyphs, the SP count-up, placement and scale options), the icon atlas with **every** icon of U 6 (draw them all now), the action-bar messages, RTL layouts with `HudDraw.visual`, the settings entry. Use `PreviewPuppet` for the 3D preview with the animator's

preview API. Tests: `HudTests`, `ScreenDataTests`, `LoadoutTests` (payload round trip), `TreeTests` (pure rules through L1's interface), `IconTests` (every ability and state has a cell). Real client: every screen in both languages at 854×480 and 1920×1080."

L6-LOOKS: "Draw and build the six Earth looks per `LOOKS.md` 0 and 1 and the shared layer kinds of L 0 (all of them, for every release): skins with `tools/art/gen_looks.py` (pixel scripts, committed), each layer's `LayerDefinition` and texture, `Looks`/`Look` registry with per-look `color`/`glow`/`metalColor`/`bloodColor`/`fireStyle`, the skin mixin (`SuitSkins` pattern), the face icons, the Element Scroll, Water Skin (9 stages) and Air Glider (tinted) item art, the Earth block textures of L 1.7. Tests: `LooksTests` (64×64 PNG, valid `SkinCheck`, every texture exists, both names, slim flags). Real client: every look front/back/side, under the armour layers, in the picker."

L7-SOUND: "Build the sound pipeline per `AUDIO.md` 1 (commit `tools/sound/sfx.py`, `render.py`, `recipes/`), render the shared and Earth events of S 2–3 (every recipe as specified, 2–3 takes where marked, mono 44.1 kHz Ogg, peaks ≤ −1 dBFS, loops seamless), the Earth theme and the choice piece of S 7 as stems, wire `PowerLoopSounds` kinds for Earth and `EntitySoundLoop`, `ElementMusic` (stems, crossfade, `SelectMusicEvent`), replace the placeholders, empty the S0 placeholder list in `SoundAssetTests`. Tests: `SoundAssetTests` (format, length, not placeholder, subtitles), `MusicTests` (stem timing). Real client: every sound at 1/8/24 blocks; the loops follow and stop; the music under a fight."

L8-DOCS: "Write the player's guide for 1.0 in `docs/wiki/` (EN and `-HE` pages wrapped in `<div dir="rtl">`), the README in both languages, `docs/DESIGN.md`'s 1.0 section from `docs/plan/DESIGN.md` with the shipped numbers, and `SUGGESTIONS.md`'s decisions as made by Eilon recorded in `docs/DECISIONS.md`. Merge last; read every lane's `docs/lanes/*.md`."

Release 1.1 — Air (כשפות אוויר)

Contents: every Air ability (D 3), the glider item, Spiritual Projection (the spirit entity and camera), the rides (scooter), Zaheer's path, the six Air looks (Jae and Karthik as drafts until pictures arrive), Air sounds and the Air theme, the Air wiki. Protocol **"2"**.

S0-1.1 (seam keeper, 1–2 agent-days; branch `v11-skel` from `main` 1.0.x): the Air payload kinds 100–199 and `SpiritPayload`/`SpiritMovePayload`/`GliderDyePayload` handlers to stubs, the `AirFxPayload` record, the `spirit`, `air_scooter`, `tornado`, `air_blade` entity classes' shells become real classes (registered in 1.0, bodies now), `SpiritCamera` seam, the `rideOverride` hook for the scooter and glider (L3 of 1.0 built the flag), the `Live` bits already exist, lang keys of 1.1, the Air sound placeholders are registered (1.0 did), config exists; protocol "2" pinned; a test that the Water tab stays hidden.

Lane	Owns	Agent-days
L1-AIR-A	Air Blast, Air Shield, Air Blades, Tornado, Air Wheel: server (`air/`), entities (`AirBladeEntity`, `TornadoEntity`), `AirClient` gestures/stances/first person for them, their effects (`AirFx`), the shield's `ShieldStyle` and renderer wiring	5–6
L2-AIR-B	Air Scooter (vehicle + ride stance + renderer), Air Bubble (+ overlay), Air Glider (item, `FALL_FLYING` physics, dye, `GliderLayer`, the gliding profile and pose, Air Blast downward), Spiritual Projection (`SpiritEntity`, `SpiritCamera`, the possess/move payloads, the meditation stance and eye offset, `SpiritOverlay`, through-wall glows and cave outlines, Deep Meditation), Zaheer's path (the pet-death gate, True Flight profile, Suffocation hold + victim clutch, Air Vacuum, the push/damage factors, the unlock moment)	7–8

Lane	Owns	Agent-days
L3-AIR-LOOKS	the six Air looks (L 2), `eyes_closed` layer (A 14.9), `air_lattice/air_swirl` textures, glider art for 16 dyes, Jae's board (suggestion, behind the Jae look), the draft badge	3–4
L4-AIR-SOUND	S 4's 50 events, the Air theme stems, loops	3–4
L5-AIR-DOCS	wiki Air pages EN/HE, README rows, DESIGN 1.1	1–2

Merge order: S0 → L1-AIR-A → L2-AIR-B → L3 → L4 → L5; joins: the shield-burst vs tornado interaction, the scooter's `rideOverride`, the glider's `BackLayer` vs robes. **Gate:** full suite both loaders (≈ 650), real-client check, a 4-bender fight with two airbenders (tornado + scooter budgets), the cross-element rows of `MatchupTests` for Air vs Earth. **Calendar ~2.5 weeks.**

Briefs: the 1.0 preamble with `1.1/v11-*`; then, **L1-AIR-A:** "Build Air Blast, Air Shield, Air Blades, Tornado and Air Wheel per `DESIGN.md` 3.1, 3.2, 3.4, 3.8 and `ANIMATION.md` 4.0–4.2, 4.4, 4.8 (gestures with exact keyframes, first person, effects with counts, sounds by event name), the `TornadoEntity` (orbiting riders, drift, 4-tick position payloads), `AirBladeEntity`, the Air Shield on Enhanced's shield pipeline (`ShieldStyle` + sphere passes + ripples + burst), `AirFx` kinds 100–149. Tests per `TECH.md` 14.1 rows Air Blast/Shield/Blades/Tornado in batches `air`, `air_pvp`, `air_anim`." **L2-AIR-B:** "Build Air Scooter, Air Bubble, Air Glider, Spiritual Projection and Zaheer's Path per `DESIGN.md` 3.3, 3.5–3.7, 3.9 and `ANIMATION.md` 4.3, 4.5–4.7, 4.9, 10.1, 10.2, 10.5, 10.6, 11.3: the scooter as a vehicle entity with the ride stance through `rideOverride`, the glider's elytra-based glide with the `GLIDER` flight profile and the glider layer, the bubble and its overlay, the spirit entity possessed by the client's camera with `SpiritMovePayload` validated server-side, the meditation stance with the eye-height offset, the spirit overlay and glows, the pet-death gate (`LivingDeathEvent` of an `OwnableEntity` whose owner is an airbender; the option for proximity), True Flight through the flight attribute, Suffocation as a `Hazards` hold with the victim's `CLUTCH` pose and air-supply drain, Air Vacuum, the push/damage factors in `AirAbilities` shared math, the `TETHERS_CUT` moment. `AirFx` kinds 150–199. Tests per the Glider/Projection/Bubble/Scooter/Zaheer rows, batches `air_rides`, `air_spirit`, `air_zaheer`."

Release 1.2 — Water + Ice + Blood (כשפות מים, קרח ודם)

Contents: the water source system and the Water Skin, every Water, Ice and Blood ability, the Water tab and gauge, the full-moon indicator and moon bells, the Water looks (Katara default), Water sounds and theme, `ChainSim` ropes in use, the puppet poses. Protocol **"3"**.

S0-1.2 (1–2 agent-days): payload kinds 200–299, the Water entity shells to classes, `WaterSkinItem` component logic seam, `SourceFinder` seam filled by L1, the Water tab unhidden behind the element, moon helpers in `McApi`, protocol "3".

Lane	Owns	Agent-days
L1-WATER-A	`WaterSource/SourceFinder` (open water, rain, the Skin, plants with drying), `WaterSkinItem`, Water Shot + Whip, Healing + Cleanse, Water Arms + Grab/Climb/Swing, the water passives and the full-moon factor, `WaterClient` for these, `WaterFx` kinds 200–239, the Water tab and gauge wiring (U 2.2d, 3.3), the moon indicator (U 3.6), `WaterArmsLayer`	6–7

Lane	Owns	Agent-days
L2-WATER-B	Water Octopus (layer + `ChainSim` tentacles + slaps/strikes/lash/grab), Water Spout (vehicle + ride + ram + freeze + Wave Ride), Ice (knives, spikes, freeze disc/feet/constructs, snow, wall, dome: temp ice blocks, `IceSpikeBlock`, the chilled/wet effects), `WaterFx` kinds 240–279	7–8
L3-BLOOD	Blood Bending (the moon rule, ES rule, the hold with puppet movement by look, fling, the two-target M10), Bending Severance (the touch, the duration formula, the severed state across logout, the healable option), the `PUPPET`/`SEVERED` victim poses switched on (data from 1.0's L3), `BloodOverlay`, the victim overlays and input locks, `WaterFx` kinds 280–299, the blood sounds' wiring	4–5
L4-WATER-LOOKS	the four Water looks (L 3) incl. Ming-Hua's armless skin and Amon's mask/hood, the Water Skin layer, `water_foam`/`ice_frost`/`wet` textures	3
L5-WATER-SOUND	S 5's 60 events, the Water theme + moon bells stem	3–4
L6-WATER-DOCS	wiki Water/Ice/Blood pages EN/HE, DESIGN 1.2	1–2

Merge order: S0 → L1-WATER-A → L2-WATER-B → L3-BLOOD → L4 → L5 → L6; joins: the octopus and arms sharing `ChainSim` parameters, Freeze acting on L2's constructs and L1's arms, the matchup rows Water vs Fire **prepared** (fire lands in 1.3: the factors are tested against a scripted "fire damage" source), the Water tab in the screen. **Gate:** full suite (≈ 900), real client incl. a full-moon night and the Nether (no moon), a 4-bender fight with two waterbenders (octopus + spout budgets). **Calendar ~3 weeks.**

Briefs: preamble `1.2`/`v12-*`; **L1-WATER-A:** "Build the water source system and Water Shot/Whip, Healing/Cleanse, Water Arms per `DESIGN.md` 4.0–4.2, 4.5 and `ANIMATION.md` 5.0–5.2, 5.5 (the 3-tick source draw chained before every gesture: use the `param` delay or implement A 14.6 chaining in `BenderAnimator`), the Water Skin item with its component, refills and the HUD gauge (`UI.md` 3.3), the plant-drying temp states, the full-moon factor and indicator (`UI.md` 3.6), `WaterArmsLayer` with the `RenderArmEvent` first-person override. Tests per the Water Shot/Healing/Water Arms rows and `SourceTests` (each source in order, the Nether, rain, plants at M7, claims), batches `water`, `water_source`, `water_anim`." **L2-WATER-B:** "Build Water Octopus, Water Spout and the three Ice cards per `DESIGN.md` 4.3, 4.4, 4.6 and `ANIMATION.md` 5.3, 5.4, 5.6, 10.3: the octopus veil + 8 `ChainSim` tentacles with slap/strike/lash/grab, the spout vehicle with the ride stance, ram, M8 freeze and Wave Ride, the ice temp blocks (`IceSpikeBlock` rise states, frosted-look ice discs, walls, domes), knives/snow projectiles, Freeze on water/feet/constructs (the construct hook is an interface L1's arms and your octopus implement), the `chilled`/`wet` effects. Tests per the Octopus/Spout/Ice rows, batches `water_shapes`, `water_ice`, `water_rides`." **L3-BLOOD:** "Build Blood Bending and Bending Severance per `DESIGN.md` 4.7 and `ANIMATION.md` 5.7, 12.4, 12.5, 2.2: the moon and ES gates, the hold as a `Hazards` object moving the victim by the caster's look with speed caps, the victim's `PUPPET` pose and input lock, fling, two targets at M10, the severance touch and formula, the severed state in `BenderSave` and the `severed` effect refusing every ability and passive, the HUD greying, the overlays and heartbeat loops, `blood_refused` on a stronger waterbender. Tests per the Blood/Severance rows, batches `blood`, `blood_moon` (sets the moon), `blood_pvp`."

Release 1.3 — Fire + Lightning (כשפות אש וברק)

Contents: every Fire ability, lightning generation and redirection, the fire passives, Azula's blue fire, the five Fire looks (Zuko default), Fire sounds and theme, the flame beam drawer and lightning FX in use, the paralysis pose, all matchups live. Protocol **"4"**.

S0-1.3 (1–2 agent-days): payload kinds 300–399, `FireballEntity` to a class, the `FLAME`/`BLUE_FLAME` beam styles registered, the `paralysed` effect's client lock seam, `HandTracker.belly` (A 14.8), protocol "4".

Lane	Owns	Agent-days
L1-FIRE-A	Fireball (+ small/great), Fire Punch (+ kick, combo, jab), Fire Whip (+ lash, wrap, cutting rocks/blades), Fire Circle (+ legs, jet flight profile), the fire passives (resistance, self-extinguish, sun), fires-where-allowed through `BendingRights.mayHurtBlock`, `FireClient` for these, `FireFx` kinds 300–349, the per-look fire palette switch (Azula) in every fire effect	6–7
L2-FIRE-B	Flamethrower + Fire Breath (two `ContinuousBeam`s per player, the `FlameBeamDrawer`, the breath lock and `HeatOverlay`), Fire Healing + Burn Out, Lightning Generation (charge hold, release, dual at M10, arc, `LightningFx`, thunder, the `paralysed` effect with `PARALYSED` pose and overlay, rods attracting, water spreading, no vanilla bolt), Lightning Redirection (the stance, the window, chance by mastery with seeded RNG, pass/fail/ground, the body path on `HandTracker.belly`, `RedirectPathOverlay`, vanilla lightning redirect, Enhanced's lightning if present via `is_lightning`), `FireFx` kinds 350–399	7–8
L3-FIRE-LOOKS	the five Fire looks (L 4) incl. the crowns and `white_lotus_robe` tints, `fire_mouth`, `heat`, `scorch` textures	3
L4-FIRE-SOUND	S 6's 55 events incl. the blue-fire pitch variants, the Fire theme + blue stem	3–4
L5-MATCHUPS	every row of D 1.7 switched on and tested across all four elements (`MatchupTests`, `CrossElementTests`, `CoexistenceTests` with the 4-bender script of A 13), balance pass on the numbers with Eilon's feedback from 1.0–1.2 play (config defaults only)	2–3
L6-FIRE-DOCS	wiki Fire/Lightning pages EN/HE, DESIGN 1.3, the matchups page	1–2

Merge order: S0 → L1-FIRE-A → L2-FIRE-B → L3 → L4 → L5 → L6; joins: water vs the beams and whip, lightning vs steel and water constructs, the blue palette in L2's beams, redirection of L2's own bolts in PvP. **Gate:** full suite (≈ 1,200), real client with Azula's blue fire in every effect, a thunderstorm, the Nether, a 4-bender fight of all four elements (the A 13 budget measured), both loaders and cross-loader, `1.3.0`. **Calendar ~3 weeks.**

Briefs: preamble `1.3/v13-*`; **L1-FIRE-A:** "Build Fireball, Fire Punch, Fire Whip, Fire Circle and the fire passives per `DESIGN.md` 5.0–5.2, 5.4, 5.5 and `ANIMATION.md` 6.0–6.2, 6.4, 6.5, 10.7 (the Fire Kick and leg sweep use the leg-z keys; the Fire Circle spins with yaw; the Fire Jet is a flight profile), `FireballEntity`, the fire whip on `ChainSim` with upward gravity, fires only through `BendingRights.mayHurtBlock` with the caps, the Azula palette switch (`Looks.fireStyle(data)` read by every effect and sound pitch). Tests per the Fireball/Punch/Whip/Circle rows, batches `fire`, `fire_blocks` (fires), `fire_pvp`, `fire_anim`." **L2-FIRE-B:** "Build Flamethrower/Fire Breath, Fire Healing, Lightning Generation and Lightning Redirection per `DESIGN.md` 5.3, 5.6–5.8 and `ANIMATION.md` 6.3, 6.6–6.8, 12.5, 2.2: two beams per player on the copied `ContinuousBeam`/`BeamRenderer` with the `FlameBeamDrawer`, the breath lock, the lightning charge as a held gesture with input lock and a server-side raycast bolt (the mod's damage type in `is_lightning`, no vanilla `LightningBolt`),

`LightningFx` with branches and flashes, delayed `thunder_distant`, the `paralysed` effect with its pose/overlay/lock, redirection as a `Hazards` window with mastery chance (seeded by game time + player for tests), the body-path effect and screen overlay, Ground It. Tests per the Flamethrower/Fire Healing/Lightning/ Redirection rows, batches `fire_beams`, `lightning`, `lightning_pvp`."

After 1.3: 1.4+ (suggestions, if Eilon says yes)

Each extra of D 8 is one lane in its own small release: 1.4 **Temples and Masters** (structures, the tutorial NPCs, bending scrolls: 3 lanes), 1.5 **Bosses** (the four spirit bosses on the copied boss framework: 4 lanes + a framework copy lane), 1.6 **Companions** (sky bison, lemur, polar bear dog, fire ferret: 2 lanes), 1.7 **Duels and dummies**, 1.8 **Later Minecraft versions** (the `r2` adapter and the matrix), 2.0 **The Avatar**.

Roles

- **Seam keeper** (one agent per release, the S0 author): keeps `v1x-skel` green, merges `request(I)` branches within the day, pins seams with tests, answers "who owns this file" questions from `docs/LANES.md`.
- **Integrator** (one agent, may be the seam keeper after S0): merges in order, resolves conflicts by "keep both sides" (everything is additive) then `./gradlew build`, does the joins, runs the lanes' batches together then the full suite on both loaders twice, does or assigns the real-client checklist, writes the "how 1.x was put together" section, bumps `mod_version`, tags.
- **Lanes**: as briefed. A lane that needs another lane's code before it merges codes against the S0 interface and, if it must, reaches the other lane's accessors **by reflection until the merge** (Enhanced's L7 of 3.8 did this); the integrator replaces reflection with direct calls in the joins.

Risks and how the plan holds them

Risk	Hold
The Fabric half and the universal jar are new ground (no Enhanced precedent)	L0-FABRIC starts first with the biggest budget; 1.0-neoforge is the fallback milestone; every later release's gate runs both loaders
Saves across loaders	`SavedData` for everything persistent (T 2.4); `SaveTests` prove byte-equality; the real-client checklist moves a world
Enhanced's platform layer changes under us (its P1)	Bending copies at a pinned Enhanced commit (888609c) and records it in `docs/LANES.md`; a later re-sync is a lane
Temp blocks and claims on servers	`BendingRights` + the loader's events + the journal + caps; `RightsTests`/`TempBlockTests`; the Fabric limitation documented
Performance of four benders	per-tick caps in every effect row, measured at every gate (A 13)
Seven Havens looks without references	drafts behind a badge; Eilon's pictures turn them final in a one-day art task
Balance numbers wrong for an 11-year-old's fun	every number in config; L5-MATCHUPS of 1.3 and a balance pass after each release from Eilon's play

Risk	Hold
Animation and effect quality (the heart of the request)	the bible's keyframes are the tests' expected numbers; the real-client check per lane; `/bending_anim` for frame-by-frame review with Eilon

SUGGESTIONS.md — הצעות לאילון

הצעות לאילון — מה צריך להחליט לפני שמתחילים לבנות

היי אילון! זה המוד שלך: **Bending** (כשפות), בעולם של אווטאר. כל מה שביקשת נמצא בתוכנית. פה יש כמה דברים שאנחנו צריכים שתחליט עליהם, ולכל אחד כתבתי מה **ברירת המחדל**: מה נעשה אם לא תגיד כלום. אפשר פשוט לענות "כן" או "לא" או לכתוב מה אתה רוצה במקום.

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1. איך בוחרים אלמנט

ברירת המחדל: מגילת האלמנטים. מכינים מגילה בשולחן העבודה (4 ניירות, שק דיו, מקל, ובפינות: אבן, נוצה, דלי מים ואבקת בלייז), לוחצים עליה קליק ימני, ונפתח מסך עם ארבע כרטיסיות: אדמה, אוויר, מים, אש. בוחרים, ויש התעוררות גדולה (כותרת, צליל, חלקיקים, והודעה לכל השרת: "אילון הפך לכשף אדמה!").

אפשרות אחרת: רוח קטנה (שועל זוהר) שמופיעה פעם אחת ליד שחקן חדש ומציעה לו לבחור. אפשר גם את שתיהן.

השאלה: מגילה? רוח? שתיהן?

2. מגן האוויר: 3.5 או 4 לבבות?

כשמגן האוויר נשבר הוא מתפוצץ לכל הכיוונים ודוחף את כולם. אמרת "ארבעה לבבות, שלושה וחצי לבבות". **ברירת המחדל: 3.5 לבבות.** זה מספר בהגדרות, אז אפשר לשנות בכל רגע.

3. המקשים

Enhanced כבר תופס את R, G, V, B, N, H, Y, Z (וגם J מיינקרפט תופס את W A S D E Q F T L P ואת המספרים. נשארו מעט מקשים נוחים ליד היד השמאלית.

ברירת המחדל:

מקש	מה הוא עושה
X	משבצת קרב 1 (התכופפות + X: היכולת השנייה של המשבצת)
C	משבצת קרב 2
Alt שמאלי	משבצת קרב 3
K	משבצת קרב 4
O	מסך הכשפות: עץ הכישורים, סידור הקרב, המראות (התכופפות + O: מחליף מראה מיד)

ארבע משבצות, לכל אחת גם התכופפות = **8 יכולות מוכנות בבת אחת**. את שאר היכולות מחליפים במסך. **השאלה:** המקשים האלה בסדר? אפשר במקום K את I, או רק 3 משבצות. (אפשר לשנות הכול ב-Options → Controls.)

4. ג’יי וקארתיק מ-*אווטאר: שבע הממלכות*

הסדרה חדשה (יצאה באוקטובר (2025, ולא הצלחנו למצוא תמונות טובות שלהם, רק תיאורים: * **ג’יי**: כשף אוויר בן 19 מ”חטיבת הכשפים” של אלורה. מדים צמודים ועתידניים בצהוב כהה, כתום כהה ואפור בהיר, עם סמל משולש כחול-כתום על החזה (כמו חץ של נוודי האוויר), חגורת ריצה עם כיסים, מכנסיים חומים רפויים, מגני רגליים אפורים, מגפיים גבוהים. יש לו **לוח מתקפל עם חור באמצע שקטנוע אוויר ממלא** (רחפן!). **לא ידוע:** השיער, הפנים, גון העור. * **קארתיק**: כשף אוויר מ”מסדר הלוטוס הלבן”, “קפטן קארתיק”, גבוה, בן ארבעים בערך, **שיער חום עם פסים לבנים** למטה, זיפים/זקן קצר, עיניים עייפות. **לא ידוע:** הבגדים והצבעים שלהם.

ברירת המחדל: נצייר אותם “בטיוטה” לפי התיאור (עם תווית “טיוטה” במסך המראות), ו**נבקש ממך תמונות** שלהם: צילום מסך מהסדרה של כל אחד מקדימה, ואם יש גם מהצד ומאחור. כשיש תמונות, מתקנים ביום. **בונוס אם תרצה:** כשלובשים את ג’יי, קטנוע האוויר נראה בתוך הלוח שלו.

5. מה כשפות המתכת יכולות להרוס

“הרס מתכת” מוערך מכשירים ממתכת ל-60 שניות (ואז הם מתקנים את עצמם, ושום דבר שבתוכם לא הולך לאיבוד). **ברירת המחדל (הרשימה):** הופר, הופר-מיינקארט, דרופר, דיספנסר, פיסטון (רגיל ודביק), אובזרבר, ריפיטר, קומפרטור, לפיד רדסטון, ידית (לבר), דלת ברזל, דלת-מלכודת ברזל, סורגי ברזל, שרשרת, פנס, פעמון, סדן, קלחת, כל המסילות, מיינקארט, כבשן, כבשן-פיצוץ, מעשנה, מוט ברק, לוחות לחיצה ממתכת, וו חוט-מעידה, אבן השחזה, חותך אבן, בלוקי ברזל/זהב/נחושת (רק נראים מעוכים), נורת נחושת, סורג נחושת, דלתות נחושת. **לעולם לא:** תיבות, חביות, קופסאות שאלקר, ביקון, שולחן קסמים. **השאלה:** להוריד או להוסיף משהו? (השרת יכול לשנות את הרשימה גם אחר כך.)

6. נקודות כישור ועלויות

כמה נקודות נותן כל מפלצת (ברירת מחדל):

מפלצת	נקודות
זומבי, שלד, עכביש, קריפר, טבוע, עכביש-מערה, סליים, פילאג’ר, פנטום…	1
אנדרמן, בלייז, מכשפה, פיגלין ברוט, גרדיאן, וינדיקייטור, קוביית מאגמה, הוגלין, ברִיז…	2
אבוקר, רוואג’ר, שאלקר, גאסט	4
אלדר גרדיאן	10
וורדן	25
ווית’ר	40
דרקון האנדר	60
שחקנים אחרים	0 (כדי שלא יהיה שווה “לחוות” חברים)
חיות, כפריים, גולמים	0

הקילה נחשבת אם ההורגת הייתה כשפות, או שפגעת במפלצת בכשפות ב-5 השניות האחרונות. **3 נקודות הראשונות של כל אלמנט מגיעות בחינם** עם האלמנט, ו**היכולת הראשונה חינם** (זריקת סלע / הדף אוויר / יריית מים / כדור אש).

כמה עולה לפתוח יכולת (ברירת מחדל): 2 עד 8 נקודות, לפי כמה היא חזקה. למשל: עמוד אדמה 2, קיר אדמה 3, שריון אדמה 4, כשפות לבה 6, כשפות דם .8 **רמת שליטה** (1 עד (10 עולה כשמשתמשים (בהצלחה) ביכולת, ואפשר גם לקנות רמות בנקודות **עד רמה 7**; רמות 10–8 רק מאימון. עץ מלא של אלמנט עולה בערך 160 נקודות (ווית’ר אחד ומאתיים זומבים). **השאלה:** זה מרגיש הוגן? מהר מדי? איטי מדי? (הכול בהגדרות.)

7. להחליף אלמנט אחר כך?

ברירת המחדל: לא. בוחרים פעם אחת, כמו גיבור ב-.Enhanced אופרטור יכול לאפס בפקודה. **אפשרות:** חפץ נדיר, **"מי רוח"** (מהבוס של המים או מנווה מדבר רוחני), שמוחק את האלמנט **וגם את כל הנקודות והרמות** ומאפשר לבחור מחדש. **השאלה:** רוצה את האפשרות?

8. "דרכו של זהיר": איזה מוות של חיית מחמד נחשב?

כשחיית המחמד שלך מתה, אתה "מנתק את הקשרים הארציים": מקבל תעופה אמיתית וחניקה, והאוויר שלך דוחף פחות ופוגע יותר. **ברירת המחדל: כל מוות של כל חיה מאולפת שלך נחשב** (זאב, חתול, תוכי, סוס מאולף... וגם ביזון שמיים אם נבנה אותו), גם אם לא היית בסביבה. **אפשרות:** רק אם היית במרחק 16 בלוקים כשזה קרה. **השאלה:** כל מוות, או רק כשאתה קרוב?

9. מספרי האיזון: הנה הגדולים, לבדיקה שלך

(הכול בלבבות; הכול בהגדרות השרת)

יכולת	נזק	קולדאון	מה גדל עם רמת השליטה
זריקת סלע	2 → 3.8 (סלע ענק (6	1.5 ש׳	ברמה 4 שני סלעים, ברמה 7 שלושה, ברמה 10 סלע ענק
שריון אדמה	שריון של 12 → 25.5 לבבות; תנועה ×0.7	20 ש׳ אחרי שנשבר	השריון חזק יותר ומאט פחות
קיר אדמה	חוסם 10 שניות	8 ש׳	2×3 → 4×7 בלוקים
חול טובעני	0.5 לשנייה, 4 → 7.6 שניות	10 ש׳	נכשל על כשף אדמה חזק ממך
שוריקן לבה	3 + בעירה	4 ש׳	שניים ברמה 8
הדף אוויר	0.5 + זריקה של 4 → 10 בלוקים	2 ש׳	רחוק יותר, קונוס רחב יותר
מגן אוויר	פיצוץ 3.5 (או (4	12 ש׳	גדול וחזק יותר
להבי אוויר	2.5 → 3.4	1.5 ש׳	2 להבים ברמה 4, 3 ברמה 8
טורנדו	1 + 0.5 לשנייה	8 ש׳	גבוה וארוך יותר, 2 ברמה 8
יריית מים	1.5 → 2.85 (×1.5 נגד כשפי אש)	1 ש׳	2 זרמים ברמה 4, 3 ברמה 8
ריפוי במים	לב כל 2 שניות → לב כל שנייה	—	מהר יותר
עמוד מים	דחיפה 4 → 7	10 ש׳	גובה 3 → 12 בלוקים
סכיני קרח	1.5 כל אחת, 3 → 7 סכינים	2 ש׳	יותר סכינים
כשפות דם	0.5 לשנייה, החזקה 4 → 10 שניות	25 ש׳	ברמה 10 גם בלי ירח מלא, ושני יעדים
שלילת כשפות	9 דקות לחלש, 2–1.5 דקות למיומן	120 ש׳	—
כדור אש	2 → 3.8 (ענק (5	1.2 ש׳	2 כדורים ברמה 5, 3 ברמה 8, ענק ברמה 10
להביור	לב כל חצי שנייה	6 שניות החזקה	ברמה :7 נשיפת אש מהפה
שוט אש	1.5 → 2.4 לכל הצלפה + שורף	4 ש׳	ארוך יותר, 2 שוטים ברמה 4
מעגל אש	2 (3 מרמה 6 עם הרגליים)	8 ש׳	רדיוס 4 → 8

יכולת	נזק	קולדאון	מה גדל עם רמת השליטה
ברק	4.5 → 5.85 + שיתוק 2 ש׳; ברמה 10 שתי ידיים: 6	20 ש׳ ((25	—
הטיית ברק	מחזיר את הברק; סיכוי 20% → 100% ברמה 9	—	—

חוקי האלמנטים: מים ×1.5 נגד אש (ומכבים אותה); אש ×1.5 נגד קרח (וממיסה אותו); וגם (הוספנו): ברק ×1.5 על מי שרטוב או במתכת, ×0.5 על שריון אדמה; קיר אדמה עוצר להבי אוויר; הדף אוויר מסיט כדורי אש. **השאלה:** משהו פה נראה לך חזק מדי או חלש מדי?

10. דברים כיפיים שלא ביקשת (רק אם תרצה; כל אחד "כן/לא")

1. **בובות אימון** (בלוק): מכים בהן בכשפות ומקבלים רמת שליטה (עד רמה 5, בחצי קצב). בובת מתכת לאימון מתכת וברק.
2. **מגילות כשפות** בתיבות של מבנים: מגילה של האלמנט שלך = 3 נקודות.
3. **מקדשים ומאסטרים**: ארבעה מבנים נדירים (שער ממלכת האדמה, מקדש אוויר על פסגה, ביתן שבט המים על ים קפוא, מקדש חכמי האש ליד הר געש), ובכל אחד מאסטר שמסביר את האלמנט בשש שורות, נותן מגילה ראשונה ומאמן.
4. **בוס לכל אלמנט**: מלך הגיריות-חפרפרות (אדמה), רוח הסערה (אוויר), "חסר הפנים" כמו קו (מים), הדרקונים ראן ושאו (אש: ריקוד, לא קרב). כל אחד 30 נקודות.
5. **חברים**: ביזון שמיים (לטוס עליו! 3 מקומות), למור מכונף (על הכתף, מביא חפצים), כלב-דוב-קוטב (לרכוב ולשמור), חמוס אש (על הכתף, מחמם ומכבה אותך).
6. **דו-קרב** `/duel bending` עם דרגות.
7. **שביט סוזין** (פעם ב-40 ימים: אש ×2 ושמיים אדומים), **תה של אירו** (משחזר רעב ונותן כשפות ×1.1 ל-10 שניות), **שולחן פאי שו** לקישוט, **סוחר הכרובים**.
8. **להיות האווטאר** (כל ארבעת האלמנטים + מצב האווטאר): **לא נבנה עכשיו**, זה רעיון לגרסה 2.0; כבר השארנו לו מקום בקוד.

11. דברים קטנים שהחלטנו, ואפשר לשנות

- **כשפי אוויר** נופלים בחצי נזק, רצים 10% מהר יותר וקופצים גבוה יותר.
- **כשפי מים** שוחים מהר יותר, וכל הכשפות שלהם ×1.25 **בלילה של ירח מלא** (יש מחוון ירח ב-HUD.)
- **כשפי אש** נפגעים מאש בחצי (ביקשת), ואש שלהם קצת חזקה יותר ביום ובנת׳ר, קצת חלשה בגשם.
- **כשפי אדמה** נדחפים פחות כשהם עומדים על אדמה (מושרשים).
- **כל בלוק שהכשפות מזיזות הוא זמני** וחוזר למקום: קיר אחרי 10 שניות, עמוד אחרי 15, לבה אחרי 12, קרח אחרי 20. אף פעם לא בשטח שמור של מישהו אחר. אז אי אפשר להרוס עם כשפות, רק לפגוע ביריבים.
- **מינג-הואה** (מראה של מים): בלי זרועות! כשזרועות המים דלוקות יש לה זרועות מים. **השאלה:** כשהן כבויות: בלי זרועות בכלל, או "גדמי מים" קצרים?
- **אמון**: עם המסכה והברדס; אפשר גם מראה שני של נואטק בלי מסכה. **השאלה:** רוצה גם את נואטק?
- **קטארה**: הבגד הכחול של העונה השלישית עם פרוות לבנה; אפשר גם את הפרקה של העונה הראשונה. **השאלה:** רוצה גם?
- **אירו**: בגלימת הלוטוס הלבן (ביקשת); אפשר גם את הבגד הירוק של המטייל. **אוזאי**: הגלימה והכתר; אפשר גם את "מלך הפניקס" בלי החולצה.
- **זהיר**: ציירנו אותו עם שיער קשור מאחור וזקן (תגיד אם אתה רוצה אותו מגולח).
- **ההשלכה הרוחנית** רואה דרך קירות ומערות; ברמה 8 היא יכולה גם לראות עפרות (כבוי כברירת מחדל: זה חזק מדי לדעתנו). **השאלה:** לדלוק או כבוי?
- **ליצור קטנה (רמה 6 בכדור אש)**: כדור אש קטן שמדליק מדורות ונרות (לא TNT ולא בלוקים).

12. איך זה ייבנה

1. **1.0**: היסודות + **כל כשפות האדמה** (וגם הפעלה ב-NeoForge וב-Fabric באותו קובץ).
2. **1.1**: כל כשפות האוויר (וגם דרכו של זהיר, ההשלכה הרוחנית, הגלשן).
3. **1.2**: כשפות מים, קרח ודם (ונרתיק המים).
4. **1.3**: כשפות אש וברק (ואש כחולה של אזולה).

כל שלב נבדק באלפי בדיקות אוטומטיות ובמשחק אמיתי בעברית ובאנגלית, לפני שהוא יוצא.

מה אנחנו צריכים ממך עכשיו: תשובות ל-12 הסעיפים (או "ברירות המחדל בסדר"), ו**תמונות של ג'יי וקארתיק**.

Bending — תוכנית בנייה (עברית)

זהו התרגום לעברית של PLAN.md, תוכנית הבנייה של המוד **Bending** (כשפות). המסמך המקורי באנגלית הוא המקור המחייב. שמות קבצים, ענפים, פקודות, מזהים ושמות מחלקות נשארים באנגלית כמו שהם בקוד. **הנחיות הנתיבים (lane) (briefs נשארות באנגלית** כי מדביקים אותן כמו שהן להנחיית הסוכנים שבונים, ולכן כאן מופיע רק תיאור קצר של כל הנחיה.

איך בונים: השיטה

השיטה היא זו של Enhanced (גרסאות 3.0 עד :(3.8

1. **שלד .S0** שלב אחד שעורך את כל הקבצים המשותפים פעם אחת: רישום פריטים וגושים, מסרים ((payloads, תכונות ((attachments, אפשרויות הגדרה, מפתחות שפה בעברית ובאנגלית, מציינים מקום לצלילים, תגיות, שלדי תוספים ((plug-ins וכל התפרים, עם מספר פרוטוקול שנעול בבדיקה.
2. **נתיבים (.lanes)** כל נתיב עובד בעץ עבודה (git worktree) ובענף משלו, יוצא מהשלד. כל נתיב **הבעלים** של החבילות שלו, וכותב בקבצים משותפים **רק בתוך האזור שלו** (`// ===== <lane>;lt& ;(( 1.0 =====`).
3. **מנהל תפרים (.seam keeper)** סוכן אחד, שהיה כותב השלד, עונה לבקשות לשינויים בקבצים משותפים, שומר על השלד ירוק, ומוסיף בדיקות שמצמידות את התפרים.
4. **מאחד (.integrator)** מאחד את הנתיבים לענף הגרסה בסדר קבוע, עושה את "החיבורים" בין הנתיבים, מריץ את בדיקות הנתיבים ואחר כך את כל הבדיקות.
5. **שער (.gate)** נתיב נכנס רק אם: הבנייה ירוקה (`portLint` בשני המסגרות), הבדיקות שלו עוברות פעמיים ברצף בשני המסגרות, שפת השפות מלאה בעברית ובאנגלית, מישהו שאינו הכותב עבר על רשימת הבדיקה במשחק אמיתי, והוא לא ערך קובץ מחוץ לבעלותו. שער השחרור: כל הבדיקות ירוקות בשני המסגרות, דוחות השוויון ((parity זהים, והקובץ המאוחד נבנה ורץ בשניהם.

הערכות זמן מופיעות בימי סוכן (יום עבודה אחד של סוכן אוטונומי) ובימים קלנדריים עם נתיבים במקביל. הערכות אלה נשענות על הקצב של .Enhanced

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סיכום הגרסאות

גרסה	תוכן	מספר פרוטוקול	זמן קלנדרי (בערך)
1.0	ליבה + כשפות אדמה (כולל Fabric ו-NeoForge)	`1""`	כ-3 שבועות
1.1	כשפות אוויר, דרכו של זהיר, ההשלכה הרוחנית, הגלשן	`2""`	כ-2.5 שבועות
1.2	כשפות מים, קרח ודם	`3""`	כ-3 שבועות
1.3	כשפות אש וברק	`4""`	כ-3 שבועות

גרסה 1.0 — ליבה + כשפות אדמה

מה יש בה: המאגר, שכבת הפלטפורמה לשני המסגרות (NeoForge ו-Fabric) וקובץ JAR אוניברסלי אחד, מודל הנתונים של הכשף, ההתקדמות, המקשים וסידור הקרב, מסך בחירת האלמנט, מסך הכשפות, ה-HUD, ערכות האנימציה והאפקטים, בלוקים זמניים וזכויות, צינור הצלילים, **כל יכולות האדמה** עם התנועות, האפקטים, הצלילים והמראות (טוף כברירת מחדל), בדיקות ובדיקה במשחק אמיתי. מספר הפרוטוקול הוא `1""`.

S0-1.0 — השלד (מנהל התפרים; 3–4 ימי סוכן; ענף `v10-skel`)

1. **מאגר.** `bending` עם התיקיות `common/`, `neoforge/`, `fabric/`, `processor/`, `tools/` ו-`docs/`. המסמכים של התוכנית נמצאים תחת `docs/plan/`. `docs/DESIGN.md` הוא התכנון החי. דפי הוויקי ב-`docs/wiki/` בעברית ובאנגלית. יש גם קבצי CI ו-`.gitattributes`.
2. **העתקת הפלטפורמה.** כל המחלקות של המסמך הטכני (`TECH.md`, סעיף 2.1 מועתקות מ-Enhanced עם שם החבילה ומזהה המוד שונים. מעבד ההערות ((processor, צינור ההגדרות, מתאם מבחני המשחק וה-`portLint` (עם כלל חדש ל-Fabric מועתקים גם הם. נקודות הכניסה לשני המסגרות הן שלדים שמתקמפלים, ובהן פונקציות שזורקות שגיאה עם הערה "נתיב L0-FABRIC ימלא את זה".
3. **מודל הנתונים.** `Element` ו-`Ability` (כל הכרטיסים של כל האלמנטים בכל הגרסאות, כדי שלא יסתדרו מחדש בהמשך), `BenderData` ו-`BenderSave` ו-`WorldSave` ו-`bending:victim`, וכל האחזקות (`Live`, `Toggle`) בתוך האזורים שלהם, בלי התנהגות.
4. **רישומים.** כל הפריטים, הגושים, ישויות הגוף, סוגי הנזק, האפקטים, אירועי הצליל ((232, סוגי החלקיקים, תגיות, רכיבי נתונים והישגים. ישויות ריקות בכל הגרסאות כדי שהסדר לא יזוז. `ModAttachments`, ו-`ModNetwork` עם **כל המסרים של 1.0**, כשהמטפלים קוראים לשלדי הנתיבים. מספר פרוטוקול `"1"`.
5. **הגדרות.** `BendingConfig` עם כל הסעיפים וכל האפשרויות של כל הארבעה אלמנטים (המספרים מ-`DESIGN.md` כברירות מחדל), ו-`BendingClientConfig`. לכל אפשרות הערה שנותנת את היחידה.
6. **שפה.** כל מפתחות 1.0 בשתי השפות: אלמנטים, יכולות עם וריאציות התכופפות, ה-HUD, המסכים, ההודעות, כתוביות לכל 232 הצלילים, פריטים, גושים, אפקטים, הודעות מוות ותרגומי ההגדרות. לכל נתיב יש אזור עם מפתח סימון (`"_": "" lane.earth"`) שמוסר בשחרור, וכלי `tools/add_lang.py` שמאחד בלוקים.
7. **שלדי שלבים.** `EarthClient/EarthServer`, `AirClient/AirServer`, `WaterClient/WaterServer` ו-`FireClient/FireServer`, כולם רושמים ולא עושים כלום. מודולי `ElementFx`, `TempBlocks`, `BendingRights`, `Hazards`, `Grip`, `AreaTargets`, `Raycasts`, `SourceFinder`, `MetalCharge` ו-`Victims` הם ממשקים עם מימושים ריקים במקומות שנתיב אחר ימלא אותם (התפרים). כל אחד נעול בבדיקה.
8. **העתקת הלקוח.** מערכת האנימציה, ערכת האפקטים, ערכת ממשק המשתמש, `PowerLoopSounds`, בסיס השכבות, `ClientSetup` עם אזור לכל נתיב, `KeyBindings`, `ClientEvents`, שלד ה-HUD, שלד מסך הכשפות עם ארבע לשוניות ריקות, ושלד מסך בחירת האלמנט. הקבועים של `GestureId`, `Stance` ו-`RunCharacter` כולם קיימים מהשלד, בתוך אזורים לכל נתיב.
9. **בדיקות.** `TestKit` עם העזרים החדשים, `SkeletonTests`, `DataTests`, `SoundAssetTests` (עם רשימת המקומות של (1.0, `NetworkTests`, `PlatformSeamTests`, `ConfigFileTests`, `AnimationMathTests`, `AnimationSignalTests` (אזורים ריקים לכל נתיב) ו-`tools/parity.py`. **השלד עובר ב-NeoForge** (בערך 120 בדיקות) ומתקמפל ב-Fabric.
10. **תיעוד.** `docs/LANES.md`: האזורים, התפרים, הכללים, הנתיבים של הגרסה וסדר המיזוג.

הנתיבים של 1.0 (במקביל מתוך `v10-skel`)

נתיב	הבעלות	ימי סוכן	תלוי ב-
L0-FABRIC	`platform/fabric/**`, בניית `fabric/`, `bending.fabric.mixins.json`, משימת `universal`, עבודת ה-CI של Fabric	6–8	כלום (הסיכון הגדול; מתחילים ראשון)

נתיב	הבעלות	ימי סוכן	תלוי ב-
L1-CORE	התנהגות הכשף (התקדמות, נקודות מקילים, רמות שליטה, פתיחה וקנייה, סידור הקרב, כוח האלמנט, טיימר השלילה, דגל זהיר), `AreaTargets`, `Raycasts`, `BendingRights`, `TempBlocks` (עם יומן ושחזור), `Grip`, `Hazards`, `Victims`, הפקודות, מטבע ה-`ElementScroll`, `compat/`	4–5	כלום
L2-EARTH	כל יכולות האדמה: השרתים, הישויות והגושים, מסרי `EarthFx`, הלקוח והתנועות, האפקטים, שכבות השריון, השכבות הנשלפות (`LayerReveal`)	7–9	ממשקי L1 שהשלד פרסם
L3-ANIM	הרחבות של מערכת האנימציה (מפתחות של רגל בציר z, קבוצות RIDE/HIT/AUTO, ארבע משבצות, מסכות ערוצים, בעלות על הגוף, העברת המצלמה), `VictimPoses` עם שלושה מיקסינים, העמדות המשותפות, סוגי הריצה, תנועות ההתעוררות, ו-`bending_anim/`	4–5	כלום
L4-FX	חלקיקי גרגר וטיפה, סוגי הרינדור של החומר (`MATTER_*`), `BendingWorldFx`, `MatterShapes`, `ChainSim`, רעידת מצלמה, עוצמת האפקטים, בסיס השכבות	3–4	כלום
L5-UI	מסך בחירת האלמנט, מסך הכשפות עם עץ הכישורים וסידור הקרב והמראות, ה-HUD המלא, אטלס הסמלים, הודעות שורת הפעולה, פריסה מימין לשמאל	5–6	ממשקי ההתקדמות של L1
L6-LOOKS	שש המראות של האדמה, שכבות המשותפות (טופנוט, כתר, שיער, זקן, גלימה, כתף, גב, ברדס, מסכה, פס זרוע), מסכות הכשף, סמלי הפנים, פריטי הגלשן והמים	4–5	כלום
L7-SOUND	ערכת הצלילים (`tools/sound/*`), צלילי המשותפים והאדמה (67 אירועים), ערכי הלולאה, נושא האדמה ומוזיקת הבחירה	4–5	כלום

נתיב	הבעלות	ימי סוכן	תלוי ב-
L8-DOCS	דפי הוויקי בעברית ובאנגלית לגרסה, README בשתי השפות, סעיף 1.0 ב-`DESIGN.md`	2–3	כל השאר (אחרון למיזוג)

סדר המיזוג (ענף `v10`): S0, ואז L0-FABRIC (כדי שכל מיזוג אחריו ירוץ בשני המסגרות), ואז L1-CORE, L3-ANIM, L4-FX, L2-EARTH, L6-LOOKS, L5-UI, L7-SOUND ו-L8-DOCS. אחר כך החיבורים: הפעלת מימושי L1 במקום השלדים של L2, קריאת שורות המצב של האדמה ב-HUD, המראות בבורר, חיבור אירועי הצליל לסוגי ה-FX, ובדיקת מיקומי הסמלים מול `Ability`. **זמן האיחוד:** 3–2 ימי סוכן.

השער: כל הבדיקות בשני המסגרות (כ-450 בדיקות), דוח השוויון ריק, רשימת הבדיקה במשחק אמיתי (`TECH.md`, סעיף 14.2) בשני המסגרות ובין המסגרות, עולם שנשמר באחד נפתח באחר ובו הכשף עדיין קיים, ארבעה אדמה-כשפים בקרב בתוך תקציב האפקטים, והתקנה יחד עם Enhanced בלי קריסה ועם שני פאנלים ב-HUD. גרסת המוד `1.0.0`, תגית `v1.0.0`. **זמן קלנדרי:** כ-3 שבועות.

מסלול חלופי: אם L0-FABRIC מתעכב, משחררים **1.0-neoforge** (ה-JAR של NeoForge לבד) עם כשפות האדמה. L0 נכנס ב-1.0.1 בלי לשנות את המשחק.

הנחיות הנתיבים של 1.0

כל הנחיה נכתבת באנגלית (בדיוק כפי שהיא ב-`PLAN.md`) ומתחילה בנוסח המשותף: שם הנתיב, ענף העבודה, מה הוא בעלים עליו, הכללים לקבצים משותפים, כללי הבדיקות, כללי הבדיקה במשחק והדרישה לקובץ `md.;lane>;docs/lanes/<` בסוף. בפועל:

- **L0-FABRIC:** לממש את חצי הפלטפורמה של Fabric (`TECH.md` 2.3) ואת ה-JAR האוניברסלי (2.4): `FabricPlatform`, הוקים, גישה לפריטים, רשת עם לחיצת יד בשלב התצורה, תכונות, רישומים, שכבות לקוח, שירותי לקוח, גשר אירועים (קריאות חוזרות של Fabric, ומיקסינים כשאין מקבילה), ומתאם מבחני משחק. הוכחה: השלד עובר ב-Fabric באותן תוצאות כמו ב-NeoForge, ולקוח של מסגרת אחת מצטרף לשרת של השנייה.
- **L1-CORE:** בניית הליבה של הכשף (`DESIGN.md` 1 ו-`TECH.md`: 5–8): שמירה, נקודות כישור על הרג, רמות שליטה, פתיחה וקנייה עם תנאים, כוח אלמנט, סידור הקרב והגדרות קבועות מראש, ניתוב המקשים עם וריאציית ההתכופפות, קירור, עלויות רעב, מטבע הבחירה, הכללים לבלוקים (`BendingRights`), בלוקים זמניים עם יומן ושחזור אחרי קריסה, ו-`Victims`/`Grip`. הבדיקות כוללות בדיקת שמירה שמעבירה את אותם בתים בין שני המסגרות.
- **L2-EARTH:** כל יכולות האדמה לפי `DESIGN.md` 2 ו-`ANIMATION.md`: 3 השרתים, הישויות, הבלוקים, הלקוח, האפקטים, השכבות. משתמשים בממשקי L1 דרך השלד (עד המיזוג, בודקים בעזרת זיוף מקומי בתוך אזור הבדיקות).
- **L3-ANIM:** הרחבות המנוע של האנימציה (`ANIMATION.md` 14.4–14.1 ו-12–11): מפתחות רגליים בציר z, קבוצות חדשות, מסכות ערוצים, בעלות על הגוף, תיקון המצלמה, `VictimPoses` עם המיקסינים שלו, העמדות המשותפות, סוגי הריצה, תנועות ההתעוררות, ושקופיות של כל תנוחות `ANIMATION.md` 1.2 במשחק אמיתי.
- **L4-FX:** תשתית האפקטים (`ANIMATION.md` 0.4, 13, 14.5, 14.12): חלקיקי גרגר וטיפה, סוגי הרינדור של החומר, `BendingWorldFx` עם ארבע קבוצות ציור, `ChainSim`, רעידת המצלמה, עוצמת האפקטים (כפל של הגדרת וניל) ושלדי `LightningFx` ו-`FlameBeamDrawer`.
- **L5-UI:** כל המסכים וה-HUD לפי `UI.md`: בחירת האלמנט, מסך הכשפות עם הלשוניות, ה-HUD עם כל שורות המצב, אטלס הסמלים עם כל הסמלים, הודעות שורת הפעולה והפריסה מימין לשמאל. בדיקות לכל מסך בשתי השפות.
- **L6-LOOKS:** שש המראות של האדמה לפי `LOOKS.md` 1–0, כל סוגי השכבות המשותפות (לכל הגרסאות), רישום `Looks`, החלת העור, סמלי הפנים ופריטי ההנחיה. בדיקות: קובץ PNG בגודל 64×64 שעובר את `SkinCheck`, כל מרקם קיים, שני שמות בשתי השפות.
- **L7-SOUND:** צינור הצלילים (`AUDIO.md`: 1 כלי `tools/sound/`) מועלה לגיט, רינדור הצלילים של האדמה והמשותפים (67 אירועים), נושאי המוזיקה כמקטעים, והחלפת הצלילים הזמניים. בדיקות לפורמט, לאורך, ולכתוביות.

- **L8-DOCS:** דפי ויקי ל-1.0 בשתי השפות, README, סעיף 1.0 ב-`DESIGN.md`, וקובץ `docs/DECISIONS.md` עם ההחלטות של אילון. נמגז אחרון, ולפני המיזוג קורא את כל `docs/lanes/*.md`.

גרסה 1.1 — כשפות אוויר

מה יש בה: כל יכולות האוויר (`DESIGN.md` (3, הגלשן כפריט, ההשלכה הרוחנית (ישות הרוח והמצלמה), קטנוע האוויר, דרכו של זהיר, שש המראות של האוויר (ג'יי וקארתיק כטיוטות עד שיגיעו תמונות), צלילי האוויר והנושא שלו, ותיעוד בוויקי. מספר פרוטוקול `""2`.

S0-1.1 (2–1 ימי סוכן): רישום סוגי ה-FX של האוויר ((199–100, מטפלי מסרי הרוח, הגלשן וצבע הצבע, מחלקות הישויות (`spirit`, `air_scooter`, `tornado`, `air_blade`) שהופכות ממבנים למחלקות ממשיות, נקודת מיקום `SpiritCamera`, נקודת `rideOverride` לקטנוע ולגלשן, מפתחות שפה של 1.1, ובדיקה שלשונית המים נשארת מוסתרת.

נתיב	הבעלות	ימי סוכן
L1-AIR-A	כלום מההתנהגות של האוויר שהופך לשרת: מכת האוויר, מגן האוויר, להבי האוויר, הטורנדו וריצת הטורנדו; הישויות `AirBladeEntity` ו-`TornadoEntity`; לקוח האוויר לתנועות, העמדות ותצוגת היד; האפקטים של `AirFx`; מגן האוויר על צינור ה-`ShieldStyle`	5–6
L2-AIR-B	קטנוע האוויר (כלי רכב, עמדת רכיבה, מעבד), בועת האוויר, הגלשן (פריט, פיזיקת `FALL_FLYING`, צבע, שכבה, פרופיל טיסה), ההשלכה הרוחנית (`SpiritEntity`, `SpiritCamera`, הזזה מאומתת בשרת, עמידת המדיטציה, שכבת העיניים, זוהר דרך קירות), ודרכו של זהיר (שער מוות החיה, פרופיל טיסה אמיתית, חניקה, ואקום, גורמי הדחיפה והנזק, והרגע של ניתוק הקשרים)	7–8
L3-AIR-LOOKS	שש המראות של האוויר (`LOOKS.md` (2, שכבת העיניים הסגורות, מרקמי `air_lattice` ו-`air_swirl`, אמנות הגלשן לשישה-עשר צבעים, סימון "טיוטה"	3–4
L4-AIR-SOUND	50 אירועי הצליל של האוויר (`AUDIO.md` (4, נושא האוויר, הלולאות	3–4
L5-AIR-DOCS	דפי ויקי לאוויר בשתי השפות, שורות ב-README, סעיף 1.1 ב-`DESIGN.md`	1–2

סדר מיזוג: S0, L1-AIR-A, L2-AIR-B, L3, L4, .L5 חיבורים: אינטראקציה בין מגן הפיצוץ לטורנדו, `rideOverride` של הקטנוע, והגלשן מול השכבות של הגלימות. **שער:** כל הבדיקות בשני המסגרות (כ-(650, בדיקה במשחק אמיתי, קרב של ארבעה כשפים עם שני אווירניים (תקציבי הטורנדו והקטנוע), והתאמת שורות `MatchupTests` של אוויר מול אדמה. **זמן:** כ-2.5 שבועות.

גרסה 1.2 — כשפות מים, קרח ודם

מה יש בה: מערכת מקורות המים ושק המים, כל יכולות המים, הקרח והדם, לשונית המים ומד המים, מחוון הירח המלא וצלצולי הירח, שלוש המראות של המים (קטארה כברירת מחדל), צלילי המים והנושא, שימוש ב-`ChainSim` לחבלים, ותנוחות הבובה. מספר פרוטוקול `""3`.

S0-1.2 (1–2 ימי סוכן): רישום סוגי ה-FX של המים ((299–200, ישויות המים, ממשק `SourceFinder` שימולא, לשונית המים שנחשפת לאלמנט, עזרי ירח ב-`McApi`, ומספר פרוטוקול `""3`.

נתיב	הבעלות	ימי סוכן
L1-WATER-A	מקורות המים (פתוחים, גשם, שק המים, צמחים עם ייבוש), שק המים כפריט, יריית המים והשוט, הריפוי והניקוי, זרועות המים עם אחיזה, טיפוס והנדנדה, מעבר השלב של מקור המים, מכפיל הירח, לשונית המים ומד המים, מחוון הירח, `WaterArmsLayer`	6–7
L2-WATER-B	תמנון המים (שכבה, שמונה שרשראות, סטירות, מהלומות, שאיבה ואחיזה), עמוד המים (כלי רכב, רכיבה, מהלומה, הקפאה בגובה 8, גל הרכיבה), הקרח (סכינים, חודים, הקפאה לדיסקה, לרגליים ולמבנים, שלג, קיר, כיפה)	7–8
L3-BLOOD	כשפות הדם (שערי הירח והכוח, האחיזה שמזיזה את הקורבן לפי מבט המטיל, ההטלה, שני יעדים ברמה (10, ושלילת הכשפות (המגע, נוסחת המשך, מצב השלילה שנשמר בין התנתקויות, אפשרות ריפוי)	4–5
L4-WATER-LOOKS	ארבע המראות של המים (`LOOKS.md` (3, כולל מינג-הואה בלי זרועות ואמון עם מסכה, שכבת שק המים, מרקמי קצף וקרח	3
L5-WATER-SOUND	60 אירועי הצליל של המים (`AUDIO.md` (5, נושא המים וצליל הירח	3–4
L6-WATER-DOCS	דפי ויקי למים, לקרח ולדם בשתי השפות, סעיף 1.2	1–2

סדר מיזוג: S0, L1-WATER-A, L2-WATER-B, L3-BLOOD, L4, L5, L6. **שער:** כל הבדיקות (כ-(900, בדיקה במשחק אמיתי כולל לילה של ירח מלא ובנת'ר (שם אין ירח), וקרב של ארבעה כשפים עם שני מים (תקציבי התמנון והעמוד).

גרסה 1.3 — כשפות אש וברק

מה יש בה: כל יכולות האש, יצירת הברק והטיית הברק, המעברים הפסיביים של האש, האש הכחולה של אזולה, חמש המראות של האש (זוקו כברירת מחדל), צלילי האש והנושא, קורן הלהבה ותצוגת הברק בשימוש מלא, תנוחת השיתוק, וכל ההתאמות החיות. מספר פרוטוקול `""4`.

S0-1.3 (1–2 ימי סוכן): רישום סוגי ה-FX של האש ((399–300, ישות כדור האש כמחלקה, סגנונות הקרן `FLAME` ו-`BLUE_FLAME`, נקודת שיתוק הלקוח, `HandTracker.belly` (`ANIMATION.md` (14.8, ומספר פרוטוקול `""4`.

נתיב	הבעלות	ימי סוכן
L1-FIRE-A	כדור האש (קטן וגדול), אגרוף האש (בעיטה, שילוב, זינוק), שוט האש (הצלפה, לולאה, חיתוך סלעים ולהבים), מעגל האש (רגליים, סילון הטיסה), המעברים הפסיביים, שריפות רק במקומות המותרים, והמיפוי הצבעוני של אזולה	6–7
L2-FIRE-B	להביור ונשיפת האש (שתי קרניים, מצייר הלהבה, נעילת הנשיפה, שכבת החום), ריפוי באש וכיבוי, יצירת הברק (החזקה, שחרור, כפול ברמה 10, קשת, `LightningFx`, רעם, שיתוק, נוכחות מוטות ברק, אין ברק וניל), הטיית הברק (העמדה, החלון, סיכוי לפי רמה עם זרע, מעבר, כשלון, הארקה, נתיב הגוף, שכבת הנתיב)	7–8
L3-FIRE-LOOKS	חמש המראות של האש (`LOOKS.md` 4), כתרי הלהבה, צבע צמר הלוטוס הלבן, מרקמי פה האש והחום	3
L4-FIRE-SOUND	55 אירועי הצליל של האש (`AUDIO.md` 6), וגרסאות הגובה לאש הכחולה, נושא האש וגזע כחול	3–4
L5-MATCHUPS	כל השורות של `DESIGN.md` 1.7 מופעלות ונבדקות בין ארבעת האלמנטים (`MatchupTests`, `CrossElementTests`, `CoexistenceTests` עם התסריט של ארבעה כשפים), ואיזון מספרים לפי משוב של אילון מ-1.2–1.0 (רק ברירות מחדל)	2–3
L6-FIRE-DOCS	דפי ויקי לאש ולברק בשתי השפות, סעיף 1.3, דף המתאמים	1–2

סדר מיזוג: S0, L1-FIRE-A, L2-FIRE-B, L3, L4, L5, L6. **שער:** כל הבדיקות (כ-1,200, בדיקה במשחק אמיתי עם האש הכחולה בכל אפקט, סערת רעמים, נת׳ר, קרב של ארבעה כשפים מכל האלמנטים (תקציב `ANIMATION.md` 13 נמדד), שני המסגרות וההעברה ביניהם, וגרסה `1.3.0`.

אחרי :1.3 גרסאות 1.4 ומעלה (הצעות, אם אילון יסכים)

כל הצעה מ-`DESIGN.md` 8 היא נתיב אחד בגרסה קטנה משלה:

- **1.4 מקדשים ואדונים:** מבנים, מאמני הכשפים, מגילות כשפות (שלושה נתיבים).
- **1.5 בוסים:** ארבעת בוסי הרוח על מסגרת הבוסים שהועתקה (ארבעה נתיבים ונתיב העתקה).
- **1.6 חברים:** ביזון השמיים, הלמור, כלב-הדוב הקוטבי וחמוס האש (שני נתיבים).
- **1.7 דו-קרב ובובות אימון.**
- **1.8 גרסאות מיינקראפט חדשות:** מתאם `r2` והמטריצה.
- **2.0 האווטאר.**

תפקידים

- **מנהל תפרים** (סוכן אחד לכל גרסה, כותב השלד ב-S0) שומר על `v1x-skel` ירוק, ממזג את ענפי הבקשות (`request(I)`) באותו יום, מצמיד תפרים בבדיקות, ועונה על שאלות "מי הבעלים של הקובץ" לפי `docs/LANES.md`.
- **מאחד** (סוכן אחד; יכול להיות מנהל התפרים אחרי S0) מאחד לפי הסדר, פותר התנגשויות ב"שמור את שני הצדדים" (הכול מוסיפים) ואז `build gradlew/.`, עושה את החיבורים, מריץ את בדיקות הנתיבים יחד ואחר כך את כל הבדיקות בשני המסגרות פעמיים, מנהל את רשימת הבדיקה במשחק, כותב את הפרק "איך נבנתה הגרסה", מעלה את גרסת המוד ויוצר את התגית.
- **נתיבים:** כמתואר. נתיב שצריך קוד של נתיב אחר לפני המיזוג, כותב נגד ממשק השלד, ואם צריך, ניגש למתודות של הנתיב האחר **בהשתקפות ((reflection עד המיזוג**. המאחד מחליף את ההשתקפות בקריאות ישירות בחיבורים.

סיכונים ואיך התוכנית מטפלת בהם

סיכון	הטיפול
חצי ה-Fabric וה-JAR האוניברסלי הם שטח חדש (אין תקדים ב-Enhanced)	L0-FABRIC מתחיל ראשון עם התקציב הגדול ביותר; 1.0-neoforge הוא מסלול חלופי; כל שער בגרסאות הבאות רץ בשני המסגרות
שמירה בין מסגרות	`SavedData` לכל מה שנשמר (`TECH.md` §2.4; `SaveTests` מוכיח שהבתים זהים; בדיקת המשחק האמיתי מעבירה עולם
מערכת הפלטפורמה של Enhanced משתנה (שלב P1 שלה)	Bending מעתיק מקומיט קבוע של Enhanced ((888609c, ומתעד אותו ב-`docs/LANES.md`; סנכרון מחדש הוא נתיב משלו
בלוקים זמניים ומגני שטח אצל שרתים	`BendingRights` + אירועי המסגרת + היומן + מגבלות; `RightsTests` ו-`TempBlockTests`; המגבלה של Fabric מתועדת
ביצועים עם ארבעה כשפים	מגבלות חלקיקים לכל שורת אפקט, נמדדות בכל שער (`ANIMATION.md` §13)
מראות של ג'יי וקארתיק בלי תמונות	טיוטות עם סימון; אילון שולח תמונות, וזה הופך לסופי במשימת אמנות של יום אחד
מספרי איזון לא מתאימים לכיף של ילד בן 11	כל מספר בהגדרות; נתיב L5-MATCHUPS ב-1.3 ואיזון אחרי כל גרסה לפי משחק של אילון
איכות האנימציה והאפקטים (הלב של הבקשה)	מפתחות ה-`ANIMATION.md` הם המספרים הצפויים בבדיקות; בדיקה במשחק אמיתי בכל נתיב; `bending_anim/` לסקירה פריים-אחר-פריים עם אילון